Tesseract Logic Calculator

Oct 10, 2026 · @Tiffany Taylor

Tiffany Taylor (Tifinchi), Boise State University. Calculations, literature search and drafting with Claude (Anthropic); section 2 states who originated each piece.

Abstract

The Tesseract Logic Calculator is the author's system for computing with a cube of side 2. Its 27 points sit at distances √k from the centre, its eight corners carry delimiters that in the electrical frame name charge, current, flux and voltage, and every derivative, inverse or change of scale is a quarter or half turn of the tesseract that holds the cube. This paper uses it to compute the structure of reality layer by layer, from the vacuum to the cosmos, and scores every result against measurement on one yardstick, beside the standard theories' result for the same quantity. The calculator rebuilds four established structures exactly: the Standard Model's generation of matter as a 16-state register, the energy–momentum relation and interval of special relativity, the classical tests of gravity and the shells of the atom. Where the standard theories take inputs it gives numbers, among them the τ mass from the electron and muon (1,776.969 MeV, −0.4σ), the top and Higgs masses (172.57 ± 0.38 and 125.20 ± 0.92 GeV), neutrino mixing from a 15° tilt ($\sin^2\theta_{12} = 0.318$, −1.5σ), the matter share $\pi^2/32$ (−1.2σ) and the vacuum's energy scale (2.02 ± 0.20 meV, +1.1σ). All 22 committed numbers sit within 1.6σ of their primary measurements. Five tests fixed before they were run, applied to both theories, find the calculator exact as notation and its numbers indistinguishable from chance: alphabets built from random ingredients match 19 of 29 outside constants as often as its alphabet does (P = 0.79), and its own numbers carry 0.6 to 4.0 bits against chance, while the standard theories' eleven predictions of independent measurements carry 102 bits. The standard theories also carry six failures, among them the vacuum's energy and the Hubble constant. Under the author's Quad-Core rule, neither the claim that the calculator is how reality works nor the claim that the standard theories are complete stands today. Nine predictions are frozen with a Bayes-factor rule, and JUNO's $\sin^2\theta_{12}$ (about 2031) is the first to decide.

Keywords: tesseract; difference of squares; root families; Koide relation; tribimaximal mixing; vacuum energy; look-elsewhere effect; pre-registration; Bayes factor; theory comparison.

1. Introduction

The universe, as the Tesseract Logic Calculator computes it, is a reactive board, a capacitance and an inductance at every point. Every state of the board sits on one of the 27 points of a cube of side 2, and every change of state is a quarter or half turn through the four cells of the tesseract that holds the cube. From that structure and two rules, balance and the three-phase start, the calculator rebuilds four established structures exactly: the Standard Model's generation of matter, the two tables of special relativity, the classical tests of gravity and the shells of the atom. Where the standard theories take inputs, it gives numbers. This paper carries out that computation layer by layer, from the vacuum to the cosmos. It scores every result on one yardstick against the measurement, and scores the standard theories' result for the same quantity beside it.

1.1 The calculator

The Tesseract Logic Calculator has four layers, all the author's in conception:

1. **A geometry.** The side-2 cube's 27 points sit at distances $\sqrt{k}$ from the centre, k = 0 to 3, with a held fourth level at 2. The difference-of-squares tables, root families and the dimension converter come from the author's trigonometry (section 3).
2. **A cipher and its mechanics.** Eight delimiters name the corners. In the electrical frame they carry charge, current, flux and voltage. The cube never moves: a derivative is a quarter turn into the next cell of a tesseract, an inverse is a half turn into the opposite cell, and a change of scale is the second turn (section 4).
3. **A physical reading.** The vacuum is a reactive board of capacitance and inductance. A balanced state carries equal shares; a state in motion sits slightly past balance, held by a load of one coupling squared over $8\pi^2$ per loading phase. Three phases cannot start together (sections 5 and 7).
4. **Numbers.** The reading commits to values where standard theory has inputs (sections 9 to 13), each fixed before it was compared and each scored in section 15.

1.2 How the computation is kept honest

Matching measured constants with small integers, π and square roots can look like insight and carry none. Eddington's 1931 count of the universe's particles, Lenz's $m_{\square}/m_e \approx 6\pi^5$ (now about 3×10^5 standard deviations from 1,836.15267343(11); Mohr et al. 2024) and Wyler's 1971 formula for α (about 4,000 standard deviations from CODATA 2022) are the standard warnings (Jentschura and Nándori 2014). Koide's 1982 relation is the standard counter-example: it has grown more precise as the τ was measured, and it still has no accepted derivation.

The judges of such matches are statistical:

- **Odds.** Good (1990) scored formulas as Bayesian odds, weighing the digits they get right against the prior improbability of their ingredients.
- **Trial factors.** Particle physics pays a trial factor for searching in many places, the look-elsewhere effect (Gross and Vitells 2010).
- **Bayes factors.** Evidence against an alternative is set on a fixed scale (Kass and Raftery 1995).
- **Pre-registration.** Fixing the analysis before the data separates prediction from postdiction (Nosek et al. 2018).

This paper follows five rules that come from those judges and from the author:

1. **Definitions first.** Every number the calculator commits to was written down, with its formula and its comparison, before it was compared. The nine predictions of section 17 are deposited with a SHA-256 fingerprint.
2. **The behavior rule.** In the author's words, the calculator uses “the alignments that do not create down-wind mismatched observeables”: a number set can sit in range while what it produces in other alignments does not, so an alignment stays only if nothing downstream of it contradicts a measurement. Successes stay with their proof, and trials that failed are left out.
3. **One yardstick.** The calculator and the standard theories are scored the same way: the same pull, the same categories of standing, and the same tests for chance (sections 15 and 16).

4. **Paying for density.** Every match is weighed against how densely the calculator's numbers cover the region where it falls (section 16).
5. **Origination.** Every piece is credited to whoever originated it: the author's unpublished work is cited as hers, prior work to its authors, and each completion Claude added is marked (added) (sections 2 and 18).

1.3 The structure in one page

Table 1 reads the computation from the bottom up. Each row names a layer of reality, what the calculator computes there, what the standard theories give for the same thing, and the measurement.

Table 1. The structure of reality as the calculator computes it, beside the standard theories.

Layer	The calculator computes	The standard theories give	Measured; pull of the calculator	Section
the board	$Z_0 = 2\alpha \cdot h/e^2 =$ 376.730 Ω; a gap of $2m_ec^2 = 1.022$ MeV; virtual particles as the board's reactive response	the same impedance (Maxwell); the same gap (Dirac 1930)	exact	5
spacetime	the sum table as $E^2 = (pc)^2 + (mc^2)^2$, the difference table as $s^2 = (ct)^2 - x^2$, the 45° line as light	special relativity	exact	6
balance and three phases	the load $\alpha/2\pi =$ 0.00116141 per loading phase; the forces cross in pairs at 1.1×10^{13}, 1.7×10^{14} and 3.0×10^{16} GeV	QED's first term (Schwinger 1948); the Standard Model's running	same	7
matter	one generation as a 16-state register; every anomaly sum zero; $\sin^2\theta_W =$ 3/8 where the first two couplings meet	the same representations, as inputs; 3/8 as a unification input	exact	8

Layer	The calculator computes	The standard theories give	Measured; pull of the calculator	Section
mass	Q = ⅔ with phase 2/9; the τ at 1,776.969 MeV; neutrinos at 0.364, 8.66 and 50.13 meV; top 172.57 ± 0.38 GeV; Higgs 125.20 ± 0.92 GeV	free parameters	−0.4σ; −0.4σ; inside the cosmological cap; +0.1σ; −0.1σ	9
mixing	$\sin^2\theta_{13} = 0.0223$ and $\sin^2\theta_{12} = 0.318$ from a 15° tilt; the unitarity-triangle angle α = 90°	free parameters	−0.6σ; −1.5σ; −1.6σ (all channels) and +0.6σ (B → ρρ)	10
gravity	light bent 1.7512″; precession 6,604.8 and 39.1 mas/yr; at most $1 - 1/\sqrt{2} = 29.3\%$ of a black hole's mass returns	the same, from general relativity	within 1σ	11
the strong scale	quark spin ⅓, gluon spin ⅙ at 2 GeV, orbital ⅙; the proton's radius as 4 reduced Compton wavelengths	computed on the lattice	−0.1σ; −0.2σ; −0.4σ; −0.8σ	12
the cosmos	a vacuum energy scale of 2.02 ± 0.20 meV; a matter share of $\pi^2/32 = 0.3084$; w = −1	ΛCDM fits the share and assumes w = −1; quantum field theory's vacuum energy is 10^{120} too large	+1.1σ; −1.2σ; +0.3σ for constant w, with a drift preferred at 2.8 to 4.2σ	13

Layer	The calculator computes	The standard theories give	Measured; pull of the calculator	Section
atoms	hydrogen's lines as the inverse difference table; shells of $2n^2$; all 118 elements placed in the two tables	Bohr's levels and QED; the periodic table	exact	14

1.4 Conventions

- **Pulls.** A pull is (measured − computed)/σ, with uncertainties combined in quadrature; for an asymmetric error the side facing the computed value is used.
- **"Same".** The calculator's number equals the standard theories' number.
- **Standing.** Every result carries one of six: exact (an identity or theorem), same, new number, reading (a re-description that changes no number), prediction (a number not yet measured well enough to decide), or exposure (a commitment the data currently press on).
- **Citations and data.** Citations are author–year. Constants are CODATA 2022 and PDG 2026 unless stated, as of 10 October 2026.

2. Origination, sources and method

The concepts, methods, mathematics and physical readings in this paper are the author's. Where the same result had been published before, the earlier work is cited at the point of use and in section 18. The derivation of the turns from her constraints, the computations, the literature search, the statistics and the drafting were done with Claude, Anthropic's AI model, as the tool, and every completion Claude added to her pages is marked (added).

2.1 Who originated what

The author. She devised and hand-compiled:

- the delimiter cipher and the cube it writes (Taylor, *ascii geometry*; *Tifinchi's Cypher Notes*);
- Tifinchi Trig, the difference-of-squares tables and root families (Taylor 2025; Taylor 2026e);
- the dimension converter, the cube cut into identical pyramids, the Balance Chart and the phase circle (*Tifinchi's Cypher Notes*);
- the element tables and the Rosetta fold (*Root Elements*; *Calc Elements*; Taylor 2026f);
- the Fibonacci options (Taylor 2026g) and the axis notebook (Taylor 2026b);
- the Quad-Core claim test and the reasoning framework around it (Taylor 2026a).

She also originated:

- **the physical tenets.**

    - The vacuum is a reactive board of capacitance and inductance, her electrical reading drawn on the emergent vacuum of Bass's papers (Bass 2015 to 2026). Bass wrote the dynamics of the vacuum's subcomponents; the circuit, inductance included, is hers.
    - Balance sits slightly past 45°, the way a balanced rock needs a little more than 45° to tip.
    - Three phases cannot start together.
    - The electron is a reaction of the board, and virtual particles are its reactive response, not particles.
    - Inductance comes from the $\sqrt{2} \rightarrow \sqrt{4}$ swing into the perpendicular dimension.
    - Gravity is a pull from what is held, with no dark-matter particle.
- **the mechanics.** The cube never moves; derivatives, inverses and changes of scale are turns of a tesseract; information folds rather than being lost; a misfit is found from its neighbours.
- **the behavior rule.** Keep only alignments that create no downstream mismatch.
- **the direction of every check,** including the demand that the calculator and the standard theories be scored without bias toward either.

Her rules are quoted verbatim where they are used (sections 3 and 4).

Claude. Claude did the following, each labelled as Claude's where it appears:

- derived the quarter turn J uniquely from her electrical frame, and the second turn K;
- computed every number, and wrote and ran the scripts and the statistical tests;
- searched and opened the literature;
- drafted the text.

Marks. Each authored piece ends with its standing against the literature searched on 9 and 10 October 2026:

- **N**, *novel*: not found in the literature searched;
- **M**, *matches*: the same result is published, and the paper says where;
- **D**, *diverges*: the author's version departs from the published one, and the paper says how.

Many pieces are composite: the algebra matches a classical result, and its use or reading is new.

In the author's words: ‘As the “author,” I want to be forthcoming about the level of AI usage in this document, and to do so I must be explicit about my neurodivergency. I came up with the methods and concepts; I understand and hand-compiled the original cube cipher math and geometry. The subsequent calculations and resources were compiled by Claude. Using AI is my method of translating how a neurodivergent mind understands into a precise language the rest of the world can hopefully read.’

Prior dissemination. Parts of this work were posted online as drafts between November 2025 and October 2026. None has been peer reviewed. This paper restates the material in full, recomputed, and supersedes those drafts.

2.2 How the author reasons: structure before sequence

The author's two research papers on cognition (Taylor 2026c; Taylor 2026d), written for a first-year writing course at Boise State University, describe the reasoning style that produced the calculator. They explain why this paper keeps her structures as drawn and builds the sequential proofs around them.

Her definition. She defines multimodal reasoning as "a neural pathway built from overlapping relationships between space, time, scale, repetition, sequence, geometry, pattern, linearity, angularity, and process". It synthesises three existing uses of "multimodal", each of which she cites:

- in cognitive science, perception across several senses at once (Lumen Learning, *Waymaker Psychology*);
- in computing, models that integrate text, images and sound (IBM);
- in business writing, a weaving of disciplines (Feng, in *International Business Times*, 2025).

Her illustration. Three students are asked to show with their hands that 2 + 2 = 4:

- the first holds up four fingers, the value;
- the second holds up two and two, the parts;
- the third, her own answer, merges two pairs by moving fingers from one hand to the other, the operation.

Her argument. Instructions such as "show the steps" fit sequential reasoners and over-constrain structural ones. Three sources carry it, cited as she quotes them:

- **Bor, Billington and Baron-Cohen (2007).** The savant Daniel Tammet showed no advantage for structured over unstructured digit sequences, whereas controls did; for him "all digit sequences have internal structure linked to his synaesthesia."
- **Kargupta et al. (2025).** Language models "deploy behavioral strategies based on learned priors rather than adapting to problem-specific demands."
- **Crompton et al. (2021).** Autistic groups passed a story on as accurately as non-autistic groups, and mixed groups less accurately.

Two sources that support her thesis directly (added).

- **Anthropic (2025).** For 36 + 59, Claude runs parallel paths inside, an approximate magnitude and an exact last digit, yet when asked how it added it describes carrying the 1. That is a measured gap between a non-sequential process and a sequential explanation.
- **Liu et al. (2025).** Chain-of-thought prompting lowers performance on tasks where verbal deliberation also hurts humans.

What it means for the mathematics (added). The 2 + 2 demonstration separates a value from its derivation, and the same split runs through her pages: $\sqrt{8}$, $2\sqrt{2}$ and $\sqrt{(2^3)}$ are one value with three derivations (section 3.2). "Too literal" then means answering at the level of the derivation when the listener asked at the level of the value. That is a precise, testable description, not a deficit.

Standing. M: the uses of "multimodal" she synthesises. **N**: multimodal reasoning as a style to be diagnosed and benchmarked, and value against derivation as the account of "too literal". Neither is tested here.

2.3 The author's decision rule: the Quad-Core test

Her reasoning framework decides whether a claim may stand: "Quad-Core evaluation runs four independent rejection checks in parallel. Stabilization is permitted only if ALL checks return NOT-REJECTED" (Taylor 2026a).

1. **Outcome.** Compare prediction with observation; on a mismatch, reject.
2. **Constraint.** Test the claim under changes of scale, time, cost, causality and context; “if claim breaks, REJECT.”
3. **Alternative.** Compare competing explanations; “if an alternative yields fewer violations under other cores, REJECT.”
4. **Consequence.** Project what follows if the claim is wrong; “if error is deferred rather than eliminated, REJECT or UNDETERMINED.”

Each core returns REJECT, NOT_REJECT or UNDETERMINED. The synthesis is REJECTED if any core rejects, UNSTABLE if none rejects but one is undetermined, and STABILIZED if all four do not reject. “No core overrides another. No majority rule.” Her definition of proof follows: “Proof is the closure of all active rejection paths under quad-core evaluation.”

Stopping rules (added). An open domain never runs out of alternatives or consequences, so the cores need a finite list and a horizon. This paper fixes both:

- **the alternatives:** the standard theories and alphabets built from random ingredients;
- **the horizon:** the data as of 10 October 2026.

Section 16 applies the rule to the calculator and, on the same terms, to the standard theories.

The framework around it. *Tifinchi's Diagnostic and Reasoning Framework* is a 7,817-line plain-text specification meant to run in any reader or model without prior conversation. Its main parts:

- eight fixed passes, each leaving a record that cannot be altered;
- the Quad-Core veto;
- curiosity, run forwards from a gap and backwards from an outcome that cannot be denied;
- transfer between domains;
- mandatory revision by “prove-don't-prune”: add proof, never delete structure;
- reduction allowed only when the short form can regenerate the full process;
- an ethics pass ordered by irreversibility, consent, asymmetry of power and truth.

Standing. M: the Outcome core is falsification (Popper 1959); the Alternative core is the method of multiple working hypotheses (Chamberlin 1890) and strong inference (Platt 1964). **N**: four parallel vetoes with an explicit unstable state and no majority, as one procedure, were not found in this form.

2.4 Data

Every number is computed from the following, as of 10 October 2026:

- **constants:** CODATA 2022, with the exact SI values of h, e and c (Mohr et al. 2024);
- **particle data:** the Particle Data Group's 2026 tables (PDG 2026);
- **neutrino mixing:** NuFIT 6.0 and 6.1 (Esteban et al. 2026) and JUNO's first result (JUNO Collaboration 2025);
- **cosmology:** Planck 2018 (Planck Collaboration 2020), DESI DR2 (DESI Collaboration 2025), ACT DR6 (ACT Collaboration 2025) and the supernova compilations cited where used;
- **the proton:** the continuum-limit lattice results of Alexandrou et al. (2026) and Zhao et al. (2025);
- **the Standard Model's running:** two-loop couplings and three-loop QCD from the matching of Buttazzo et al. (2013), updated to PDG 2026.

2.5 One yardstick for every theory

The calculator and the standard theories are scored by the same rules.

1. **The pull.** For any computed value, the pull is (measured − computed)/σ, with the uncertainties of both combined in quadrature.
2. **What counts as a prediction.** A theory predicts a quantity when its value follows from the theory and measured inputs other than that quantity. When the quantity is itself an input, a free parameter, the theory scores *input* and earns no pull. The Standard Model's 19 parameters (26 with neutrino masses), and ΛCDM's densities, are inputs in this sense.
3. **Failures count the same.** A standard prediction that misses is scored like a missed number of the calculator. Examples are the vacuum energy, and the Higgs mass a vacuum sitting exactly on its edge would have.
4. **Standing.** Every result carries one of the six standings of section 1.4.
5. **Chance.** The same tests for chance, calibration on an outside list, the precise constants and the look-elsewhere effect, are applied to both (section 16).
6. **Downstream.** An alignment stays only if nothing downstream of it contradicts a measurement: the author's behavior rule, applied to every theory alike.

3. The author's mathematics

The calculator is built from the author's own mathematics, written between November 2025 and October 2026 in her notes, notebooks and papers (Taylor 2025; Taylor 2026b; Taylor 2026e; Taylor 2026f; Taylor 2026g; *ascii geometry*; *Tifinchi's Cypher Notes*; *Root Elements*; *Calc Elements*). It is stated here in full, so that nothing in the computation that follows depends on an unpublished page. Her words are quoted where they carry a rule. Where she stated a result without a proof, or left a chart unfinished, the proof or the completion is given and marked (added); everything so marked is Claude's. The scripts appA.py (SHA-256 e3623e8a8e92c85fe252461d55b46b9d5bdc86f5535da09184c07952808638ae) and ledger118.py (SHA-256 7d19eddd7647a8e2c316d2a40c9ac81de47d922cec1bc50e9894b66ee5ed62bb) reproduce the completions and the element ledger of section 14.5. Each subsection ends with its standing against the literature, marked as in section 2.1.

3.1 One notation across the pages

Her pages reuse letters. This paper fixes them as follows.

Table 2. Symbols.

Symbol	In this paper	Other uses in her pages	How they relate
a, b	the long leg and the short leg	the same everywhere	—
c	the hypotenuse, $c^2 = a^2 + b^2$	—	—

Symbol	In this paper	Other uses in her pages	How they relate
d	the virtual leg, $d^2 = a^2 - b^2$	d = a/b − 1 in the cube engine of her reasoning framework	the engine's d is the area ratio N/M (section 3.11.2)
h	the leftover, a − b	written e on the Rosetta page	—
e	the sliver, $e^2 = b(a - b)$: twice the thin triangle of section 3.2	$e^2 = (½b)^2$ on the same 3-4-5 page; e = b/2 on the phase circle's first triangle; Euler's number in the cube engine	only the area construction locks the triangle (section 3.2); Euler's number is written *e*
f	the diagonal of the square on the long leg, a√2	the third physical edge on the converter page; f = 1 − b/a in the cube engine	the engine's f is N/S; the converter's third edge is written z
g	the diagonal of the square on the short leg, b√2	—	—
M, N, S	the triangle ½ab, the wedge ½a·h, the half-square M + N	m, n and o are the face diagonals √8 on the converter page	M, N and S are kept for areas
K	the inscribed radius, a/2	—	—
P_0 to P_3	the converter's stages	r and L in *Small Pattern, Big Deal* (Taylor 2025)	the same quantities
’	one corner delimiter	written ‘ at the start of the Bottom string	the same corner; a parser that kept both would count nine corners
*	the fourth corner of the Top face	÷ on the square page	the same corner

3.2 The difference of squares as a measure of imbalance

Her starting observation: “Instead of adding the squares of legs to find the hypotenuse $\sqrt{(a^2+b^2)}$, I looked at the difference between them $\sqrt{(a^2-b^2)}$ … The Pythagorean theorem asks for the size of the space a triangle occupies. The difference-of-squares asks for the size of the imbalance between the legs.”

For a right triangle with legs $a \geq b$ she writes four quantities:

- **c**, the hypotenuse: $c^2 = a^2 + b^2$.
- **d**, the virtual leg: $d^2 = a^2 - b^2 = (a - b)(a + b)$. It is a leg of the "virtual triangle" whose hypotenuse is a and whose other leg is b.
- **h**, the leftover: $h = a - b$, what is left of the long side after the short side is taken off.
- **M**, the triangle's area: $M = \frac{1}{2}ab$.

She draws each square as a grid of dots and reads its identities off the drawings:

1. **The gnomon.** Take a $b \times b$ square out of an $a \times a$ square; the L-shaped remainder slides into an $h \times (a + b)$ rectangle, so $d^2 = h(a + b)$.
2. **The tilted square.** Four copies of the triangle inside a square of side $a + b$ leave a tilted inner square of area c^2, so $(a + b)^2 = c^2 + 4M$.
3. **Sum and difference.** $c^2 - d^2 = 2b^2$ and $c^2 + d^2 = 2a^2$, so $c^2 = 2a^2 - d^2$. Halved, $\frac{1}{2}c^2 - \frac{1}{2}d^2 = b^2$, a difference of two half-squares, which are the areas of 45° triangles. Doubled, $2c^2 - 2d^2 = (2b)^2$.
4. **The ladder.** d^2, a^2 and c^2 are always b^2 apart: three squares in equal steps.
5. **Path against chord.** At 45° the diagonal of the square on a is $f = a\sqrt{2} = (\sin 45° + \cos 45°)a$. In general $\sin \theta + \cos \theta = (a + b)/c$ is the ratio of the two-leg path to the straight path, and it equals $\sqrt{2}$ only at 45°.

Worked example: the 3-4-5 triangle.

- **The squares.** With $a = 4$, $b = 3$, $c = 5$ the dot squares hold 16, 9 and 25.
- **The virtual leg.** $d^2 = 16 - 9 = 7 = (4 - 3)(4 + 3)$, drawn as an L of seven dots straightened into a strip.
- **The identities.** $c^2 - d^2 = 18 = 2 \cdot 9$, and $c^2 = 2 \cdot 16 - 7 = 25$.
- **The half-areas.** $a^2/2 = 8$, $b^2/2 = 9/2$, $c^2/2 = 25/2$ and $d^2/2 = 7/2$; $\sin A = 4/5$ and $\sin B = 3/5$.

The lock. In the square of side a cut by its 45° diagonal, she splits the half above the diagonal, of area $a^2/2$, into three pieces:

- the corner triangle $b^2/2$, with legs b on the two outer sides;
- a thin triangle $e^2/2$ between the hypotenuse c and that corner triangle;
- the remainder, $d^2/2 - e^2/2$, between the hypotenuse and the diagonal.

For 3-4-5 her "e breakdown" cuts the thin triangle, whose sides are $a - b = 1$, $b\sqrt{2} = 3\sqrt{2}$ and $c = 5$, along its altitude 3/5 into two right triangles:

- their areas are $\frac{1}{2}(4/5)(3/5) + \frac{1}{2}(3/5)(21/5) = 12/50 + 63/50 = 3/2$, so $e^2 = 3$;
- the remainder is then $7/2 - 3/2 = 2$.

Her rules: "d&e give the original container information and lock a triangle's angles into position", and "If $(d^2/2 - e^2/2)$ is present, the angles of the inner triangle can't be configured any other way. Scaling info is retained."

Proof (added). Place the square with corners (0, 0) and (a, a), the corner triangle at $(a - b, a)$, (a, a), $(a, a - b)$, and the hypotenuse from (0, a) to $(a, a - b)$. The thin triangle then has area $\frac{1}{2}b(a - b)$, so $e^2 = b(a - b)$, $d^2 - e^2 = a(a - b)$, and $d^2 - 2e^2 = (a - b)^2$. Hence

```
a-b=\sqrt{d^{2}-2e^{2}},\qquad b=\frac{e^{2}}{\sqrt{d^{2}-2e^{2}}},\qquad
a=\frac{d^{2}-e^{2}}{\sqrt{d^{2}-2e^{2}}},\qquad
\frac{a}{b}=\frac{d^{2}-e^{2}}{e^{2}}
```

The pair (d^2, e^2) fixes the ratio a/b, which sets the angles, and also the size, which is the container: for (7, 3) it returns a − b = 1, b = 3 and a = 4. Her lock holds, with a proof.

Two notes on her letters. The same page also writes $e^2 = (½b)^2$, which gives 9/4 rather than 3 for b = 3; the area construction gives 3 and is the one used here. On another page she writes e for the leftover a − b; this paper writes h for that.

The 4 × 4 square on a 3 × 3 grid.

- **The grid.** Since 4 = 3 × 4/3, a 4 × 4 square is a 3 × 3 grid of cells of side 4/3. Each cell has area $16/9 = (1 + ⅓)^2 = 1 + 2 \cdot ⅓ + ⅓^2$, so "length & area embed the original container (3×3)."
- **The box rule.** In the 3 × 3 box, $n^2 = 2n + (n^2 − 2n)$ with $n^2 − 2n = ⅓$ of the area, true only at n = 3.

Keep the factors. Her principle, from two pages: "simplifying too soon loses info" and "simplifying root calculations also loses some information."

- **Fractions.** $(5/3)^2 = 9/9 + 12/9 + 4/9$ keeps the parts 1 and ⅔ visible.
- **Roots.** A triangle with a = b = 2 has c = √8, which "can be 2√2 or $\sqrt{2^3}$". The second form "can now be used to scale the entire root family", and $\sqrt{(2^3)}$ is the volume of a cube of edge √2: "lengths and area combo configured as cube."

SSA and the inverse grid. "d&e create an SSA situation there all right triangles have 4 potential answers (dual, null, & 1 solution), and since all non-right triangles can be split into 2 right triangles SSA is always possible with the container info." Classical trigonometry gives two sides and a non-included angle no triangle, one, or two (the ambiguous case); her four potential answers are these outcomes with the dual pair counted twice. She also notes that "a √2 × √2 grid is inverse of 1 × 1 grid": the half-diagonal of a √2 square is 1, and that of a unit square is √2/2.

Standing.

- **M.** The gnomon is Euclid's *Elements* II.5. The tilted-square proof is the rearrangement drawn in Zhao Shuang's third-century commentary on the *Zhoubi Suanjing*. The equal-step ladder of squares is Fibonacci's congruum (*Liber Quadratorum*, 1225), and Fermat proved its step is never a square, so a, b, c and d are never all whole. The triangle area is Euclid I.41.
- **N.** Treating the difference of squares as a triangle's imbalance and tabulating it, the e construction and the (d, e) lock were not found in the literature searched.
- **Closest prior framing.** Wildberger's rational trigonometry (2005) also works in squared lengths and avoids square roots. Mansfield and Wildberger (2017) read the Old Babylonian tablet Plimpton 322 as "a completely different form of trigonometry that does not need the familiar modern concept of angles", built on exact ratios of sides.
- **D.** None in the mathematics.

3.3 The Tifinchi Trig Method: a triangle becomes motion

She built a teaching sequence that plots a sine wave from one triangle before any ratio is named: “Tifinchi Trig is NOT intended to replace the Unit circle. Its purpose is to help students plot motion and understand the concepts of the Unit circle prior to angle ratio conversions.”

Section 1: one 45° triangle onto a sine wave.

1. **Compare.** A triangle compares two pieces of information. Collapse the legs of the 45° triangle together and “neither has lost or gained anything”.
2. **Plot and normalise.** Plot the base triangle with angles 45°, 45° and 90° and legs a = b = 3, with the 45° vertex B at (0, 0), A = (3, 3) and C = (3, 0). Multiply by 3 to go to Tifinchi Trig and divide by 3 to go to the unit circle; in general divide by the hypotenuse. The hypotenuse is $3\sqrt{2}$, and scaling changes only the whole number while the root family stays.
3. **Rotate.** Turn the triangle 90° about the origin instead of flipping it between quadrants. The legs swap roles, which is “the 2D version of Manhattan steps”.
4. **Plot the wave.** The four 90° positions are enough to plot a full sine wave from 0° to 360°.

Section 2: precision before new angles.

1. **Half and double angles.** These come from rotation: moving the triangle back by half gives new points. Radians are written over 4 and then reduced, so the quarter step shows. Rotating in 45° steps gives eight positions per cycle, “which occurs in analyzing sound, compressing images, binary data”.
2. **The 30° and 60° triangles.** They keep their own sizes, the 30° smaller and the 60° larger, and are normalised to add points to the 45° wave.
3. **The unit circle.** Then it is taught: “the Tifinchi Trig method has been normalizing the Unit Circle values all along.”

Her reason for 3: “3 is the smallest integer that locks increments of 45 degrees into integers for the coordinates.” As stated this needs a qualification (added): any whole leg gives whole-number corners at the four 90° positions, and no whole leg gives them at the in-between 45° positions. What singles out 3 in her system is a sector: a 40° ($2\pi/9$) slice of the radius-3 circle has area $\frac{1}{2} \cdot 3^2 \cdot (2\pi/9) = \pi$, the area of the whole unit circle, which ties the two scales together. The other link is the one-third grid of section 3.2.

Standing.

- **M.** The sine as the shadow of a turning radius goes back to Euler's *Introductio* (1748); the radian to Cotes (1722).
- **N.** The sequence itself, rotation before ratio with the 30° and 60° triangles normalised onto the 45° wave, was not found as a published method. It shares rational trigonometry's choice to postpone angles (Wildberger 2005).

3.3.1 Why 3, a ladder of angles, and the gap between the paths (added)

Why 3 keeps appearing. Two reasons, both on her pages, back her choice of 3 (section 3.3).

- **The grid.** 3 is the only square grid whose diagonal and the cells on either side of it are equal in number: $n = n(n - 1)/2$ only at $n = 3$. This is her "3×3 box only".
- **The coordinates.** 3 is the smallest leg that makes the corners of the 30° and 60° triangles fraction-free: $(3, \sqrt{3})$ and $(3, 3\sqrt{3})$. Leg 1 gives $\sqrt{3}/3$. Her phase circle uses exactly $(3, \sqrt{3})$.

A ladder of difficulty. Angles whose coordinates stay in one root family compose without new roots:

1. Steps of 90° need only whole numbers.
2. 45° needs $\sqrt{2}$.
3. 30° and 60° need $\sqrt{3}$.
4. 15°, 75° and 105° mix $\sqrt{2}$ and $\sqrt{3}$: $\cos 75° = (\sqrt{6} - \sqrt{2})/4$.
5. 22.5° nests one root inside another: $\cos 22.5° = \sqrt{(2 + \sqrt{2})}/2$.

Her teaching sequence puts 22.5° before 30° and 60° (section 3.3), while the ladder makes 22.5° the harder step. Her pyramid page's $7\pi/12$, which is 105°, sits on the mixing step.

The gap between the paths is the container's surface.

- **The ratio.** The ratio of the two-leg path to the straight one is itself a sine wave: $(a + b)/c = \sin\theta + \cos\theta = \sqrt{2}\sin(\theta + 45°)$. It is 1 on the axes and $\sqrt{2}$ at 45°. Her "90° out of phase" describes the order of the moves, legs in sequence against one rotation, not this ratio, which is shifted by 45°.
- **In two dimensions.** The squared gap is $(a + b)^2 - c^2 = 2ab = 4M$.
- **In three dimensions.** It is $(a + b + c)^2 - (a^2 + b^2 + c^2) = 2(ab + bc + ca)$, the surface area of the box: 6 for the unit cube.

So the squared gap between the step path and the straight path measures the container's surface: a measured form of her "container information".

3.4 The two tables and root families

Her term: "Root Families, a term I coined to explain the pattern I noticed, uses the exact same formula $\sqrt{(a^2-b^2)}$ to ask 'If we calculate the difference of many possibilities, is there a pattern?'"

Definition. Write each cell $\sqrt{N}$ with N in prime powers, such as $\sqrt{20} = \sqrt{(2^2 \cdot 5)}$, rather than simplified. The part left after every squared factor is removed (here 5) names the family. "A 'Root Family' is a batch of triangles whose results share the same root, sometimes even the same answer from a different shape."

Three examples from her charts.

- **The $\sqrt{3}$ family** of the difference table: (2,1) $\sqrt{3}$, (4,2) $\sqrt{12}$, (6,3) $\sqrt{27}$, (7,1) $\sqrt{48}$, (8,4) $\sqrt{48}$ and (10,5) $\sqrt{75}$. The line $a = 2b$ is the 30-60-90 triangle $(b : b\sqrt{3} : 2b)$. The cell (7,1) gives the same $\sqrt{48}$ as (8,4) with a different shape $(1 : 4\sqrt{3} : 7)$: "not simplifying actually helps us see the full structure they originate from."
- **The zero family.** "The 45 degree triangle (where a=b so $\sqrt{(a^2-b^2)}=\sqrt{0}$) is the first real 'special' triangle. It belongs to the zero difference family, the pure symmetry family (no loss, no gain on either leg)."
- **Prime factor families.** (3,1) gives $\sqrt{8} = \sqrt{(2^3)}$; (5,3) gives $\sqrt{16} = \sqrt{(2^4)}$; (4,2) gives $\sqrt{12} = \sqrt{(2^2 \cdot 3)}$.

Her summary: "triples classify shape existence and root families classify shape behavior."

Tables 3 and 4 give her charts for legs 1 to 10, in her factored form. Where N is at most 118, the cell also carries the element with atomic number N, as on her *Root Elements* page.

Table 3. Sum table, $\sqrt{(a^2 + b^2)}$.

a \ b	1	2	3	4	5	6	7	8	9	10
1	$\sqrt{(2)}$ He									
2	$\sqrt{(5)}$ B	$\sqrt{(2^3)}$ O								
3	$\sqrt{(2\cdot5)}$ Ne	$\sqrt{(13)}$ Al	$\sqrt{(2\cdot3^2)}$ Ar							
4	$\sqrt{(17)}$ Cl	$\sqrt{(2^2\cdot5)}$ Ca	$\sqrt{(5^2)}$ Mn	$\sqrt{(2^5)}$ Ge						
5	$\sqrt{(2\cdot13)}$ Fe	$\sqrt{(29)}$ Cu	$\sqrt{(2\cdot17)}$ Se	$\sqrt{(41)}$ Nb	$\sqrt{(2\cdot5^2)}$ Sn					
6	$\sqrt{(37)}$ Rb	$\sqrt{(2^3\cdot5)}$ Zr	$\sqrt{(3^2\cdot5)}$ Rh	$\sqrt{(2^2\cdot13)}$ Te	$\sqrt{(61)}$ Pm	$\sqrt{(2^3\cdot3^2)}$ Hf				
7	$\sqrt{(2\cdot5^2)}$ Sn	$\sqrt{(53)}$ I	$\sqrt{(2\cdot29)}$ Ce	$\sqrt{(5\cdot13)}$ Tb	$\sqrt{(2\cdot37)}$ W	$\sqrt{(5\cdot17)}$ At	$\sqrt{(2\cdot7^2)}$ Cf			
8	$\sqrt{(5\cdot13)}$ Tb	$\sqrt{(2^2\cdot17)}$ Er	$\sqrt{(73)}$ Ta	$\sqrt{(2^4\cdot5)}$ Hg	$\sqrt{(89)}$ Ac	$\sqrt{(2^2\cdot5^2)}$ Fm	$\sqrt{(113)}$ Nh	$\sqrt{(2^7)}$		
9	$\sqrt{(2\cdot41)}$ Pb	$\sqrt{(5\cdot17)}$ At	$\sqrt{(2\cdot3^2\cdot5)}$ Th	$\sqrt{(97)}$ Bk	$\sqrt{(2\cdot53)}$ Sg	$\sqrt{(3^2\cdot13)}$ Ts	$\sqrt{(2\cdot5\cdot13)}$	$\sqrt{(5\cdot29)}$	$\sqrt{(2\cdot3^4)}$	
10	$\sqrt{(101)}$ Md	$\sqrt{(2^3\cdot13)}$ Rf	$\sqrt{(109)}$ Mt	$\sqrt{(2^2\cdot29)}$ Lv	$\sqrt{(5^3)}$	$\sqrt{(2^3\cdot17)}$	$\sqrt{(149)}$	$\sqrt{(2^2\cdot41)}$	$\sqrt{(181)}$	$\sqrt{(2^3\cdot5^2)}$

Table 4. Difference table, $\sqrt{(a^2 - b^2)}$.

a \ b	1	2	3	4	5	6	7	8	9	10
1	0									
2	$\sqrt{(3)}$ Li	0								
3	$\sqrt{(2^3)}$ O	$\sqrt{(5)}$ B	0							
4	$\sqrt{(3\cdot5)}$ P	$\sqrt{(2^2\cdot3)}$ Mg	$\sqrt{(7)}$ N	0						
5	$\sqrt{(2^3\cdot3)}$ Cr	$\sqrt{(3\cdot7)}$ Sc	$\sqrt{(2^4)}$ S	$\sqrt{(3^2)}$ F	0					

a \ b	1	2	3	4	5	6	7	8	9	10
6	√(5·7) Br	√(2^5) Ge	√(3^3) Co	√(2^2·5) Ca	√(11) Na	0				
7	√(2^4·3) Cd	√(3^2·5) Rh	√(2^3·5) Zr	√(3·11) As	√(2^3·3) Cr	√(13) Al	0			
8	√(3^2·7) Eu	√(2^2·3·5) Nd	√(5·11) Cs	√(2^4·3) Cd	√(3·13) Y	√(2^2·7) Ni	√(3·5) P	0		
9	√(2^4·5) Hg	√(7·11) Ir	√(2^3·3^2) Hf	√(5·13) Tb	√(2^3·7) Ba	√(3^2·5) Rh	√(2^5) Ge	√(17) Cl	0	
10	√(3^2·11) Es	√(2^5·3) Cm	√(7·13) Pa	√(2^2·3·7) Po	√(3·5^2) Re	√(2^6) Gd	√(3·17) Sb	√(2^2·3^2) Kr	√(19) K	0

What the tables hold. The author's chart of origins sets each fact beside its source, and every fact below was checked cell by cell.

1. **Which numbers enter the sum table.** N is a sum of two squares exactly when every prime of the form 4k + 3 divides it to an even power (Girard 1625; Fermat 1640; proved by Euler 1747–55). So 45 = 3^2·5, 90 and 117 appear, but no cell carries a single 3, 7 or 11.
2. **Repeats in the sum table.** 50 = (7,1) = (5,5), 65 = (8,1) = (7,4) and 85 = (9,2) = (7,6) are different triangles with one hypotenuse. A product of two sums of two squares is a sum of two squares in two ways (Diophantus; Brahmagupta; Fibonacci 1225).
3. **Which numbers enter the difference table.** $a^2 - b^2 = (a - b)(a + b)$ has two factors of equal parity, so it is never 2 more than a multiple of 4: 2, 6, 10 and 14 never appear. Its repeats (15, 24, 32, 45, 48) come from different factor pairs, as 48 = (7,1) = (8,4).
4. **Whole-number cells are Pythagorean triples.** In the sum table, 25 at (4,3) and 100 at (8,6); in the difference table, 9 at (5,4), 16 at (5,3), 36 at (10,8) and 64 at (10,6). The difference table finds the same triangles from the hypotenuse side: (5,4) → 3 is the 3-4-5 that (4,3) → 5 finds (Plimpton 322; Euclid X.29, Lemma 1).
5. **Numbers in neither table** (6, 14, 22, 30, …) need three squares. N is a sum of three squares unless $N = 4^a(8b + 7)$ (Legendre 1798; Gauss 1801). Every number that needs four squares is odd or a multiple of 4, so the difference table reaches it. Her reading: the gap between the tables is a "3D tier", reached by the dimension converter of section 3.7, and a 4D tier is never needed.
6. **Lines from the corner.** Cells with the same ratio a : b share one angle (Euclid VI.4). The line a = b is the 45° line: the √2 family in the sum table, the zero family in the difference table. The line a = 2b is the 26.57° √5 line of the sum table and the 30-60-90 √3 line of the difference table. Her chart adds: "Root families do NOT share angles". A family groups the irrational part of a length, not a shape.
7. **One complex square** (added). $(a + bi)^2 = (a^2 - b^2) + 2ab \cdot i$, with modulus $a^2 + b^2$. The difference table, a third table of 2ab, and the sum table are one squaring, and squaring doubles the angle: $(2 + i)^2 = 3 + 4i$, the 3-4-5. These are the three quadratic forms of Wildberger's chromogeometry (2008).

Power and signals. Her chart reads the tables as a power calculator.

- **The sum table** gives a total from two parts: apparent power from real and reactive power, $S = \sqrt{(P^2 + Q^2)}$, and impedance, $|Z| = \sqrt{(R^2 + X^2)}$.
- **The difference table** gives a missing part from the total and one part: $Q = \sqrt{(S^2 - P^2)}$, $X = \sqrt{(Z^2 - R^2)}$, and the damped frequency $\omega_d = \sqrt{(\omega_0^2 - \gamma^2)}$, so $\omega_0 = 5$ and $\gamma = 3$ give 4.
- **Clean readings** are the whole-number cells. P = 8 and Q = 6 give S = 10 and a power factor of 0.8, the standard generator rating; P = 3 and Q = 4 give a power factor of 0.6.
- **Each line from the corner** has one power factor: 0.707 at a = b and 0.894 at a = 2b in the sum table.

The sources are Heaviside (1886, the word impedance), Kennelly (1893, impedance as a right triangle) and Steinmetz (1893, complex numbers for alternating current).

Fibonacci in the tables.

- **Option 1.** Run the recurrence $F_n = F_{n-1} + F_{n-2}$ in either direction: “This formula can be reversed to expand outward instead of winding inward.” Run backwards, it passes 0 and alternates sign.
- **Option 3.** Read the sequence off the two tables:
  - $1^2 - 0^2 = 1$, $1^2 + 1^2 = 2$, $2^2 - 1^2 = 3$, $2^2 + 1^2 = 5$, $3^2 - 1^2 = 8$, $3^2 + 2^2 = 13$;
  - $5^2 - 2^2 = 21$, $5^2 + 3^2 = 34$, $8^2 - 3^2 = 55$, $8^2 + 5^2 = 89$, $13^2 - 5^2 = 144$, $13^2 + 8^2 = 233$.
- **Option 2** is a fill-in chart, laid out with its entries computed in section 3.4.1.

In standard notation, Option 3 is two identities:

- **Sum cells** with consecutive Fibonacci legs give the odd-place terms: $F_n^2 + F_{n+1}^2 = F_{2n+1}$.
- **Difference cells** with legs two places apart give the even-place terms: $F_{n+1}^2 - F_{n-1}^2 = F_{2n}$. Here $h = F_n$ and $a + b = L_n$, a Lucas number, so $d^2 = F_n L_n$.

The sum cells $\sqrt{5}$, $\sqrt{13}$, $\sqrt{34}$ and $\sqrt{89}$ hide whole triangles: 5, 13, 34 and 89 are the hypotenuses of (3,4,5), (5,12,13), (16,30,34) and (39,80,89). One chart therefore holds both of Fibonacci's main contributions, the sequence and the congruum.

The (2,1) cell holds the golden ratio (added): with $\varphi = (1 + \sqrt{5})/2$,

- $\varphi + 1/\varphi = \sqrt{5}$ and $\varphi - 1/\varphi = 1$;
- $\varphi^2 + \varphi^{-2} = 3$ and $\varphi^2 - \varphi^{-2} = \sqrt{5}$.

Standing.

- **M.** All of the number theory is classical, as cited. Root families are the square classes of number theory: a length $\sqrt{N}$ with $N = k^2 m$ lies in the field $\mathbb{Q}(\sqrt{m})$. The Fibonacci identities are standard (Lucas 1878; Vajda 1989).
- **N.** The name “root family”, its use to group triangles by their imbalance, the tier reading of the gap between the tables, and the power-calculator reading of the chart were not found in the literature searched. Her own claim, that “this geometric framing does not appear in any standard literature”, agrees with that search.
- **D.** None.

3.4.1 Fibonacci: the second option

Her *Fib Fam Fun* (Taylor 2026g) gives three methods for "the SAME Fibonacci Sequence". Options 1 and 3 are in section 3.4. Option 2 is a chart that fills itself: "Use a Fibonacci Chart to find a Fibonacci sequence from the Hypotenuse and Difference tables. Start with $n_1 = 1$." Table 5 lays it out as she does, with the entries computed.

Table 5. Option 2.

Row	Difference cell	Sum cell
$n_1 = 1$	$n_1^2 - (n_1 - 1)^2 = 1^2 - 0^2 = n_1 = 1$	$n_1^2 + n_1^2 = n_2 = 2$
$n_2 = 2$	$n_2^2 - n_1^2 = n_3 = 3$	$n_2^2 + n_1^2 = n_4 = 5$
$n_3 = 3$	$n_3^2 - n_1^2 = n_5 = 8$	$n_3^2 + n_2^2 = n_6 = 13$
$n_4 = 5$	$n_4^2 - n_2^2 = n_7 = 21$	$n_4^2 + n_3^2 = n_8 = 34$
$n_5 = 8$	$n_5^2 - n_3^2 = n_9 = 55$	$n_5^2 + n_4^2 = n_{10} = 89$
$n_6 = 13$	$n_6^2 - n_4^2 = n_{11} = 144$	$n_6^2 + n_5^2 = n_{12} = 233$

What the chart is (added). Her n_k is F_{k+1}. Each row takes one term and returns two terms twice as far along:

- $n_k^2 - n_{k-2}^2 = n_{2k-1}$, that is $F_{k+1}^2 - F_{k-1}^2 = F_{2k}$;
- $n_k^2 + n_{k-1}^2 = n_{2k}$, that is $F_{k+1}^2 + F_k^2 = F_{2k+1}$.

Six rows reach the twelfth term, so the chart doubles the index at every row. These are the doubling identities (Lucas 1878; Vajda 1989), which compute F_k in about $\log_2 n$ steps rather than n. The chart's row labels are its own earlier answers, so it needs nothing but its start, $n_1 = 1$. Of Option 1 she writes that the formula "can be reversed to expand outward instead of winding inward"; Option 2 expands outward through the two tables.

Two more facts from the same chart (added):

- **The line from the corner.** The cells (F_{k+1}, F_k) lie on lines whose angle at the long leg tends to $\arctan(1/\varphi) = 31.7175°$. Its double is $\arctan 2 = 63.4349°$.
- **Inverted legs.** $1/F_k^2 + 1/F_{k+1}^2 = F_{2k+1}/(F_k F_{k+1})^2$, exactly.

Standing.

- **M.** The identities.
- **N.** Option 2 as a chart that fills itself through the two tables.

3.4.2 One complex number per cell

For $z = a + bi$:

- **The sum table.** $z\bar{z} = a^2 + b^2$.

- **The difference table and a third.** $z^2 = (a^2 - b^2) + 2ab \cdot i$: the difference table, and a third table of 2ab, which is 4M on the Rosetta page. This holds in all 55 cells, and $(a^2 + b^2)^2 = (a^2 - b^2)^2 + (2ab)^2$. Squaring doubles the angle: (2, 1) gives 3-4-5, (3, 2) gives 5-12-13, and (8, 5) gives 39-80-89.
- **The turn.** Multiplying by i is a 90° turn in one step.

Three behaviours.

- A constant sum rides a circle or sphere: rotation and bound orbits.
- A constant difference rides a hyperbola: relativity and escape.
- A zero difference, the 45° line, is the boundary between them: light, the parabolic orbit, the turnaround.

Inverted legs.

- **Root families survive.** The legs (1/a, 1/b) return both tables' roots divided by ab, so root families survive inversion in all 55 cells.
- **The angle mirrors across 45°.** In circuit terms:
  - a resistance and a reactance in series give $|Z| = c$;
  - in parallel they give the altitude ab/c, since $1/a^2 + 1/b^2 = 1/(ab/c)^2$;
  - for (4, 3), 5 Ω in series and 2.4 Ω in parallel, with power factors 0.8 and 0.6, and $0.8^2 + 0.6^2 = 1$.
- **Repeats split.** The value repeats of the sum table (50, 65, 85) split under inversion: families are the structure, and the repeats a coincidence of scale.
- **The flip through the unit circle.** $1/\bar{z} = z/|z|^2$ keeps the direction and inverts the size. Adding a point at infinity makes the plane a sphere, on which the unit circle is the fixed equator while the centre and infinity swap.

Inside out is negative. Flipping every sign has determinant $(-1)^n$:

- a rotation by 180° in two dimensions ($i \times i = -1$);
- a mirror in three, which is not one of the cube's 24 rotations;
- a rotation again in four (section 4.4).

The weak force tells a process from its mirror image (Wu et al. 1957).

The cube against its inside-out octahedron. The side-2 cube's distances pair with the octahedron's under $r \longrightarrow 1/r$: 1 with 1, √2 with 1/√2, √3 with 1/√3, and the four-dimensional corner's 2 with ½. Each pair multiplies to 1, and each figure has 6, 12 and 8 points with the centre, 27 in all. The tesseract's 16 corners and 8 axis points form the self-dual 24-cell (section 4.6).

Standing.

- **M.** Gauss's complex numbers, chromogeometry's three forms (Wildberger 2008), circuit duality, the Riemann sphere and Wu's experiment.
- **N.** Their reading as one cell, its inverse and its inside-out.

3.5 The elements in the two tables

Her *Root Elements* page tags every whole-number cell of Tables 3 and 4 up to 118 with the element of that atomic number. With the rules of section 3.4, every element can be placed, which extends her page to all 118:

- **Squares (10):** H, Be, F, S, Mn, Kr, In, Gd, Tl, Fm.
- **Sum table only (12):** He, Ne, Ar, Fe, Se, Sn, Ce, W, Pb, Th, Cf, Sg. All are 2 more than a multiple of 4.
- **Both tables (28):** B, O, Al, Cl, Ca, Cu, Ge, Rb, Zr, Nb, Rh, Te, I, Pm, Tb, Er, Hf, Ta, Hg, At, Ac, Bk, Md, Rf, Mt, Nh, Lv, Ts.
- **Difference table only (50):** every other odd Z and multiple of 4. A prime $4k + 3$ has only the gnomon route $((Z + 1)/2)^2 - ((Z - 1)/2)^2$, so lawrencium is $52^2 - 51^2$ and gold $40^2 - 39^2$.
- **Three squares only (18):** C, Si, Ti, Zn, Sr, Mo, Pd, Xe, Sm, Dy, Yb, Pt, Rn, Pu, No, Ds, Fl, Og. Each is a sum cell plus one leg, the converter's step:
    - C 6 = B(2,1) + 1^2; Si 14 = Al(3,2) + 1^2; Ti 22 = Ar(3,3) + 2^2; Zn 30 = Cu(5,2) + 1^2;
    - Sr 38 = Se(5,3) + 2^2; Mo 42 = Nb(5,4) + 1^2; Pd 46 = Rh(6,3) + 1^2; Xe 54 = Sn(5,5) + 2^2;
    - Sm 62 = Pm(6,5) + 1^2; Dy 66 = Sn(5,5) + 4^2; Yb 70 = Pm(6,5) + 3^2; Pt 78 = W(7,5) + 2^2;
    - Rn 86 = Pm(6,5) + 5^2; Pu 94 = At(7,6) + 3^2; No 102 = Cf(7,7) + 2^2; Ds 110 = At(7,6) + 5^2;
    - Fl 114 = Cf(7,7) + 4^2; Og 118 = Ts(9,6) + 1^2.

With differences allowed no element needs four squares, so her 4D tier is empty, checked to Z = 20,000.

On her page as printed, with legs 1 to 10:

- 75 elements sit in the two tables;
- 32 more are reached with a third leg of 10 or less;
- 11 (V, Ga, Ag, Lu, Au, Fr, U, Am, Lr, Rg, Cn) need legs beyond 10.

What the tiers do not do. The tier is a property of the integer Z, set by Z mod 4 and its prime factors, and it carries no chemistry; this was checked three ways (added).

- **Madelung exceptions.** The twenty exceptions to the filling rule fall 10 to 20% into every tier, as a random 17% would.
- **Unstable elements.** The elements with no stable isotope fall 10 to 36% into every tier.
- **Noble gases.** They split three sum-only (He, Ne, Ar), one square (Kr) and three three-square (Xe, Rn, Og). That split is the parity of the shell structure, not chemistry. The period lengths $2n^2$ are the diagonal cells (n, n), and xenon is krypton plus one 18-electron period, $54 = 6^2 + (3,3)$.

Standing.

- **M.** The theorems (Fermat and Euler; Legendre and Gauss; Lagrange 1770; Euclid II.5) are classical.
- **N.** The labelling, and the observation that the four-square tier is empty once differences are allowed, a short corollary of Legendre's theorem, were not found stated.
- **D.** None; the tables predict no chemistry, as checked.

3.6 Dual solutions, and Manhattan against Euclidean distance

Dual solutions. “Have you ever wondered why the equation $x^2 = 1$ has two solutions (±1), while both 1^2 and $\sqrt{1} \times \sqrt{1}$ are taught as producing only +1? … using only the positive branch accidentally hides the deeper structure. Every root calculation inherently contains the possibility of two real solutions or no real solution.”

She gives two demonstrations:

- **The paper strip.** A strip coloured on one side can be joined with the colour outside, with it inside, or with a half twist into a Möbius strip, “the symmetrically balanced, one-answer solution (because it has one continuous surface, the color ‘side’ appears evenly everywhere)”.
- **The $\sqrt{2}$ grid.** Draw a triangle on graph paper, set the grid's scale to $\sqrt{2}$, and ask what did and did not change.

In a later paper: “For every root, we can surmise the possibility of a positive or negative answer. This means every triangle has a counterpart that forms a perfect square when paired along the hypotenuse.” (Added: two copies joined along the hypotenuse make the a × b rectangle, a square when a = b.)

Manhattan and Euclidean distance. “Manhattan Distance measures the distance by following how many orthogonal steps it took to get to a new location and Euclidean distance measures the shortest path to that same location (the hypotenuse). These two descriptions produce the same final location, but they are 90° out of phase. You cannot take the Manhattan forward-and-sideways step at the same time, because orthogonal motions add sequentially. Euclidean rotation captures what Manhattan has to do in two moves.”

- **Each dimension adds a phase.** “Each time you add a perpendicular dimension, you add a 90° phase to Manhattan measurements. Each new phase adds a new squared term, producing the next root family.”
- **Loops.** A single triangle cannot make the two walkers line up, but “it is possible to make a minimum set of three perfect 45 degree triangles loop back if they are 90 degrees out-of-phase with each other”.
- **Phase over boundary.** “phase and frequency could be the primary path of three dimensions rather than the boundaries (length, width, and height).”

Computed with these ideas (added):

- **The average cost.** A straight unit step at angle θ costs $|\cos\theta| + |\sin\theta|$ in legs. Averaged over all directions that is $4/\pi = 1.2732$ in two dimensions and $3/2$ in three.
- **The worst case.** The worst case is the diagonal: $\sqrt{2}$, $\sqrt{3}$ and 2 in two, three and four dimensions (the Cauchy–Schwarz inequality).
- **The number of paths.** An a × b grid is crossed by $C(a + b, a)$ shortest leg-paths, all equally long in the legs' own measure.

Standing.

- **M.** The taxicab metric is Minkowski's, and its geometry is set out by Krause (1975). The principal-root convention is standard. The one-sided strip is Möbius's (published 1865).

- **N.** Reading the two metrics as 90° out of phase, and each added dimension as an added phase, was not found.
- **D.** None.

3.7 The dimension converter and the root ladder

Her converter: “Dimension Converter (Nested Pythagorean)”,

```
\sqrt{(a^{2}+b^{2})+f^{2}}=\sqrt{a^{2}+(b^{2}+f^{2})}
```

and: “Nested Pythagorean usually is used for triplets but this also gives more information.”

On the side-2 cube (a = b = f = 2):

- **The faces.** Each face diagonal is $\sqrt{8} = 2\sqrt{2} = \sqrt{(2^3)}$, and she labels the three m, n and o: “motion (x)”, “motion (y)” and “motion (z)”.
- **The ladder.** “1D = 2 = line, 2D = $2\sqrt{2}$ = plane, 3D = $2\sqrt{3}$ = cube.” The triangle with legs 2 (a line) and $2\sqrt{2}$ (a plane) has hypotenuse $\sqrt{12} = 2\sqrt{3}$ (a cube).
- **Her two rules.** “a, b, f are physical lengths with 2 stages of perpendicular turns”, and “m, n, o are summaries of distance moved (so motion).”

Its radians form. A point on a circle of radius r at angle θ is (r cos θ, r sin θ). Adding a perpendicular axis a gives $L = \sqrt{(a^2 + r^2(\cos^2\theta + \sin^2\theta))} = \sqrt{(a^2 + r^2)}$, so motion on a circle plugs into three dimensions whole.

The ladder of roots. “Root families are the intrinsic dimensional ladder”:

- the square $\sqrt{(1^2 + 1^2)} = \sqrt{2}$;
- the cube $\sqrt{(1^2 + (\sqrt{2})^2)} = \sqrt{3}$;
- and the 2 × 1 difference triangle $\sqrt{(2^2 - 1^2)} = \sqrt{3}$ lands on the same rung.

Run axis by axis, $P_1 = \sqrt{(x^2 + y^2)}$, $P_2 = \sqrt{(P_1^2 + z^2)}$ and $P_3 = \sqrt{(P_2^2 + w^2)}$, with w a fourth space axis, so the unit diagonals are $\sqrt{2}$, $\sqrt{3}$ and $\sqrt{4} = 2$, “a whole number, twice the edge”. Measured from the centre of a cube of half-side K the rungs are $R_? = K\sqrt{k}$: K, $K\sqrt{2}$, $K\sqrt{3}$ and 2K, to a face, an edge midpoint, a corner and a 4D corner. The last equals the edge.

Time as a difference. Her chart sets the spacetime interval in the same frame: $D^2 = (ct)^2 - P_2^2 = (ct + P_2)(ct - P_2)$. A moving observer multiplies one factor by k and divides the other by k, so the product is unchanged. Reading D as her virtual leg d, with ct as the long leg, is her interpretation; the algebra is Minkowski's (1908) and the factor-k argument is Milne's and Bondi's.

Standing.

- **M.** The n-dimensional distance formula follows Grassmann (1844), Cayley (1845) and Schläfli (1852).
- **N.** Reading m, n and o as summaries of motion, the converter as the step that adds a phase, and the interval's D as the virtual leg were not found.
- **D.** None.

3.7.1 The 2D-to-3D converter, and the leg it must be given

Her rule: going from two dimensions to three always needs one leg the two-dimensional data cannot supply. Each case below is proved; Table 6 at the end separates what is exact from what is a reading. The

script appC.py (SHA-256 b19a7d28343721a5145212bb242ff9a8e0467218a09d40864f85759dc9707109) recomputes every number here and in sections 3.4.2, 3.7.2, 4.4.1 and 4.9.1 with the PDG 2026 inputs, and each symbolic statement was proved by reducing the difference of its two sides to zero.

1. A two-dimensional view loses exactly the diagonal. The Clarke matrix

```
K=\tfrac{2}{3}\begin{pmatrix}1&-\tfrac12&-\tfrac12\\[2pt]0&\tfrac{\sqrt3}{2}&-\tfrac{\sqrt3}{2}\end{pmatrix},\qquad K\,(1,1,1)^{\mathsf T}=(0,0)^{\mathsf T}
```

has rows that sum to zero, so it sends (1, 1, 1) to the origin. It has rank 2, so its kernel is exactly that line. Any amount γ on the diagonal gives the same two-dimensional point: γ = 0, +0.3 and −0.5 all return (0.7, −0.4) for the same input. So the third leg must be supplied from outside: a three-wire system forces γ = 0, and a four-wire system lets γ flow in the neutral.

2. The Scott-T makes three phases from two. The main winding carries V∠0° with a centre tap. The teaser, from that tap, carries V∠90° scaled by √3/2, so

- V_A = j(√3/2)V, V_B = −V/2, V_C = V/2;
- V_AB = V/2 + j(√3/2)V, V_BC = −V, V_CA = V/2 − j(√3/2)V.

Each line voltage has magnitude √(¼ + ¾)V = V, and since ½ + j√3/2 = e^(jπ/3), each is the last turned by 120°: a balanced three-phase set. The teaser ratio √3/2 is the unit cube's centre-to-corner distance.

3. Three-phase power is flat (section 7.2), and **third harmonics live on the diagonal.** Phase n sits at −2πn/3, so its third harmonic sits at −2πn ≡ 0: the three third harmonics are in phase, pure (1, 1, 1), invisible to the Clarke view and added in the neutral. With 30% third harmonic in each phase:

- the circle keeps radius 1.000000;
- the diagonal swings ±0.300 at three times the frequency;
- the neutral peaks at 0.900;
- from a balanced source the diagonal carries zero average power: it oscillates both ways and delivers nothing.

4. Three roots.

Ratio	Value	Proof
Peak over RMS of one phase	√2	the mean of sin² over a period is ½
Line over phase voltage	√3	|1 − e^(−j2π/3)|² = (3/2)² + (√3/2)² = 3
Single-phase power, peak over average	2 = √4	power goes as sin²: peak 1, average ½

5. Four dimensions is the special case. The centre-to-corner distance of the unit n-cube is √n/2, which equals the edge only at n = 4.

6. The Higgs mechanism fills the third leg. The Higgs field has four real components, and its vacuum sits on a 3-sphere of radius v at ⟨H⟩ = (0, v/√2), with v = (√2 G_F)^(−1/2) = 246.220 GeV.

- **The mass matrix.** From $|(gT^aW^a + g'YB)\langle H\rangle|^2$, for (W_1, W_2, W_3, B):

$$M^{2}=\frac{v^{2}}{4}\begin{pmatrix}g^{2}&0&0&0\\0&g^{2}&0&0\\0&0&g^{2}&-gg'\\0&0&-gg'&g'^{2}\end{pmatrix}$$

  Its eigenvalues are $g^2v^2/4$ twice (W^+ and W^-), $(g^2 + g'^2)v^2/4$ (the Z) and 0 (the photon).
- **Why the photon gets nothing.** The W_3–B block is $(v^2/4)(g, -g')^T(g, -g')$: rank one, determinant zero. Its zero direction is $A = (g'W_3 + gB)/\sqrt{(g^2 + g'^2)}$. The reason is that the vacuum's charge is $Q = T_3 + Y = -½ + ½ = 0$.
- **The triangle.** $m_Z^2 = m_W^2 + (g'v/2)^2$ exactly: a right triangle with legs $m_W = 80.3625$ GeV and $g'v/2 = 43.094$ GeV, and hypotenuse $m_Z = 91.1880$ GeV. Its angle, 28.20°, is the weak mixing angle. At tree level $g = 0.65277$, $g' = 0.35005$ and $e = 0.30849$.
- **The count, 12 = 12.** Before: four carriers with two legs each (8) and four Higgs components (4). After: three massive carriers with three legs each (9), the photon with two, and one Higgs boson. SU(2) × U(1) has four generators and keeps one, the charge, so three break. By Goldstone's theorem each broken direction is one Higgs angle on the 3-sphere, and each becomes the third leg of W^+, W^- or Z. The radial direction left over is the Higgs boson.

7. The filled leg rides the hyperbola, and the Higgs boson keeps it bounded.

- **The polarisation.** $\varepsilon_L = (p, 0, 0, E)/m$ has $\varepsilon_L \cdot \varepsilon_L = (p^2 - E^2)/m^2 = -1$ and $\varepsilon_L \cdot k = 0$. With rapidity η, $E/m = \cosh\eta$ and $p/m = \sinh\eta$. The two transverse legs turn on a circle (helicity ±1).

W energy	p/m, E/m	$\cosh^2 - \sinh^2$
100 GeV	0.741, 1.244	1.000000000
1,000 GeV	12.403, 12.444	1.000000000
10,000 GeV	124.432, 124.436	1.000000000

- **The amplitude.** With $V = \lambda(H\dagger H - v^2/2)^2$ and $H = (w^+, (v + h + iz)/\sqrt{2})$, the couplings are $w^+w^-z^2 \to \lambda$, $hw^+w^- \to 2\lambda v$, $hz^2 \to \lambda v$, and $m_h^2 = 2\lambda v^2$. Then

$$\mathcal M(w^{+}w^{-}\to zz)=-2\lambda-\frac{(2\lambda v)^{2}}{s-m_h^{2}}=-\frac{m_h^{2}}{v^{2}}\,\frac{s}{s-m_h^{2}}$$

  For $s \ll m_h^2$ it grows as s/v^2, the case without a Higgs boson. For $s \gg m_h^2$ it levels at -2λ. By the equivalence theorem (Cornwall, Levin and Tiktopoulos 1974; Chanowitz and Gaillard 1985) it equals longitudinal $W^+W^- \to ZZ$ at high energy.
- **Unitarity.** The partial wave $a_0 = \mathcal{M}/16\pi$ must stay within ±½.

√s	a_0 without a Higgs boson	a_0 with it
300 GeV	+0.0295	−0.00622
1,000 GeV	+0.3282	−0.00522
3,000 GeV	+2.9534, broken	−0.00515
10,000 GeV	+32.8159, broken	−0.00514

  - **Without a Higgs boson**, unitarity breaks at $\sqrt{(8\pi)}\cdot v = 1{,}234.4$ GeV.

- **With one**, a_0 levels at $-2\lambda/16\pi = -0.00514$, with $\lambda = m_H^2/2v^2 = 0.12914$.
- **The bound.** Lee, Quigg and Thacker (1977) bound the Higgs mass at $\sqrt{(16\pi/3)}\cdot v$ = 1,007.9 GeV; it is measured at 125.13 GeV.

The level -2λ uses the same λ that the running of section 9.3 follows, which turns negative near 5×10^{10} GeV where the gauge terms are as large. That link is qualitative.

Table 6. Exact against interpretation.

Her picture	Standard result	Status
2D to 3D must be given the other leg	the Clarke kernel is the diagonal (1, 1, 1)	exact, proved
Fill it in phases	the Scott-T, with its $\sqrt{3}/2$ teaser	exact, proved
The diagonal oscillates both ways	third harmonics are pure (1, 1, 1) and carry zero average power	exact, proved
The action circle absorbs at $\sqrt{2}$	peak over RMS is $\sqrt{2}$	the identity is exact; “absorbs” is a reading
The physical holds at $\sqrt{4}$	single-phase peak over average is 2; the four-dimensional centre-to-corner equals the edge	the identities are exact; “holds” is a reading
From the legs, the straight path looks zigzag	the legs are at most $\sqrt{n}$ times the straight path; C(a + b, a) equal staircases (section 3.6)	exact, proved
Carriers go from 2D to 3D	two legs become three; 12 = 12	exact, standard
The leg is filled by phases	the three Goldstone angles of the 3-sphere	exact, standard
The photon stays unfilled	the rank-one block; the vacuum's charge is 0	exact, proved
The third leg rides the hyperbola	$\varepsilon_L = (\sinh \eta, \cosh \eta)$	exact, proved
The Higgs boson keeps the leg bounded	$\mathcal{M} \to -2\lambda$	exact at tree level

Standing.

- **M.** Every proof is standard: the Clarke transform (Clarke 1943), the Scott-T (Scott, Westinghouse, 1890s), Goldstone's theorem, the Higgs mechanism (Englert and Brout 1964; Higgs 1964; Guralnik, Hagen and Kibble 1964), the equivalence theorem, and the unitarity bound.
- **N.** Her rule that every step up a dimension must be given its leg from outside, read across power, geometry and the Higgs mechanism.

3.7.2 The dimension steps

From the centre. For a cube of side s with inscribed radius K = s/2, $R_1 = K$ and $R_{n+1} = \sqrt{(R_n^2 + K^2)}$, so $R_n = K\sqrt{k}$ (section 3.7). Each step swings round the centre by an angle φ with $\tan \varphi = K/R_n = 1/\sqrt{k}$:

- 1D to 2D: 45°;
- 2D to 3D: 35.26°;
- 3D to 4D: 30°.

The 3D-to-4D step triangle, $(K\sqrt{3}, K, 2K)$, is the 30-60-90 triangle. The four-dimensional corner's radius is $2K = s$, equal to the edge, and it runs $K(2 - \sqrt{3})$ past the three-dimensional corner. Each cubic cell of a tesseract has its centre at distance K from the tesseract's centre, on the inscribed sphere.

Three phased cubes. She draws cubes of sides 1, √2 and √3.

- **Her diamond.** Cutting each corner at half the edge (her corner pyramids) leaves a cuboctahedron, her "3D diamond", whose radius equals its edge, $s\sqrt{2}/2$:

Side	Radius = edge of the diamond	Corner radius of the cube
1	√2/2 = 0.7071	√3/2 = 0.8660
√2	1: the diamond sits on the unit sphere	√6/2 = 1.2247
√3	√6/2 = 1.2247	3/2

- **The Balance Chart.** The corner radii √3/2, √6/2 and 3/2 are the cube column of her Balance Chart (Table 7).
- **Radius equal to edge.** The shapes whose radius equals their edge are the hexagon in two dimensions, the cuboctahedron in three, and the tesseract and the 24-cell in four. The 24-cell, with vertices (±1, ±1, 0, 0), is the four-dimensional form of her diamond. The tesseract is where the cube family itself reaches radius = edge, since √4/2 = 1.

From a corner.

- **Step angles.** The step angle is $\tan \varphi_n = 1/\sqrt{(n - 1)}$: 45°, 35.26°, 30° and 26.57°.
- **Cubes meeting at a corner.** In n dimensions, n cubes of one dimension less meet at a corner. Their far diagonal ends form a regular simplex of edge √2 with centre $((n - 1)/n)(1, \ldots, 1)$. In four dimensions this takes four cubes, not three, and the centre (¾, ¾, ¾, ¾) is 3/2 from the corner.
- **The step triangles.** Each recurs on her pages:

Step	Triangle	Where it recurs
1D to 2D	(1, 1, √2)	the Rosetta half-square S
2D to 3D	(√2, 1, √3)	the converter page
3D to 4D	(√3, 1, 2)	the Rosetta page's hidden 30-60-90

Step	Triangle	Where it recurs
4D to 5D	(2, 1, √5)	the Rosetta triangle M at the key cell

Standing.

- **M.** The radii of n-cubes, the cuboctahedron's radius equal to its edge, the tesseract and the 24-cell (Coxeter 1973).
- **N.** The translation into her K and her phased cubes.

3.8 Centroids, the balance chart and the inside-out flip

The four facts. “A unit square in the first quadrant (1,1) has a centroid located at ½(√2) from the origin. A square (√2,√2) has a centroid located at 1 from the origin. A cube (1,1,1) has a centroid located at ½(√3) from the origin. A cube (⅔(√3),⅔(√3),⅔(√3)) has a centroid located at 1 from the origin. Geometrically this means if you flip a square or cube inside out, its new dimensions are root families scaled down!” With them: “You can multiply Root Families by grid-spacing to get derived dimensional diagonals (1, 1√2, 1√3, 2).”

Why the flip gives the grids (added). Inversion through the unit sphere, r ⟶ 1/r, sends a corner of the unit k-cube, at distance √k, to 1/√k. A cube of half-side s has its level-k points at s√k, so s√k = 1/√k gives s = 1/k. The half-sides of the flipped cubes are 1, ½, ⅓ and ¼, the grids of the calculator (section 7.1). Her square of side √2 and her cube of side ⅔√3 are the cases k = 2 and k = 3: the square and the cube whose centres sit at distance 1 from a corner.

Her Balance Chart. It lists the centroid distances by root level. Distances are measured from a corner, with the side length across the top and her own forms kept.

Table 7. Tifinchi's Balance Chart.

Column	Shapes	Side 1	√2	√3	2	3
Centroid distance, a√2/2	triangle, square and circle	√2/2	1	√3√2/2	2√2/2	3√2/2
Centroid distance, a√3/2	cube and sphere	√3/2	√2√3/2	3/2	2√3/2	3√3/2
⅓ centroid distance, a√2/(2·3)	triangle and square	√2/(2·3)	⅓	√2·√3/(2·3)	√2/3	√2/2
⅓ centroid distance, a/3	triangle and square; circle	⅓	√2/3	√3/3	⅔	1

She tags the columns with index labels ([6 to 3], [6 to 8], [6 to 27], [6 to 7], [23 to 8], [17 to 26]), kept here as written. Three columns, perimeter or circumference, ½ area and ⅓ area, are headed but left blank.

The chart lists the balance points of her square-on-the-long-leg figure (added):

- a√2/2 is the radius of the circle through a square's corners, and a√3/2 the cube's;
- the centroid of the half-square triangle sits a/3 from each leg and a√2/6 from the diagonal.

Standing.

- **M.** A triangle's centroid lies a third of the way along each median (Archimedes, *On the Equilibrium of Planes* I). Inversion in a sphere is classical.
- **N.** The chart that ties the root ladder to balance points, and the flip-and-scale reading of the grids, were not found.

3.8.1 The Balance Chart completed (added)

Her chart, Table 7, heads three columns that it leaves blank: perimeter or circumference, ½ area and ⅓ area. Table 8 fills them using figures from her own pages, so each entry has a source in her work:

- **Perimeter.** The square, the half-square triangle S, and the two circles of the Rosetta page: the inscribed circle of radius K, and the circle through the corners of radius f/2.
- **½ area.** The half-square S = s²/2 cut by the diagonal (Rosetta line ⑧), and in three dimensions the bi-4 prism of her pyramid page.
- **⅓ area.** Three figures:
    - the diagonal band of the 3 × 3 grid: her box rule, true only at n = 3 (section 3.2), splits that grid into three equal bands, the diagonal and the cells on either side of it;
    - the 120° sector of her phase circle (section 3.10);
    - in three dimensions, the tetrahedron left when four corner pyramids are cut away, a third of the cube (section 3.9).

Table 8. The three blank columns, filled. Sides as in her chart.

Column	Figure	Rule	Side 1	√2	√3	2	3
Perimeter	square	4s	4	4√2	4√3	8	12
Perimeter	half-square triangle S	s(2 + √2)	2 + √2	2 + 2√2	2√3 + √6	4 + 2√2	6 + 3√2
Circumference	inscribed circle, radius K = s/2	πs	π	π√2	π√3	2π	3π
Circumference	circle through the	π√2 · s	π√2	2π	π√6	2π√2	3π√2

Column	Figure	Rule	Side 1	√2	√3	2	3
	corners, radius f/2						
½ area	half-square S	$s^2/2$	½	1	3/2	2	9/2
⅓ area	diagonal band of the 3 × 3 grid	$s^2/3$	⅓	⅔	1	4/3	3
⅓ area	120° sector of the inscribed circle	$\pi s^2/12$	π/12	π/6	π/4	π/3	3π/4
Surface	cube	$6s^2$	6	12	18	24	54
Surface	inscribed sphere	πs^2	π	2π	3π	4π	9π
½ volume	bi-4 prism	$s^3/2$	½	√2	3√3/2	4	27/2
⅓ volume	inscribed tetrahedron, edge s√2	$s^3/3$	⅓	2√2/3	√3	8/3	9
Volume	inscribed ball	$\pi s^3/6$	π/6	π√2/3	π√3/2	4π/3	9π/2

Four readings follow (added):

1. **The √2 square's corner circle is the unit circle.** At side √2 the circle through the corners has circumference 2π. This is her fact that the square (√2, √2) has its centre at distance 1 from a corner (section 3.8).
2. **The √3 cube's third.** At side √3 the tetrahedron holds √3, the unit cube's body diagonal, and the band holds 1: a third of the √3 square is the unit square.
3. **Fixed shares.** The inscribed ball fills π/6 of the cube at every size; at side 2 it is the unit ball, 4π/3. The four-dimensional ball's share of the side-2 tesseract is $\pi^2/32$ (section 4.6).
4. **The balance point moves from ⅓ to ¼.** A triangle's centroid sits a third of the way up each median, and a tetrahedron's a quarter of the way up from each face. So the chart's ⅓ column becomes ¼ for the cube's tetrahedron.

Standing.

- **M.** Each entry is classical.

- **N.** The columns as filled are a reading of her chart, marked as added.

3.9 The cube cut into identical pieces

Her page “Pyramid & Pyramid & Pyramid (cube cut into identical pieces)” draws three cuts, each with its net:

- **bi-4**, “bisect 4 corners”, two per cube: the plane through two opposite edges halves the cube into two triangular prisms.
- **tri-3**, “trisect 3 corners”, four per cube: the corner pyramid cut off through the three corners next to one corner, with three right angles at its tip.
- **tri-4**, “trisect 4 corners”, six per cube, “height ½ of cube”: the square pyramid on each face with its apex at the centre.

The tetrahedron. With a = 1 and c = √2, the cube minus four corner pyramids leaves the regular tetrahedron of edge c. She draws the four cuts one at a time, with the faces each cut leaves on the net.

Volumes (added). Each corner pyramid holds $a^3/6$, so four remove $2a^3/3$ and leave the tetrahedron with $a^3/3$, exactly a third of the cube; each face pyramid also holds $a^3/6$. The face pyramids and the prisms tile the cube with identical pieces. The corner pyramids come off four at a time and leave the tetrahedron.

The ladder. The page ends with √8/2 → √5/2 → √2/2, its arrows marked √3, and with √2 · √2 = √4, √2 + √2 = √8, (√2 + √2)/2 = √8/2 and √4 · √2/2 = √8/2. In half-units the squares 8, 5, 2 are the sum-table cells (2,2), (2,1) and (1,1), and each step removes $3 = 2^2 - 1^2$, the arrows' √3 (added).

A correction (added). The page labels a segment of the tetrahedron √5/2 = 1.118 and builds it from legs 1 and ½. In the tetrahedron of edge √2, opposite edges are 1 apart and half an edge is √2/2, not ½. The segment from a vertex to the midpoint of the opposite edge is therefore √(1 + ½) = √6/2 = 1.225, and the height above a face is 2/√3 = 1.155.

Standing.

- **M.** A prism divides into three equal pyramids (Euclid XII.7). The tetrahedron inscribed in a cube, and its third of the volume, are classical (Kepler's *stella octangula*, 1619).
- **N.** The labelling by corners cut and pieces per cube, and the step-by-step net, are hers.
- **D.** The √5/2 label, corrected above.

3.9.1 One object across four pages (added)

The regular tetrahedron {+, ×, ”, .} of the delimiter cube (section 3.12) appears on four of her pages:

- **Root Patterns.** Its “minimum set of three perfect 45 degree triangles” that “loop back if they are 90 degrees out-of-phase” is the loop + → × → ” → +. That loop crosses Top, Right and Front, three mutually perpendicular faces, along three face diagonals of length √2, and closes one face of the tetrahedron.
- **The converter page.** Its m, n and o are three face diagonals √8 on mutually perpendicular faces of the cube of edge 2. They close the same triangle at side 2√2.
- **The pyramid page.** Its tetrahedron is what is left after four corner cuts. The four corners cut away, −, *, ’ and ,, form the dual tetrahedron.

- **The electrical frame of section 4.2.** It puts charge, current, flux and voltage on the tetrahedron's four corners.

The volumes on the pyramid page fill the Balance Chart's ½ and ⅓ columns in three dimensions (Table 8):

- the bi-4 prism holds ½ of the cube;
- each corner pyramid and each face pyramid holds ⅙;
- the tetrahedron holds ⅓.

3.10 The phase circle

On a circle of radius r = 3 with an inscribed square she marks “(⅓) phase = 120°”, “πr distance along curve” and a “wedge” between the arc and the square's side. She then sets three right triangles side by side.

Table 9. Her three triangles on the radius-3 circle.

	Initial potential (Q1)	Container potential (Q2)	Change
Angles A, B, C	60°, 30°, 90°	45°, 45°, 90°	45°, 45°, 90°
Legs and hypotenuse a, b, c	3, √3, 2√3	3, 3, 3√2	√3, √3, √6
d, e	√6, √3/2	0, 0: “total potential retains container”	0, 3
Wedge area	left blank	left blank	left blank

Checks (added). $d_1^2 = 9 - 3 = 6$, $c_1^2 = 12$, $c_2 = 3\sqrt{2}$ and $c_3 = \sqrt{6}$ all hold. Two relations tie the triangles together:

- the change triangle's hypotenuse is the initial triangle's virtual leg, $c_3 = d_1 = \sqrt{6}$;
- its legs are the initial triangle's short leg, $\sqrt{3}$.

The change is the 45° triangle built on the initial triangle's imbalance, and the container is the 45° triangle on the radius. For the container, the wedge between the arc and one side of the inscribed square has area $r^2(\pi - 2)/4 = 2.5686$ at r = 3. The e column follows this page's own construction, which differs from the one in section 3.2.

Standing.

- **N.** The potential, container and change triangles on one circle are hers.
- **M.** A third of a cycle is 120°, the displacement of three-phase power; the triangles themselves are classical.

3.10.1 The phase circle completed (added)

Table 9 leaves the wedge areas blank, and its e column follows no single rule. Table 10 completes both.

The wedges. Her figure shades the region between the arc and one side of the inscribed square. Lay each triangle's hypotenuse as a chord of the radius-3 circle, and each triangle gets a wedge: the segment between chord and arc, of area ½r²(θ − sin θ). For the container this is the value given in section 3.10. Table 10 also gives a second reading of "wedge": container minus triangle, which is N.

Table 10. The wedges on the radius-3 circle.

Triangle	Hypotenuse as a chord	Central angle	Wedge	Container minus triangle, N
Initial potential (60°, 30°, 90°; 3, √3, 2√3)	2√3	arccos ⅓ = 70.53°	(9/2)(arccos ⅓ − 2√2/3) = 1.2967	(9 − 3√3)/2 = 1.9019
Container potential (45°, 45°, 90°; 3, 3, 3√2)	3√2	90°	9(π − 2)/4 = 2.5686	0
Change (45°, 45°, 90°; √3, √3, √6)	√6	arccos ⅔ = 48.19°	(9/2)(arccos ⅔ − √5/3) = 0.4307	0

The initial triangle's hypotenuse subtends the angle whose cosine is ⅓. That is the angle between two body diagonals of a cube, 70.53°, the supplement of the tetrahedral 109.47°: the 30-60-90 triangle of the √3 family, laid on the radius-3 circle, cuts the cube's own angle. The other chords of her figure give:

- the ⅓ phase, 120°: a segment of 3π − 9√3/4 = 5.528;
- the 60° chord, the side of the inscribed hexagon: 3π/2 − 9√3/4 = 0.815.

Table 11. The e column under each definition.

Triangle	Written	Area construction, e² = b(a − b) (section 3.2)	Leftover h = a − b	b/2, from her e² = (½b)²
Initial	√3/2	√(3√3 − 3) = 1.482	3 − √3 = 1.268	√3/2
Container	0	0	0	3/2
Change	3	0	0	√3/2

Of the definitions in Table 11, only the area construction locks the triangle. With it, (d², e²) = (6, 3√3 − 3) gives tan A = d²/e² − 1 = √3, so A = 60° exactly. The written column fits b/2 in its first row and no rule in its third, so the construction's values are used.

Standing.

- **N.** The three triangles are hers; the chord reading and the e values are added.
- **M.** Circle segments are classical.

3.11 The Rosetta fold

The operation. Her Rosetta page reads one operation through several mathematical scripts, as the Rosetta Stone carries one text in three:

```
R(x,y)=(x+y,\;x-y),\qquad R^{2}=2I
```

R is a fold across the 22.5° line with a $\sqrt{2}$ stretch. Applied twice it doubles everything, $\sqrt{2} \times \sqrt{2} = 2$: her $\sqrt{2}$ grid.

Where it acts.

- **On the legs.** (a, b) → (a + b, a − b), the Manhattan path and the leftover h, and then to (2a, 2b).
- **On the squares.** $(a^2, b^2) \to (c^2, d^2) \to (f^2, g^2) = (2a^2, 2b^2)$, where $g = b\sqrt{2}$ is the diagonal of the square on the short leg.
- **Two ladders.** R leaves two ladders of evenly spaced squares: $d^2 \to a^2 \to c^2$ in steps of b^2, and $g^2 \to c^2 \to f^2$ in steps of d^2.

The seven scripts.

1. **Area.** Every area is half of one length times another: M = ½ab, N = ½a·h, a sector ½(arc)(r), a circle ½(C)(r) (Archimedes, *Measurement of a Circle*).
2. **Trigonometry.** $\sin 45° = \cos 45° = \sqrt{2}/2$ is R's fold, and $\sin 45° + \cos 45° = \sqrt{2}$. Also $(a + b)^2 = c^2 + 4M$, $h^2 = c^2 - 4M$ and $d^2 = h(a + b)$.
3. **Complex numbers.** $(a + bi)^2 = d^2 + 4M \cdot i$, with size c^2; the example (4, 2) squares to (12, 16, 20) = 4 × (3, 4, 5).
4. **Circle and balance.** $K = a/2$ is the inscribed radius and $f/2 = K\sqrt{2}$ the radius through the corners, the Balance Chart's first column. The centroid of the half-square sits a/3 from each leg and $a\sqrt{2}/6$ from the diagonal, its second column.
5. **Dimension.** The converter adds one squared term per axis, $1 \to \sqrt{2} \to \sqrt{3}$. The unit diagonal $(\sqrt{3}/3)(1, 1, 1)$ makes 54.74° with each axis.
6. **Tier.** c^2 is the sum tier, d^2 the difference tier and P_2 the three-square tier, all on one page; the example (4, 2) is the (2,1) cell doubled.
7. **Origami.** d is the swing fold: pin a corner and swing the point at distance a onto the line b away, and it lands d across.

Six alignments at a = 2b. Six things line up only when the long leg is twice the short one:

1. M = N.
2. The point where b meets h lands on the inscribed circle.
3. The virtual triangle (b, d, a) is $1 : \sqrt{3} : 2$, the 30-60-90.
4. b, g, d = $b(1, \sqrt{2}, \sqrt{3})$: the edge, face diagonal and body diagonal of the cube on the short leg.
5. The squared triangle is 3-4-5.
6. It is the (2,1) cell of both tables, $\sqrt{5}$ and $\sqrt{3}$.

Standing.

- **M.** R is the unnormalised Hadamard matrix (Sylvester 1867), and u = ct + x, v = ct − x are the light-cone coordinates of relativity (Dirac 1949). Each script is classical, as cited.

- **N.** The translation key, one figure read in seven scripts, and the six alignments at a = 2b were not found.

3.11.1 The Rosetta proofs, line by line

Her page *Tifinchi Rosetta Trig Proofs* (Taylor 2026f) numbers each line. Table 12 gives each line's derivation and first source. The last column says how her use differs from the source's, where it does. Lines marked def are definitions. Throughout, a is the long leg and b the short one; if b > a the two are swapped.

Table 12. The Rosetta proofs.

Line	Statement	Derivation	Origin	Her use, where it differs
①	$a^2 + b^2 = c^2$	four copies of the triangle in a square of side a + b leave a tilted square of area c^2, so $(a + b)^2 = c^2 + 4M$	Plimpton 322 (c. 1800 BCE); Euclid I.47, with the converse I.48; Zhao Shuang's diagram on the *Zhoubi Suanjing* (3rd century)	a rearrangement proof, which keeps M in view; Euclid's own proof uses rectangles and congruent triangles
②	$d^2 = \lvert a^2 - b^2 \rvert$	take a b × b square from an a × a square; the L-shaped rest slides into an h × (a + b) rectangle	Euclid II.5 (the gnomon); I.47 read backwards ($b^2 + d^2 = a^2$)	the side of the square equal to the gnomon becomes a length, a leg of the virtual triangle (b, d, a), so the imbalance can be drawn and folded
③	M = ½ab	the hypotenuse halves the a × b rectangle	Rhind Papyrus, problem 51 (c. 1550 BCE); Euclid I.41	—
def	$h = \lvert a - b \rvert$	the long side less the short side	absolute-value bars: Weierstrass (1841)	—
④	$c^2 - d^2 = 2b^2$	$(a^2 + b^2) - (a^2 - b^2)$	the pattern: Fibonacci's congruum (*Liber*	Fibonacci's three squares are whole numbers (1, 25, 49); here d^2, a^2 and c^2 are b^2 apart

Line	Statement	Derivation	Origin	Her use, where it differs
			Quadratorum, 1225)	for any lengths. Fermat showed the step is never a square
⑤	$2c^2 - 2d^2 = (2b)^2$	④ doubled	doing the same to both sides: al-Khwarizmi (c. 820)	done to make a shape: the right side becomes one square
⑥	$\frac{1}{2}c^2 - \frac{1}{2}d^2 = b^2$	④ halved	al-Khwarizmi (c. 820)	halving turns c^2 and d^2 into half-squares, the areas of 45° triangles
⑦	$c^2 + d^2 = 2a^2 = 4(M + N)$	the sum of the two definitions; $M + N = a^2/2$ by ⑧	Fibonacci (1225)	as in ④
⑧	$S = M + N$	the diagonal f halves the square; M and N fill one half	Euclid I.34	—
⑨	$N = \frac{1}{2}a \cdot h = \frac{1}{2}a^2 - M$	N has base h and height a	Euclid I.41	—
⑩	$f = a\sqrt{2} = (\sin 45° + \cos 45°)a$, at 45° only	① with both legs a	YBC 7289 (c. 1800–1600 BCE); the sine: Aryabhata (499); the unit circle: Euler (1748)	$\sin\theta + \cos\theta$ is the ratio $(a + b)/c$ of the two-leg path to the straight one, $\sqrt{2}$ only at 45°, where it equals $1/\cos 45°$
def	D, c, t	interval, speed of light, time	Minkowski (1908)	—
def	U, V, W	unit radius 1, circumference 2π, area π	area = ½ × circumference × radius: Archimedes, *Measurement of a*	—

Line	Statement	Derivation	Origin	Her use, where it differs
			Circle; the symbol π: Jones (1706)	
⑪	K = ½a·U	the inscribed circle has diameter a	Euclid IV.8	—
⑫	½V = π	V = 2π	Archimedes; Jones (1706)	—
⑬	W = ½ · 3² · (2π/9) = π	a sector of angle θ has area ½r²θ	Archimedes, *Measurement of a Circle*, Prop. 1; areas go as squares of size: Euclid XII.2; the radian: Cotes (1722), named by Thomson (1873)	two circles are compared: the radius-3 circle has nine times the area, so its 40° sector equals the whole unit circle. This ties her normalise-to-3 method to the unit circle
⑭	tan 45° = 1	the two legs are equal	tangent tables: Habash al-Hasib (c. 860); the name: Fincke (1583)	—
def	g, the diagonal	opposite corners joined	Euclid I.34, where it is the "diameter"	—
⑮	(x, y, z, w)	four mutually perpendicular axes, w a space axis	coordinates: Descartes (1637); more than three dimensions: Grassmann (1844), Cayley (1845), Schläfli (1852); time as the fourth axis: Minkowski (1908)	she uses two different fourth axes and keeps them apart: w adds (P_3, the four-dimensional c), and ct subtracts (D, the four-dimensional d)

Line	Statement	Derivation	Origin	Her use, where it differs
⑯	(1, 1): diagonal $\sqrt{2}$	① with legs 1 and 1	YBC 7289	—
⑰	$(\frac{1}{2}\sqrt{2}, \frac{1}{2}\sqrt{2})$: diagonal 1	$\sqrt{(\frac{1}{2} + \frac{1}{2})} = 1$, the point (cos 45°, sin 45°)	Euler (1748)	—
⑱	(1, 1, 1): diagonal $\sqrt{3}$	① twice	Euclid XIII.15	Euclid fits the cube in a sphere; here it is nested Pythagoras
⑲	$(\frac{1}{3}\sqrt{3}, \frac{1}{3}\sqrt{3}, \frac{1}{3}\sqrt{3})$: diagonal 1	$\sqrt{(\frac{1}{3} + \frac{1}{3} + \frac{1}{3})} = 1$	unit vectors: Hamilton (1843)	the cube's diagonal shrunk to length 1; each entry is a direction cosine, cos 54.74°, the case ⑰ one dimension up
⑳	R, the distance from a centre to a point	definition	the radius: Euclid I, Def. 15	a ladder of radii from one centre, to a face, an edge, a corner and a four-dimensional corner, each the radius of a nested sphere
def	$P_0 = x$	one axis	Descartes (1637)	—
㉑	$P_1 = \sqrt{(x^2 + y^2)}$	$x \perp y$, so ①	Descartes (1637)	—
㉒a	$P_2 = \sqrt{(P_1^2 + z^2)}$	$z \perp P_1$, so ① with legs P_1 and z	three-dimensional coordinates: Clairaut (1731)	—
㉒b	$P_3 = \sqrt{(P_2^2 + w^2)}$	$w \perp$ the whole xyz space	Cayley (1845); Schläfli (1852)	—
㉒c–e	(1, 1, 1, 1): $P_1 = \sqrt{2}$, $P_2 = \sqrt{3}$, $P_3 = \sqrt{4} = 2$	substitution	YBC 7289; Euclid XIII.15; the word tesseract: Hinton (1888)	the four-dimensional diagonal is a whole number, twice the edge

Line	Statement	Derivation	Origin	Her use, where it differs
㉒f–g	(⅓√3, ⅓√3, ⅓√3): $P_1 = \sqrt{(2/3)}$, $P_2 = 1$	substitution	Hamilton (1843)	confirms ⑲
㉓	$P_n = \sqrt{(P_{n-1}^2 + x_n^2)}$	induction on ㉒a	Grassmann (1844); Cayley (1845); Schläfli (1852)	—
㉔	$D^2 = (ct)^2 - P_2^2$	factor as $(ct + P_2)(ct - P_2)$; a moving observer multiplies one factor by k and divides the other by k	Minkowski (1908); the factor k: Milne (1935), named by Bondi (1964)	the k argument covers motion along one line. Reading D as the page's d, with ct as the long leg, is hers; the algebra is correct
㉕a	$R(k+1) = \sqrt{(R(k)^2 + K^2)}$	each step out from the centre adds one perpendicular K	regular figures in any dimension: Schläfli (1852)	the P chain run from the centre at half scale, $R_n = K\sqrt{k}$: it climbs the dimensions without leaving the cube
㉕b–e	$R_1 = K$, $R_2 = K\sqrt{2}$, $R_3 = K\sqrt{3}$, $R_4 = 2K$	① repeated	Euclid IV.8, I.47 and XIII.15; Hinton (1888)	the distance to the four-dimensional corner equals the edge
✓	a = 4, b = 2: 20 + 12 = 32 = 2 · 16 = 4 · 8	—	the author (2026)	her example

Her chart for the two tables, markers ㉖ to ㊿, is set out in section 3.4.

Standing. M for every line, with the sources given. **N** for the page as one translation key (section 3.11).

3.11.2 The other pages, read through the fold

Several pages written before or apart from the Rosetta page turn out to be parts of its figure. Each relation below was checked in code (added).

- **The 3-4-5 page's squares.** The coloured squares at the foot of the page are an earlier version of the Rosetta figure:

    - red $a^2/2$ is S;
    - green $(d^2 - e^2)/2$ is N;
    - blue $b^2/2$ and yellow $e^2/2$ together make M, so $M = \frac{1}{2}b^2 + \frac{1}{2}b \cdot h$.

  Her identities $c^2 - d^2 = 2b^2$ and $c^2 = 2a^2 - d^2$ are lines ④ and ⑦ at (4, 3).
- **The lock run backwards.** With the leftover h, $a = (d^2/h + h)/2$ and $b = (d^2/h - h)/2$, which is R inverted; checked for (4, 3), (4, 2) and (7, 3). The areas fix the angle: $\tan A = a/b = S/M$ and $N/M = \tan A - 1$, while S sets the size. That is her "scaling info is retained".
- **The √2 grid page.** "A $\sqrt{2} \times \sqrt{2}$ grid is inverse of 1×1 grid" is R: a 45° turn with a $\sqrt{2}$ stretch carries the unit grid onto the $\sqrt{2}$ grid, and a second gives the doubled grid.
    - Her SSA count refines to this: (d, h) fix the triangle once whenever $h > 0$.
    - At $h = 0$, the 45° triangle, the size is lost; this is her null case.
    - Her dual is the choice of which leg is longer, which the absolute-value bars remove.

  So there is one solution, one null case and one choice of branch.
- **The converter page.** The cube of edge 2 has edge 2, face diagonal $2\sqrt{2} = \sqrt{8}$ and body diagonal $2\sqrt{3} = \sqrt{12}$. These are the key cell's b, g and d at $a = 4$, $b = 2$, and they line up only because $a = 2b$.
- **The pyramid page's ladder.** Its squares ½, 5/4 and 2 are the ladder $g^2 \to c^2 \to f^2$, in steps of $d^2 = \frac{3}{4}$, for the triangle (1, ½): the key cell at quarter scale.
    - The point (1, ½) sits at 26.57°, not on R's fold line at 22.5°, which meets the edge at height $\sqrt{2} - 1 = 0.414$. Halving the angle does not halve the side.
    - Doubling 26.57° gives 53.13°, the angle of the 3-4-5 triangle.
- **The cube engine of her reasoning framework.**
    - Its variables are Rosetta ratios: $d = a/b - 1 = N/M$ and $f = 1 - b/a = N/S$.
    - Its budget triangle (b, a − b) has legs b and h and area $\frac{1}{2}b \cdot h = d^2/2 - N$, the 3-4-5 page's yellow piece.
    - Its pyramid lift (b, b, a) has length $\sqrt{(g^2 + a^2)}$: the converter with $P_1 = g$.
    - Its inscribed and circumscribed spheres, of radius ½ and $\sqrt{3}/2$, are K and f/2 one dimension up.
- **The cognition papers.**
    - The 2 + 2 demonstration ends at (4, 0), the sum and the difference: R(2, 2), the a = b cell. With unequal hands a move of fingers keeps only the sum.
    - Tammet's product $53 \times 131 = 6943 = 92^2 - 39^2$ is quarter-squares multiplication, $4ab = (a + b)^2 - (a - b)^2$. In the figure, $M = [(a + b)^2 - (a - b)^2]/8$.
- **Three related triangles.** *Small Pattern, Big Deal* calls (3, 1), (5, 3) and (4, 2) related by their prime families. They lie in different families (√2, the whole number 4, √3). What they share is $h = 2$, so $d^2 = 2(a + b)$ and $d = 2\sqrt{(\text{mean leg})}$. In general a fixed leftover h gives $d^2 = h(2b + h)$.
- **A sphere in a cube.** *Root Patterns* (Taylor 2026e) says the points of a sphere in a cube match the root families. The distances from the centre to the face centres, edge midpoints and corners are $(s/2)(1, \sqrt{2}, \sqrt{3})$; in two dimensions they are K and f/2.
- **What the ascii notation cannot yet write.** It writes the Rosetta square and its diagonal f, with / and \. It has no mark for the point that splits an edge into b and h, which c, M and N need, or for the centre, which K needs. Two more delimiters, an edge point and a centre, would let it encode the whole page.

3.12 The delimiter cipher

Her page *ascii geometry* builds the cube from a square:

“First a square by rotating edges `|+_-|  |-_×|  |×_÷|  |÷_+|` Now a square with diagonals `|+/+_-\-|  |-/-_×\×|  |×/×_÷\÷|  |÷/÷_+\+|` Understanding the Diagonals create this shape X where lines of the square are repeated relative to their vertices delimiter (+ - × ÷). The delimiters are in this order because to rotate, the vertex must move in the same relation to the other vertices. All Edges are the same length with each other. All diagonals are equal lengths to each other. Understand which information is being repeated. the vertices are not separate instances. They are literally the same point, but we are expanding information around them to gain higher definition of the space and consistency under rotation. Edges with the same delimiters are the same, if the delimiters trade places, the image is now reversed. the square logic is a face of the cube.”

The six faces, verbatim:

```
Top          +|+_-|-|-_×|×|×_*|*|*_+|+
Front        +|+_’|’|’_”|”|”_-|-|-_+|+
Right Side   -|-_”|”|”_,|,|,_×|×|×_-|-
Left Side    *|*_.|.|._’|’|’_+|+|+_*|*
Back         ×|×_*|*|*_.|.|._,|,|,_×|×
Bottom       ‘|’_”|”|”_,|,|,_.|.|._’|’
```

Each string walks a face's four corners and returns to its start: a corner is written x|x, and “_” joins two corners along an edge. The Bottom string opens with ‘, read as ’. The strings fix every corner (Table 13, read off by Claude).

Table 13. The eight corners. Coordinates are on the unit cube, with Top at z = 1 and Front at y = 0.

Delimiter	(x, y, z)	Neighbours	Opposite corner
+	(0, 0, 1)	−, *, ’	,
−	(1, 0, 1)	+, ×, ”	.
×	(1, 1, 1)	−, *, ,	’
*	(0, 1, 1)	+, ×, .	”
’	(0, 0, 0)	+, ”, .	×
”	(1, 0, 0)	−, ’, ,	*
,	(1, 1, 0)	×, ”, .	+
.	(0, 1, 0)	*, ’, ,	−

What the strings hold.

- **Two tetrahedra.** The corners with an odd coordinate sum, {”, ., +, ×}, form one regular tetrahedron, with every edge a face diagonal. The four even corners form its dual.
- **Orientation.** Read by the right-hand rule, Top, Front, Right and Left are written outward and Back and Bottom inward.

- **The turns** (checked by script, section 4). Front and Back are the two four-cycles of a quarter turn about the Front–Back axis. Back and Bottom are the point-inverted Front and Top, as written. Left is the inverted Right, reversed.

Standing.

- **M.** Her reversal rule is the orientation of an edge, [a, b] = −[b, a], in algebraic topology, reached independently. Eight symbols on the corners of a cube have a precedent in Leibniz's 1703 reading of the eight trigrams as the binary numbers 0 to 7.
- **N.** The delimiter notation, its face strings, and its rule that a repeated vertex is one point "expanded" for definition under rotation were not found.

3.13 A notebook of axes and derivatives (February 2026)

Her notebook *Rosetta Octants* (Taylor 2026b; 17 pages, 24 February 2026) is a Rosetta of another kind: the axes and rates of several fields, set side by side. Its contents:

- **Five axis systems on one cube.** Drawn from (0, 0, 0) to (1, 1, 1), with the signs of every octant marked:
  - "Quadrant system (x=width, [y=length or height])";
  - "Octant/Cartesian system (Math)";
  - "Render system (Blender/CAD)";
  - "Gaming system (Video Games) (x=right, y=forward, z=up)";
  - "Sphere/Circle system (2D: x=r cosθ, y=r sinθ) (3D: x=r cosθ sinϕ, y=r sinθ sinϕ, z=r cosϕ)".
- **Definitions in one list.** Height, the hypotenuse $\sqrt{(a^2 + b^2)}$, magnitude in two and three dimensions, the dot product $x_1y_1 + x_2y_2 + x_3y_3 = |a||b| \cos \theta$, area l·w, volume l·w·h, velocity as length over time and acceleration as length over time squared. One line reads: "time always grows to the right (on x axis) on a graph".
- **Forces as directions of growth**, drawn on a tire:
  - gravity "grows toward Earth's core";
  - the normal force "in opposite direction of objects' contacting surfaces";
  - tension "toward a physical tether";
  - friction "away from objects' momentum direction";
  - the centripetal force "toward the center of rotation and is opposed by inertia/centrifugal direction".
- **The derivative as positions over instants.**
  - "d/dx when x is Δ instants"; "dx/dy when y is Δ positions".
  - "Δ in instants ≠ one instant".
  - "acceleration = Δ positions/Δ instants".
  - "To find a position at an instant, the position and instant must both change … An object cannot change position without changing the instant."
  - The chain rule appears as nested sets: f′(g(x)) is the "change in 2 positions − set of positions inside the set / change between first set's instants".
- **A teaching example.** A race track with a hairpin:
  - f(x) is track length over start-to-finish time, the average speed;

    - f′ is the change in speed over the run;
    - f″ is how that change varies over several runs;
    - "prime = speed, 2nd = accel, 3rd = jerk, 4th = snap";
    - the hairpin is an inner function g.

  In her words: "GET GOOD! How to figure out where you've been slowing down using derivatives."
- **Newton's laws**, with her glosses.
- **Keeping dimensions in ratio.** A rock is thrown up at 5 m/s on a planet without air and ends 20 below its start at 15 m/s.
    - $v_f^2 = v_0^2 + 2a\Delta y$ gives $225 = 25 + 40a$, so $a = 5$.
    - Her ratio method reaches the same 5 by halving 20 : 10 : 5.

Checks (added).

- **Units.** The worked example is right if the 20 is in metres. As written, 20 ft with speeds in m/s, the acceleration would be 16.4 m/s².
- **The halving chain.** It gives the right answer here because the time taken is $\Delta y/\bar{v} = 20/10 = 2$ s, so each step divides by 2. In general the time is $2\Delta y/(v_0 + v_f)$, and $a = (v_f - v_0)$ divided by it.
- **The first law.** Her gloss, "constant speed + Fnet = zero Is constant non-zero acceleration (INERTIA)", conflicts with the law: zero net force means constant velocity and zero acceleration (Newton 1687).
- **The sheared rectangles.** Two pages explore the derivative of x^2 as the change of an area, with sheared rectangles of sides b and b′ ("(x)(x) : x + x", a product beside its derivative), without reaching a stated result. The standard picture is a square growing by two strips, $d(x^2) = 2x\,dx$.
- **Conventions.** Axis conventions differ between programs: Blender puts z up, and many game engines put y up. So a chart like hers is a working translation table, which is how she uses it.

Why it matters here. The notebook separates the derivative over instants from the derivative over positions eight months before the calculator, where J is the derivative of time and K the derivative of scale (section 4.4).

Standing.

- **M.** All of its physics is standard.
- **N.** The chart of five axis systems as one translation table, and the split between instants and positions as the seed of the time and scale turns, are hers; the second is a reading.

3.14 A classifier of regimes

On 2 October 2026 the author shared *Revisions 2.0*, with two accompanying pieces, *Lateral Structures* and *Stuff*. Together they define one sign rule for every domain:

```
\Delta_{\rm master}=n-\frac{1}{\Lambda^{2}}\sum_i
x_i^{2}-\log\frac{1}{\varepsilon}-D_{\rm domain}
```

Its classes. $\Delta > 0$ gives "2 viable configurations (dual)", $\Delta = 0$ gives "1 critical configuration (boundary / turning)", and $\Delta < 0$ gives "0 viable configurations (forbidden)". In her words: "We are not changing any physical law. We're giving a single, dimensionless classifier that you can map existing laws into."

The same 2/1/0 structure elsewhere. It appears in:

- the quadratic discriminant: two real roots, one, or none;
- the damped oscillator's $c^2 - 4mk$: under-damped, critically damped or over-damped;
- the WKB sign of $E - V$: oscillatory, a turning point, or evanescent;
- the spacetime interval: spacelike, lightlike or timelike;
- the saddle-node bifurcation.

Its disclaimer holds, and the sum and difference tables with their zero line are its case n = 2.

Three fixes (added).

1. **Tie it to the cube.** $\Delta = n - \Sigma\xi_i^2$ vanishes at $|\xi| = \sqrt{n}$, the corner of the unit n-cube, so Λ should be the cube's edge and the critical surface the sphere through its corners. Its oscillator examples set Λ to the orbit's radius, which gives $\Delta = n - 1$ on the orbit rather than 0. Either drop n or set $\Lambda = r/\sqrt{n}$.
2. **Weight the domains.** D_domain adds unlike dimensionless ratios, such as α, E/k_BT and GM/Rc^2, each with weight 1. Each needs its own justified weight.
3. **Count oscillatory solutions.** In the forbidden region of the Schrödinger equation there are still two solutions, one growing and one decaying. The count 2/1/0 is of oscillatory solutions.

Standing.

- **M.** Each of the regime classifications it collects is standard.
- **N.** One classifier with adapters for each domain.
- **D.** The three points fixed above.

4. The machine

The calculator never moves its cube. Every derivative, inverse or change of scale is a turn of the tesseract the cube belongs to, so each datum keeps the corner it came from, and each new datum names the corner and the circle it travelled. This section states the author's rules, then the turns that satisfy them; the turns were derived by Claude and checked by script. It goes on to her four structures in closed form and ends with the method of use that sections 5 to 14 follow.

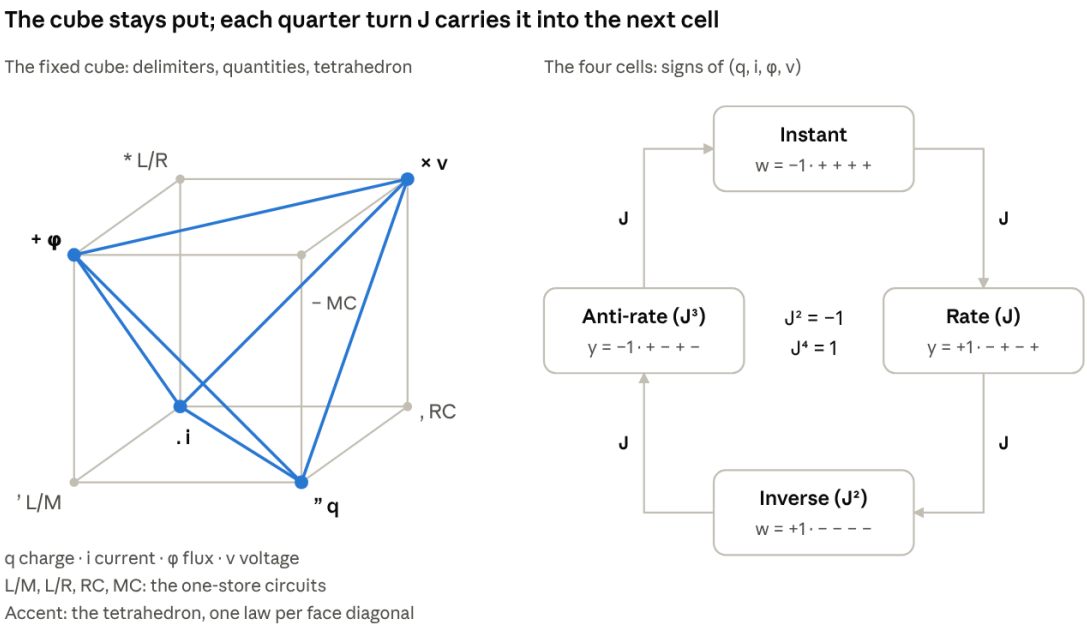


Figure 1 · the cube in its tesseract: eight corners, four cells, one quarter turn

Figure 1 shows the frame and the ring. Charge, current, flux and voltage sit on the cube's inscribed tetrahedron, with the one-store circuits on the other four corners. One quarter turn J carries the cube into the next cell, and two give the inverse, $J^2 = -1$. Neighbouring cells share one face: instant and rate share the Back face, rate and inverse the turned Front face, inverse and anti-rate the turned Back face, and anti-rate and instant the Front face.

4.1 The author's rules

These rules were given in the author's words while the calculator was built, and are quoted as written.

1. **Provenance.** A cube where “when we invert or rotate the cube we can keep track of exactly which component the new data originated from.”
2. **The fixed cube.** “the cube itself doesnt move...inverses become tesseracts...sphere has pi euler inversion...anything in motion starts as a single vertice on the cube...rather than changing size of the cube for derivatives, we are switching to another layer of tesseract but the data points should still remain anchored in the process...same for dimension inversions...for the pois, there should already observed data that lines up in ever intansantanous and motion of the cipher.”
3. **Time and scale.** “now you can also use this cipher applied as derivatives and inverses of time and scale without ever moving the cube.”
4. **Folding, not reversing.** “in the t-bar flip time isnt reversed neg or inverse on the mag isn't reverse...the same as rt2 to rt4 causes higher resolution, it decreases in reverse. info isnt lost just folded”
5. **Neighbours.** “if components don't fit, we should be able to find them by their adjascent data to the vertice....meaning we should be able to sort out the correct direction, flow, scale, time, motion, distance, and resolution or literally anything in unverse despite unknown dynamics because those work in ranges across the full structure”

6. **The vertex lock.** “if a data point is a vertice...then the 3 side lengths and subsequent 3 data points from should also line up.”
7. **The sphere.** Action rides the curve and inside is no action; the T-bar flip rotates between action (√2) and no action (√4).
8. **The reaction.** Put electrons “as a reaction in the fabric rather than a particle, and align the inductance and other components to their correct geometry.”
9. **Behavior.** “anomalous BEHAVIOR plays the most key role in the calculator, because while a number set may be ‘in range’ the practical real effects of what the numbers produce in every alignment do not. So we use the alignments that do not create down-wind mismatched observeables.” And: “We leave in success with proof, and leave out trials that failed.”

4.2 The frame on the corners

Four structures on the tetrahedron. Her four structures sit on the four tetrahedron corners, and they are the inverse and the derivative of scale and of time (section 4.7 gives each in closed form):

Corner	Structure	Formula	Operation	Electrical frame
”	Inversion	$x \mapsto \rho^2 x/r^2$	the inverse of scale	charge q
.	Dimension:derivative	S□₋₁ = dV□/dr	the derivative of scale	current i
+	Euler:π	e^(iπ) = −1	the inverse of time: half a cycle forward	flux φ
×	T-bar	T = 2π dJ/dE	the derivative of time	voltage v

What fixes the electrical frame. Three faces force it. Left is the load (inductance, φ = Li), Back the derivative (resistance, v = Ri), and Top the rate (Faraday's law, v = dφ/dt). Current is the corner Left and Back share, flux the one Left and Top share, voltage the one Back and Top share, and charge the last.

One law per face. Each face carries exactly one tetrahedron edge, on its diagonal, and each such edge is one circuit law:

Face	Her string	Edge	Law
Top	+ − × *	φ to v	Faraday: v = dφ/dt
Front	+ ’ ” −	φ to q	memristance: dφ = M dq
Right	− ” , ×	q to v	capacitance: q = Cv
Left	* . ’ +	i to φ	inductance: φ = Li
Back	× * . ,	v to i	resistance: v = Ri

Face	Her string	Edge	Law
Bottom	’ ” , .	q to i	i = dq/dt

The dual corners are the one-store circuits, each lacking the variable opposite it:

- * lacks charge (L/R);
- , lacks flux (RC);
- ’ lacks voltage (L/M);
- − lacks current (MC).

In the variable frame each is half the sum of its three variables minus the missing one; for example L/M = ½(q + i + φ − v).

Standing.

- **M.** The tetrahedron of q, i, φ and v with six laws on its edges is the bond-graph “tetrahedron of state” (Paynter, as described in the bond-graph literature) and Chua's memristor diagram (Chua 1971); Vongehr (2012) draws it with six relations.
- **N.** Placing it inside the cube, with the laws on face diagonals and the dual tetrahedron as the one-store circuits, was not found. The nearest are a relational graph of circuit elements (Shang et al. 2015) and a six-element star (Wang 2021).

4.3 The quarter turn J

Put the side-2 cube at w = −1, so its corners are (x, y, z, −1) with x, y, z = ±1. It is one of the eight cubic cells of the tesseract.

The frame fixes J (Claude's derivation). Lifted to w = −1, the corners carrying q, i, φ and v are four mutually perpendicular vectors of length 2. Two conditions remain:

- each store turns into its rate, q → i and φ → v;
- two quarter turns give the inverse, $J^2 = -1$.

They leave exactly one map:

```
J:\;(x,\,y,\,z,\,w)\;\mapsto\;(z,\,-w,\,-x,\,y),\qquad J^{2}=-1,\qquad J^{4}=1,\qquad \det J=+1
```

J is a signed permutation of the axes, so it carries the 16 corners and the 32 edges of the tesseract onto themselves.

What J does.

- **Two planes at once.** It turns the (x, z) plane a quarter turn about the Front–Back axis and, at the same time, turns the Front–Back axis y into the held axis w. This is her dimension inversion: after one quarter turn, the axis that was held is a cube axis.
- **A ring of four cells.** It carries the cube from w = −1 to the cell y = +1, then to w = +1, then to y = −1, and back; these are her layers instant, rate, inverse and anti-rate. The other four cells form a second ring, x = −1 → z = +1 → x = +1 → z = −1.
- **A circle for each corner.** Turning by an angle θ, cos θ + J sin θ, moves each corner on a great circle of radius 2. Four circles pass through the 16 corners, four corners each:

- the charge circle q → i → −q → −i;
- the flux circle φ → v → −φ → −v;
- two circuit circles, L/M → L/R → −L/M → −L/R and MC → RC → −MC → −RC.

A moving quantity starts at one corner, and that corner's circle is its path. Table 14 gives the four cells, and Table 15 follows each corner through the four layers.

Table 14. The four cells.

Layer	Cell	Signs of (q, i, φ, v)	Holds	Face shared with the next layer
Instant	w = −1, the cube	+ + + +	the stores q, φ on Front; the rates i, v on Back	Back: i, v, L/R, RC
Rate (J)	y = +1	− + − +	i, v with −q, −φ	turned Front: −q, −φ, −L/M, −MC
Inverse (J^2)	w = +1	− − − −	every corner turned over	turned Back: −i, −v, −L/R, −RC
Anti-rate (J^3)	y = −1	+ − + −	−i, −v with q, φ	Front: q, φ, L/M, MC

Table 15. Each corner through the four layers.

Corner	Frame	Instant	Rate (J)	Inverse (J^2)	Anti-rate (J^3)	Circle
”	charge	q	i, at .	−q	−i	charge
.	current	i	−q	−i	q	charge
+	flux	φ	v, at ×	−φ	−v	flux
×	voltage	v	−φ	−v	φ	flux
’	L/M	L/M	L/R, at *	−L/M	−L/R	first circuit
*	L/R	L/R	−L/M	−L/R	L/M	first circuit
,	RC	RC	−MC	−RC	MC	second circuit
−	MC	MC	RC, at ,	−MC	−RC	second circuit

Her strings already hold J.

- **The derivative is written in.** J sends the Front string + ’ ” − onto the Back string × * . , letter for letter, starting points included. Front holds the stores and Back the rates, so one quarter turn takes memristance, dφ = M dq, to its rate form v = Mi, a resistance law, as Chua (1971) noted for the memristor.

- **The inverse is written in as its shadow.** $J^2 = -1$ has determinant +1 in four dimensions: a rotation by π in two planes at once, not a mirror. From inside the cube it looks like the point inversion, of determinant −1, and her Back and Bottom strings are the point-inverted Front and Top.
- **For matter the half turn is CPT.** It turns the electrical variables over (C) and spacetime end for end (PT). In the proof of the CPT theorem, spacetime inversion is connected to the identity in the complex Lorentz group, though not in the real one (Jost, as described by Greenberg 2006). Her "sphere has pi euler inversion" is that step, reached independently.

4.4 Time and scale: the second turn and the eight cells

J and K. J carries " to . and + to ×, turning each inverse into its derivative. A second quarter turn K carries " to + and × to ., trading scale for time.

Table 16. The turns in coordinates (checked by script).

Turn	(x, y, z, w) ↦	Order	Carries the cube
J	(z, −w, −x, y)	4	across the Back face
K	(w, z, −y, −x)	4	across the Left face
JK	(−y, x, −w, z)	4	across the Top face
−1	(−x, −y, −z, −w)	2	into the opposite cell
T	(−w, z, y, −x)	2	across the Right face, keeping × and "

What K does.

- **Scale becomes phase.** $K^2 = (JK)^2 = -1$ and JK = −KJ. On ln z = ln r + iθ, the log of a size and a phase, K's move is $z \to z^{i}$: the log of the size becomes the phase, and the phase becomes minus the log of the size. Two such turns give $z \to 1/z$.
- **Cauchy–Riemann.** For analytic quantities the two derivatives are tied the same way, $\partial/\partial\theta = i \cdot r\,\partial/\partial r$.
- **The eight cells are the quaternion group.** J and K generate {±1, ±J, ±K, ±JK}, whose eight elements carry the cube into the eight cells, one each.
- **Inverting one pair.** Turning over the two variables a face does not hold carries the cube across that face and leaves the face in place: (q, i) crosses Top, (φ, v) Bottom, (q, φ) Back, (i, v) Front, (q, v) Left and (i, φ) Right. Inverting scale alone crosses Top, time alone crosses Bottom, and both together is −1.

The size flip is a turn. For the electron, J multiplies by the time scale $\omega = m_e c^2/h$ and K by the scale $R_0 = h/e^2$, so its four corners are e, eω, eR_0 and $e\omega R_0$, which are q, i, φ and v.

- **Two radii set both scales.** $\omega = c/(2\pi\sqrt{(r_e a_0)})$ comes from their geometric mean, the reduced Compton wavelength ƛ̄. $R_0 = (Z_0/2)\sqrt{(a_0/r_e)}$ comes from their ratio, $1/\alpha^2$.
- **K is the flip.** K carries the Right face, the capacitor at r_e = 2.818 fm, onto the Left face, the inductor at $a_0 = ƛ̄^2/r_e$ = 52.92 pm: the flip through the Compton sphere.

- **The flip keeps the circuit.** Every capacitor of radius r paired with an inductor of radius $\bar{\lambda}^2/r$ has $\sqrt{(LC)} = h/m_e c^2$, and impedance $\sqrt{(L/C)} = (Z_0/2)(\bar{\lambda}/r)$: 25,812.807 Ω at r_e, 188.365 Ω at $\bar{\lambda}$ and 1.37457 Ω at a_0.

The cube keeps its size, and the Right string comes back as the Left string read the other way round: the reversed image her rule gives when delimiters trade places.

Gravity's half turn is both inverses. In isotropic coordinates, $\rho \to (GM/2c^2)^2/\rho$ keeps the Schwarzschild space metric, turns the lapse over and exchanges the two exteriors, with the horizon as its fixed sphere (Popławski 2010). It is the scale flip and the time inverse together. In Kruskal's coordinates (Kruskal 1960) the quarter turn $(T, X) \to (X, -T)$ carries region I to II, II to III, III to IV and IV back to I, and the half turn is an exact isometry swapping I with III (checked by script).

One circle alone is not a turn. A quarter turn of the time circle alone (+ → ×), or of the scale circle alone (" → ."), keeps the four variables but is not a symmetry of the tesseract: it swaps the circuits with the cell centres of the 24-cell (checked). J turns both circles at once, as Cauchy–Riemann ties the two derivatives.

4.4.1 Four dimensions: the interval, proper time, rates and the T-bar flip

1. The converter extends to four dimensions (section 3.7). Its unit diagonal is 2, and split into two planes it is $\sqrt{((\sqrt{2})^2 + (\sqrt{2})^2)} = 2$: the $\sqrt{2}$ of each plane squares out.

2. Time enters as a difference. For motion along x, put $u = ct + x$ and $v = ct - x$, so $s^2 = (ct)^2 - x^2 = uv$: the Rosetta $d^2 = h \times$ (Manhattan path), with R's two coordinates.

- **The boost.** A boost at speed βc gives $u' = \gamma(1 - \beta)u = u/k$ and $v' = \gamma(1 + \beta)v = kv$, with $k = \sqrt{((1 + \beta)/(1 - \beta))}$. One coordinate stretches and the other shrinks by the same factor, so their product, the difference of squares, stays fixed while the sum $(ct)^2 + x^2$ changes.
- **A check.** ct = 5, x = 3 and β = 0.6 give k = 2 and $s^2 = 16$ before and after.
- **Light.** s = 0 means ct = x: the 45° zero-difference family.

3. Proper time is the antiderivative of the difference root.

```
(c\,d\tau)^{2}=(c\,dt)^{2}-dP^{2}\;\Rightarrow\;\tau=\int\sqrt{1-v^{2}/c^{2}}\;dt
```

Ten seconds at 0.6c give τ = 8 s.

4. Rates pass through the converter level by level. Differentiating $P_n^2 = P_{n-1}^2 + x_n^2$ gives

```
\dot P_{n}=\frac{P_{n-1}\dot P_{n-1}+x_{n}\dot x_{n}}{P_{n}}\qquad(P_{n}\neq 0)
```

On the diagonal, with every $x_i = s(t)$, $P_n = \sqrt{n} \cdot s$, so every derivative and every integral is scaled by $\sqrt{n}$: by exactly 2 in four dimensions.

5. The T-bar flip rate in sums and differences of squares. A box with sides x > y > z spinning at rate Ω about its middle axis.

1. **Moments.** A uniform box has $I_1 = m(y^2 + z^2)/12 = m\,c^2_yz/12$, $I_2 = m\,c^2_xz/12$ and $I_3 = m\,c^2_xy/12$, so $I_1 < I_2 < I_3$.
2. **Gaps.** $I_2 - I_1 = m\,d^2_xy/12$ and $I_3 - I_2 = m\,d^2_yz/12$.
3. **Euler's equations without torque.** $I_1\dot{\omega}_1 = (I_2 - I_3)\omega_2\omega_3$, $I_2\dot{\omega}_2 = (I_3 - I_1)\omega_3\omega_1$, $I_3\dot{\omega}_3 = (I_1 - I_2)\omega_1\omega_2$.

4. **Near the middle axis.** With ω = (small, Ω, small), $\dot{\omega_2} \approx 0$ and $\ddot{\omega_1} = [(I_3 - I_2)(I_2 - I_1)/(I_1 I_3)]\Omega^2\omega_1 = \lambda^2\omega_1$. Since $\lambda^2 > 0$ a small wobble grows exponentially and the body flips. About the long and short axes the same product is negative, and they only wobble.
5. **In her quantities.** The factors m/12 cancel, and with $\cos 2\theta = (1 - \tan^2\theta)/(1 + \tan^2\theta) = d^2/c^2$ for a face of angle θ,

   ```
   \lambda=\Omega\sqrt{\frac{d^{2}_{xy}}{c^{2}_{xy}}\cdot\frac{d^{2}_{yz}}{c^{2}_{yz}}}=\Omega\sqrt{\cos 2\theta_{xy}\,\cos 2\theta_{yz}}
   ```
6. **As one difference of squares.** $(I_3 - I_2)(I_2 - I_1) = h^2 - \delta^2$, with $h = (I_3 - I_1)/2$ and $\delta = I_2 - (I_1 + I_3)/2$.
7. **The cube.** At x = y = z every d^2 is 0, so $\lambda = 0$: no flip.
8. **The path.** In the body frame $\dot{L} = L \times \omega$, so $L \cdot \dot{L} = 0$ and $|L|$ is constant: a sphere. The energy is also constant: an ellipsoid. The motion follows their intersection.

For the box (3, 2, 1): $\lambda^2 = (5/13)(3/5) = 3/13$, so $\lambda = 0.4804\Omega$. Integration of Euler's equations gives a growth rate of 0.4808Ω. Started 1% off the middle axis, the middle component of ω reverses 8 times in 200 time units ($\Omega = 1$). Two corrections to her picture:

- the flip is about the middle principal axis, not about the diagonal f;
- the exact flip period needs an elliptic integral (section 4.7.4).

Standing.

- **M.** The converter (Pythagoras), the interval, light-cone coordinates and boosts, proper time, Euler's equations and the intermediate-axis theorem.
- **N.** The translation into her terms:
  - the boost as R's two coordinates, with $s^2 = h \times$ Manhattan;
  - light as the zero-difference family;
  - the flip rate written wholly in face sums and differences, as a product of $\cos 2\theta$ and as $h^2 - \delta^2$.

4.5 Motion reversal and folding

Motion reversal, not time reversal. The operation physics calls time reversal turns over current and flux and keeps charge and voltage, which is the map T of Table 16. T is a tesseract symmetry of order 2 and determinant +1. It carries the cube across the Right face, keeps the capacitance law, and turns J into $J^{-1} = J^3$: the same cycle met the other way round, three quarter turns forward. The clock runs forward throughout, as the T-bar's flip turns a body upside down without running time backwards. A negative or an inverse is a turn, not a reversal (rule 4).

Folding. A coarser view folds the information rather than losing it; the $\sqrt{2} \to \sqrt{4}$ swing raises resolution and the reverse lowers it. Two applications follow:

- **Lightning.** Hill (1968) measured channel turns averaging 17.3°, Gaussian about zero, with a mean turn that grows only as the section length to the power 0.1, and Fourier amplitudes falling as $1/\sqrt{n}$, a 1/f power spectrum. That is the staircase of section 6.4 with legs that fold as the view coarsens: the same mean turn at every length and a 1/f spectrum. The earlier reading, with legs of fixed length, predicted a $1/\sqrt{L}$ fall that Hill's data do not show.
- **The proton's spin legs** (section 12.2).

4.6 Circles, the torus and the spheres in the cube

- **Her torus.** J turns the (x, z) and (y, w) planes together, so every corner keeps |(x, z)| = |(y, w)| = √2. All 16 corners and all four circles lie on the torus of two √2 circles on the √4 sphere, the torus she described, with area $(2\pi\sqrt{2})^2 = 8\pi^2 = 78.957$. That is the same $8\pi^2$ that divides the load per phase (section 7.2); the equality is a fact, and its physical tie is a reading.
- **Linked circles.** The four circles are fibres of the Hopf fibration that J defines: great circles that stay a fixed distance apart all the way round, 2√2 for the charge and flux circles.
- **The 24-cell.** Add the eight points where the axes meet the √4 sphere, and the 24 points form a 24-cell, each with eight nearest neighbours. In the variable frame they split into three 16-cells: the variables ±q, ±i, ±φ, ±v; the circuits ½(±q ± i ± φ ± v) with an odd number of minus signs; and the cells, with an even number. J keeps each set.
- **The spheres in the cube.** They have radius 1 through the six face centres ("no action inside"), √2 through the twelve edge midpoints (the action circles), √3 through the eight corners (the delimiters), and 2 through the sixteen tesseract corners (the held 4D corner). The ball fills $\pi/6 = 0.5236$ of the 3D cube and $\pi^2/32 = 0.3084$ of the side-2 tesseract.
- **The sphere meets each law at its middle.** The inscribed sphere touches each face at its centre, the midpoint of that face's tetrahedron edge, so it touches every circuit law halfway between its two variables.

Standing of the mechanics.

- **M.** Each part is known:
  - a time derivative multiplies a phasor by iω, a quarter turn (Steinmetz 1893);
  - quarter turns with $J^2 = -1$ in four dimensions are isoclinic rotations, products with unit quaternions (Hamilton 1843);
  - the inverse as a rotation is the complex-Lorentz step in the proof of the CPT theorem;
  - Kruskal's four regions; the Clifford torus and the Hopf fibration (Hopf 1931); the 24-cell as three 16-cells (Coxeter 1973).
- **N.** Their assembly as a cipher with a fixed cube, cells as layers and data anchored at corners, the turn fixed uniquely by the electrical frame, the string property and the time-and-scale cells were not found in the literature searched.

4.7 The four structures in closed form

Section 4.2 puts the author's four structures on the four corners of the tetrahedron: inversion (the inverse of scale), dimension:derivative (the derivative of scale), Euler:π (the inverse of time) and the T-bar flip (the derivative of time).

Here each structure is computed exactly and set beside published observations that realise it. One rate ties them together: the T-bar's flip rate λ is an inverted oscillator. Classically it sets a logarithmic hold time; at a horizon it sets the temperature $\hbar\lambda/2\pi k$, observed at 0.35 nK with no free parameters. The scripts are:

- numbers.py, SHA-256 b78bd1e947eead172a6a1152a55e899d4fd793ebd1514a06220718640d301523;
- inversion.py, SHA-256 9f97f4f463914c37f34aac114f69f0c95a4de25dd68d5114ec4e07004aafb7fc;

- tbar.py, SHA-256 115e0b7f11b52083d2da34ba71b8c9db1637a50fc78ac7ced36cc4e7e35fb118.

The observation papers of Table 17 were read in full from their arXiv texts on 8 October 2026.

Table 17. Twelve observations that realise the four structures.

Structure	Observation	What realises the structure	Result
Inversion	Maxwell's fish eye (Ma et al. 2011)	the field is the source minus its inversion image, E(z) − E(1/z̄), focused at the antipode of a virtual sphere	outlets 0.2λ apart resolved, below the 0.5λ diffraction limit
Inversion	the λ-point in orbit (Lipa et al. 2003), against the bootstrap (Chester et al. 2020)	a conformal, inversion-symmetric critical point; the theory is built on that symmetry	measured ν = 0.6709(1), bootstrap 0.67175(10): 6.1σ apart, open
Dimension:derivative	torsion balance (Lee et al. 2020)	potential r^(2−n) with n = 3	$1/r^2$ holds from 3.0 mm down to 52 μm; any force of gravitational strength has range below 38.6 μm (95%)
Dimension:derivative	GW170817 (Pardo et al. 2018)	wave amplitude falls as d^(−(D−2)/2)	D = 4.02 +0.07/−0.10, +0.2σ from 4
Euler:π	graphene transistors (Schmitt et al. 2023)	pair creation ∝ e^(−πV⍰/V), the Schwinger form	G_0 = 0.18 ± 0.02 mS against 0.186 mS predicted, −0.3σ
Euler:π	superfluid bubbles (Zenesini et al. 2024; Cominotti et al. 2025)	decay time ∝ e^(E_c/kT), the instanton	exponential in the barrier, with the finite-temperature instanton dependence
Euler:π	an analogue black hole (Muñoz de Nova et al. 2019)	T = ħg/2πck at the sonic horizon	thermal at the predicted 0.35 nK, with no free parameters
Euler:π	an Unruh simulation (Hu et al. 2019)	T = ħA/2πck for an accelerated frame	temperature and entropy as Unruh predicts

Structure	Observation	What realises the structure	Result
T-bar	a phone's gyroscope (Wheatland et al. 2021)	one 360° turn with a 180° flip; the spin's sign changes as e^(iπ) = −1	ω_x goes from about +17 to −17 rad/s
T-bar	asymmetric-top molecules (Hamraoui et al. 2018)	time near the separatrix grows as the logarithm of one over the distance	rotation number $\alpha - \beta$ ln(distance); the quantum value converges to the classical one, computed from molecular constants
T-bar	an inverted oscillator in a condensate (Ji et al. 2026, preprint)	the middle axis as an inverted oscillator: stretch one way, squeeze the other	10.6(1.3) dB below vacuum; the ellipse angle matches with no free parameters; time reversal restores the state
T-bar	LIGO's fourth run (Capote et al. 2025)	a squeeze ellipse of fixed area turned a quarter turn across the band	up to 5.2 dB (Hanford) and 6.1 dB (Livingston) less quantum noise

Two related papers are theory or proposal, not observation: quantum persistent flips (Ma et al. 2020) and flips in magnetars (Wada and Shimoda 2024).

4.7.1 The full cube inversion overlap

The overlap shrinks level by level and closes exactly at the corner sphere. Inverted in its circumsphere, the cube lands wholly outside itself and keeps only its corners.

The structure. Inversion in a sphere of radius ρ sends each face plane $x_i = 1$ to a sphere through the centre, of radius $\rho^2/2$, centred at $(\rho^2/2)e_i$. Each level of the cube gives a radius at which the images pass through its points:

- at $\rho = 1$, the six images have radius ½ and sit inside the cube (the flower);
- at $\rho = \sqrt{2}$, they have radius 1, centred on each face centre and passing through its edge midpoints;
- at $\rho = \sqrt{3}$, they have radius 3/2 and pass through each face's corners.

```
x\mapsto\frac{\rho^{2}x}{|x|^{2}},\qquad O_n(\rho)=\Big\{x\in[-1,1]^n:\
|x|^{2}>\rho^{2}\max_i|x_i|\Big\}
```

The overlap O is the part of the cube outside all 2n image balls, and the inversion maps it onto itself: it is the cube's inversion-invariant core. Along a direction u with $m = \max|u_i|$ it runs from $r = \rho^2 m$ to $r = 1/m$,

so it exists only while $\rho^2 m^2 < 1$. Since $m^2 \geq 1/n$, it closes at $\rho^2 = n$. Table 18 gives its volume as a share of the cube.

$$\mathrm{Vol}\,O_n(\rho)=\frac{1}{n}\int_{\rho^{2}m^{2}<1}\Big(m^{-n}-\rho^{2n}m^{n}\Big)\,d\Omega,\qquad m=\max_i|u_i|$$

Table 18. The share of the cube (2^n) that survives its own inversion. Here A = arctan(1/√2) = 35.26°, the angle between an edge direction (√2) and a corner direction (√3).

Dimension	ρ = 1 (faces)	ρ = √2 (edges)	ρ = √3	ρ = 2
2D	$(3 - \pi/2)/4$ = 0.357301	0 (the corners in 2D)	—	—
3D	$23/24 - (5\sqrt{2}/16)A$ = 0.686327	$2/3 + \pi/2 - (5\sqrt{2}/2)A$ = 0.061414	0 (the corners)	—
4D	$191/192 - \pi^2/256 - 13\sqrt{3}\pi/864$ = 0.874366	$11/12 + \pi^2/16 - 13\sqrt{3}\pi/54$ = 0.223551	0.0078342 (no closed form found)	0 (the corners)

The flower volumes. At ρ = 1 they are:

- $F_2 = 1 + \pi/2$;
- $F_3 = 1/3 + (5\sqrt{2}/2)A$;
- $F_4 = 1/12 + \pi^2/16 + 13\sqrt{3}\pi/54$.

Their rational parts, 1, 1/3 and 1/12, are 2/n! in all three cases. Every closed form matches direct integration to 25 digits, and Monte Carlo with 4×10^7 points matches every entry within 1.3σ.

Observation: the inversion as an instrument. Maxwell's fish eye has index $n = 2/(1 + r^2/a^2)$, so waves in a plane move as if on a sphere (the stereographic map). With a mirror at r = a, the inversion's fixed circle, the exact field is the source term minus its inversion image, $E(z) - E(1/\bar{z})$. Ma et al. (2011) imaged 3 cm microwaves and resolved two outlets 0.2λ apart, below the 0.5λ diffraction limit.

4.7.2 Dimensions:derivatives

Going up two dimensions costs exactly one full turn. The step ratio at n = 2, 3 and 4 is π, 2π/3 and π/2: the half-turn, the third-turn and the quarter-turn.

The structure. The boundary is the derivative of the volume, and two dimension steps multiply by 2π/n:

$$S_{n-1}=\frac{dV_n}{dr}\Big|_{r=1}=n\,V_n,\qquad V_n=\frac{2\pi}{n}\,V_{n-2}\;\Rightarrow\;S_{n-1}=2\pi\,V_{n-2}$$

n	Ball V_n	Sphere S_{n-1}	Step V_n/V_{n-2} = 2π/n	Move
2	π	2π	π (180°)	$-1 = e^{i\pi}$
3	4π/3	4π	2π/3 (120°)	$\omega = e^{2\pi i/3}$

n	Ball V□	Sphere S□₋₁	Step V□/V□₋₂ = 2π/n	Move
4	π²/2	2π²	π/2 (90°)	i = e^(iπ/2)

The Kelvin weight. In n dimensions, inversion keeps harmonic fields harmonic only with the Kelvin weight |x|^(2−n). So a point source's field is the inverted constant: ln r in two dimensions, 1/r in three, 1/r² in four. In four dimensions the propagator is 1/(2π²x²), and one loop carries $S_3/(2\pi)^4 = 1/8\pi^2$, the load denominator of section 7.2:

```
u^{*}(x)=|x|^{2-n}\,u\!\left(\frac{x}{|x|^{2}}\right),\qquad
\frac{1}{S_{3}}\,x^{2}}=\frac{1}{2\pi^{2}x^{2}},\qquad
\frac{S_{3}}{(2\pi)^{4}}=\frac{1}{8\pi^{2}}
```

The Gaussian. It carries the same spheres: ∫e^(−x²)dx = √π, and $(\sqrt{\pi})^n = \tfrac{1}{2}\Gamma(n/2)S_{n-1}$. The Gaussian is the fixed point of the Fourier transform, as the unit sphere is the fixed set of the inversion.

Observation: the three-dimensional weight down to 52 µm. A torsion balance with 18-fold and 120-fold attractors found Newton's 1/r² from 3.0 mm down to 52 µm (Lee et al. 2020). Any extra force of gravitational strength must have a range below 38.6 µm (95%). That is below the dark-energy length, about 85 µm in the authors' definition, (ħc/ρ_d)^(1/4).

Observation: the dimension count in a wave amplitude. In D spacetime dimensions a wave's strain falls as d^(−(D−2)/2), set by the sphere's area. GW170817's wave distance against its light distance gives D = 4.02 +0.07/−0.10 with the SH0ES H_0 and 3.98 +0.07/−0.09 with Planck's (Pardo et al. 2018): +0.2σ and −0.3σ from 4. Any leakage scale must exceed about 20 Mpc.

Hyperscaling. Critical exponents carry d as well: α = 2 − dν. With the bootstrap's ν, the orbit experiment's α reads d = 2.9962. This restates the open O(2) misfit of section 4.7.1; it is not evidence of a fractional dimension.

4.7.3 Euler:π

Every rate that has to cross a barrier comes out as e raised to minus a turn over a coupling. The π-side number is a turn (π, 2π, or 8π² = 4π · 2π), and the coupling divides it.

The structure. e^(iπ) = −1, so iπ = ln(−1): the half-turn reached through the rate. Barrier rates take the same form with the i removed.

Process	Weight	Turn on top	Value
Schwinger pair creation	e^(−πE□/E), $E_□ = m^2c^3/e\hbar = 1.32 \times 10^{18}$ V/m	π	0.016 at 10^{18} V/m; 9×10^{-19} at 10^{17} V/m
Instanton (tunnelling between vacua)	e^(−8π²/g²) = e^(−2π/α)	8π²	7.5×10^{-24} for α_s(m_Z) = 0.1180; 2.9×10^{-162} for electroweak B + L, e^(−4π/α_W)

Process	Weight	Turn on top	Value
Dimensional transmutation	$\Lambda = \mu\, e^{(-2\pi/(b_0 \alpha))}$	2π	Λ_{QCD} with five flavours: 87.8 MeV at one loop, 208.7 MeV at two; world value about 0.21 GeV
Higgs vacuum bounce	$e^{(-S)}$, $S = 8\pi^2/(-3\lambda)$	$8\pi^2$	$\lambda_{min} = -0.01075$ (PDG 2026 masses, section 9.3.2) gives S = 2,448
Horizon temperature	$T = \hbar\kappa/2\pi ck$ (Hawking), $\hbar a/2\pi ck$ (Unruh)	2π, one turn of imaginary time	6.2×10^{-8} K for one solar mass; 4.0×10^{-20} K at 1 g

The load and the hold are one number read twice. The load of section 7.2 is $\delta = g^2/8\pi^2 = \alpha/2\pi$, and the instanton weight is $e^{(-1/\delta)}$. For the Higgs bounce, the load $(-\lambda/8\pi^2)$ times S is exactly ⅓: an identity of the formula, not a result.

Euler's other sense meets this one in the T-bar. Mardešić et al. (2020) show that the tennis-racket twist comes from a pole of a Riemann surface, by the Picard–Lefschetz formula. A pole contributes $2\pi i$ times its residue, and the twist becomes a perfect π in the limit of an ideal asymmetric body. The same pole puts a logarithm of the start offset into the rotation angle, with the residue as its coefficient: the T-bar's logarithmic hold law (section 4.7.4).

Observation: Schwinger pairs in graphene. In graphene the gap and the Fermi velocity replace mc^2 and c, which brings the Schwinger field to laboratory scale. Across six ballistic transistors the zero-bias extrapolation collapsed onto $G_0 = 0.18 \pm 0.02$ mS (Schmitt et al. 2023). The universal one-dimensional Schwinger value is $1.20 \times 4e^2/h = 0.186$ mS, so the pull is -0.3σ.

Observation: false-vacuum bubbles. In a ^{23}Na ferromagnetic superfluid, bubbles nucleate with a time that grows exponentially with the barrier, $\tau = A(E_c/kT)^{(-1/2)}\, e^{(E_c/kT)}$, as instanton theory and simulations predict (Zenesini et al. 2024). A follow-up varied the temperature and found the finite-temperature instanton scaling (Cominotti et al. 2025).

Observation: a horizon at 0.35 nK. In a flowing condensate, the sonic horizon's surface gravity predicts $T_H = 0.35$ nK. The measured spectrum of Hawking and partner pairs matches the thermal curve at that temperature with no free parameters (Muñoz de Nova et al. 2019). The horizon's rate is therefore $\lambda = 2\pi kT/\hbar = 288\ s^{-1}$.

Observation: an accelerated frame. A modulated condensate emits coherent pairs of atoms whose distribution is Boltzmann for a local observer, with temperature and entropy as Unruh predicts (Hu et al. 2019). The pair operators mix by cosh and sinh, a hyperbolic turn: the same transformation as a squeeze.

4.7.4 T-bar flip time:action

The period is the derivative of the action, and near the middle axis the time per flip grows as $(2/\lambda)\ln(1/\varepsilon)$. The same λ is an inverted oscillator, which is what the quantum experiments see.

The structure. J is the action on the sphere of angular momentum: |L|/2π times the solid angle an orbit encloses.

```
T=2\pi\,\frac{dJ}{dE},\qquad J=\frac{|L|}{2\pi}\,\Omega_{\rm enclosed},\qquad t_{\rm flip}=\frac{2}{\lambda}\ln\frac{1}{\varepsilon}+\text{const}
```

On the 3:2:1 box, 2π dJ/dE and the simulated period agree to every printed digit at four energies: 11.90892, 17.09192, 24.91104 and 32.87000 time units at 0.5, 0.9, 0.99 and 0.999 of the way to the separatrix. The flip rate is λ = Ω√(cos 2θ_xy cos 2θ_yz) (section 4.4.1), which is √(3/13) = 0.48038 for this box at Ω = 1.

Start offset ε	Time between flips	λ × time	ln(1/ε)
10^{-4}	44.725	21.485	9.210
10^{-5}	54.312	26.091	11.513
10^{-6}	63.899	30.696	13.816
10^{-7}	73.462	35.290	16.118

Each decade adds 4.60 to λ × time, which is 2 ln 10, so the slope is exactly 2/λ. A full return is two flips, (2/λ) ln(1/S), with S ∝ ε² the energy distance from the separatrix. That is the classical period Ma et al. (2020) give, (2/v) ln[4v²/((A₃ − A₁)S)].

The separatrix. It is two great circles through the middle axis. For a box with sides x > y > z, the lune around the major axis has half-angle θ□, set by differences of fourth powers, which are products c² · d² of the sum table and the difference table:

```
\sin^{2}\theta_s=\frac{B-A}{C-A}=\frac{y^{4}-z^{4}}{x^{4}-z^{4}}=\frac{c_{yz}^{2}\,d_{yz}^{2}}{c_{xz}^{2}\,d_{xz}^{2}}
```

For the 3:2:1 box, sin²θ□ = 15/80 = 3/16. The lune's dihedral angle is 51.318° and its solid angle 4 arcsin(√3/4) = 1.79133 sr. That is the geometric phase Hamraoui et al. (2018) give at the separatrix, −4 arcsin√((B − A)/(C − A)).

The quantum side. Near the middle axis the motion is an inverted oscillator with rate λ. A parabolic barrier passes a wave with a thermal-looking probability (Kemble 1935), at the temperature ħλ/2πk:

```
P(\Delta E)=\frac{1}{1+e^{-2\pi\Delta E/\hbar\lambda}},\qquad k_{B}T_{\lambda}=\frac{\hbar\lambda}{2\pi}
```

That is Hawking's formula with κ/c replaced by λ. The classical hold, (1/λ) ln(1/ΔE) per pass, and the quantum temperature are one rate read two ways. The chaos bound λ_L ≤ 2πkT/ħ (Maldacena, Shenker and Stanford 2016) states the same link, and black holes saturate it.

Observation: a phone in the air. A phone's gyroscope samples all three spin components at 100 Hz or more (Wheatland et al. 2021).

- Tossed about its intermediate axis, the phone turns 360° about x and flips 180° about y, and ω_x goes from about +17 to −17 rad/s.
- Tossed about either other axis, it only wobbles.

From the paper's measured moments, $\lambda = 13.1\ s^{-1}$ at 17 rad/s, so the flip builds on a 0.08 s scale inside a half-second toss.

Observation: the logarithmic law in molecules. For water and other asymmetric tops, the classical rotation number near the separatrix is $\alpha - \beta \ln(\text{distance})$, and the quantum rotation number converges to it (Hamraoui et al. 2018). Near the separatrix the quantum value is close to 2, the racket effect's mark in the spectrum. This is computed from measured molecular constants, not measured anew.

Observation: the inverted oscillator itself. Radio-frequency dressing flips a condensate's trap into an inverted oscillator with $\omega_I = 2\pi \times 0.9$ kHz (Ji et al. 2026). Tomography shows one quadrature stretched and the other squeezed 10.6(1.3) dB below vacuum, with the ellipse angle as predicted and no adjustable parameters, and a time-reversed loop restores the state. In temperature units, $\hbar\omega_I/2\pi k = 6.9$ nK. This is a June 2026 preprint.

Observation: the squeeze turned a quarter turn. LIGO's squeezed vacuum is an ellipse of fixed area: one quadrature's noise down, the other's up. A 300 m filter cavity turns the ellipse with frequency, from amplitude squeezing at low frequency to phase squeezing at high frequency: a quarter turn (Kimble et al. 2001). In the fourth observing run this cut quantum noise by up to 5.2 dB at Hanford and 6.1 dB at Livingston (Capote et al. 2025).

Two theory papers extend the flip.

- Ma et al. (2020) predict that tunnelling keeps quantum flips coherent where classical averages wash out.
- Wada and Shimoda (2024) propose flips in magnetars with ellipticity near 10^{-9}, one flip per about 5.7×10^9 spin periods, as a source of bursts.

4.7.5 How the four connect

The four structures close into one loop, and each neighbouring pair shares one exact object:

1. **Inversion and dimension:derivative** share the Kelvin weight r^(2−n).
2. **Dimension:derivative and Euler:π** share $S_3 = 2\pi^2$, which gives the one-loop $1/8\pi^2$.
3. **Euler:π and the T-bar** share one rate λ, through $T = \hbar\lambda/2\pi k$, and the flip e^(iπ) = −1.
4. **The T-bar and inversion** share the fixed sphere, which is where a horizon sits: the calculator's sphere of no action.

The third and fourth ties carry the finding. The inversion's fixed sphere is where a horizon sits, and the T-bar's λ sets its temperature through Euler's 2π. The first two close the loop through the dimension count: n sets the inversion's weight, and the four-dimensional sphere's $2\pi^2$ sets the $8\pi^2$ in every load and every tunnelling exponent.

4.7.6 What fits, what scrapes, and cautions

What fits.

- **The step.** $2\pi/n$ lands exactly on the three moves at n = 2, 3 and 4: −1, ω and i. Nothing is fitted.
- **The inversion ladder** closes at the corners: there is overlap while $\rho^2 < n$ and none at $\rho = \sqrt{n}$.
- **The separatrix angle** is a difference of fourth powers, $(y^4 - z^4)/(x^4 - z^4)$: the sum table times the difference table, inside a spinning body.

- **λ is a temperature at a horizon**, and the horizon is the inversion's fixed sphere. The analogue horizon at 0.35 nK, with no free parameters, is the cleanest laboratory case of that link.
- **The racket's twist and its logarithm** come from one residue, so Euler's number and Euler's equations give the same half-turn.
- **$8\pi^2$** is the load as a ratio and the hold as an exponent: one sphere times one turn on both sides of the flip.

What scrapes.

- **The three-dimensional flower has no π.** Its transcendental is A = arctan(1/√2), the angle between the √2 and √3 directions, while two and four dimensions carry π and π^2. Odd dimension speaks in angles between levels, and even dimension in turns.
- **No closed form.** The four-dimensional overlap at ρ = √3 (0.78% of the cube) has none in π, √2, √3, A or the Clausen values tried.
- **The three-dimensional overlap share** sits on Planck alone but 3σ off the combined fits (section 4.7.1).
- **The 6.1σ O(2) misfit** is the one place where an inversion-symmetric calculation and a measurement disagree. The calculator does not say which is right.

Cautions. None of the new numbers here counts as a prediction; the observations confirm the structures, not new values.

- **The look-elsewhere effect.** The overlap and flower shares join about a dozen cube shares already near 0.31 and 0.69. Some number in a set that size lands in a ±0.0073 window about one time in five.
- **Patterns on few points.** The rational parts 2/n! hold for n = 2, 3 and 4 only. One exact three-dimensional identity, the √2 overlap share = $F_2 - F_3$, fails in four dimensions and is set aside as a coincidence.
- **Identities, not results.** Load × bounce = ⅓, T = 2π dJ/dE and the barrier temperature $\hbar\lambda/2\pi k$ are exact mathematics. They organise the observations; they do not predict them.
- **Analogues.** The condensate horizon, the Unruh simulation and graphene's Schwinger pairs run the same equations in condensed matter. They test the structure (the 2π, the exponent), not the vacuum effects themselves.
- **Preprint, theory, computation.** Ji et al. (2026) is not yet refereed. Ma et al. (2020) and Wada and Shimoda (2024) are theory, and Hamraoui et al. (2018) compute from measured molecular constants.
- **Leading order.** The Higgs bounce uses the running coupling of section 9.3 with the tree-level bounce formula.

Standing.

- **M.** Each closed form rests on standard mathematics, and each observation is published.
- **N.** The four structures as the author's corners of scale and time; the overlap ladder closing at $\rho^2 = n$; the separatrix as a product of the two tables; and one λ as both the T-bar's hold and a horizon's temperature, which is a reading.

4.8 The method

To use the calculator, anchor each part of a datum at a corner of the fixed cube, follow each moving part round its corner's circle, and match every position on the circle to a measurement. A rate takes one quarter turn and an inverse two. Sections 5 to 14 apply the method layer by layer; worked examples sit with the layer they belong to: the electron in section 5.2, a generation step in section 8.2, gravity's four regions in section 11.2 and the proton in section 12.2.

1. **Write the cube.** Copy the eight delimiters and the six face strings exactly (section 3.12). Mark the tetrahedron {”, ., +, ×} and the dual corners.
2. **Choose the layer's frame.** Say what sits on the corners, faces or axes in this layer, and which axis is held. A tetrahedron layer puts four quantities on ”, ., + and ×; an axis layer puts three components on x, y and z. In spacetime layers, time is the held axis.
3. **Fix the corners from faces already named.** Two adjacent faces share exactly one tetrahedron corner, so three mutually adjacent named faces fix three corners, and the fourth is the one left.
4. **Anchor every datum.** Write which corner, face or axis carries each part. Data never leave their corners.
5. **Turn; do not move.** For a rate, take each part one quarter turn along its corner's circle (Table 15); for the inverse, two; for the anti-rate, three. A mirror in three dimensions becomes a half turn through the held axis, and a size flip is the turn K. Nothing is resized.
6. **Read provenance.** Each corner in the new cell came from exactly one corner of the cube, along one circle. Name every new datum by that corner and circle.
7. **Check.** The turn must keep all 32 edges, four quarter turns must return every corner, and the Front string must land on the Back string as written.
8. **Choose each alignment by its behavior.** Where a number can be stated in more than one alignment, a scale, a scheme or a frame, carry each alignment to every observable it touches down-wind. Leave out any that misses a measurement by more than 2σ, and keep the best-matched (rule 9). A miss that every alignment shows, the standard one included, is inherited and stays on the record. The observables used to choose an alignment are not counted as evidence for it (sections 17.1 and 17.2).
9. **Match every position.** For each point of interest, find the measurement at its instant and at each turn. Where none exists, record the empty cell and the measurement that would fill it. Compute the pull for every number the layer commits to, and keep misses visible. Before counting a misfit against the model, check its neighbours (section 4.9).

4.9 Finding a misfit by its neighbours

The author's rule 5 says that a datum that does not fit can be found from the data at the corners next to it, without the dynamics that made it.

1. **List the neighbours.** A corner has three neighbours along the cube's edges, one along w in the next cell, and the faces it shares. Take the measured data at each.
2. **Take a range from each.** The edge or face relation turns each neighbour's datum into a range for the misfit: its sign, cell, scale or rate. In the electrical frame the relations are identities, such as qφ = h, i = qω and v = φω.

3. **Intersect.** If the misfit lies inside every range, it sits at the wrong corner, cell, sign or resolution: move it. If not, check the neighbours themselves, their calibration and selection, before counting the misfit against the model.

Table 19. Eight misfits, read by their neighbours.

Misfit	Neighbours read	Result
The electron's ring: moment μ_B, angular momentum ħ	the next cell on its circle is the inverse; spin ½ repeats after 4π	located: spin ħ/2 with g = 2 (section 5.2)
DES-SN5YR's Ω□ = 0.352 ± 0.017, +2.6σ from π²/32	the other supernova samples, through the supernovae they share	located: calibration. Recalibrated, DES gives 0.330 ± 0.015, +1.4σ (Popovic et al. 2025)
Curvature Ω_k = +0.0023 ± 0.0011 (DESI with the CMB)	the supernovae with the CMB give −0.0063 ± 0.0037	on the matter-share edge; open
w = −1 against an evolving w	the acceleration, q_0 = −0.537 ± 0.026 with constant w (Popovic et al. 2025), against the calculator's −0.53736	narrowed to the evolving fits; open
Σmν below the floor; E(z) at z = 0.35 to 0.55	the same matter-share edge	narrowed; open
$\sin^2\theta_{12}$, −1.5σ	its neighbour in the 15° frame, $\sin^2\theta_{13}$, at −0.6σ	local; JUNO decides
The proton's ring, 4 μ_N against 2.793 μ_N	its spin pieces, and the neutron	narrowed by folding from 43% to 7% (section 12.2)
Lithium, ×2.95	D/H agrees with the same baryon density (Fields et al. 2020)	local to ^{7}Li

Of the eight misfits in Table 19, the neighbours:

- locate two outright: the electron's spin and DES's matter share;
- show two to be local to themselves: $\sin^2\theta_{12}$ and lithium;
- put three on one edge: the matter share;
- narrow one: the proton's moment, by folding.

Neighbours locate. They give a value only where an identity links them, as qφ = h does in the electrical frame.

4.9.1 Bands and steady clocks

Her rule is that a misfit can be placed by the range it must lie in, without the dynamics that made it (section 4.9). These checks show where a band can be computed and where an instant can also be placed from a steady clock.

- **Single-phase power.** At 1,000 VA and power factor 0.8 the average is 800 W, with no component at the line frequency and a swing at twice the frequency of exactly 1,000 VA, which splits into P = 800 W and Q = 600 var. Squaring doubles the frequency, an octave.
- **A ringing circuit.** L = 0.1 H, C = 100 µF and R = 37.95 Ω ring at $\sqrt{(\omega_0^2 - \gamma^2)}$ = 252.97 rad/s. Placing instants on the decaying phasor reproduces a step-by-step simulation.
- **A pendulum at 120°.** Its period follows from the band alone, $T = 4\sqrt{(L/g)}\,K(\sin 60°)$ = 2.754 s for L = 1 m, against 2.006 s for small swings. It reaches −120° at clock angle 180° by moving forward: a negative is not a reversal.
- **Kepler.** The clock is swept area. At eccentricity 0.6, a sixth of the period puts the planet at 130.22° from perihelion.
- **The T-bar.** The rotation per cycle is 2ET/|L| less the signed solid angle that L sweeps on its sphere in the body (Montgomery 1991): a time part and a sphere part, her action and no-action.
- **Chaos.** For a double pendulum the band (its energy) can be computed and the instant cannot. Regular systems have steady-clock coordinates, the action-angle variables of Liouville and Arnold, and chaotic ones do not.
- **1/f noise.** 400 RC relaxations with time constants spread evenly in logarithm from 10^{-4} to 10^{2} s give a spectrum of slope −0.99 between 0.1 and 100 Hz (McWhorter 1957).
- **Three phases.** Three phases on three axes keep $|v| = \sqrt{(3/2)}\,V_{peak}$ and ride a great circle; at 120 V, 10 A and power factor 0.8 the power is a flat 2,880 W. Line voltages are $\sqrt{3}$ times phase: 120 → 207.8, 277 → 479.8 and 347 → 601.0 V. Equal RMS currents and equal average power can hide different motions: a balanced set traces a circle with flat power and no neutral current, while an unbalanced set with the same averages traces an ellipse, its power swings, and current flows in the neutral.

Standing.

- **M.** All standard.
- **N.** Bands against instants as the working rule for placing data (section 4.9).

4.10 The layers in full: orientation and the forty positions

This section gives the tables behind the method, layer by layer: each layer's frame in the four cells, the measurement at every one of the forty positions on its circles, and six orientations the literature settles. Section 15.3 places the published tensions in the same cells. All values are those of PDG 2026 and of the 2025 and 2026 measurements cited.

4.10.1 The orientation of each layer

The layers share the turns but not the meaning of a corner. Table 20 gives, for each layer, what sits in each of the four cells: the instant (the cube), the rate (one quarter turn), the inverse (the half turn) and the anti-rate (three quarters).

Table 20. The layer frames in the four cells.

Layer	Instant	Rate	Inverse	Anti-rate
fabric	” charge, . current, + flux, × voltage; a law on each face; one-store circuits on the dual corners	each store's rate: i and v, with −q and −φ	every variable turned over, −q, −i, −φ, −v: charge conjugation	−i and −v, with q and φ
electron	charge e; $C_e = 4\pi\varepsilon_0 r_e$ on the Right face	current eω and moment μ_B; $L_e = \pi\mu_0 a_0$ on the Left face	the positron	the positron's moment
gravity, four regions	region I, outside the horizon, with time on the held axis	region II, inside: time and radius trade places	region III, the other outside: an exact symmetry, with positive mass	region IV, the white hole
gravity, motion	position on an orbit	velocity	acceleration, $-\omega^2 r$ on a circle	jerk
leptons	$\sqrt{m_e}$, $\sqrt{m\mu}$ and $\sqrt{m\tau}$ on the 45° cone, the generations a third of a turn apart	the moments	antileptons: roots negated, masses and Q unchanged	the antileptons' moments
neutrinos	one root flipped, a half turn through the held axis: 59.2 meV in all	oscillation at $\Delta m^2/2E$	antineutrinos	antineutrino oscillation
Higgs vacuum	two phases, y_t^2 and g^2, at 120°: top 172.57 ± 0.38 GeV	the decay widths	the antitop	the antitop's width
strong scale	½ΔΣ, ΔG and L at ⅙ each; $2\pi r_{\square} = 4\lambda_{\square}$, one Compton wavelength per layer	the proton's moment	the antiproton	the antiproton's moment

Layer	Instant	Rate	Inverse	Anti-rate
cosmos	Ω□, Ω_Λ and Ω_k on x + y + z = 1, with Ω□ = π²⁄32	the expansion rate H	the acceleration: $q_0 = -0.537$, changing sign at z = 0.649	the jerk, j = 1
T-bar	spin about the middle axis, moments 5, 10 and 13 for the 3:2:1 box	the flip rate, λ = 0.4804Ω	the body upside down after half a cycle	the second flip, continuing the same loop: upright again, the clock running forward
rates	a well: oscillation at ω	ω → iλ: the barrier, growth rate λ (λ = κ/c at a horizon)	the energy turned over, H → −H: P(−ΔE) = 1 − P(ΔE)	—
time and scale	” the inverse of scale, . the derivative of scale, + the inverse of time, × the derivative of time	each inverse turned into its derivative: ” → ., + → ×	all four turned over; inverting scale alone crosses the Top face instead, and time alone the Bottom face	three quarters forward: each derivative reaches its inverse, . → ”, × → +
the cube's own numbers	1	i	−1	−i

One half turn moves every layer at once. It turns the electrical variables over, carries region I to region III, and turns matter into antimatter: together, CPT, the one inversion relativistic quantum field theory guarantees (section 8.3). The rates layer's quarter turn is the same move in time. Turned into imaginary time, a horizon's growth rate λ becomes a circle whose period sets its temperature, T = ħλ/2πk (section 4.7.3).

4.10.2 The measurement at every position

Every point of interest has a measured value at its instant, and 33 of the 40 positions on ten circles hold a published measurement. For matter the half turn is the CPT image, so the inverse and anti-rate positions belong to the antiparticle. Pulls are (measured − model)/σ; at the inverse positions the model is equality with the particle.

Table 21. Observed values at each position.

Point of interest	Instant	Rate	Inverse	Anti-rate
electron, charge circle (”)	charge e, exact in SI	moment: the ring gives μ_B; measured 1.001159652180595(13) μ_B (Fan et al. 2023)	the positron: charge equal and opposite within 4 × 10^{-8}, mass equal within 8 × 10^{-9} (PDG)	the positron's g-factor equals the electron's within (−0.5 ± 2.1) × 10^{-12}, −0.2σ (Van Dyck, Schwinberg and Dehmelt 1987)
electron, flux circle (+)	h/e = 4.1357 × 10^{-15} Wb, the Aharonov–Bohm period of gold rings (Webb et al. 1985)	$m_e c^2/e$ = 510,998.95 V, the rest energy 0.51099895069(16) MeV (CODATA 2022)	the positron's flux: not measured	the positron's rest energy: equal within 8 × 10^{-9} (PDG)
hydrogen	1S–2S at 2,466,061,413,187,035(10) Hz (Parthey et al. 2011)	the two moments coupled: the hyperfine line at 1,420,405,751.768(2) Hz (Hellwig et al. 1970)	antihydrogen: 1S–2S equal within about 2 × 10^{-12} (Ahmadi et al. 2018); Lamb shift within 11% and fine structure within 2% (ALPHA Collaboration 2020)	the antihydrogen hyperfine line, 1,420.4 ± 0.5 MHz, 0.0σ (Ahmadi et al. 2017)
charged leptons, the cone	the τ at 1,776.93 ± 0.09 MeV against 1,776.969, −0.4σ (PDG 2026)	the moments: a_e (Fan et al. 2023); aμ against the 2025 prediction, +0.6σ (Muon g−2 Collaboration 2025)	antilepton masses: e± within 8 × 10^{-9}; τ± within 2.8 × 10^{-4} (Abe et al. 2007)	antilepton moments: e±, −0.2σ; μ±, $(g^+ - g^-)/g$ = (−0.11 ± 0.12) × 10^{-8}, −0.9σ (PDG)
neutrinos, one root flipped	Σmν < 0.064 eV (DESI with the microwave background) against 0.0592 eV: inside, but the effective mass is −3.4σ (section 9.2)	solar neutrinos: Δm^2_{21} = 6.10 +0.95/−0.81 × 10^{-5} eV^2 (Super-Kamiokande Collaboration 2024)	the antineutrino mass: m($\bar{\nu}_e$) < 0.45 eV at 90% (KATRIN Collaboration 2025)	reactor antineutrinos: Δm^2_{21} = 7.50 ± 0.12 × 10^{-5} eV^2 (JUNO Collaboration 2025), −1.5σ from solar

Point of interest	Instant	Rate	Inverse	Anti-rate
proton	r = 0.84075(64) fm against 4ƛ, −0.8σ (CODATA 2022)	the moment, 2.79284734462(82) μ_N (Schneider et al. 2017); a charge at c round 4ƛ would give 4 μ_N	the antiproton: charge-to-mass ratio equal within 16×10^{-12} (Borchert et al. 2022)	the antiproton's moment, −2.7928473441(42) μ_N, −0.1σ (Smorra et al. 2017)
top and Higgs	top 172.60 ± 0.27 (direct) against 172.57 ± 0.38 GeV, +0.1σ, and 172.1 ± 0.6 (pole), −0.7σ; Higgs 125.13 ± 0.11 against 125.20 ± 0.92 GeV, −0.1σ (PDG 2026; section 9.4)	the Higgs width, 3.2 +2.4/−1.7 MeV, against 4.1 in the Standard Model (CMS Collaboration 2022); no number of the calculator's	the antitop: m(t) − m(t̄) = −0.15 ± 0.21 GeV, −0.7σ (CMS Collaboration 2017)	the antitop's width: not measured
gravity, motion	the orbit's shape: S2's Schwarzschild precession, f = 1.10 ± 0.19, +0.5σ from general relativity (GRAVITY Collaboration 2020)	velocities: the rotation curves of 175 galaxies (Lelli, McGaugh and Schombert 2016)	accelerations: g† = 1.20×10^{-10} m/s² (McGaugh, Lelli and Schombert 2016); no number of the calculator's	the jerk: not checked
gravity, four regions	region I: γ − 1 = $(2.1 \pm 2.3) \times 10^{-5}$ (Bertotti, Iess and Tortora 2003); shadows of 42 ± 3 and 51.8 ± 2.3 μas match Kerr (Event Horizon Telescope Collaboration 2019, 2022)	region II: only its face with region I, the horizon, whose area grew at 4.4σ in GW250114 (LIGO, Virgo and KAGRA Collaborations 2025)	region III: unobservable	region IV: unobservable
cosmos	Ω = π²/32 against DESI,	H_0 from 67.4 ± 0.5 to 73.50 ± 0.81 km/s/Mpc, the	the acceleration: $q_0 = -0.537 \pm 0.026$ with	the jerk: j = 1 in flat ΛCDM; not checked

Point of interest	Instant	Rate	Inverse	Anti-rate
	−1.3σ, and Planck, +0.9σ	Hubble tension; no number of the calculator's	constant w (Popovic et al. 2025), and −0.51 ± 0.024 from Pantheon+ below z = 0.8 (Brout et al. 2022): +0.0σ and +1.1σ; first seen as $q_0 < 0$ at 2.8σ and 3.9σ (Riess et al. 1998) and $\Lambda > 0$ at 99% (Perlmutter et al. 1999)	

The inverse layer is CPT, and every comparison found agrees. The half turn turns the electrical variables over (C) and spacetime end for end (PT). Relativistic quantum field theory guarantees this one inversion. The key step in Jost's proof is the author's rule: total inversion cannot be reached by turning in real spacetime, but can be in the complexified picture (Greenberg 2006).

- **The agreement.** Every comparison in Table 21 agrees within 1.5σ, and all but two (the neutrinos and antihydrogen's fall) within 1σ.
- **The tightest test.** The tightest test found, the neutral kaon pair, puts the masses of K^0 and its antiparticle equal within 6×10^{-19} (PDG).
- **One tension.** Solar neutrinos and reactor antineutrinos give values of Δm^2_{21} 1.5σ apart. CPT requires them to be equal. The calculator's own Δm^2_{21}, 7.49×10^{-5} eV², was taken from oscillation fits, so it is not a test.

The empty positions. Seven positions hold no measurement.

- Four are unmeasured or unchecked:
  - the positron's flux, which an Aharonov–Bohm measurement with positrons would fill;
  - the antitop's width;
  - the jerk of an orbit;
  - the jerk of the cosmic expansion.
- Three are the inside of a black hole, regions II to IV, which no observer outside can see.

4.10.3 The six items that needed an orientation

Table 22 gives, for each of the six items, what the literature settles, the cell it lands in, the data and what is still open.

Table 22. Orientation from the literature, by cell.

Item	What the literature settles	Cell	Data	Still open
the proton's radius, 4ƛ□	Trinhammer and Bohr (2019) write it as πr⍰ = 2λ□: the charge circumference 2πr⍰ holds four Compton wavelengths h/(m⍰c)	instant: one Compton wavelength per layer of the ring	r⍰/ƛ□ = 3.9977 ± 0.0030 (CODATA 2022), −0.8σ from 4; the lattice's r⍰ = 0.820 ± 0.014 fm (Djukanovic et al. 2024) gives 3.90 ± 0.07, −1.5σ; a charge at c on that circle would give 4 μ_N, against the measured 2.793 μ_N	not derived; the relation is prior art
1/f noise	switching times τ = τ₀e^(E/kT), with a broad, flat spread of activation energies, give S(f) ∝ 1/f (Dutta and Horn 1981)	rate: barrier-crossing rates, spread flat	the test, as restated by Müller et al. (2009): α(T) = 1 − [∂ln S/∂ln T − 1]/ln(2πfτ₀)	each system's spread of barriers
ball lightning	the light oscillates at 99.4 Hz through the stable stage, with a 35 kV line about 20 m away. In each cycle the air lines (N I, O I) rise and fall with the light, while the soil lines (Si I, Fe I, Ca I) stay nearly constant (Cen, Yuan and Xue 2014)	the half turn: light follows power, which repeats every half cycle of the line, so it flickers at 2f; the hold is in the soil vapour (section 11.3)	99.4 Hz, 0.6% below 2f = 100 Hz, with no uncertainty given for the flicker or the line	a formula for the hold
held mass	the radial-acceleration relation fixes the extra pull by the visible mass with	inverse of orbital motion: an acceleration	g† ÷ (cH₀/2 π) = 1.15 for Planck's H₀ and 1.06 for SH0ES's	what the mass is made of; the coincidence is not derived

Item	What the literature settles	Cell	Data	Still open
	one scale, g† = 1.20 ± 0.02 ± 0.24 × 10⁻¹⁰ m/s² (McGaugh, Lelli and Schombert 2016); Milgrom (2020) notes 2πa₀ ≈ cH₀			
lightning's 17.3°	Hill's Table 1 gives 13 measurements of 9 photographs, with mean turns of 15.0° to 23.4°; their average is 17.3°, and 16.3° after about 6° of measuring noise is removed (Hill 1968; Cooray 2018; Brignone et al. 2022). Idone and Orville's (1988) data give 17.0°	motion from the dimension corner (.): the law 64.65°/√s, with s legs folded into every resolved section, whatever its length (section 4.5)	the mean turn stays near 17° from 5.5 to 68 m sections: an exponent of +0.11 ± 0.05 from Hill's Table 1, 2σ from the 0 of folded legs and 11σ from the −0.5 of legs fixed in length	a derivation of s
the black hole's centre	GW250114: the final horizon area exceeds the initial areas at 4.4σ, and two ringdown modes fit Kerr at 4.1σ	region II, the rate cell of gravity's four regions	the horizon, its face with region I, holds; nothing inside is observed	whether the centre ends or turns (section 13)

Lightning against Hill's data. Hill's full paper settles three things the staircase says about the turns. The folding itself is in section 4.5.

- **Shape.** His Figure 1, about 200 turns of the Norinder 4 stroke, shows the signed turn in the photograph's plane as one Gaussian centred on zero. Unsigned, that is the half-normal the staircase predicts, with zero the most likely turn. A Gaussian at 17° in the plane would leave a dip at zero, which the figure does not show. Read as a three-dimensional deflection seen side-on, it also peaks at zero, so a photograph cannot separate the two.

- **Scale.** With folded legs every resolved section holds the same s legs, so the mean turn is the same at every section length.
    - Hill finds the turns “largely independent” of section length, and fits the section length to the power 0.1.
    - This paper's fit to his Table 1 gives +0.11 ± 0.05.
    - Legs fixed in length would make the turn fall as $1/\sqrt{L}$, which his data exclude at 11σ.
    - Within one photograph the Norinder 4 stroke gives 16.5° with 105 cells and 15.0° with 210, where fixed legs need 23.3°.
- **Spectrum.** Folded legs at every scale make the direction a 1/f signal, whose neighbouring averages differ by the same amount at every length (a flat Allan variance), while fixed legs make it white. Hill's Fourier amplitudes fall as $1/\sqrt{n}$, which is a 1/f power spectrum.
- **The value is still a fit.** s = 14.0 at Hill's 17.3°, or 15.7 at his 16.3°. Hill himself reads the constant turn as noise from the breakdown process or the inhomogeneous air, and finds no characteristic wavelength.

5. The board: what space is made of

Computed on the calculator's ring, the vacuum is a transmission line. Its capacitance ε_0 and inductance μ_0 per length give the speed c, its impedance is $Z_0 = 2\alpha \cdot h/e^2 = 376.730\ \Omega$, and it holds a gap of $2m_ec^2 = 1.022$ MeV before anything acts. A reaction of the board carries four corners, charge, current, flux and voltage, joined by identities: for the electron $q = e$, $\varphi = h/e$, $i = \omega e$ and $v = m_ec^2/e$, so $q\varphi = h$ and $qv = \varphi i = m_ec^2$. Photons, electrons and atoms are the board's reactions, and what quantum electrodynamics calls a virtual particle is its reactive response. Every number in this section is the standard one; what the calculator adds is one circuit in which they are a single device, and a bookkeeping of the electron's spin by its two stores.

5.1 The board

Standard. The vacuum's constants make a transmission line (Table 23).

Table 23. The board's numbers.

Board quantity	Value	Standard name
capacitance per length	$\varepsilon_0 = 8.854 \times 10^{-12}$ F/m	vacuum permittivity
inductance per length	$\mu_0 = 1.2566 \times 10^{-6}$ H/m	vacuum permeability
signal speed $1/\sqrt{(L'C')}$	$c = 299{,}792{,}458$ m/s	the speed of light
base ohms $\sqrt{(\mu_0/\varepsilon_0)}$	$Z_0 = 376.730\ \Omega = 2\alpha \cdot h/e^2$	the impedance of free space; the physics in it is α, and the ohms are an SI convention
gap before action	$2m_ec^2 = 1.022$ MeV	the pair-creation threshold (Dirac 1930)

Board quantity	Value	Standard name
exciton	positronium: 6.80 eV binding, a 511 keV glow on recombination	positronium; the annihilation line
breakdown field	1.32×10^{18} V/m	the Schwinger limit
kinetic inductance of the weak current	screening depth $\hbar c/m_W = 2.46 \times 10^{-18}$ m	the W mass; the London depth of a superconductor is the same mechanism

How it works.

- **Electric and magnetic are a quarter turn apart.** E ↔ H is the quarter turn (the duality $J^2 = -1$). In the near field $Z_E Z_H = Z_0^2$, with $Z_0/\sqrt{2}$ at −45° and $\sqrt{2}\cdot Z_0$ at +45° where kr = 1.
- **Light is the board ringing on its 45° line.** For a plane wave Faraday's law gives E = cB, so $u_E = \tfrac{1}{2}\varepsilon_0 E^2$ equals $u_B = B^2/2\mu_0 = E^2/(2\mu_0 c^2) = \tfrac{1}{2}\varepsilon_0 E^2$, using $c^2 = 1/(\mu_0 \varepsilon_0)$. On a transmission line the same statement is V/I = Z_0 everywhere: the electric store $\tfrac{1}{2}C'V^2$ equals the magnetic store $\tfrac{1}{2}L'I^2$ exactly when $Z_0^2 = L'/C'$, which defines the line's impedance.
- **The number in the ohms is α.** $Z_0 = 2\alpha \cdot h/e^2 = 2 \times 0.0072974 \times 25{,}812.8\ \Omega$: the vacuum's impedance is two fine-structure constants' worth of the quantum of resistance.

The reading. Space is a board with a capacitor and an inductor at every point, and everything else is what the board does when driven.

- **The sea.** Before anything acts, the board is a semiconductor: a filled sea 1.022 MeV below the conduction states. A photon above the gap lifts an electron out and leaves a hole, which is the positron.
- **Inductance.** Inductance is the store in the perpendicular dimension, the $\sqrt{2} \rightarrow \sqrt{4}$ swing of section 4.6.

Virtual particles as the board's reactive response. Around any charge the field has three parts:

- terms in $1/(kr)^3$ and $1/(kr)^2$ whose energy flow is reactive: stored in one quarter cycle, returned in the next;
- a term in 1/(kr) whose flow is real and leaves;
- the boundary between them at kr = 1.

What quantum electrodynamics calls virtual-photon exchange is, in the circuit, capacitive and inductive storage; what it calls a real photon is the mode that carries power away. Three sources agree with this reading:

- W. E. Lamb's “Anti-photon” (Lamb 1995);
- Hobson's “there are no particles, there are only fields” (Hobson 2013);
- Jaffe (2005), who shows the Casimir force can be computed without reference to zero-point energy, as Bass and Krzysiak (2020) also note.

Whose board. The circuit is the author's. The vacuum it is drawn on is Steven D. Bass's emergent vacuum (Bass 2015, 2020, 2024):

- the Standard Model as the long-range tail of a statistical system near a critical point;
- gauge symmetries that emerge, and particles as stable collective excitations of unknown degrees of freedom;
- a cosmological constant that is the small residual left after condensate and zero-point energies cancel.

Bass does not write the vacuum as a circuit.

The electron's three lengths are a flip. The classical radius $r_e = \alpha\lambda\!\!\!^{-}_e = 2.8179$ fm, the reduced Compton wavelength $\lambda\!\!\!^{-}_e = 386.16$ fm and the Bohr radius $a_0 = \lambda\!\!\!^{-}_e/\alpha = 52.918$ pm stand in the ratio $\alpha : 1 : 1/\alpha$. So $r_e a_0 = \lambda\!\!\!^{-}_e{}^2 = 1.49119 \times 10^{-25}$ m² exactly: the atom's radius and the charge's radius are each other's inversion through the Compton sphere.

Standing.

- **M.** Every number is standard. The reactive reading of virtual particles agrees with Lamb, Hobson and Jaffe, and the residual is Bass's.
- **N.** Drawing Bass's vacuum as a capacitive–inductive circuit, term by term, and reading $r_e a_0 = \lambda\!\!\!^{-2}$ as an inversion were not found.

5.2 Worked example: the electron on the ring

1. **Frame.** The electrical frame puts q on ”, i on ., φ on + and v on ×.
2. **Anchor.**
   - The Right face (q, v) holds the electron's capacitance, $C_e = e^2/(m_e c^2) = 4\pi\varepsilon_0 r_e = 3.1354 \times 10^{-25}$ F.
   - The Left face (φ, i) holds its inductance, $L_e = h^2/(e^2 m_e c^2) = \pi\mu_0 a_0 = 2.0891 \times 10^{-16}$ H.
3. **The four corners.** With q = e, the turn gives:
   - $i = \omega e = 19.796$ A, with $\omega = m_e c^2/h = 1.2356 \times 10^{20}$ circuits a second;
   - $\varphi = L_e i = h/e = 4.1357 \times 10^{-15}$ Wb;
   - $v = q/C_e = m_e c^2/e = 510{,}998.95$ V.

   All six faces then read off. Resistance v/i and memristance φ/q are both $h/e^2 = 25{,}812.807\ \Omega$, the von Klitzing constant. The products are $qv = \varphi i = m_e c^2$ and $q\varphi = h$.
4. **One quarter turn.** The charge goes to the current corner: one elementary charge per Compton period $h/(m_e c^2)$, a charge moving at c round the Compton circle of radius $\lambda\!\!\!^{-} = 386.16$ fm.
   - Its moment is $i \cdot \pi\lambda\!\!\!^{-2} = e\hbar/2m_e = \mu_B$ exactly, checked to 30 digits.
   - The measured moment is 1.00115965218059(13) μ_B (Fan et al. 2023). The 0.116% excess is $\alpha/2\pi$ at first order, which the calculator files on the Left face as the load.
5. **The radii stay put.** At the same quarter turn the energy passes from the capacitor, at $r_e = 2.818$ fm, to the inductor, at $a_0 = 52.92$ pm. The two radii multiply to $\lambda\!\!\!^{-2} = 1.4912 \times 10^{-25}$ m² exactly, so nothing is resized.
6. **Two quarter turns.** Every sign turns: −q, −i, −φ, −v. For the electron this is the positron.
7. **Observed at every position.**
   - The positron's charge is equal and opposite to the electron's within 4×10^{-8}, and its mass equal within 8×10^{-9}.
   - Its g-factor equals the electron's within $(-0.5 \pm 2.1) \times 10^{-12}$ (Van Dyck, Schwinberg and Dehmelt 1987, via PDG).

    - On the flux circle, h/e is the Aharonov–Bohm period of gold rings (Webb et al. 1985), and $m_e c^2/e$ is the rest energy, 0.51099895069(16) MeV (Mohr et al. 2024).
8. **Provenance.** The Bohr magneton came from the charge corner by one quarter turn, and the positron by two.

A tank caveat. $\sqrt{(L_e C_e)} = h/m_e c^2$ is one circuit's time, since the corners carry one charge and one flux quantum per circuit. A standard LC tank built from L_e and C_e, whose $1/\sqrt{(LC)}$ counts radians, rings at 1.9665×10^{19} Hz, 2π below the ring's 1.23559×10^{20} circuits a second.

The spin's half: the electron as a reaction. The ring above carries angular momentum $\hbar$, twice the electron's spin, so as it stands it reproduces the moment but not the spin. The author's rule supplies the half: the electron is a reaction of the board, not a mass on a ring.

- **Two stores.** Its rest energy sits in two stores. Half, $\frac{1}{2}qv = \frac{1}{2}m_e c^2 = 255.499475$ keV, is held on the capacitor; this equals $e^2/8\pi\varepsilon_0 r_e$, the energy of a charge on a sphere of radius r_e. The other half, $\frac{1}{2}\varphi i = \frac{1}{2}m_e c^2$, moves in the inductor: her share rule of action on the curve and none inside.
- **The half goes round.** Only the moving half goes round the Compton circle, so the angular momentum is $(\frac{1}{2}m_e c^2/c)\lambda\!\!\!^{-} = \hbar/2$.
- **The whole charge goes round.** The whole charge makes every circuit, so the moment stays μ_B, and g = 2.
- **The 4π rule.** A state turned through 2π lands in the inverse cell: $e^{(-i2\pi J/\hbar)} = -1$ only for $J = \hbar/2$. This is the 4π periodicity seen in neutron interferometers (Rauch et al. and Werner et al., 1975, as analysed by Sassoli de Bianchi 2017).

The Dirac field reaches the same factor by another split: its charge circulates twice as fast as its mass (Sebens 2019). Ohanian (1986, as described by Sebens) treats spin and moment as flows of energy and charge in the electron's wave field. The measured g cannot tell the two splits apart.

Three limits (added):

1. **Classical fields give $\hbar/3$.** Classical field theory puts angular momentum in the field, $\varepsilon_0 E \times B$. A charge e on a sphere of radius r_e with moment μ_B at its centre holds $\mu_0 e\mu_B/(6\pi r_e) = \hbar/3$ in its field, and $\hbar/2$ only at $2r_e/3$, the radius Blinder (2002) fits.
2. **The rate matters.** The half uses the ring's rate. At the tank's rate the moving half would carry $\pi\hbar$.
3. **g = 2 exactly.** The excess, $a_e = 0.00115965218059$, is the next layer: QED's series, $\alpha/2\pi$ = 0.0011614097 at one loop, 0.0011596374 with two loops and 0.0011596522 with three.

Standing.

- **M.** The factor of two is known.
- **N.** The split by stores was not found in a short search. It is bookkeeping, not a field calculation.

5.3 Oscillation and holding: photons, electrons and atoms as reactions of the board

Agreeing with Bass, in circuit terms. Bass's vacuum (section 5.1) says what the particles are:

- “the quarks and leptons as well as the gauge bosons and Higgs boson would be the stable collective long-range excitations of some more primordial degrees of freedom that exist above the scale of emergence”;

- the electron quasiparticle “develops a mass gap from interaction with the condensate”, as in a superconductor;
- a photon in a superconductor becomes a massive plasmon, “a non-relativistic precursor” of the Higgs mechanism;
- the zero-point energy of the massless photon is zero, and the net zero-point energy of the gapped excitations is cancelled against the condensate's;
- the Casimir force, “sometimes claimed as experimental evidence for zero-point energies, can also be calculated without reference to” them (Bass and Krzysiak 2020, after Jaffe 2005).

The calculator agrees with each of these sentences and reads each as a circuit element: the board is ε_0 and μ_0, its gap is $2m_ec^2$, its stable excitations are its resonant reactions, and its condensate is the held store. The one change is the name of what QED calls a virtual particle (section 5.1). The energy-time relation $\Delta E \cdot \Delta t \lesssim \hbar$ is then the statement that a reactance returns what it stores within a cycle.

The photon is the board ringing.

- **One quantum.** A photon is one quantum of the board's ringing on its 45° line, E = hf, so the action quantum per cycle (section 7.2) is the energy per cycle of the mode.
- **Two legs.** The photon carries the two transverse legs, its two polarisations, and no third, which is why it is massless.
- **Near and far.** Around any charge the field's $1/(kr)^3$ and $1/(kr)^2$ terms carry reactive power in and out, and the $1/(kr)$ term carries real power away. The boundary is kr = 1. The Coulomb force is the near field at zero frequency: an exchange that never leaves, the board's capacitance between two charges. At the electron's Compton frequency the boundary sits at $\lambda\!\!\!^{-}_e$ = 386 fm, the electron's reactive zone.

Table 24. The photon's observations.

Observation	Number	Source	The circuit reading
the photon has no third leg	$m_\gamma < 10^{-18}$ eV; the $1/r^2$ law holds to 10^{-16}	PDG; Williams, Faller and Hill 1971	two legs on the 45° line, the held leg empty
light scatters off light	$\sigma(\gamma\gamma \to \gamma\gamma)$ = 78 ± 13 ± 7 nb at 8.2σ; 107 ± 24 ± 13 nb, agreeing with the next-to-leading-order prediction	ATLAS Collaboration 2019; CMS Collaboration 2025	the board's reactance is non-linear: the Euler–Heisenberg term is the vacuum's Kerr effect, a reaction with no particle emitted
two photons make a pair	$\sigma(\gamma\gamma \to e^+e^-)$ = 263.5 ± 1.8 ± 17.8 μb, leading-order QED; the polarisation-dependent angular modulation	CMS Collaboration 2025; STAR Collaboration 2021	breakdown of the gap: two photons above $2m_ec^2$ lift an electron out of the sea and leave the hole

Observation	Number	Source	The circuit reading
a magnetar's field makes the vacuum birefringent	X-ray polarisation 46 ± 4% at 2–8 keV and 59 ± 5% at 2–3 keV from 1E 1547.0−5408, inferred through atmosphere and geometry models	Stewart et al. 2026; earlier 4U 0142+61, Taverna et al. 2022	at 10^{10} T the board is a crystal: the same cubic reactance, read as a refractive index
the breakdown field is not yet reached	$E_S = m_e^2c^3/e\hbar = 1.32 \times 10^{18}$ V/m; critical field 4.41×10^9 T	Schwinger 1951	the board's dielectric strength

The electron is the board held above its gap.

- **The pair.** A photon or a collision above the gap lifts one excitation out and leaves one hole: the electron and the positron, each carrying half the gap, 511 keV, as its held leg.
- **The charge.** It is the U(1) phase coupling that the Hopf projection hides (section 14.2), and $e^2 = 2\alpha h/Z_0 = 2.5670 \times 10^{-38}$ C², checked.
- **The spin.** Spin ½ is the half-turn: one full turn of the electron's phase is two turns of the field's, the double cover whose parity is the start rule (section 7.3).
- **The moment and α.** The measured g/2 = 1.00115965218059(13), to 0.13 parts per trillion, is the most precisely known property of any particle (Fan et al. 2023). With QED's series it reads α^{-1} = 137.035999166(15). That sits 2.2σ from rubidium's 137.035999206(11) (Morel et al. 2020) and 3.9σ from caesium's 137.035999046(27) (Parker et al. 2018), and the two atomic values disagree with each other at 5.5σ: the board's one number read three ways, with the electron's in the middle.
- **No size.** The electron has no measured size. Its structure is absent down to 10^{-18} m, and its electric dipole moment is below 4.1×10^{-30} e·cm (Roussy et al. 2023), round to the limit of the measurement.
- **Three lengths, one flip.** It has three lengths, and they are one flip (section 5.1):
    - the classical radius, where the board's electric store around the charge equals the rest energy;
    - the reduced Compton wavelength, below which the board answers a probe with pairs;
    - the Bohr radius, where the atom's orbit sits.

The muon. It is the same reaction 206.768 times heavier, and its load shows the board reacting with its hadrons: about 60 parts per million of its anomaly is the board's hadronic polarisation (58.7 ppm with the 2020 data-driven value, 60.4 ppm with the 2025 lattice value).

- **The measurement.** The final Fermilab value is $a_\mu = 116\,592\,070.5 \pm 14.8 \times 10^{-11}$ (127 ppb; Muon g−2 Collaboration 2025).
- **The predictions.** It agrees with the 2025 lattice-based prediction, $116\,592\,033 \pm 62 \times 10^{-11}$, at 0.6σ, while the 2020 prediction built on e^+e^- data sits 5.7σ below it (Muon g−2 Theory Initiative 2025).

The open discrepancy is between two readings of the board's hadronic reaction, not between the muon and QED.

Table 25. The virtual particle, row by row. Nothing here changes a number of QED's. What the calculator owns is the vocabulary, and the claim that the reactance which stores the Coulomb field is the inductance that loads the mass tables (sections 7.2 and 9.1).

QED's name	What is measured	The circuit reading
virtual-photon exchange, the Coulomb field	the $1/r^2$ law and the photon's zero mass	the capacitive near field at zero frequency: reactive power that never leaves
vacuum polarisation, the Uehling term	α runs from 1/137.036 at zero momentum transfer to about 1/129 at m_Z (PDG), and the running is measured directly in small-angle Bhabha scattering at LEP (OPAL Collaboration 2006); it is 205.0 of the 206.0 meV QED part of muonic hydrogen's Lamb shift, and −27.1 MHz of hydrogen's	the sea polarises like a dielectric: bound pairs align with the field and screen the charge, so the board's permittivity depends on the probe's frequency, which is dispersion
self-energy, the Lamb shift	2S sits 1057.8 MHz above $2P_{1/2}$ in hydrogen, to 3 kHz (Bezginov et al. 2019); Bethe's 1947 one-loop estimate 1040 MHz	the board's ringing shakes the electron, with a root-mean-square jitter $\sqrt{(2\alpha/\pi \cdot \ln(1/\alpha))}\,\lambda\!\!\!^{-}_e$ = 58 fm (Welton 1948), and lifts the S state at the bottom of the Coulomb well
the anomalous moment	$g/2 - 1$ = 0.00115965218059(13)	the load per action cycle, $\alpha/2\pi$, then the board's smaller reactions
the Casimir force	$P = \pi^2\hbar c/240d^4$ = 13 Pa at 100 nm and 1.3 mPa at 1 µm; measured to 5% (Lamoreaux 1997), 1% (Mohideen and Roy 1998) and 0.2% (Decca et al. 2007)	a reactive load of the board's modes between two conductors, computable from the plates' charges with no vacuum energy (Jaffe 2005)
light-by-light scattering and vacuum birefringence	Table 24	the cubic reactance of the board
pair creation	Breit–Wheeler at the LHC; the Schwinger field unreached	breakdown of the gap: above $2m_ec^2$ the reaction is no longer

QED's name	What is measured	The circuit reading
		returned within the cycle, and the real pair leaves
zero-point energy	its naive sum is 10^{120} times the dark energy; the measured residual is 2.24 meV	the stored energy of the gapped reactions, zero for the massless photon, cancelled against the condensate's; the residual, measured at 2.24 meV, is read as $v^2/M_{23} = 2.02 \pm 0.20$ meV (section 13)
Hawking radiation	unobserved; its analogue at a sonic horizon in a condensate seen, thermal at 0.35 nK (Steinhauer 2016; Muñoz de Nova et al. 2019; section 4.7.3)	the board's reaction at a horizon

The atom is the board's resonance with a nucleus. Every length and energy in hydrogen is one rung of the ladder of α:

- **Lengths.** r_e, $\lambda\!\!\!^{-}_e$ and a_0.
- **Energies:**
  - $m_e c^2 = 511.0$ keV;
  - $\alpha \cdot m_e c^2 = 3.729$ keV, the Bohr orbit's momentum times c;
  - $\alpha^2 \cdot m_e c^2 = 27.211$ eV, the Hartree, of which the binding is half, 13.606 eV, by the virial theorem;
  - the fine structure, $\alpha^4 \cdot m_e c^2/32 = 10.95$ GHz, against the measured 10.97 GHz for the n = 2 doublet;
  - the hyperfine line, $\alpha^4 \cdot m_e c^2 \cdot (m_e/m_?) \cdot (4/3) g_? = 1421.2$ MHz at leading order and 1418.8 MHz with the reduced-mass factor, against the measured 1420.405751768 MHz (Hellwig et al. 1970; Essen et al. 1971);
  - the Lamb shift, at $\alpha^5 \cdot m_e c^2$ times a logarithm.

Each power of α is one more turn of the per-cycle coupling. The whole spectrum is measured to parts in 10^{15}: the 1S–2S interval is 2,466,061,413,187,035 ± 10 Hz (Parthey et al. 2011). The hidden symmetry that makes the shells squares is Fock's four-dimensional sphere, in Baez's sharper form of 2025 (section 14.5): the atom is a field on the 3-sphere, the held torus and the Hopf map in one more form, and the periodic table is its spectrum.

Positronium, the board's own exciton. It is the same spectrum at half the Rydberg, because its two halves weigh the same: binding 6.803 eV, and its 1S–2P line at 243.0 nm, twice Lyman-α.

- **Cooling.** It was laser-cooled from 610 K to 1 K along the beam in 2024 with a chirped pulse train on that line (Shu et al. 2024), after AEgIS reached 170 K (Glöggler et al. 2024).
- **CPT.** Its decays test CPT, with no violation found at a statistical precision of 10^{-4} (Moskal et al. 2021).

- **The glow.** It ends as two 511 keV photons, the board's glow.

The Lamb shift, worked.

- **Welton's picture.** It is the circuit's: the board's zero-point ringing shakes the electron, with mean-square displacement $\langle(\delta r)^2\rangle = (2\alpha/\pi)\lambda\!\!\!^{-2} \ln(1/\alpha)$. A shaken electron feels the Coulomb cusp smoothed by $(1/6)\langle(\delta r)^2\rangle\nabla^2 V$, which for the 2S state gives $\Delta E = \alpha^5 m_e c^2 \ln(1/\alpha)/(6\pi) = 667$ MHz.
- **The full value.** Bethe's calculation with the proper logarithm gives 1,040 MHz, and the measured n = 2 interval is 1,057.8 MHz, to 3 kHz. The same measurement gives $r_p = 0.833 \pm 0.010$ fm (Bezginov et al. 2019).
- **The polarisation term.** About −27 MHz of the measured shift is the board's polarisation, the Uehling term.
- **Muonic hydrogen.** There the orbit sits 207 times closer, and the polarisation is the main term: the measured shift is 202.37 meV, the one-loop vacuum polarisation alone is 205.03 meV, and the proton's size subtracts 3.7 meV (Antognini et al. 2013).
- **The running of α.** From 1/137.036 at rest to about 1/129 at the Z mass, it is the same polarisation read as dispersion.

The turn and the hold. Everything in the calculator that moves does so in one of two ways, the odd and even loops of section 7.3:

- **The odd ring.** An odd ring of inversions cycles: the clock, the start, the turn, the π side of the pair.
- **The even ring.** An even ring holds: the latch, the √4 level, the store. When the store is space itself, its expansion is exponential, the e side.

The two are joined by the quarter turn, $V = L\,dI/dt$: a turning phase feeds a store and a store feeds a turn, and the $\sqrt{2} \rightarrow \sqrt{4}$ swing is the one cycle. A held state driven by a turning one answers at twice the frequency, because it couples to the square, $\cos^2\omega t = (1 + \cos 2\omega t)/2$. That covers single-phase power, ball lightning's 99.4 Hz, the inflaton-driven curvature $R \propto ½ + (3/2)\cos 2mt$ (section 11.3), and the pulse of any store that is squeezed and released.

The expansion itself does not oscillate (section 13): the store holds and the phases turn. A biological pulse, a pacemaker cell firing when its stored charge crosses a threshold (van der Pol and van der Mark 1928), is the same logic in a membrane. The calculator gives it the mechanism, not a number.

5.4 The circuit beside Bass's vacuum

The reactive-board circuit is the author's own electrical reading, inductance included. It is built on Steven D. Bass's emergent-vacuum work, which supplies the subcomponent dynamics and not the circuit. Section 5.3 quotes his statements; Table 26 sets each piece of the circuit beside its counterpart.

Table 26. The circuit and its counterparts in Bass's papers.

The circuit	Bass's vacuum	Where
the board, a static cloud everywhere	the vacuum is the many-body ground state of degrees of	Bass 2025 (arXiv:2508.15289); Bass 2026 (arXiv:2606.19403)

The circuit	Bass's vacuum	Where
	freedom above the emergence scale, $M \approx 10^{16}$ GeV	
particles are reactions, not particles or waves	particles are “the stable long range excitations” of that ground state; spin-1 excitations make it a gauge theory	the same
virtual particles are sub-reactions	zero-point energy, $\frac{1}{2}\hbar\omega$ per mode, bosons counting + and fermions −; zero for massless particles, so only gapped reactions load the vacuum	Bass and Krzysiak 2020, equations 4 and 5
semiconductive before action, a gap	the superconductor parallel: quasiparticles “develop a mass gap from interaction with the condensate”; in particle physics the Higgs and QCD condensates play that role	Bass 2026
inductance	screening currents expel the magnetic field, and the photon becomes a massive plasmon, “a non-relativistic precursor” of the Higgs mechanism. In circuit terms the W mass is the vacuum's kinetic inductance: screening depth $\hbar c/m_W = 2.46 \times 10^{-18}$ m, cutoff $m_W c^2/h = 1.94 \times 10^{25}$ Hz	Bass 2026; the circuit numbers are the author's
the hum: balanced phases cancel and leave a share	zero-point and condensate energies cancel, keeping the vacuum the same everywhere, as the Gibbs–Duhem relation does in quantum liquids; what is left is of order v^2/M	Bass 2024; Bass 2026
action only where the phases have shifted	gauge symmetries “dissolve” above M; below it, an effective theory	Bass 2020 (arXiv:2004.05489); Bass 2026

The circuit	Bass's vacuum	Where
the Higgs just past the edge, held	the vacuum is near-critical, with "possible new critical phenomena"	Bass 2024
positronium as the board's exciton	his J-PET positronium experiments	J-PET Collaboration 2025 (arXiv:2511.20098) and 2026 (arXiv:2607.19495)

What the circuit adds. It adds explicit electrical numbers that Bass's papers do not state:

- the base ohms, $Z_0 = 376.73\ \Omega$;
- the reactance split;
- the three-phase balance;
- the cube geometry of the $\sqrt{2} \rightarrow \sqrt{4}$ torus swing.

The emergence scale, with the factor in front taken as 1.

- **The scale the hum needs.** $v^2/\mu_vac = 2.7 \times 10^{16}$ GeV.
- **The crossings.** The forces cross only in pairs, in the two-loop running with PDG 2026 inputs:
  - $\alpha_1 = \alpha_2$ at 1.1×10^{13} GeV;
  - $\alpha_1 = \alpha_3$ at 1.7×10^{14} GeV;
  - $\alpha_2 = \alpha_3$ at 3.0×10^{16} GeV.
- **Which crossing.** The author's rule says action starts where phases overlap, and only two can overlap at once. Coming down from where nothing has shifted, the first overlap is weak = strong. Taking that crossing as Bass's M gives the hum within 1.1σ (section 13).
- **What makes it exact.** $\alpha_s = 0.1171$, -1.0σ from PDG.
- **The neutrinos.** If the lightest neutrino is also v^2/M, it is 2.0 meV and the sum is 61 meV, under the cap of about 64 meV from DESI's second release with the microwave background. This is untested. The measured splittings need factors of 4 to 25 in front, so only the lightest neutrino could sit at v^2/M.
- **The same lever as the balance test.** At $\alpha_s = 0.1171$ the three-phase rule still holds:
  - a top of 172.21 ± 0.38, -0.15σ against the pole mass;
  - a Higgs of 125.91 ± 0.93, -0.8σ.

  Both leads hinge on α_s and the top mass, so α_s to ± 0.0002 and a top threshold scan test both.
- **Near the edge.** Bass puts the vacuum "within about 1.3σ of being stable". With PDG 2026 inputs the calculator gets 1.6σ with the pole mass and 3.6σ with the direct mass (section 9.6). Bass explains why the Higgs sits near the edge, a critical point where the symmetries emerge. The author's rule says how far past it: the 0.955% three-phase offset.

5.5 Beside the standard theories

On this layer the standard theories compute every digit, and the calculator reproduces each of them as an identity or a reading. It adds no number here; it changes the account of what a virtual particle is.

Table 27. The board on one yardstick.

Quantity	Standard theory	The calculator	Measured	Standing
impedance of free space	$\sqrt{(\mu_0/\varepsilon_0)}$ = 376.730 Ω (Maxwell)	$2\alpha \cdot h/e^2$, the board's base ohms	376.730 Ω (Mohr et al. 2024)	exact; same
pair threshold	$2m_ec^2$ = 1.022 MeV (Dirac 1930)	the board's gap	pairs made by light on light (STAR Collaboration 2021; CMS Collaboration 2025)	same
the electron's g/2	QED's series in α	the load α/2π as the first term, then QED's series	1.001159652180599(13) (Fan et al. 2023)	same; the calculator supplies the first term's geometry and no digit
the electron's spin	ħ/2 from the Dirac equation	half the rest energy moving round the Compton circle: $(\tfrac{1}{2}m_ec^2/c)\lambda\!\!\!^{-} = \hbar/2$	ħ/2	reading
virtual particles	internal lines of perturbation theory	the board's reactive response, stored and returned within a cycle	the Casimir force computed without zero-point energy (Jaffe 2005)	reading; no number
hydrogen's Lamb shift at n = 2	QED, 1,057.8 MHz with the proton's radius	the board's jitter and polarisation (Welton 1948)	1,057.8 MHz, to 3 kHz (Bezginov et al. 2019)	same
the electron's three lengths	r_e, $\lambda\!\!\!^{-}_e$ and a_0 in the ratio α : 1 : 1/α	an inversion through the Compton sphere, $r_e a_0 = \lambda\!\!\!^{-}_e{}^2$	exact	reading
light on light; vacuum birefringence	the Euler–Heisenberg term	the board's non-linear reactance	8.2σ (ATLAS Collaboration 2019); 46 ± 4% polarisation (Stewart et al. 2026)	same

Quantity	Standard theory	The calculator	Measured	Standing
the muon's anomaly	116 592 033 ± 62 × 10^{-11} with the 2025 lattice hadronic term	the same load, with the board's hadronic polarisation	116 592 070.5 ± 14.8 × 10^{-11} (Muon g−2 Collaboration 2025)	same; +0.6σ

The score on this layer.

- **The standard theories** compute every quantity in the table from first principles, and every one agrees with its measurement within its errors.
- **The calculator** gives the same numbers, by identity. Its own content here is the vocabulary: the board, its reactive response and the two stores. It also contributes the inductive reading of Bass's vacuum (section 5.4).
- **Origination. M** for every number; **N** for the circuit reading of Bass's vacuum, the store split of the electron's spin and the reading of $r_e a_0 = ƛ_e^2$ as an inversion.

6. Spacetime: the two tables, light and the views

On the calculator, spacetime is the pair of tables. The sum table is energy and momentum, $E^2 = (pc)^2 + (mc^2)^2$; the difference table is the interval, $s^2 = (ct)^2 - x^2$; and zero difference, the 45° line, is light. Relativity's three curves, the circle, the hyperbola and the line where they touch, are the three behaviours of one complex square (section 3.4.2). Every view of the cube loses one direction, and a straight step at an angle costs a staircase of legs. Every number in this section is the standard one. What the calculator adds is the single figure that holds them.

6.1 Energy and momentum: the sum table

Write the energy as the hypotenuse of a right triangle whose legs are pc and mc^2. At a fixed energy the states lie on a circle in the (pc, mc^2) plane; a particle at rest sits on the mass axis, and light sits on the momentum axis. The rest mass comes back from the difference table: $(mc^2)^2 = E^2 - (pc)^2$, so every rest mass in the particle tables is a cell of the difference table.

Table 28 sets the three behaviours of one square, the circle, the hyperbola and the 45° line, beside the physics each carries and the measurement that checks it.

Table 28. The three behaviours of one square, in physics.

Table	Curve	In physics	Check
sum, $a^2 + b^2 = c^2$	a circle at fixed c	$E^2 = (pc)^2 + (mc^2)^2$ at fixed energy; bound orbits	the kinematics of every collision
difference, $a^2 - b^2 = d^2$	a hyperbola at fixed d	the interval $s^2 = (ct)^2 - x^2$ under boosts; the rest mass from E and p	time dilation at γ = 29.33, to 0.2% (Bailey et al. 1977)

Table	Curve	In physics	Check
zero difference, a = b	the 45° line	light: s = 0 and m = 0	the photon's mass below 10^{-18} eV (PDG 2026)

Worked: a muon in a storage ring. At γ = 29.33 a muon of 105.658 MeV has $E = \gamma mc^2$ = 3,098.96 MeV and $pc = \sqrt{(E^2 - (mc^2)^2)}$ = 3,097.16 MeV. Its clock runs on the difference: proper time is $d\tau = dt\sqrt{(1 - \beta^2)} = dt/\gamma$, so the laboratory lifetime is $\gamma\tau_0$ = 29.33 × 2.19698 μs = 64.44 μs. The CERN storage ring measured that dilation to 0.2% (Bailey et al. 1977).

Standing. M: special relativity (Einstein 1905; Minkowski 1908). **N**: reading E, pc and mc^2 as the cells of the author's two tables (section 3.4).

6.2 Light: the 45° line

Zero difference is where the circle and the hyperbola touch: equal legs, s = 0, no mass. It is also the line of equal shares (section 7.2): a light wave carries equal electric and magnetic energy at every instant (section 5.1).

- **No third leg.** The photon's mass is below 10^{-18} eV (PDG 2026).
- **The same line for gravity.** Gravitational waves and light from GW170817 arrived together, so their speeds agree to about 3 parts in 10^{15} (LIGO Scientific and Virgo Collaborations 2017).
- **No preferred frame.** The line has no ether wind (Michelson and Morley 1887). A board with physical properties but no state of motion is Einstein's own description of space after relativity (Einstein 1920), and it is the board of section 5.

Standing. M: every statement. **N**: light as the zero-difference line of the author's tables, shared with the 45° balance of section 7.2.

6.3 Time as the held axis

The calculator holds time as the fourth axis, and a rate is a quarter turn into it (section 4.4). The interval factors, $(ct)^2 - x^2 = (ct + x)(ct - x)$: a boost stretches one factor and shrinks the other by the same amount, so the product holds (at ct = 5, x = 3 and β = 0.6, $s^2 = 16$ before and after; section 4.4.1). Proper time is the root of the difference. Four dimensions is also where the unit diagonal reaches exactly 2, the held level $\sqrt{4}$ (section 3.7.2), and gravitational waves are damped as they would be in four dimensions (section 4.7.2).

Standing. M: the algebra and the measurements. **N**: time as the held axis of the author's fixed cube, reached by a quarter turn.

6.4 The view down a diagonal, and the staircase

A view loses one direction. A projection from three dimensions to two has a one-dimensional kernel, its line of sight; for the Clarke matrix, K·(1, 1, 1) = 0.

- **Thirteen lines of sight.** The cube's 26 non-centre points lie on 13 lines through the centre: 3 face axes, 6 edge axes and 4 corner axes.

- **Shadows.** Down a unit direction n the unit cube casts a shadow of area |n_x| + |n_y| + |n_z|, which is 1, √2 and √3 on the three kinds of axis: the root ladder as shadow areas. The mean over all directions is 3/2 (Cauchy 1841; Monte Carlo 1.49993).
- **Corner-on.** The view down a corner axis is the Clarke frame and the tribimaximal matrix (section 10.1).

The sequence tester reads the tilt in volts. Two equal loads and a capacitor with X_C = R, in star with the star point floating, put (3 ± √3)/√10 of the phase voltage across the two loads. Their ratio is 2 + √3 = 3.732 = cot 15°, and which load is bright tells ABC from ACB, the order no single phase carries.

The staircase. A straight unit step at angle θ costs |cos θ| + |sin θ| in legs, on average 4/π in two dimensions and 3/2 in three. A staircase of s legs on a 45° path changes direction between segments by (2/√π)/√s radians on average, 64.65°/√s. Hill's 17.3° for lightning (Hill 1968) corresponds to s ≈ 14 legs per segment; s is fitted, not derived.

Standing.

- **M.** Cauchy's mean-projection theorem, Fortescue's sequences and the lamp-and-capacitor tester are standard.
- **N.** The 13 lines of sight with shadows 1, √2 and √3, and the tester's ratio as cot 15°, were not found.

6.5 Beside the standard theories

On this layer too the standard theories give every number, and the calculator gives the same ones. Its additions are the single figure and one identification: the tester's ratio as the cotangent of the mixing tilt.

Table 29. Spacetime on one yardstick.

Quantity	Standard theory	The calculator	Measured	Standing
energy and momentum	$E^2 = (pc)^2 + (mc^2)^2$ (special relativity)	the sum table	the kinematics of every collision	same
time dilation	$\gamma = 1/\sqrt{(1 - \beta^2)}$	the root of the difference table	0.2% at γ = 29.33 (Bailey et al. 1977)	same
light and gravity on one line	equal speeds in general relativity	both on the 45° line	about 3 parts in 10^{15} (LIGO Scientific and Virgo Collaborations 2017)	same
the number of dimensions	3 + 1, assumed	the held fourth level at √4	four, from the damping of GW170817 (Pardo et al. 2018)	same

Quantity	Standard theory	The calculator	Measured	Standing
a cube's mean shadow	a quarter of its surface, 3/2 (Cauchy 1841)	the root ladder's shadows 1, √2 and √3, mean 3/2	Monte Carlo 1.49993	same (mathematics)
streamer width at altitude	similarity in the air's density (Ebert et al. 2010)	the air's scale set by g	0.20 ± 0.02 mm·bar (Briels et al. 2008)	same
the sequence tester	(3 + √3)/(3 − √3) from circuit theory	cot 15°, the mixing tilt read in volts	3.73 on a meter	same number; the identification is new

Origination. M for every number. **N** for the thirteen lines of sight with shadows 1, √2 and √3, the tester's ratio as cot 15°, and the segment law in closed form.

7. Balance and three phases

Two rules run the calculator's physics. **Balance:** a state is balanced when its two shares are equal, the 45° line, and a state in motion sits slightly past the line, held there by a load of one coupling squared over $8\pi^2$ per loading phase, per action cycle. **Three phases:** three equal phases 120° apart sum to zero, and a start cannot use all three; the third rides the first two. Computed, the first rule gives the escape speed as √2 times the orbital speed, flat three-phase power and, at its first order, the electron's anomalous moment $\alpha/2\pi$. The second gives the forces crossing in pairs, at 1.1×10^{13}, 1.7×10^{14} and 3.0×10^{16} GeV, never at one point. This section states the cube's levels, moves and grids that the rules act on, then the two rules.

7.1 Levels, moves and grids

The count. A point with k non-zero coordinates chooses which k of the three axes are active, in C(3, k) ways, and a sign for each, in 2^k ways: $C(3, k)\cdot 2^k = 1, 6, 12$ and 8, summing to $27 = 3^3$. Distance squared is the number of unit coordinates, so the level counts the active axes, which is all that “levels count phases” says. The tesseract's corner, at 2, is the held fourth level.

The moves are the roots of 1 (checked by enumeration):

- inside-out, $p \to -p$, is -1;
- a third of a turn about a body diagonal, $(x, y, z) \to (z, x, y)$, is ω;
- a quarter turn about a face axis is i.

Together they generate the cube's 48 symmetries, every permutation of the axes times every sign pattern, which crystallographers call m3m.

The moves are bit operations.

- Inside-out, $p \to -p$, is NOT on all three bits.
- The 120° turn about a body diagonal is a cyclic shift of the bits.
- The quarter turn about a face axis is a swap with one NOT.

The count of 48 is forced. A symmetry of the cube sends face centres to face centres, so it sends each axis to plus or minus some axis: a permutation of the axes (6) times a sign on each (8). Every signed permutation is a symmetry, so there are no others, and enumeration confirms that the three moves generate all 48. Each move is the smallest operation of its order that the cube allows: $(-1)^2 = \omega^3 = i^4 = 1$.

Levels and chords. The levels are the squared chords of the moves: $|1 - e^{i\theta}|^2 = 2(1 - \cos\theta) = 1, 2, 3, 4$ at $\theta = 60°, 90°, 120°, 180°$.

Grids. The flip $r \to 1/r$ makes the grids 1, ½, ⅓ and ¼ (section 3.8). The share one leg carries of the k-cube's diagonal is $\cos^2\theta = 1/k$, and the corners project onto the diagonal at the fractions j/k with binomial counts: (1, 2, 1), (1, 3, 3, 1) and (1, 4, 6, 4, 1).

The unit cube inside the side-2 cube. Put the side-2 cube on $[0, 2]^3$, with its centre at (1, 1, 1). The unit cube's corners then sit as follows:

- the corner with no coordinate equal to 1 is a corner of the big cube;
- the three with one 1 are its edge midpoints (level $\sqrt{2}$);
- the three with two 1s are its face centres (level 1, the four-fold symmetry points);
- (1, 1, 1) is its centre.

So k ones sit at level $\sqrt{(3 - k)}$: the grading (1, 3, 3, 1) read backwards, the same complement $k \to 3 - k$ as the inside-out of section 8.1.

The tetrahedron as a third. Cutting a corner of the side-2 cube along the three face diagonals that meet at the neighbouring corners removes a tetrahedron with three perpendicular edges of length 2, of volume $(1/6) \cdot 2 \cdot 2 \cdot 2 = 4/3$. Four alternate corners remove 16/3 of the cube's 8 and leave 8/3, exactly ⅓ of the cube: a regular tetrahedron of edge $2\sqrt{2}$.

Three curves of physics. The sum, the difference and the 45° line are $E^2 = (pc)^2 + (mc^2)^2$ (a circle in (pc, mc^2) at fixed E), the interval $s^2 = (ct)^2 - x^2$ (a hyperbola under boosts), and the light cone $s = 0$.

Standing.

- **M.** The symmetry group is textbook crystallography, and the gradings (1, 3, 3, 1) and (1, 4, 6, 4, 1) are those of the Clifford algebras of three- and four-dimensional space. Gauss (1831) called +1, −1 and $\sqrt{-1}$ the direct, inverse and lateral units.
- **N.** Reading levels as active phases, and the moves as roots of 1 on the cube's own axes, were not found.

7.2 Balance and the load

Equal shares, with their proofs. A light wave carries $u_E = u_B$ at every instant (section 5.1), and three more cases carry equal shares:

- **Escape is $\sqrt{2}$ × orbit.** On a circular orbit $GMm/r^2 = mv_c^2/r$, so $v_c^2 = GM/r$. Escape needs kinetic energy equal to the well's depth, $\frac{1}{2}mv_e^2 = GMm/r$, so $v_e^2 = 2GM/r$. On the circle the kinetic energy is half the potential's size and at escape they are equal; the speed ratio is $\sqrt{2}$ because energy goes as the square of speed. This is the virial theorem (Clausius 1870).
- **Three-phase power is flat.** With voltages $\cos(\theta - 2\pi n/3)$ and currents $\cos(\theta - 2\pi n/3 - \varphi)$, each phase's power is $\frac{1}{2}[\cos\varphi + \cos(2\theta - 4\pi n/3 - \varphi)]$. Summed over n = 0, 1, 2 the three 2θ terms

are 120° apart and cancel, $1 + \omega + \omega^2 = 0$, leaving $(3/2)\cos\varphi$. With one phase nothing cancels the $\cos 2\theta$ term, and $\cos^2\theta = (1 + \cos 2\theta)/2$ peaks at twice its average: single-phase power pulses at 2f.

- **The Koide identity.** Put $\sqrt{m_k} = \mu[1 + a\cos(\delta + 2\pi k/3)]$. Over the three phases $\Sigma\cos(\delta + 2\pi k/3) = 0$ and $\Sigma\cos^2(\delta + 2\pi k/3) = 3/2$ for every δ, so $\Sigma\sqrt{m} = 3\mu$, $\Sigma m = \mu^2(3 + 3a^2/2)$ and $Q = (1 + a^2/2)/3$.
    - The waveform's DC power is 1 and its AC power $a^2/2$, so $a = \sqrt{2}$ is DC = AC.
    - The angle between $(\sqrt{m_0}, \sqrt{m_1}, \sqrt{m_2})$ and (1, 1, 1) has $\cos\theta = 1/\sqrt{(3Q)}$, so $Q = \frac{2}{3}$ is $\theta = 45°$.

  One ratio is read three ways: as a power split, as a quadratic form and as an angle.

Slightly past, and held. Nature sits just past the line and is held there:

- static friction exceeds sliding friction (Amontons 1699; Coulomb 1785);
- a balanced rock outlasts the earthquake (Housner 1963; Brune 1996);
- stick-slip governs faults (Brace and Byerlee 1966);
- the Higgs vacuum's quartic coupling dips just below zero, with a lifetime of order 10^{139} years (Andreassen, Frost and Schwartz 2018).

The load. The size of "slightly" is the load per loading phase,

$$\delta=\frac{g_i^{2}}{4\pi\cdot 2\pi}=\frac{\alpha_i}{2\pi}$$

the coupling squared over the face sphere's solid angle 4π and one full turn 2π. For the electron it is 0.00116141, the first term of its anomaly (Schwinger 1948).

The couplings run by the same unit per e-fold of scale: $1/\alpha_i(\mu) = 1/\alpha_i(m_Z) - (b_i/2\pi)\ln(\mu/m_Z)$, with b = 41/10, −19/6 and −7 read off the register of section 8.1.

Why the denominator is $8\pi^2$.

- **The geometry.** The unit four-dimensional ball has volume $\pi^2/2$, so the unit 3-sphere has area $2\pi^2$, its derivative at r = 1.
- **The loop.** A one-loop integral in four dimensions carries $\int d^4k/(2\pi)^4$, whose angular part is $2\pi^2/(2\pi)^4 = 1/8\pi^2$. The first correction to any one-phase quantity is one closed loop of that phase, and in four dimensions a closed loop costs exactly this angular share; what stands in front is the problem's own factor, which for the electron's moment is 1.
- **In SI.** $\alpha = e^2/(4\pi\varepsilon_0\hbar c)$ and $h = 2\pi\hbar$, so $\alpha/2\pi = e^2/(4\pi\varepsilon_0 hc) = 0.00116141$, checked to twelve digits: charge squared, per sphere, per quantum of action per cycle (h, not ħ; the 2π between them is the turn).
- **Schwinger's series.** $a_e = \alpha/2\pi - 0.328479(\alpha/\pi)^2 + 1.181241(\alpha/\pi)^3 - 1.9122(\alpha/\pi)^4 + \ldots = 0.00116141 - 0.00000177 + 0.00000001 - \ldots = 0.001159652$, against the measured 0.00115965218. The first term is one loop of the one loading phase, the calculator's unit; the rest is QED's series.
- **e and π as a pair.** The exponential is the function equal to its own rate, so for a phasor $d/d\theta$ $e^{(i\theta)} = i\,e^{(i\theta)}$: the rate is the phasor turned a quarter turn, which is the inductive law $V = L\,dI/dt$. At $\theta = \pi$ the phasor is −1, so $i\pi = \ln(-1)$: the half-turn is π on the angle side and is reached through e on the rate side, as $\sqrt{2}$ and $1/\sqrt{2}$ are exchanged by the flip.

Where the load applies. The unit sphere touches the cube at the six face points, the level the flip fixes, so a load is a property of one phase, and a state with several phases combines one-phase loads at 120° (section 9.1).

- **The same law in a circuit.** A current already flowing resists change, V = L dI/dt, 90° ahead. Inertia is F = m dv/dt, with momentum as flux and ½mv² as ½LI².
- **Linked loops.** Two linked flux loops share one flux: Gauss's linking integral and mutual inductance. The cube's two √2 circles linked once on the √4 sphere (section 4.6) are that geometry.

Where it stays slight. Only where $\alpha_i/2\pi$ times its count is small: 0.00116 for the electron and about 1% for the top. At the strong scale it is not small, and that is where the proton's statics sit far past the grid (section 12.1).

Standing.

- **M.** The equal-share half is the virial theorem, equipartition and resonance. The load's size is QED's first term, and the running is the renormalisation group.
- **N.** The claim that the same per-cycle unit sets the mass tables, where standard physics has no rule, was not found. It is tested in section 9.1 and section 16.

7.3 Three phases and the start rule

On the cube. Down the body diagonal from ’ to ×, the other six corners form a hexagon. It is the six-step sequence of a variable-frequency drive, with ’ and × as the zero vectors.

- **The Clarke transform** (Clarke 1943) is the cube seen down its body diagonal, with its zero row (1, 1, 1)/√3.
- **As logic.** Three inverters in a ring cannot settle and oscillate, with their nodes 120° apart; two in a ring hold. The start rule (the third phase rides the first two) is that contradiction read as a rule.

One phase rotation. A 120° turn about that axis is one phase rotation, and three phases on three axes keep |v| = √(3/2)·V_peak constant (section 4.9.1).

The line voltage. It is √3 times the phase voltage: $1 - e^{-i2\pi/3} = 3/2 + i\sqrt{3}/2$ has squared modulus 9/4 + 3/4 = 3, and the two phasors with their difference form the 30-60-90 triangle.

The ring oscillator, timed. Three inverters in a ring each have delay t_d:

- a rising edge at node 0 becomes a falling edge at node 1 after t_d, a rising edge at node 2 after 2t_d, and a falling edge back at node 0 after 3t_d;
- so node 0 is inverted every 3t_d and the period is 6t_d;
- node k lags by k·t_d, which is k × 60° of the period, and is inverted k times, another k × 180°: in all k × 240° ≡ −k × 120°.

A loop of n inverters returns a node's level to itself after n inversions. With n even the level agrees with itself and the loop can rest; with n odd it contradicts itself and must keep changing.

The start rule as a theorem. In space, turns do not commute. A quarter turn about x followed by a quarter turn about y has trace 0, so it is a 120° turn, about the body diagonal (1, 1, −1). In quaternions (Hamilton 1843),

$$\frac{1+i}{\sqrt2}\cdot\frac{1+j}{\sqrt2}=\frac{1+i+j+k}{2}=\cos 60^{\circ}+\sin 60^{\circ}\,\frac{i+j+k}{\sqrt3}$$

The quaternion that performs the third turn is the tesseract's corner (1, 1, 1, 1)/2 placed on the unit sphere. Once two perpendicular quarter turns have happened, the third phase is their product and is not chosen.

The covers. Two 24s and two 48s meet here and are not the same:

- the cube's 24 rotations (S_4) and its 48 symmetries with reflections;
- the 24 Hurwitz units, the binary tetrahedral group 2T: ±1, ±i, ±j, ±k and (±1 ± i ± j ± k)/2, the eight axis units with the tesseract's sixteen normalised corners, closed under multiplication, the vertices of the 24-cell;
- the 48 unit quaternions of the binary octahedral group 2O.

The 48 quaternions double-cover only the 24 rotations and cannot represent a reflection. The 24 Hurwitz units double-cover the 12 rotations that keep the two inscribed tetrahedra in place. $(1 + i)/\sqrt{2}$ and $(1 + j)/\sqrt{2}$ are quarter turns, outside 2T because each swaps the two tetrahedra, and their product $(1 + i + j + k)/2$ is a third turn inside it. Odd times odd is even, and $2O/2T \cong Z_2$ is the rule in one line.

The couplings cross in pairs. To one loop each $1/\alpha_i$ is a straight line in $\ln \mu$ with slope $-b_i/2\pi$, starting at m_Z from $1/\alpha_1 = 59.0$ (in the $\sqrt{(5/3)}$ normalisation), $1/\alpha_2 = 29.6$ and $1/\alpha_3 = 8.45$. Three lines meet at one point only if the intercept gaps are proportional to the slope gaps, and the Standard Model's numbers give two different ratios:

- $(1/\alpha_1 - 1/\alpha_2)/(b_1 - b_2) = 4.05$;
- $(1/\alpha_2 - 1/\alpha_3)/(b_2 - b_3) = 5.51$.

At one loop the pairs cross at 1.0×10^{13}, 2.4×10^{14} and 1.0×10^{17} GeV; at two loops, with PDG 2026 inputs:

Pair	Crossing scale	$1/\alpha$ there
hypercharge = weak	1.1×10^{13} GeV	42.3
hypercharge = strong	1.7×10^{14} GeV	40.4
weak = strong	3.0×10^{16} GeV	46.2

The three come within 3.2 in $1/\alpha$ of each other, at 1.7×10^{14} GeV. Supersymmetric models add particles precisely to change the slopes until the two ratios match (Amaldi, de Boer and Fürstenau 1991).

$\sin^2\theta_W = 3/8$ where the first two phases cross.

- $\sin^2\theta_W = g'^2/(g^2 + g'^2)$, and hypercharge is normalised as $g_1^2 = (5/3)g'^2$, the 5/3 being $\mathrm{Tr}\, Y^2/\mathrm{Tr}\, T_3^2 = (10/3)/2$.
- Where $\alpha_1 = \alpha_2$, $g^2 = (5/3)g'^2$, so $\sin^2\theta_W = 1/(8/3) = 3/8$: the fraction of the generation's charge squared that sits in the weak slots, $2/(2 + 10/3)$.

It runs from there to 0.23122 at the Z.

Standing.

- **M.** The power engineering is Steinmetz (1893), Fortescue (1918), Park (1929) and Clarke (1943). The non-commuting turns and Hamilton's rule i·j = k are classical. The running and the pairwise crossing are the Standard Model's: grand unification needs new particles to make the couplings meet (Georgi and Glashow 1974; Amaldi, de Boer and Fürstenau 1991).
- **N.** Reading the pair crossings as the start rule, and the start rule as the parity of the double cover, were not found.

7.4 Beside the standard theories

Every number the two rules produce on this layer is a standard number. The calculator's own claims here are two: that the load which gives the electron its first anomalous term also places masses (tested in section 9.4), and that the pairwise crossing of the couplings is the start rule rather than a defect to be repaired.

Table 30. Balance and three phases on one yardstick.

Quantity	Standard theory	The calculator	Measured	Standing
the cube's symmetries	the 48 operations of O_h (crystallography)	generated by the three moves, the roots −1, ω and i	by enumeration	same (mathematics)
escape against orbit	√2, from the virial theorem	equal shares, the 45° line	orbital mechanics	same
three-phase power	constant, since 1 + ω + ω² = 0 (Steinmetz 1893)	the balanced hexagon of the cube's corners	every three-phase grid	same
the electron's first anomalous term	α/2π = 0.00116141 (Schwinger 1948)	one loading phase's g²/(4π·2π)	a_e = 0.00115965218059(13) in full (Fan et al. 2023)	same; the geometry of 8π² is the calculator's
how far past balance a mass sits	no rule	α_i/2π per loading phase, combined at the phases' angles	the top and the Higgs (section 9.4)	new rule
whether the couplings meet	not with the Standard Model's particles; with new ones in grand unification	in pairs, because three phases cannot start together	crossings at 1.1 × 10^{13}, 1.7 × 10^{14} and 3.0 × 10^{16} GeV; no superpartner seen	same running; the reason is the calculator's

Quantity	Standard theory	The calculator	Measured	Standing
$\sin^2\theta_W$ where $\alpha_1 = \alpha_2$	3/8, a unification input	the weak slots' share of the charge squared, 2/(2 + 10/3)	runs to 0.23122 at the Z	same
the one-loop running counts	41/10, −19/6 and −7	read off the 16-state register (section 8.1)	the same numbers	same
odd and even rings of inverters	an odd ring oscillates, an even ring latches	the start rule and the held level	every ring oscillator and latch	same

Origination. M for every number. **N** for the load applied to the mass tables, the three-phase start as the reason the couplings cross in pairs, and the start rule as the parity of the binary octahedral cover.

8. Matter: one generation as a 16-state register

One generation of matter, computed on the tesseract, is a 16-state register: five slots, three colour slots worth −⅓ each and two weak slots worth +½ each, with an even number filled. Charge comes in thirds and halves because the grids are the inverse levels squared; every anomaly sum vanishes because the sums are the parity checks of the register's code; and hypercharge's normalisation $\sqrt{(5/3)}$, with $\sin^2\theta_W = 3/8$ where the first two couplings meet, is read off the same table. Antimatter is the register turned inside out in all five slots. These are the Standard Model's own representations, reached by a different road; the calculator adds the reasons and no new state.

8.1 The register

The register. One generation is five slots: three colour slots worth $-\frac{1}{3}$ each and two weak slots worth $+\frac{1}{2}$ each, with an even number filled. That leaves four free slots and $2^4 = 16$ states, the tesseract's corners.

- **The charges.** $Y = \frac{1}{2}$(weak slots filled) $- \frac{1}{3}$(colour slots filled), $T_3 = \frac{1}{2}$(first weak slot − second), and $Q = Y + T_3$.
- **The states.** The 16 are: ν^c ($Y = 0$); e^+ ($Y = 1$); the quark doublet, three colours each of d ($Y = 1/6$, $Q = -\frac{1}{3}$) and u ($Q = \frac{2}{3}$); $\bar{u}$ (three, $Y = -\frac{2}{3}$); $\bar{d}$ (three, $Y = \frac{1}{3}$); and the lepton doublet e ($Q = -1$) and ν ($Q = 0$), both with $Y = -\frac{1}{2}$.

What the table gives.

1. **Why sixteen.** Five slots filled or empty give $2^5 = 32$ words, and the parity condition halves them to 16. Without it the 32 would hold the generation and its mirror, the 16-bar of odd parity.
2. **Charge in thirds and halves.** With n_c colour slots filled and weak slots w_1 and w_2:
    - $Y = -n_c/3 + (w_1 + w_2)/2$;
    - $T_3 = (w_1 - w_2)/2$;
    - $Q = -n_c/3 + w_1$.

Every charge is a count of weak slots minus a third of the colour count, so colour triplets carry ⅓ or ⅔ and singlets 0 or 1. The positive charges on the cube's eight corners, 0, ⅓, ⅔ and 1, are the T and V slots of Harari and Shupe (1979).

3. **Eight bits in both directions.** Complementing the four free slots sends $n_c \to 3 - n_c$ and $w_1 \to 1 - w_1$, so $Q \to n_c/3 - w_1 = -Q$ for all 16, while Y and T_3 are not negated: e^+ ($Y = 1$, $T_3 = 0$) goes to the e of the doublet ($Y = -½$, $T_3 = -½$).
    - **Inside the generation.** This is the tesseract's $p \to -p$ on its four free coordinates, determinant +1, a rotation inside the generation.
    - **Charge conjugation proper.** $Y \to -Y$ and $T_3 \to -T_3$ is the inside-out of all five slots. It flips the parity and lands in the 16-bar, a reflection.
    - **Gravity.** Neither map touches the three-dimensional cube's face sphere, which is the calculator's reading of why antimatter falls as matter does (section 11.2).
4. **Every balance sum vanishes.**
    - $\Sigma Y^3 = 6(1/6)^3 + 3(-2/3)^3 + 3(1/3)^3 + 2(-1/2)^3 + 1^3 = 1/36 - 8/9 + 1/9 - 1/4 + 1 = 0$.
    - The $SU(2)^2$ sum over doublet members is $6(1/6) + 2(-1/2) = 0$.
    - The $SU(3)^2$ sum over triplets and anti-triplets is $6(1/6) + 3(-2/3) + 3(1/3) = 0$.

    Each is a one-loop triangle with three gauge phases at its corners, whose non-zero value would break a conservation law. Here they are the parity checks of the even-parity code of length 5 (minimum distance 2): one flipped bit leaves the generation.
5. **The hypercharge root.** $\mathrm{Tr}\, Y^2 = 6(1/36) + 3(4/9) + 3(1/9) + 2(1/4) + 1 = 10/3$, and $\mathrm{Tr}\, T_3^2 = 8 \times (½)^2 = 2$.
6. **The running counts.** For a gauge phase with group G the one-loop coefficient is

    ```
    b=-\frac{11}{3}C_2(G)+\frac{2}{3}\sum_{\rm Weyl}T(R)+\frac{1}{3}\sum_{\rm scalar}T(R)
    ```

    for the phase loading itself, the moving spin-½ states it loads, and the held scalar.
    - **U(1)**, in the $\sqrt{(5/3)}$ normalisation: $(3/5)[(2/3) \cdot 3 \cdot (10/3) + (1/3) \cdot 2 \cdot (1/4)] = 4 + 1/10 = 41/10$.
    - **SU(2):** $-22/3 + (2/3) \cdot 12 \cdot (1/2) + (1/3) \cdot (1/2) = -19/6$.
    - **SU(3):** $-11 + (2/3) \cdot 12 \cdot (1/2) = -7$.

    A spin-½ state in motion has two helicities and loads a phase with weight ⅔ per Weyl state, a held scalar loads with ⅓, and a non-abelian phase loads itself with $-11/3$ because its own charge circulates with the opposite sign: asymptotic freedom.
7. **The missing leg.** The W and Z carry a third, longitudinal polarisation that the photon does not (section 3.7.1). It is seen directly: ATLAS observed WZ events with both bosons longitudinally polarised at 7.1σ (ATLAS Collaboration 2023).

A computer, in the exact sense.

- **Register.** A corner is a 3-bit word and a tesseract corner a 4-bit word.
- **Instruction set.** The 48 operations are every bit permutation times every bit complement, all reversible (Bennett 1973; Fredkin and Toffoli 1982).
- **Code and clock.** The register is a parity code, and the ring oscillator is its clock.
- **Clock rate.** The fastest orthogonal step takes $h/4E$ (Margolus and Levitin 1998).

The lineage of the computer reading. Zuse (1969), Feynman (1982), Fredkin and Toffoli (1982), Wheeler's it-from-bit (1989), Lloyd (2000, 2002) and 't Hooft (2016). What the cube adds is the specific machine:

- the register;
- the instruction set;
- the code;
- the clock;
- a tri-state bus (above, on or below the line: 1, Z, 0);
- the substrate, a semiconductor with a $2mc^2$ gap and Dirac's holes as positive carriers.

Standing.

- **M.** This is the Standard Model's generation exactly, and the slot-and-parity construction is the binary code of the SO(10) spinor: Fritzsch and Minkowski (1975), Wilczek and Zee (1982), and the five-slot exposition of Baez and Huerta (2010).
- **N.** The calculator reaches it by another road and adds no state. Its additions are readings: charges as the grids of phase counts, the anomaly conditions as a code's parity checks, antimatter as inside-out, and the register as a reversible machine.

8.2 Worked example: a generation step

5. **Frame.** The lepton layer puts $\sqrt{m_e}$, $\sqrt{m\mu}$ and $\sqrt{m\tau}$ on x, y and z. The zero axis '–× is their common level, and the roots sit on a 45° cone about it.
6. **Motion, not a turn of the cube.** The data point moves a third of a turn round the fixed zero axis, (x, y, z) → (z, x, y), which advances the phase $\delta + 2\pi k/3$ while the cube stays. One phasor taken three times,

   ```
   \sqrt{m_k}=\mu\left[1+\sqrt{2}\cos\!\left(\delta+\tfrac{2\pi
   k}{3}\right)\right],\qquad \delta=\tfrac{2}{9}
   ```

   gives 1 : 206.77 : 3477.5 against the measured 1 : 206.768 : 3477.37.
7. **Inverse.** The half turn negates all three roots, and the masses and Koide's Q are unchanged. Antileptons have their leptons' masses: $e\pm$ within 8×10^{-9} and $\tau\pm$ within 2.8×10^{-4} (Abe et al. 2007).
8. **Mirror.** The neutrino triplet comes from flipping one root (section 9.1). With the cube fixed, the flip is a half turn in the plane of that root and the held axis, so the neutrinos are reached by turning through the fourth dimension.

8.3 The inverse cell: particle against antiparticle

One half turn is CPT, which relativistic quantum field theory guarantees, so the inverse cell must hold every particle's image exactly. Every comparison found agrees within 1.5σ:

- **The positron.** Its g-factor matches the electron's within $(-0.5 \pm 2.1) \times 10^{-12}$ (−0.2σ; Van Dyck, Schwinberg and Dehmelt 1987).
- **The antimuon.** Its g-factor matches within −0.9σ and its lifetime within +0.3σ (PDG muon listing).
- **The antiproton.** Its moment is −2.7928473441(42) μ_N (Smorra et al. 2017), −0.1σ from the proton's; its charge-to-mass ratio matches within 16 parts per trillion (Borchert et al. 2022).

- **The antitop.** The top–antitop mass difference is -0.15 ± 0.21 GeV (CMS Collaboration 2017), -0.7σ.
- **The antitau.** Its mass matches within 2.8×10^{-4} (Abe et al. 2007).
- **Antihydrogen.** Its 1S–2S line matches hydrogen's to a couple of parts in 10^{12} (Ahmadi et al. 2018), its hyperfine splitting gives 1,420.4 ± 0.5 MHz (Ahmadi et al. 2017), and its Lamb shift agrees within 11% (ALPHA Collaboration 2020). It falls down (-1.2σ; Anderson et al. 2023).
- **Neutrino against antineutrino.** Solar neutrinos give $\Delta m^2_{21} = 6.10\ ^{+0.95}_{-0.81} \times 10^{-5}$ eV² (Super-Kamiokande Collaboration 2024), against the reactor antineutrinos' $(7.50 \pm 0.12) \times 10^{-5}$ eV² (JUNO Collaboration 2025): -1.5σ. The calculator and CPT both require equality, so this is the one inverse-cell tension to watch.

8.4 Beside the standard theories

The register is the Standard Model's generation exactly, including the right-handed neutrino of the SO(10) spinor, so on this layer the two give the same states and the same numbers. The difference is in what each takes as given: the Standard Model takes the hypercharges as inputs and requires the anomalies to cancel, while the calculator derives both from the slot rule.

Table 31. Matter on one yardstick.

Quantity	Standard theory	The calculator	Measured	Standing
the states of one generation	15 Weyl states, 16 with a right-handed neutrino (the 16 of SO(10))	the 16 corners of the tesseract: five slots, even parity	15 seen; the right-handed neutrino unobserved	same as SO(10); both carry the right-handed neutrino
charges in thirds and halves	the hypercharges, as inputs	the grids of the levels $\sqrt{3}$ and $\sqrt{2}$	every quark and lepton charge	exact
anomaly cancellation	required for consistency, arranged by the hypercharges	the parity checks of the register's code	the theory's consistency	exact
hypercharge's normalisation	$\mathrm{Tr}\, Y^2 \div \mathrm{Tr}\, T_3^2 = (10/3) \div 2 = 5/3$, a unification input	read off the register	$\sin^2\theta_W$ runs from 3/8 to 0.23122 at the Z	exact
antimatter	the CPT conjugate	the five-slot inside-out, the mirror 16	the positron's g within 2×10^{-12}, the antiproton's moment at -0.1σ, antihydrogen's 1S–2S line within a few parts in 10^{12}	same

Quantity	Standard theory	The calculator	Measured	Standing
particle against antiparticle	CPT requires equality	the half turn requires equality	solar against reactor Δm^2_{21} at -1.5σ	same; the one tension to watch
how many generations	three, an input	three phases, with the fourth level held	$N_\nu = 2.9963 \pm 0.0074$ from the Z width	reading
the lepton mass ratios	free Yukawa couplings	one phasor taken three times: 1 : 206.77 : 3,477.5	1 : 206.768 : 3,477.37	new number (section 9.1)

Origination. M: the generation is the Standard Model's (Georgi and Glashow 1974), the 16 of SO(10) (Fritzsch and Minkowski 1975; Wilczek and Zee 1982), and the five-slot construction is Baez and Huerta's (2010). **N**: the anomaly sums as the parity checks of an even-parity code, antimatter as the five-slot inside-out, and the running counts read off the register.

9. Mass: the balance line and the vacuum's edge

The calculator computes masses with its balance rule: a family of three masses is three phases, balanced when its DC power equals its AC power. The charged leptons sit on that line to a few parts in a million: $Q = 0.6666645 \pm 0.0000051$ against $⅔$, an angle of $44.99991° \pm 0.00022°$ to the diagonal, with the τ predicted from the electron and muon at 1,776.969 MeV against $1{,}776.93 \pm 0.09$. The same equation, with one root on the flipped branch, gives the neutrino masses: 0.364, 8.66 and 50.13 meV, normal ordering. The Higgs vacuum sits just past its own edge, held by the three-phase load $\delta_3 = 0.955\%$, which places the top at 172.57 ± 0.38 GeV and the Higgs at 125.20 ± 0.92 GeV. The Standard Model has no mass relation: every one of these masses is, for it, an input. The lepton relation and its phase are inherited from Koide, Foot and Brannen; the reason, the neutrino reading and the load are the calculator's.

9.1 The balance line in the lepton masses

The charged leptons. The PDG 2026 masses give $Q = 0.6666645 \pm 0.0000051$, an angle of $44.99991° \pm 0.00022°$ to the diagonal. From the electron and muon alone, $Q = ⅔$ is a quadratic in $\sqrt{m_\tau}$ whose heavy root gives $m_\tau = 1776.969$ MeV, against 1776.93 ± 0.09. The phase is $\delta = 0.2222248 \pm 0.0000063$ rad $= 2/9$ at 0.4σ, and $3\delta = Q$.

The charged leptons, step by step.

- **The family.** The square roots of the PDG 2026 masses are 0.714842, 10.279026 and 42.153648. Their sum is $S = 53.147515$, so $\mu = S/3 = 17.715838$ $\sqrt{\text{MeV}}$, and $\Sigma m = 1883.0994$, so $Q = 1883.0994/53.147515^2 = 0.6666645$. From Q, $a = \sqrt{(2(3Q - 1))} = 1.414209$, and $\cos\theta = 1/\sqrt{(3Q)}$ gives $\theta = 44.99991°$.
- **The phase.** $(\sqrt{m_e}/\mu - 1)/a = (0.040350 - 1)/1.414209 = -0.678577$, whose arccosine is 132.73253°. The electron is one phase step from the reference, so $\delta = 132.73253° - 120° = 12.73253° = 0.2222248$ rad.

- **The errors.** They come almost entirely from m_τ (±0.09 MeV), propagated by Monte Carlo: Q ± 0.0000051, θ ± 0.00022° and δ ± 0.000006 rad.

The τ from e and μ alone. Write s = √m_e + √m_μ = 10.993868, t = m_e + m_μ = 106.169374 and x = √m_τ.

- **The quadratic.** Q = ⅔ means (s + x)² = (3/2)(t + x²), that is ½x² − 2sx + (3t/2 − s²) = 0, so x = 2s ± √(6s² − 3t) = 21.98774 ± 20.16637.
- **The heavy root.** It gives m_τ = 1776.969 MeV, against 1776.93 ± 0.09 (−0.4σ), and lies between the two precise methods (Table 68).
- **The light root.** At 3.32 MeV it is the other place a third phase could sit, and nothing does: the balance equation is quadratic in the third root.

The domain. The balance rule's quantity is the pole mass of a free excitation of the board that is a Yukawa leg of v. That covers the leptons and the Higgs sector. Quarks, whose masses run, and hadrons, whose mass is 99% field energy, are outside it: at the conventional scales (PDG 2026) the down and up families give Q = 0.732 and 0.845, and the heavy triple c, b, t gives 0.663, values that move with the scale.

Standing.

- **M.** The relation is Koide's (1982). The 45° reading is Foot's (1994). The three-phase form with phase near 2/9 is Brannen's (2006), and Żenczykowski (2012) found the phase indistinguishable from 2/9.
- **History.** Koide's relation of 1982 gave 1,777 MeV for the τ when the measured mass was 1,784 ± 3 MeV. The measurement moved to him.
- **N.** The reason, balance read as DC power equal to AC power by the same rule as light and the orbit, was not found, nor the phase read as the balance ratio shared over three phases, 3δ = Q.

9.2 The neutrino masses

The neutrinos. With all three roots positive, the equation cannot reach 45°. With the lightest root on the flipped branch, it crosses Q = ⅔ once:

- **normal ordering.** m_1, m_2, m_3 = 0.364, 8.66 and 50.13 meV, summing to 59.2 meV;
- **inverted ordering.** The solution sums to 102.6 meV, excluded by the 64 meV cap from DESI's second data release with the microwave background (DESI Collaboration 2025).

The neutrinos, both orderings. Only the splittings are measured: $\Delta m^2_{21} = 7.49 \times 10^{-5}$ eV², and $\Delta m^2_{31} = 2.513 \times 10^{-3}$ eV² for the normal ordering ($|\Delta m^2_{32}|$ the same for the inverted).

- **All roots positive.** Q(m_1) falls from 0.5851 at $m_1 \to 0$ to ⅓ as the three masses become degenerate, and never reaches ⅔.
- **Why a root may flip.** Every root has a ± partner, because the equation is in √m, the physical mass is the square, and the sign is the relative phase of the state's amplitude: exactly the freedom the flip gives every root.
- **The normal ordering.** With the lightest root on the flipped branch, Σ√m falls and Q rises, crossing ⅔ once at m_1 = 0.3642 meV: 0.364, 8.662 and 50.131 meV, sum 59.157 meV.
- **The inverted ordering.** The same branch crosses ⅔ at m_3 = 2.934 meV: 49.464, 50.216 and 2.934 meV, sum 102.61 meV.

- **The other sign patterns.** They never cross ⅔: with $\sqrt{m_2}$ flipped Q stays between 1.66 and 3.38, and with $\sqrt{m_3}$ flipped it stays above 3.

Cosmology caps the sum at 64 meV (DESI DR2 with the microwave background, 95%), which excludes the inverted solution and leaves the normal one, 5 meV under the cap and 0.4 meV above the normal ordering's floor of 58.8 meV.

What can and cannot see these masses.

- **Too close to the floor.** The sum sits so close to the floor that no planned cosmological or β-decay measurement can tell this solution from $m_1 = 0$.
- **Double beta decay.** Its effective Majorana mass, $m_{\beta\beta}$ = 1.3 to 4.1 meV depending on the Majorana phases, is below the reach of the planned searches:
  - KamLAND-Zen's complete data cap $m_{\beta\beta}$ at 28 to 122 meV, with $T_{½} > 3.8 \times 10^{26}$ yr (KamLAND-Zen Collaboration 2024);
  - GERDA, MAJORANA and LEGEND-200 combined give $T_{½} > 1.9 \times 10^{26}$ yr (LEGEND Collaboration 2025);
  - LEGEND-1000 is designed to reach about 10 meV.
- **Direct mass.** The β-decay cap, $m_\nu < 0.45$ eV (KATRIN Collaboration 2025), is a thousand times above the calculator's masses.

What the data can decide is the ordering and the cap.

The cap is pressing. DESI's own analysis of its DR2 data lets the cosmological mass run negative as a bookkeeping device, and finds $\Sigma m_{\nu,\mathrm{eff}} = -0.101\ ^{+0.047}_{-0.056}$ eV in ΛCDM, 3.0σ below the normal-ordering floor (Elbers et al. 2025).

- **The limit.** Its profile likelihood's 95% limit, 0.053 eV, is under the floor.
- **What would remove the tension.** It goes away only if w drifts (-0.044 ± 0.084 eV with DESY5 in w_0w_aCDM) or if the optical depth is higher than Planck's (1.7σ at $\tau = 0.074$).
- **How it is read.** DESI reads a negative value as a systematic or as new physics unrelated to neutrinos, and Craig, Green, Meyers and Rajendran (2024) read it the same way.

The calculator has committed to both ends, $w = -1$ and the floor, so this is one test with two faces. If the preference for a vanishing mass is a systematic, both stand; if it is physics, one of the two gives, since the neutrino masses carry no free parameter.

Standing. M: the solution with one flipped root is Brannen's (2006) and Rodejohann and Zhang's (2011), recomputed here with the current splittings. **N**: the flip as the reason the solution exists.

9.3 The Higgs on the board

This section gives the top and the Higgs from the vacuum's edge and one load in full; the scorecard of section 15.1 carries their rows.

Inputs. All numbers use PDG 2026:

- top 172.60 ± 0.27 GeV (direct) and 172.1 ± 0.6 GeV (pole, from cross sections);
- Higgs 125.13 ± 0.11 GeV;
- W 80.3625 ± 0.0077 GeV;

- α_s(m_Z) = 0.1180 ± 0.0009.

Code. The running is two-loop in all couplings (Luo, Wang and Xiao 2003) with three-loop QCD (Chetyrkin and Zoller 2012). It starts from the MS-bar couplings at μ = M_t, matched as in Buttazzo et al. (2013), with their theory errors on the matching: ±0.0003 in λ(M_t) and ±0.0005 in y_t(M_t). At M_t = 172.60 GeV the matched couplings are:

- g_1 = 0.46285 (GUT-normalised);
- g = 0.64759;
- g_3 = 1.16515;
- y_t = 0.93303;
- λ = 0.12603.

The scripts are listed in section 20.1.

The bias and the grips.

- **The bias.** The Higgs field's value v = (√2 G_F)^(−1/2) = 246.22 GeV is the board's bias. It sets every reaction's gap.
- **The grips.** A particle's grip on the Higgs is its mass over the bias, y = √2·m/v: 0.000607 for the muon, 0.0102 for the τ and 0.991 for the top at its pole mass.
- **The Higgs itself.** The Higgs boson is the board's own amplitude reaction, with mass √(2λ)·v. At tree level λ = $m_H^2/2v^2$ = 0.12914.

Table 32. Rules for the Higgs mass, each fixed before computing.

Rule	Says	Result
Balance (the author's)	the Higgs sits just below its exact edge, 128.55 ± 0.96 GeV	125.13 is on the predicted side, at 45.77°; a side, not a value
Held below its W-pair gap	m_H < 2m_W = 160.73 GeV	yes; this is why one W is always virtual in H → WW*
Held at √4, λ = 1/8 (a reading tested here, not the author's)	m_H = v/2 = 123.11 GeV	a miss: the measured mass is 1.64% higher, 18σ
The three-phase load δ_3 (section 9.4)	from the measured top, m_H = 125.20 ± 0.92 GeV	−0.1σ (section 9.4)

The board sets every other gap through v. The Higgs's own gap still needs λ, and of the rules in Table 32 only the δ_3 rule of section 9.4 supplies a number for it.

The decays, as reactions of the board. Each channel is classed by how it opens:

- **Open.** The pair lies below m_H, so both partners are real.
- **Held.** The pair gap lies above m_H, so one partner is virtual.
- **Sub-reaction.** There is no tree-level coupling, so the decay runs through a loop.

Table 33. The Higgs decays computed as reactions, at m_H = 125.00 GeV, beside the standard values.

Decay	Kind	Computed	Standard
b b̄	open (pair gap 8.4 GeV)	59.1%	58.2%
W W*	held: gap 160.7 GeV, one W virtual	20.9%	21.4%
g g	sub-reaction: top and bottom loops	7.78%	8.18%
τ τ	open (gap 3.6 GeV)	6.41%	6.27%
c c̄	open (gap 2.5 GeV)	2.93%	2.89%
Z Z*	held: gap 182.4 GeV, one Z virtual	2.60%	2.62%
γ γ	sub-reaction: W and top loops	0.228%	0.227%
μ μ	open (gap 0.21 GeV)	0.0227%	0.0218%
total width		4.04 MeV	4.07 MeV

How it was computed. The formulas behind Table 33 are the standard ones (Djouadi 2008). In order:

- **The quark channels.** They use MS-bar masses run to m_H at three loops, m_b(m_H) = 2.79 GeV and m_c(m_H) = 0.62 GeV, with the QCD correction to third order in α_s.
- **The held channels.** Both vector bosons are integrated over their Breit–Wigner shapes.
- **The loops.** They use the spin-½ and spin-1 loop functions A_{1/2} and A_1, with a QCD factor of 1.81 for g g.
- **The standard column.** It is the LHC Higgs Working Group's (de Florian et al. 2016).
- **The fit.** Every channel lands within 5%. The rest is the electroweak and higher-order QCD corrections left out, plus Z γ (0.15%), which is not computed.

Measured.

- **The rates.** The overall rate is 1.002 ± 0.057 of the prediction (CMS Collaboration 2022) and 1.05 ± 0.06 (ATLAS Collaboration 2022).
- **The width.** From off-shell production, 3.2 +2.4/−1.7 MeV (CMS Collaboration 2022, Nature Physics) and 4.5 +3.3/−2.5 MeV (ATLAS Collaboration 2023).
- **The couplings.** They follow the masses from the muon to the top.

The board reading of the decays is the Standard Model's, renamed. It owns the vocabulary of open, held and sub-reaction channels, and no number.

9.3.1 Why λ runs

Quantum effects make couplings depend on the energy μ at which they are probed. At one loop:

```
16\pi^2\,\frac{d\lambda}{d\ln\mu}=-6y_t^{4}+\tfrac{3}{8}\left[2g^{4}+\left(g^{2}+g'^{2}\right)^{2}\right]+24\lambda^{2}+\lambda\left(12y_t^{2}-9g^{2}-3g'^{2}\right)
```

The terms are a tug-of-war:

- the top pulls λ down ($-6y_t^4$);
- the force carriers push it up;
- λ's own term and the mixing terms follow λ.

Table 34. The one-loop pieces, with the couplings run at two loops. The first three columns are in units of $1/16\pi^2$. The last is their sum over $16\pi^2$, the change in λ per e-fold of energy. From the top mass to the Planck mass is 38.80 e-folds.

Energy	Top pulls	Carriers push	Self and mixing	Net per e-fold
10^3 GeV	−3.153	+0.234	+0.671	−0.01424
10^{10} GeV	−0.484	+0.167	+0.000	−0.00201
10^{18} GeV	−0.143	+0.137	+0.015	+0.00006

In Table 34, the top's pull weakens as y_t runs down, from 0.851 at 10^3 GeV to 0.393 at 10^{18} GeV. Near the Planck scale the pull and the push nearly cancel, which is the near-criticality of section 9.4.

Why it falls. At the top mass $\lambda(M_t) = 0.126$ and $y_t(M_t) = 0.933$ in the matching of Buttazzo et al. (2013); at tree level $\lambda = m_H^2/2v^2 = 0.12914$. So the $-6y_t^4$ term (−4.5) outweighs $24\lambda^2$ (+0.4) and λ falls.

9.3.2 What the running shows

Table 35. λ against energy, two-loop running with three-loop QCD.

Energy (GeV)	10^3	10^5	10^7	10^9	10^{11}	10^{13}	10^{15}	10^{17}	10^{19}
measured masses (top 172.60, Higgs 125.13)	0.0965	0.0493	0.0233	0.0078	−0.0015	−0.0069	−0.0097	−0.0107	−0.0105
exact edge (top 170.935, Higgs 125.13)	0.0980	0.0542	0.0305	0.0166	0.0083	0.0035	0.0010	0.0001	0.0002

- **Where it goes.** With the measured masses λ crosses zero at 3.9×10^{10} GeV and reaches its minimum, −0.01075, at 3.3×10^{17} GeV. At 1.22×10^{19} GeV it is −0.0104.

- **Metastable.** The vacuum is metastable. The complete calculation of its lifetime, with the scale-invariant instantons treated in full, gives 10^{139} years for the masses of 2017 (Andreassen, Frost and Schwartz 2018).
- **Code check.** At the inputs of Buttazzo et al. (2013), with top 173.34, Higgs 125.15 and α_s = 0.1184:
    - this code gives $\lambda(M_{Pl}) = -0.0141$, against their -0.0143;
    - it puts absolute stability at $m_H > 129.77$ GeV, against their 129.6 ± 1.5 GeV.

9.4 The edge and the balance rule

The top and the Higgs. The Higgs quartic coupling runs negative near 10^{10} GeV, so the vacuum is metastable (Degrassi et al. 2012; Buttazzo et al. 2013). The edge, where its minimum is exactly zero, sits at m_t = 170.93 ± 0.38 GeV for m_H = 125.13. Two candidate rules are tested against it:

- **Slightly past, held.** The calculator puts nature past the edge by the three-phase load of the top's two phases, its Yukawa coupling and the weak coupling, combined at 120°:

  $$\delta_3=\frac{\left|\,y_t^{2}+g^{2}e^{i120^{\circ}}\right|}{8\pi^{2}}=0.955\%$$

  This gives a top of 172.57 ± 0.38 GeV against 172.60 ± 0.27 from direct measurements and 172.1 ± 0.6 as a pole mass, and a Higgs of 125.20 ± 0.92 GeV against 125.13 ± 0.11.
- **Exactly on the edge.** The proposal that nature sits exactly on the edge (Froggatt and Nielsen 1996; Shaposhnikov and Wetterich 2010), which is not the author's, gives a Higgs of 128.55 ± 0.96 GeV, 3.5σ above the measured 125.13, and a top of 170.94 ± 0.38 GeV, 3.6σ below the direct mass and 1.6σ below the pole mass from cross sections.

The edge. The edge is the set of pairs (m_t, m_H) for which the minimum of λ over all energies is exactly zero. On the stable side λ never reaches zero; on the metastable side it crosses zero twice, as in Table 35.

- **For the measured Higgs.** The edge top is m_t = 170.935 ± 0.375 GeV. Its error budget:
    - α_s, ±0.354;
    - the matching, ±0.113 (λ ±0.072, y_t ±0.087);
    - the Higgs mass, ±0.055.
- **For the measured top.** The edge Higgs is m_H = 128.548 ± 0.964 GeV. Its error budget:
    - α_s, ±0.746;
    - the top mass, ±0.564;
    - the matching, ±0.228.
- **The slope.** Along the edge the Higgs moves 2.02 GeV per GeV of top at m_t = 171.0, and 2.10 at m_t = 172.6.

The balance rule, the author's. At exactly 45° a balanced block is poised. It takes slightly more to break static balance into a kinetic turn, the way a balancing rock stands until it is pushed a little past its tipping point. The everyday physics behind the rule:

- static friction exceeds sliding friction, which is why things stick and then slip (Amontons 1699; Coulomb 1785);
- earthquakes are this stick-slip on faults (Brace and Byerlee 1966);

- precariously balanced rocks in California have stood through thousands of years of earthquakes, because a rocking block needs more than its static tipping push to fall (Housner 1963; Brune 1996).

For the vacuum, the rule says the measured pair sits just past the edge on the metastable side: a top above 170.94 and a Higgs below 128.55. Both measurements fall on those sides.

- **As angles.** tan θ = 172.60/170.935 gives 45.28° read on the top, and tan θ = 128.548/125.13 gives 45.77° read on the Higgs. They are one offset in two units, because the edge moves about 2 GeV of Higgs per GeV of top.
- **As λ.** λ dips to −0.0108 against 0.126 at the top mass: the balancing rock that has not tipped.

How far past 45°, from measured contacts. At 45° the push equals the running resistance. Starting needs the static resistance instead, so the start angle is tan θ = μ_static/μ_kinetic.

Table 36. Start angles from measured friction (OpenStax College Physics, Table 5.1).

Contact	Start / run	Start angle
Teflon on steel	0.04 / 0.04 = 1.00	45.0°
bone with joint fluid	0.016 / 0.015 = 1.07	46.8°
shoes on wood	0.9 / 0.7 = 1.29	52.1°
rubber on wet concrete; waxed wood on wet snow	1.40	54.5°
rubber on dry concrete	1.0 / 0.7 = 1.43	55.0°
wood on wood; metal on wood; oiled steel	1.67	59.0°
dry steel; shoes on ice	2.00	63.4°
ice on ice	3.33	73.3°
steel on ice	20	87.1°
the Higgs vacuum	top 1.0097; Higgs 1.0273	45.3° to 45.8°

- **Always past.** Every measured contact in Table 36 starts at or above 45°, as the rule says. How far past depends on the surfaces, from 0° to 42° past.
- **The vacuum.** It is the slightest of all, 0.3° to 0.8° past, in the class of the slickest contacts.

The load. The rule that gives a size reads the offset as a load. The anchor was fixed before computing: the electron's magnet sits past its balance, g = 2, by its first sub-reaction, α/2π = 0.00116141 (measured 0.00115965). Here α is the board's conductance, the base ohms times the conductance quantum over 4 = 1/137.036 (section 7.2).

- **The perpendicular-conductivity rule.** A held system's offset is the coupling of the reaction carrying its perpendicular current, at its own mass scale, over 2π, carried by that reaction's mass.
- **The vacuum's conductor.** For the vacuum the conductor is the top.
- **The overlap version.** The start cannot use all three phases, because three phases add to zero. So the start rides the sum of the two previous phases, 120° apart:
    - the conductor (the top, $y_t^2 = 0.8705$);
    - the magnetic spin (the weak field, $g^2 = 0.4194$).
    - Gravity's part, $(m_t/\bar{M}_{Pl})^2 = 5.0 \times 10^{-33}$ at the top's mass, is the tiny third.

The load, worked. With $y_t = 0.9330$ and $g = 0.6476$ at $M_t = 172.60$ GeV, $|0.8705 + 0.4194(-½ + i\sqrt{3}/2)| = |0.6608 + 0.3632i| = 0.7540$, over $8\pi^2 = 78.957$, so $\delta_3 = 0.955\%$.

- **The phase read from the masses.** Solved for the angle that lands the top exactly, the phase is 117.9°, with a 1σ range of 87° to 149°. Three-phase's 120° sits in the middle of it.
- **Against the pole mass.** The rule's 172.57 is 0.7σ above the pole mass from cross sections, 172.1 ± 0.6.
- **The honest count.** δ_3 is about the twentieth candidate load tried, and while the edge is known to ±0.38 GeV any top load from 0.70% to 1.24% passes at 1σ. So the closeness is a lead, not evidence, and section 16 scores it that way.

What decides it. The top's direct mass carries a scheme ambiguity of about 0.5 GeV, since it is a quark measured through simulations (Hoang 2020). An e^+e^- threshold scan measures a mass with a clean definition, to about ±0.035 GeV (a 2025 study of the scan; arXiv:2503.18713).

- **The edge then.** With α_s known to ±0.0002 and the Higgs to ±0.05 GeV, the rule's top prediction tightens to ±0.14 GeV, with the matching's ±0.11 then the largest part.
- **Telling the loads apart.** Three-phase (172.57) and the nearest rival load, the top's coupling alone, $y_t^2/8\pi^2$ (172.82), are 0.25 GeV apart, 1.7σ at that precision.
- **The rule's failure.** A pole mass at or below the edge would put the vacuum on the stable side and break the balance rule outright.

Standing. N: the balance reading, slightly past the edge and held, is the author's; the δ_3 load at 120° is new, but it was about the twentieth candidate tried, so it is a lead and not evidence (section 16). **D**: "slightly past the edge" diverges from the proposals that put nature exactly on it (Froggatt and Nielsen 1996; Shaposhnikov and Wetterich 2010), which miss today by 3.5σ on the Higgs.

9.5 Gravity at the boundary

Gravity sets a boundary. Above the scale where gravity takes over, it adds a term $A\cdot\lambda$ to λ's running, and the prediction needs $A > 0$ (Shaposhnikov and Wetterich 2010):

$$\frac{d\lambda}{d\ln\mu}=A\,\lambda+B \quad\Longrightarrow\quad \lambda(t)=-\frac{B}{A}+\left(\lambda_0+\frac{B}{A}\right)e^{A(t-T)}$$

Flowing down from any starting value λ_0 at a high scale T, with $A > 0$, gives $\lambda = -B/A$ at the boundary. Here B is the Standard Model running at $\lambda = 0$ at the reduced Planck mass, 2.435×10^{18} GeV. It is $B = +4.17 \times 10^{-6}$ per e-fold: tiny, because the Standard Model is near-critical there.

Table 37. The boundary for several strengths of gravity.

Gravity's strength A (per e-fold)	λ at the boundary	Higgs it predicts
1.0	-4.2×10^{-6}	128.55 GeV
0.3	-1.4×10^{-5}	128.54 GeV
0.1	-4.2×10^{-5}	128.54 GeV
0.03	-1.4×10^{-4}	128.51 GeV
needed for 125.13	−0.0107	$A = 3.9 \times 10^{-4}$

- **The prediction.** The gravity boundary of Table 37 gives m_H = 128.55 ± 0.96 GeV, 3.5σ high: the exact edge to 0.01 GeV, because λ's minimum sits near the Planck scale.
- **Validation.** Shaposhnikov and Wetterich used m_t = 171.3 GeV and predicted 126 GeV, "with only a few GeV uncertainty". This code at m_t = 171.3 gives 125.86 GeV.
- **What would land it.** The prediction moves 2.09 GeV per GeV of top. The top that lands it on 125.13 is 170.945 GeV.
- **The double root.** A double root at the Planck mass, λ = 0 and β_λ = 0 together (the multiple-point principle of Froggatt and Nielsen 1996), fixes both masses:
    - m_t = 174.29 ± 0.57 GeV, −2.7σ;
    - m_H = 132.13 ± 0.47 GeV, −14.5σ.
    - At 1995 precision the same principle gave 173 ± 5 and 135 ± 9 GeV, which today sit at −0.1σ and −1.1σ.

Gravity and the decay of the vacuum. The flat-space bounce action is $S_0 = 8\pi^2/(3|\lambda|)$. Gravity's leading correction is $\Delta S = 256\pi^3/[45(R\,\bar{M}_{Pl}\,\lambda)^2]$, with bounce size R = 1/μ (Isidori, Rychkov, Strumia and Tetradis 2008). Each is minimised against the survival requirement for 13.8 billion years, 4 ln(τ_U μ).

Table 38. The bounce at leading order.

	Action	Its scale	Survival margin	Lifetime at leading order	Already decayed if
flat space	2,448	4.6×10^{17} GeV	1,900	about 10^{216} years	m_t > 178.11 GeV
with gravity	2,559	1.0×10^{16} GeV	2,027	about 10^{230} years	m_t > 178.31 GeV

Table 38 shows that gravity makes the vacuum slightly more stable and moves the cheapest bubble to a larger size. Isidori et al. reach the same conclusion: gravitational corrections "only induce a minor shift" on the instability border. The leading-order lifetimes compare the two columns; they are not the lifetime. The complete calculation, with the fluctuation determinant and the scale-invariant instantons, gives 10^{139} years for the masses of 2017 (Andreassen, Frost and Schwartz 2018).

The load times the bounce is exactly ⅓: $(|\lambda|/8\pi^2) \times 8\pi^2/(3|\lambda|) = 1/3$, an identity of the formula and not a result (section 4.7.3).

Gravity's direct effect. At collider energies gravity's direct effect is $(m_H/M\bar{}_{Pl})^2 = (125.13/2.435 \times 10^{18})^2 = 2.6 \times 10^{-33}$. Below the Planck scale it changes nothing measurable in the Higgs, and acts only through the boundary. The Higgs vacuum's own energy, and the 10^54.7 by which it overshoots the dark energy, are in section 13.

Table 39. What the Standard Model itself predicts, against measurement. Where the calculator restates the Standard Model, it predicts exactly this well, and no better.

Prediction	Predicted	Measured	Pull
W mass, global electroweak fit	80.356 ± 0.005 GeV	80.3625 ± 0.0077 GeV	+0.7σ
top mass, global electroweak fit	172.85 ± 0.55 GeV	172.60 ± 0.27 GeV	−0.4σ
top mass before its discovery, 1994 fits	about 145 to 185 GeV	176 ± 13 GeV (CDF 1995); 199 ± 30 GeV (DØ 1995)	inside
Higgs mass before its discovery, 2011 fit without direct searches	91 +30/−23 GeV	125.13 ± 0.11 GeV	+1.1σ
Higgs mass ceiling from unitarity, $\sqrt{(16\pi/3)}\cdot v$	below 1,007.9 GeV	125.13 GeV	passes
Higgs couplings to W and Z scale with mass	$\kappa_V = 1$	1.023 ± 0.026	+0.9σ
Higgs couplings to fermions scale with mass	$\kappa_F = 1$	0.94 ± 0.05	−1.2σ
Higgs self-coupling, trilinear $3m_H^2/v$ = 190.8 GeV	$\kappa_\lambda = 1$	−1.24 to 6.49 at 95% (CMS 2022)	inside

Sources for Table 39: the fit values from PDG's electroweak review (2024); Gfitter (Baak et al. 2012); Lee, Quigg and Thacker (1977); CDF Collaboration (1995) and DØ Collaboration (1995); the coupling modifiers from PDG (2024). The W and top fit values are relations among many measurements rather than independent forecasts.

9.6 The top-mass measurements

There are two kinds of top-mass measurement:

- **Direct.** The mass is fitted to simulations, and it is the simulation's mass setting.
- **Pole.** The mass is extracted from a rate computed with a cleanly defined mass.

The two may differ by about 0.5 GeV (Hoang 2020).

A pole mass worked out from the cross section. ATLAS's Run 2 review gives the inputs (ATLAS Collaboration 2025, Physics Reports):

- **The prediction.** 834 +21/−30 (scale) ± 21 (PDF and α_s) pb at m_t = 172.5 GeV, falling 23 pb per GeV of mass.
- **The measurement.** 829.0 ± 1.0 ± 13.2 ± 8.0 pb with 140 fb^{-1}.
- **The mass.** $m_t(pole) = 172.5 + (834 - 829)/23 = 172.72$ +1.46/−1.73 GeV, with the experiment ±0.67 and the theory +1.29/−1.59.
- **What it would take.** Reaching ±0.5 GeV needs the prediction to ±12 pb, 1.4%.

Table 40 sets every top-mass result, direct and pole, against the rule and against the exact edge.

Table 40. Top-mass results against the rule (172.57 ± 0.38) and the exact edge (170.94 ± 0.38).

Result	Kind	Value (GeV)	Against the rule	Against the edge
PDG 2026 average	direct	172.60 ± 0.27	+0.1σ	+3.6σ
PDG 2026 average	pole	172.1 ± 0.6	−0.7σ	+1.6σ
ATLAS Run 1 combination	direct	172.69 ± 0.48	+0.2σ	+2.9σ
ATLAS and CMS Run 1 combination	direct	172.52 ± 0.33	−0.1σ	+3.2σ
ATLAS Run 2, soft muon, 36 fb^{-1}	direct	174.41 ± 0.81	+2.1σ	+3.9σ
ATLAS Run 2, dilepton, 140 fb^{-1} (preliminary)	direct	172.21 ± 0.80	−0.4σ	+1.4σ
ATLAS cross section, 36 fb^{-1}	pole	173.1 +2.0/−2.1	+0.3σ	+1.0σ
ATLAS cross section, 140 fb^{-1} (worked out above)	pole	172.72 +1.46/−1.73	+0.1σ	+1.0σ
ATLAS, top pair plus one jet, 13 TeV (2025)	pole	170.7 ± 1.5	−1.2σ	−0.2σ

Sources: the averages, PDG 2026; the ATLAS Run 1 combination, ATLAS Collaboration (2019, Eur. Phys. J. C 79, 290); the ATLAS and CMS Run 1 combination, ATLAS and CMS Collaborations (2024, Phys. Rev. Lett.

132, 261902); the soft-muon result, ATLAS Collaboration (2023, JHEP 06, 019); the cross section at 36 fb^{-1}, ATLAS Collaboration (2020, Eur. Phys. J. C 80, 528); the 140 fb^{-1} cross section, the Run 2 review (ATLAS Collaboration 2025, Physics Reports); the top pair plus one jet, ATLAS Collaboration (2025, JHEP 12, 023). The preliminary dilepton result could not be traced to a public record and is kept as reported.

- **The direct results** sit up to 3.9σ above the edge and within about 2σ of the rule.
- **The pole results** cannot yet tell the two apart.
- **The 2012 bound.** Alekhin, Djouadi and Moch (2012) found a pole mass of 173.3 ± 2.8 GeV from the cross section, and said full stability needs m_H ≥ 129.8 ± 5.6 GeV. This code at 173.3 GeV gives 130.02 GeV.

The peaks rebuilt from public collision data. ATLAS released 10 fb^{-1} of its 2016 proton collisions at 13 TeV for education, with matching simulations (ATLAS Collaboration 2020, open data).

- **The method.** Heavy particles fall apart before reaching the detector, so the pieces are added up, $m^2 = E^2 - p^2$, and real decays pile up into a peak on a smooth background.
- **The calibration.** Measured mass = the simulation's true mass + (data peak − simulation peak).

Table 41. The Higgs peak.

Channel	Selection	Events	Simulation shift	Calibrated mass	Signal
two photons	two tight, isolated photons; p_T above 35% and 25% of the mass; 105 to 160 GeV	76,562	+0.05 GeV	124.43 +1.00/−1.05 GeV	about 357 events, 2.5σ local
four leptons	four tight, isolated electrons or muons; total charge 0; same-flavour pairs; p_T above 25, 15 and 10 GeV	228	+0.96 GeV	124.06 +0.76/−0.80 GeV	about 13 events, 4.1σ against flat

- **Combined.** Weighted by uncertainty, the two channels of Table 41 give 124.20 ± 0.62 GeV, statistical only.
- **The energy scale.** The Z checks it: data minus simulation is +3 ± 8 MeV in 867,054 electron pairs and −26 ± 6 MeV in 1,068,835 muon pairs. These are offsets of +0.003% and −0.029%, worth at most 0.04 GeV at 125 GeV. Photons are assumed to follow electrons.

The top peak.

- **The selection.**
  - One tight, isolated electron or muon with $p_T > 30$ GeV.
  - Missing momentum above 30 GeV and a transverse W mass above 30 GeV.
  - At least four jets with $p_T > 30$ GeV and $|\eta| < 2.5$, exactly two of them b-tagged.
- **The W candidate** is the light-jet pair with mass closest to 80.4 GeV. The top candidate is that pair plus the b-jet giving the larger top p_T.
- **The W constraint.** The two light jets are scaled so that their pair mass is exactly 80.4 GeV.
- **The template.** The simulation (Powheg-Box with Pythia 6, $m_t = 172.5$ GeV, single top added) is shifted to match the data, with free normalisation and a straight-line background. Table 42 gives the steps and the result.

Table 42. The top mass from the open data.

Step	Value
events selected	138,027 (the simulation expects 133,761 from top pairs and single top)
W peak, data − simulation (exactly two light jets)	+0.60 ± 0.16 GeV: jets 0.74 ± 0.20% high in data
top peak, W-constrained, data − simulation	+0.53 ± 0.18 GeV (χ^2/ndf = 26.1/27)
jet-scale correction (b-jets 0.65 GeV per 1%; light jets 0.01 GeV per 1%)	−0.48 GeV
top mass	172.54 ± 0.23 (stat) ± 0.92 (syst) GeV

The uncertainties.

- **The total, 0.94 GeV.**
  - statistics, 0.23;
  - the b-jet scale against light jets, assumed up to 1%, 0.65;
  - signal modelling, 0.64, using the sizes of ATLAS's 140 fb^{-1} dilepton analysis (matching 0.40, radiation 0.17, colour reconnection 0.27, recoil 0.39);
  - fit method, 0.10;
  - calibration slope, 0.01.
- **The cross-checks.**
  - The four data periods give shifts of +0.12 ± 0.72, +0.46 ± 0.38, +0.78 ± 0.33 and +0.44 ± 0.28 GeV ($\chi^2 = 1.0$ for 3 degrees of freedom).
  - The raw three-jet mass gives 172.90 ± 0.45 (stat).
  - Leaving the b-jets uncorrected gives 173.03.

Where the rebuild sits.

- **The Higgs, 124.20 ± 0.62:**
  - −1.5σ against PDG 2026;
  - −0.9σ against the rule's 125.20 ± 0.92;
  - −3.8σ against the exact edge.
- **The top, 172.54 ± 0.94:**

- −0.1σ against the direct average;
- +0.4σ against the pole average;
- 0.0σ against the rule;
- +1.6σ against the edge.

The rebuild's limits.

- **Not for publication.** The release is educational; CERN says it “is not suited for scientific publications”.
- **One simulation.** There is only one top simulation, so the calibration slope is assumed and the modelling uncertainty is borrowed.
- **The b-jet scale** is assumed to follow the light jets within 1%. It is the largest uncertainty.
- **Direct, not pole.** The top is a direct-type mass.
- **The Higgs errors** are statistical only, and the two-photon bump is 2.5σ.
- **The sample.** The selections are simple cuts on 7% of ATLAS's Run 2 data.

Telling 171.0 from 172.5 at 3σ needs about ±0.5 GeV on a pole-type mass, so the threshold scan stays the decider.

9.7 The W–Z triangle and the filled leg

The converter's rule, that going from two dimensions to three needs one leg the two-dimensional data cannot supply (section 3.7.1), is the Higgs mechanism's. The Higgs mass matrix in the (W_3, B) block has rank one, so one combination, the photon, gets no mass, and the Z gets $(g^2 + g'^2)v^2/4$ while the W gets $g^2v^2/4$. Hence $m_Z^2 = m_W^2 + (g'v/2)^2$ exactly: a right triangle whose legs are the W mass and $g'v/2$.

Table 43 reads the W and Z masses as a right triangle: the Z on the hypotenuse, the W on one leg and the hypercharge part on the other.

Table 43. The W–Z triangle from the PDG 2026 masses.

Side	Value	Source
hypotenuse, m_Z	91.1880 ± 0.0020 GeV	PDG 2026
leg, m_W	80.3625 ± 0.0077 GeV	PDG 2026
leg, $g'v/2 = \sqrt{(m_Z^2 - m_W^2)}$	43.094 GeV	computed
angle, $\theta_W = \arccos(m_W/m_Z)$	28.20°, so $\sin^2\theta_W = 0.22334$ on shell	computed

The count closes: four carriers with two legs each and four Higgs components become three carriers with three legs, a photon with two, and one Higgs boson, 12 = 12 (Goldstone's theorem). This is the Standard Model at tree level, read as the converter's filled leg.

Standing. M: the mass matrix and the count are standard. **N**: the W and Z as the converter's third leg, filled by a phase.

9.8 Standing, beside the standard theories

The Standard Model predicts none of the masses on this layer; each is a free parameter. The one standard-adjacent rule that does fix a mass, a vacuum sitting exactly on its edge, misses the Higgs by 3.5σ. The calculator's numbers sit within 0.7σ of their measurements. Two facts temper that, and section 16 weighs both: the lepton relation and its phase are inherited, and the δ_3 load was chosen from about twenty candidates.

Table 44. Mass on one yardstick.

Quantity	Standard theory	The calculator	Measured	Pull of the calculator	Standing
Koide's Q	an input: three free Yukawa couplings	⅔, from balance	0.6666645 ± 0.0000051	−0.4σ	the value inherited (Koide 1982); the reason new
the τ from e and μ	no prediction	1,776.969 MeV	1,776.93 ± 0.09 MeV	−0.4σ	inherited
the lepton phase δ	no prediction	2/9, with 3δ = Q	0.2222248 ± 0.0000063	+0.4σ	inherited (Brannen 2006); the reading new
the neutrino masses	none in the minimal Standard Model; three free parameters when extended	0.364, 8.66 and 50.13 meV, Σ = 59.2 meV, normal ordering	Σ < 64 meV (95%); the ΛCDM effective mass −0.101 +0.047/−0.056 eV	inside the cap; −3.4σ against the effective mass	prediction; the exposure is shared with w = −1
the top mass	an input	172.57 ± 0.38 GeV	172.60 ± 0.27 direct; 172.1 ± 0.6 pole	+0.1σ; −0.7σ	new number; a lead
the Higgs mass	an input; exactly on the edge, 128.55 ± 0.96 GeV (−3.5σ)	125.20 ± 0.92 GeV	125.13 ± 0.11 GeV	−0.1σ	new number; follows from the top

Quantity	Standard theory	The calculator	Measured	Pull of the calculator	Standing
the vacuum's stability	computed: λ falls to −0.011, metastable, lifetime about 10^{139} years	the same running; nature slightly past the line, held	no direct measurement	—	same running; the reason new
the W–Z triangle	$m_Z^2 = m_W^2 + (g'v/2)^2$ at tree level	the converter's filled leg	91.1880 and 80.3625 GeV	—	same
the Higgs decays	computed (Djouadi 2008)	the same widths, computed as reactions of the board	the measured couplings (PDG 2026)	—	same

Origination. M: the running, the matching, the edge, the bounce, the decay widths and the field and eddy-current bounds, each with its source given above. **N**: the balance reading, slightly past the edge and held; the δ_3 load at 120°; the board's vocabulary for the decays. **D**: slightly past the edge, against exactly on it.

10. Mixing: the Clarke frame tilted 15°

Neutrinos change flavour because the states with a definite mass are turned against the states with a definite flavour. The calculator computes that turn as the electrician's Clarke frame, the cube seen down its body diagonal, left by one rotation of 15°, the ladder's step from $\sqrt{2}$ to $\sqrt{3}$. The frame alone is tribimaximal mixing; the 15° tilt gives $\sin^2\theta_{13} = 0.02233$ and $\sin^2\theta_{12} = 0.3181$, against 0.02200 ± 0.00056 and $0.3096\ ^{+0.0057}_{-0.0073}$. Quarks mix by a different rule, by magnitude rather than frame, and the unitarity triangle's angle α is taken as 90°. The Standard Model has none of these angles as predictions; each is an input.

10.1 The Clarke frame, tilted

The frame. The orthonormal Clarke transform and the tribimaximal matrix are one matrix:

$$K=\begin{pmatrix}2/\sqrt6 & -1/\sqrt6 & -1/\sqrt6\\ 0 & 1/\sqrt2 & -1/\sqrt2\\ 1/\sqrt3 & 1/\sqrt3 & 1/\sqrt3\end{pmatrix},\qquad U_{TBM}=\begin{pmatrix}\sqrt{2/3} & 1/\sqrt3 & 0\\ -1/\sqrt6 & 1/\sqrt3 & -1/\sqrt2\\ -1/\sqrt6 & 1/\sqrt3 & 1/\sqrt2\end{pmatrix}$$

- **The match.** The rows of K, the α axis, the β axis and the zero axis (the body diagonal), are the columns of U_{TBM} up to sign and order: U_{TBM} is K^T with the second and third rows of K swapped and one sign flipped.
- **The angles.** $\sin^2\theta_{12} = |U_{e2}|^2 = \tfrac{1}{3}$, $\sin^2\theta_{23} = |U_{\mu 3}|^2/(1 - |U_{e3}|^2) = \tfrac{1}{2}$ and $\sin\theta_{13} = |U_{e3}| = 0$.
- **Why the same frame.** It is the frame adapted to three equal phases 120° apart: one axis along the sum (1, 1, 1)/√3, the other two spanning the plane in which the three-phase vector turns.

Power engineering needs it to separate the rotating part from the zero sequence; neutrino physics needs it when the mass states sit symmetrically among three flavours. The cube's rotation group S_4 is the symmetry of that arrangement (Lam 2008).

The tilt. Rotate in the plane of the second and third columns, keeping the first column $(2, -1, -1)/\sqrt{6}$ fixed: $U = U_{TBM} R_{23}(\theta)$. Then $U_{e3} = -\sin\theta/\sqrt{3}$, so $\sin^2\theta_{13} = \sin^2\theta/3$, while $|U_{e1}|^2 = \frac{2}{3} = \cos^2\theta_{12}\cos^2\theta_{13}$ is unchanged:

$$\sin^2\theta_{12}=1-\frac{2}{3(1-\sin^2\theta_{13})}=\frac{1-3\sin^2\theta_{13}}{3(1-\sin^2\theta_{13})}$$

- **The measured tilt.** From $\sin^2\theta_{13} = 0.02200 \pm 0.00056$ it is $\theta = 14.9° \pm 0.2°$.
- **The rival, TM2.** Keeping the diagonal fixed instead fixes $|U_{e2}|^2 = \frac{1}{3}$ and forces $\sin^2\theta_{12} = 1/(3(1 - \sin^2\theta_{13})) = 0.341$. That is 3.6σ above JUNO's first result and 5.5σ above the NuFIT 6.1 value, so the data pick the α axis: the axis that survives is the one the start fixes first.
- **The ladder's angle.** The step from $\sqrt{2}$ to $\sqrt{3}$ is $45° - 30° = 15°$, the step between the 45° and 30° lines of the (2, 1) cell. Its nearest strict rival, $\arctan(1/4) = 14.04°$, sits 4σ from the measured tilt. Among the forty strictly defined cube angles, 15° is the only one within 3σ of the measured tilt, a chance of 0.12. Among a generous set of 1,422 angles, 32 lie within 3σ, so a generous reading is not selective. Table 45 sets the tilted frame against the data.

Table 45. The tilted frame against the data.

Quantity	The exact frame	Tilted 15° (the calculator)	Measured	Pull
$\sin^2\theta_{13}$	0	0.022329	0.02200 ± 0.00056 (NuFIT 6.0, as used by Esteban et al. 2026)	−0.6σ
$\sin^2\theta_{12}$	⅓	0.318107	0.3092 ± 0.0087 (JUNO's first 59 days); 0.3096 +0.0057/−0.0073 (NuFIT 6.1); 0.3085 ± 0.0073 (Capozzi et al. 2026)	−1.0σ; −1.5σ; −1.3σ
$\sin^2\theta_{23}$	½	correlated with δ_{CP}	0.561 ± 0.014 (NuFIT); 0.56 +0.03/−0.05 (T2K and NOvA 2025)	not fixed by the frame
$\cos\delta_{CP}$	—	+0.27 in the upper θ_{23} octant (0.561),	best fit $-0.47\pi = -85°$ (T2K and NOvA 2025)	consistent; DUNE, Hyper-Kamiokand

Quantity	The exact frame	Tilted 15° (the calculator)	Measured	Pull
		−0.27 in the lower (0.44)		e and the ordering decide

The CP phase. In the tilted frame the Dirac phase is tied to the atmospheric angle (Albright and Rodejohann 2009):

```
\cos\delta_{CP}=-\frac{1-5\sin^2\theta_{13}}{2\sqrt2\,\sin\theta_{13}\sqrt{1-3\sin^2\theta_{13}}\,\tan2\theta_{23}}
```

- **The octants.** The sign of tan $2\theta_{23}$ flips across 45°, which is the whole difference between the octants; the frame gives θ_{23} no value of its own.
- **The data.** The first joint T2K and NOvA fit puts the best-fit phase at −85°, against the frame's ±75° at $\sin^2\theta_{23} = 0.56$ and ±90° at maximal mixing (T2K and NOvA Collaborations 2025). The frame says the phase is near-maximal, and the data lean that way, with CP conservation excluded at 3σ only if the ordering is inverted.

What decides it. JUNO began taking data in August 2025. Its first 59 days give $\sin^2\theta_{12} = 0.3092 \pm 0.0087$ and $\Delta m^2_{21} = (7.50 \pm 0.12) \times 10^{-5}$ eV², already 1.6 times more precise than every earlier measurement combined (JUNO Collaboration 2025).

- **The global fits.** Those that fold JUNO in moved the centre towards the calculator, from 0.307 to 0.309.
- **The first-column pattern stands.** The confrontation of discrete-symmetry patterns with JUNO leaves the first-column-fixed pattern, this one, standing. It projects JUNO's six-year 3σ band at 0.3022 to 0.3118 if the centre holds, which would put 0.318 outside it (Ding, Li, Lu and Petcov 2025).
- **A second rival.** A rigid rotation of the untilted frame about (2, 1, 2)/3 by three quarters of the Cabibbo angle gives $\sin^2\theta_{12} = 0.301$ and $\sin^2\theta_{13} = 0.0216$ (Sharma and Katoch 2026).
- **The full run.** At ±0.0015:
    - a result at 0.307 excludes the tilted frame at 7σ;
    - a result at 0.318 excludes the rival at about 11σ.
- **The untilted frame is already out.** $\theta_{13} = 0$ has been excluded since Daya Bay (2012), and is 38σ away today.

Quark mixing: magnitude, not frame.

- **Two sectors.** The lepton angles contain no mass: 35.3°, 45° and 0° come from the frame alone, then the tilt. The quark angles contain nothing but mass. The up and down quarks share the frame, it cancels in $V = U_u^\dagger U_d$, and what is left is one planar triangle per pair of generations, with legs $\sqrt{m}$.
- **The Cabibbo angle.** It is the down triangle's: $\tan\theta_C = \sqrt{m_d/m_s}$ (Gatto, Sartori and Tonin 1968; Weinberg 1977). Behind it is a 2 × 2 mass block with a zero in the light slot. The off-diagonal entry is $\sqrt{m_d\, m_s} = 20.9$ MeV and $m_d = (20.9\text{ MeV})^2/m_s$: the seesaw shape again.
- **Two perpendicular triangles.** The up triangle's leg is $\sqrt{m_u/m_c} = 0.0449$ (2.6°) at 2 GeV. The angle between the two legs, fitted from $|V_{us}|$, is 83.5° to 86° ± 2°, so $\sin^2\theta_C \approx m_d/m_s +$

m_u/m_c (Fritzsch 1977). Fritzsch and Xing (2000) tie that angle to the unitarity triangle's α and predict a right-angled triangle; the calculator takes α = 90° as its value. Table 46 sets the two-leg values against the measurements.

- **Where the legs fail.** Every leg that includes the top fails:
  - V_cb = 0.0407 ± 0.0013 against at best 0.077 from √(m_s/m_b) and √(m_c/m_t);
  - V_ub = 0.00389 ± 0.00016 against √(m_u/m_t) = 0.0028.
- **Why the top is different.** The top is the one quark whose magnitude is the Higgs scale itself, y_t = 0.9914 ± 0.0016, held there by the infrared fixed point of its Yukawa coupling (Pendleton and Ross 1981; Hill 1981). Its mass is set by v, not by a texture of legs, so the leg rule holds where the legs are small beside v and stops where the leg is v.

Table 46. Quark mixing by legs.

Quantity	Two-leg value	Measured	Pull	Standing
sin θ_C	√(m_d/m_s) = 0.2249 ± 0.0014 (PDG 2026 masses); 0.2230 ± 0.0013 (lattice ratios)	|V_us| = 0.2243 ± 0.0009	−0.4σ; +0.8σ	standard (Gatto, Sartori and Tonin 1968); not counted as the calculator's
angle between the up and down legs	90°	83.5° ± 2.1° (PDG 2026 masses); 85.9° ± 2.0° (lattice ratios)	−3.1σ; −2.0σ	right angle only within 2 to 3σ
unitarity-triangle angle α	90°	84.1 +3.7/−3.0° (all channels, PDG 2026); 92.6 +4.5/−4.7° (B → ρρ alone, Belle II Collaboration 2025); 180° − β − γ = 91.0 ± 2.8° (HFLAV's γ) or 89.4 ± 4.0° (PDG's γ = 68 ± 4°)	−1.6σ; +0.6σ; +0.4σ; −0.2σ	a prediction; the ρρ channel brackets 90° from above and the all-channel average from below
V_cb, V_ub	no leg value	0.0407 ± 0.0013; 0.00389 ± 0.00016	—	the boundary: every leg that includes the top misses by 1.4 to 1.9 times

Standing.

- **M.** Tribimaximal mixing, the TM1 sum rule, the Gatto–Sartori–Tonin relation and the right-angled unitarity triangle are published.
- **N.** Three things were not found: the identification of the 1943 Clarke transform with the tribimaximal matrix; the 15° tilt taken from the ladder; and the reason the first column survives (the first two phases fix the α axis before the third rides their sum).
- **D.** A rival tilt, a rigid rotation of the untilted frame by three quarters of the Cabibbo angle (Sharma and Katoch 2026), gives 0.301. JUNO's full run separates the two at about 11σ.

10.2 Beside the standard theories

The Standard Model takes all four lepton mixing parameters and all four quark mixing parameters as inputs, so on this layer it predicts nothing to compare. The fair comparison is with the other frames that flavour symmetries propose, scored on the same data: the untilted frame is out at 38σ, the frame that keeps the diagonal fixed is out at 5.5σ, and the tilted Clarke frame and one rival remain for JUNO to separate.

Table 47. Mixing on one yardstick.

Quantity	Standard theory	The calculator	Measured	Pull of the calculator	Standing
$\sin^2\theta_{13}$	an input	0.022329, the 15° tilt	0.02200 ± 0.00056 (NuFIT 6.0)	−0.6σ	prediction
$\sin^2\theta_{12}$	an input	0.318107	0.3096 +0.0057/−0.0073 (NuFIT 6.1); 0.3092 ± 0.0087 (JUNO alone)	−1.5σ; −1.0σ	prediction; JUNO decides at ±0.0016 by about 2031
$\sin^2\theta_{23}$	an input	not fixed by the frame	0.561 ± 0.014 (NuFIT)	—	no number
δ_CP	an input	tied to the θ_{23} octant: ±75° at $\sin^2\theta_{23} = 0.56$	best fit −85° (T2K and NOvA Collaborations 2025)	consistent	reading
other frames on the same data	tribimaximal ($\theta_{13} = 0$); TM2 (0.341); a rigid rotation (0.301)	—	as above	−38σ; −5.5σ; the rigid rotation open	the alternatives, scored alike

Quantity	Standard theory	The calculator	Measured	Pull of the calculator	Standing
sin θ_C	an input	√(m_d/m_s) = 0.2249 ± 0.0014	|V_us| = 0.2243 ± 0.0009	−0.4σ	standard (Gatto, Sartori and Tonin 1968); not counted
the unitarity triangle's α	180° − β − γ = 91.0 ± 2.8° with HFLAV's γ, by unitarity	90°	84.1 +3.7/−3.0° (all channels); 92.6 +4.5/−4.7° (B → ρρ)	−1.6σ; +0.6σ	prediction, taken after Fritzsch and Xing (2000)
V_cb and V_ub	inputs	no leg value: the top's leg is v	0.0407 ± 0.0013; 0.00389 ± 0.00016	—	the rule's boundary

Origination. M: tribimaximal mixing (Harrison, Perkins and Scott 2002), the TM1 sum rule (Albright and Rodejohann 2009), the Gatto–Sartori–Tonin relation and the right-angled unitarity triangle (Fritzsch and Xing 2000). **N**: the identification of the 1943 Clarke transform with the tribimaximal matrix, the 15° tilt from the ladder, and the reason the first column survives. **D**: the rigid-rotation rival (Sharma and Katoch 2026), which JUNO's full run separates from the tilt at about 11σ.

11. Gravity, held states and black holes

Computed on the calculator, gravity is a pull along the perpendicular, the time leg, and anything that also moves along the space leg feels both. Light, on the 45° line, has equal legs, so it bends twice what the time leg alone gives: 1.7512″ at the Sun's edge. A gyroscope in orbit precesses by the time leg plus half the space leg, 6,604.8 mas/yr, and frame dragging adds 39.1 mas/yr. The black hole is the flip, inversion through a sphere whose fixed sphere is the horizon; what falls in is held, and at most $1 - 1/\sqrt{2} = 29.3\%$ of a spinning hole's mass can come back. Held states, from positronium to toponium to ball lightning, are states parked at the √4 level. Every classical number here is general relativity's; the calculator's own content is the reading of the interior as the held foundation and of held states as one family.

11.1 The two legs of the 45° line, the flip and the held level

Two legs. Gravity is read as a pull along the time leg; light, on the 45° line, takes both legs: 1.7512″ of bending at the Sun's edge, 6,604.8 mas/yr of geodetic precession and 39.1 mas/yr of frame dragging for Gravity Probe B, each derived below. Table 48 sets each of gravity's readings beside its route, its prediction and the measurement.

The bending's factor of two. In the weak field $ds^2 = -(1 + 2\Phi/c^2)c^2dt^2 + (1 - 2\gamma\Phi/c^2)dx^2$, with $\Phi = -GM/r$, and a light ray is deflected by $(1 + \gamma)\cdot 2GM/(c^2b)$.

- **The two terms.** The first comes from the time part (Newton's particle falling in the potential), the second from the curvature of space.
- **The numbers.** With $GM_\odot = 1.32712 \times 10^{20}$ m³⁄s² and $b = 6.957 \times 10^{8}$ m, $2GM/(c^2 b) = 4.245 \times 10^{-6}$ rad = 0.8756″, and with γ = 1 the full 1.7512″.
- **Why light collects both.** A slow body's path is almost all time, so it feels only the time leg; light moves equally along both, the 45° line.

The gyroscope.

- **Geodetic.** On a circular orbit of radius r a gyroscope precesses at Ω_geo = (½ + γ)(GM)^(3/2)/(c²r^(5/2)): the ½ is the time leg and the γ the space leg. For Gravity Probe B this is 6,604.8 mas/yr.
- **Frame dragging.** It is $\Omega_{LT} = GJ/(2c^2 r^3)$, averaged over the polar orbit, with $J = I\omega = 8.034 \times 10^{37} \times 7.292 \times 10^{-5}$ kg m²⁄s. That gives 40.8 mas/yr for the polar orbit, of which the component along the guide star IM Pegasi (declination 16.8°) is 40.8 × cos 16.8° = 39.1; Gravity Probe B's own prediction was 39.2.

Table 48. Gravity's readings, each with its route.

Reading	Route	Predicted	Measured	Standing
light bending, historical and radio	time leg plus space leg	1.7512″	1919 eclipse 1.98 ± 0.12″ (Dyson, Eddington and Davidson 1920); radio γ = 0.99983 ± 0.00045 (Shapiro et al. 2004)	the same as Einstein; Newton and Soldner gave half
a squared field	z^2 doubles the angle	gravitational waves at twice the orbital frequency, in two modes 45° apart, at light speed	binary pulsar decay 0.9983 ± 0.0016 of the prediction (Weisberg and Huang 2016); GW170814 favours tensor modes; GW170817's speed equals light's within 3 parts in 10^{15}	the same as Einstein
carriers on the 45° line	zero difference	photon and graviton massless	photon below 10^{-18} eV; graviton at most 2.42 ×	the same as standard

Reading	Route	Predicted	Measured	Standing
			10^{-23} eV/c^2 (LIGO, Virgo and KAGRA, GWTC-3)	
weight and inertia	one m on both legs	equal	MICROSCOPE: $\eta(\mathrm{Ti}, \mathrm{Pt}) = (-1.5 \pm 2.3 \pm 1.5) \times 10^{-15}$ (Touboul et al. 2022)	the same as standard
the horizon	the flip's fixed sphere	$2GM/c^2$ (2.95 km for the Sun); photon sphere at 1.5 r_s; Hawking temperature 6.17×10^{-8} K for one solar mass	horizons imaged (Event Horizon Telescope Collaboration 2019, 2022); no white hole seen	exact in the mathematics; no exit
what returns	the irreducible mass is the held part	at most $1 - 1/\sqrt{2} = 29.3\%$ of the mass, as spin energy	M87*'s jet power of at least 10^{42} erg/s from a magnetically arrested flow (Event Horizon Telescope Collaboration 2021)	the same as Christodoulou (1970) and Blandford and Znajek (1977)

The black hole is the flip. In isotropic coordinates ($G = c = 1$)

$$ds^2=-\Big(\frac{1-u}{1+u}\Big)^{2}dt^2+(1+u)^4\left(d\rho^2+\rho^2 d\Omega^2\right),\qquad u=\frac{M}{2\rho}$$

- **The inversion.** Under $\rho \to M^2/4\rho$, that is $u \to 1/u$, the lapse is unchanged and the spatial factor maps into itself, since $\rho = M/2u$ and $d\rho = -M\,du/2u^2$.
- **The fixed sphere.** $u = 1$ is $\rho = M/2$, the Schwarzschild radius $r = \rho(1 + u)^2 = 2M$: the horizon, where the clock rate is zero.
- **The two copies.** The inversion maps the outside onto the inside of the fixed sphere, so the solution contains two copies joined at the horizon: Einstein and Rosen's bridge. The second copy is not an exit:
    - the bridge pinches off before anything crosses (Fuller and Wheeler 1962);
    - a black hole formed by collapse has no white-hole region;
    - none has ever been seen.
- **Inside.** The coefficient $1 - r_s/r$ changes sign, so the inward direction becomes the time direction, and the centre is reached in finite proper time.

- **Kerr's ring.** For a spinning hole the singularity is a ring, and the passage through it is sealed by the inner horizon's instability, mass inflation (Poisson and Israel 1990).

In the cube the torus sits on the black-hole side, the perpendicular step from √2 moves into the held √4 dimension, and the horizon is the fixed sphere.

What returns, and how much.

- **The horizon.** A spinning hole has horizon radius $r_+ = M + \sqrt{(M^2 - a^2)}$, with a = J/M, and horizon area $A = 4\pi(r_+^2 + a^2) = 8\pi M r_+$.
- **The irreducible mass.** $M_{irr}^2 = A/16\pi = M^2(1 + \sqrt{(1 - a^2/M^2)})/2$ (Christodoulou 1970). By Hawking's area theorem A never decreases, so neither does M_{irr}.
- **How much comes back.** Only $M - M_{irr}$, the rotational energy, can be drawn out: $1 - 1/\sqrt{2} = 29.29\%$ at maximal spin, and 15 to 25% at the spins measured for M87* and the X-ray binaries ($a/M \approx 0.9$ to 0.99).
- **The mass formula is the sum table.** $M^2 = M_{irr}^2 + (J/2M_{irr})^2$ (Christodoulou and Ruffini 1971) has the held mass on one leg and the rotation on the other. At maximal spin the two squares are equal: the 45° line, with the held share at the $1/\sqrt{2}$ level.
- **The route out is magnetic.** Field lines threading the horizon are dragged by its rotation and carry away a Poynting flux $P_{BZ} \approx (\kappa/4\pi c)\Phi^2\Omega_H^2$, with $\Omega_H = ac/(2r_+)$ and $\kappa \approx 0.05$ (Blandford and Znajek 1977; Tchekhovskoy, Narayan and McKinney 2010). It is the square of the flux through the horizon times the square of the horizon's turning rate, which is why only a spinning hole drives a jet.
- **The jet's 45°.** The jet leaves at 0.9 to 0.99c along the spin axis, on the light cone: the 45° of the (ct, r) diagram, not the jet's opening angle on the sky, which is a few degrees.

Held pairs. Two equal masses bound by a coupling g have reduced mass m/2 and ground-state binding $mg^2/4$.

- **Positronium.** $m_e\alpha^2/4 = 510{,}999\ \text{eV} \times (1/137.036)^2/4 = 6.803\ \text{eV}$, with a lifetime (125 ps for para-positronium) a million times its binding period.
- **Toponium.** With the colour factor, $g = (4/3)\alpha_s$ at the Bohr scale, $\alpha_s \approx 0.14$, its binding is $m_t(4\alpha_s/3)^2/4 \approx 1.5$ GeV, against the top's own width of 1.32 GeV: barely held. This is the threshold enhancement CMS (8.8 pb) and ATLAS (9.3 pb) see (CMS Collaboration 2025; ATLAS Collaboration 2026).

Held states driven at twice their drive frequency, ball lightning and the Higgs after inflation, are worked in section 11.3.

Beside general relativity and its rivals.

- **Where it sides with Einstein.** The calculator sides with Einstein against:
  - Newton's half angle of 1801;
  - Le Sage's push (1748), which would heat and slow every body;
  - Kaluza and Klein's extra circle, given no extra particles at the LHC and gravity normal down to 52 μm (section 4.7.2).
- **What it recovers.** It recovers the Einstein–Rosen bridge as its inversion, with the second sheet as the held side, Kerr's ring as the torus, and Wheeler's “charge without charge” as its torus.

- **The published alternative.** White-hole remnants (Rovelli and Vidotto 2018) are the published alternative to no exit, and none has been observed.
- **What it adds.** No number to the classical tests. Its own statements are the matter share, held mass that moves with the galaxies, and the fork at the centre (section 13).

Standing.

- **M.** Every number above is general relativity's or the paper's own source's.
- **N.** The two-leg reading, the black hole as the flip whose interior is held, and the “ends or turns” fork at the centre (section 13) were not found in this form.

11.2 Worked example: gravity's four regions

9. **Frame.** Time sits on the held axis w and radial distance on y. The full black-hole geometry has four regions in Kruskal's coordinates (Kruskal 1960): the outside (I), the inside (II), a second outside (III) and the white hole (IV).
10. **One quarter turn.** J turns the (y, w) plane, so radial distance and time trade places: (T, X) → (X, −T) carries region I to II, II to III, III to IV and IV back to I. This is the dimension inversion at a horizon, where the radial direction becomes time. The quarter turn is not a symmetry of the geometry; it carries each region to the next.
11. **Two quarter turns.** (T, X) → (−T, −X) is an exact isometry that swaps I with III and II with IV (Röken 2025). Mass stays positive in region III, so the inverse of gravity is not repulsion.
12. **Observed at every position.**
    - In region I, light's two legs are equal: $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ (Bertotti, Iess and Tortora 2003). The shadows of M87* (42 ± 3 μas) and Sgr A* (51.8 ± 2.3 μas) match Kerr black holes (Event Horizon Telescope 2019, 2022).
    - The face between I and II, the horizon, grew in area in GW250114 at 4.4σ (LIGO, Virgo and KAGRA Collaborations 2025).
    - Regions II to IV send nothing to region I, so they stay empty cells.
13. **The inverse of matter falls down.** One half turn turns the electrical variables over (C) and spacetime end for end (PT), which together are CPT. Antihydrogen falls at (0.75 ± 0.13 ± 0.16) g (Anderson et al. 2023), 1.2σ below 1 g and opposite to repulsion.

11.3 Held states: ball lightning and the Higgs after inflation

Both systems are held states that answer at twice the drive frequency, because each couples to the square of an oscillating field.

The record. Cen, Yuan and Xue (2014) filmed a ball at 3,000 frames per second from 0.902 km:

- 1.64 s of light;
- an apparent diameter of about 5 m in the steady stage, from 160 to 1,080 ms;
- light pulsing at 99.4 Hz, a period of 10.06 ms;
- a 35 kV transmission line about 20 m away.

The authors suggest the pulsing is “possibly induced by the second-harmonic effect associated with the high-voltage transmission lines”.

Why 50 Hz gives 100 Hz light. $\cos^2\omega t = (1 + \cos 2\omega t)/2$. Anything that answers the square of a 50 Hz field pulses at 100 Hz and peaks at twice its average, the single-phase power ratio, $\sqrt{4}$. The observed 99.4 Hz means a drive at 49.7 Hz, close to the 50 Hz of China's grid.

How fast the glow must respond. A glow with response time τ shows the 2f pulsing with depth $m = 1/\sqrt{(1 + (2\omega\tau)^2)}$. The paper gives no depth, so Table 49 gives the response each depth would require.

Table 49. Response time needed for each visible depth, for a 50 Hz drive.

Visible pulsing depth	Glow must respond within
90%	0.77 ms
50%	2.76 ms
30%	5.06 ms
10%	15.84 ms

Can the line feed it? The field from one 35 kV conductor 10 m up was computed with its image in the ground, taking a conductor radius of 1 cm and a phase voltage of $35/\sqrt{3}$ kV rms. At 20 m sideways it is 133 to 150 V/m peak for a ball 2 to 10 m up. A single conductor overstates the field, since three phases partly cancel.

- **The electric bound.** A conducting sphere of volume V in a field $E_0\, e^{i\omega t}$ has interior field $3E_0/(3 - is)$, with $s = \sigma/\omega\varepsilon_0$. It absorbs $P = \tfrac{1}{2}\sigma|E_{in}|^2 V = (9/2)\omega\varepsilon_0 E_0^2 V \cdot s/(9 + s^2)$. This is greatest at $s = 3$:

  $$P_{\max}=\tfrac{3}{4}\,\omega\varepsilon_0E_0^{2}V\quad\text{at}\quad\sigma=3\,\omega\varepsilon_0$$

- **The magnetic bound.** It uses the exact eddy-current response of a conducting sphere (Landau and Lifshitz, Electrodynamics of Continuous Media, §59). The imaginary part of its polarisability peaks at 0.533 when the radius is 2.41 skin depths. The line current is not published.

Table 50. The most power the line can deliver to the ball, 5 m up, maximised over the ball's conductivity.

Source	25 cm core	whole 5 m glow
the electric field, 133 to 150 V/m	3.0 to 3.8 × 10^{-7} W	2.4 to 3.1 × 10^{-3} W
the magnetic field, 100 A peak in the line	5.4 × 10^{-7} W	4.3 × 10^{-3} W
the magnetic field, 1,000 A peak in the line	5.4 × 10^{-5} W	0.43 W

- **Under a watt.** At 1,000 A rms, the magnetic bound in the last row of Table 50 doubles to 0.87 W for the whole glow, still under a watt. A 35 kV line rarely carries that much.
- **Against a bulb.** A 100 W household bulb draws about 34,000 times the electric-field bound for the whole glow.

- **The conclusion.** The line modulates the ball but cannot feed it. The energy was stored in the ball, consistent with the soil elements in its spectrum (Si, Fe, Ca).
- **Kapitsa's idea.** Kapitsa's feeding by microwaves from the storm (1955) is a different source, untested here.

The Higgs after inflation: the same 2f drive. After inflation the inflaton oscillates, $\varphi = \Phi \cos mt$. With $\rho = \tfrac{1}{2}\dot{\varphi}^2 + \tfrac{1}{2}m^2\varphi^2$ and $p = \tfrac{1}{2}\dot{\varphi}^2 - \tfrac{1}{2}m^2\varphi^2$, the curvature is:

$$R=\frac{\rho-3p}{M^{2}}=\frac{m^{2}\Phi^{2}}{M^{2}}\left(\tfrac12+\tfrac32\cos 2mt\right)$$

Here M is the reduced Planck mass. So R pulses at twice the inflaton frequency and dips negative for part of each cycle, as E^2 does in the ball. A Higgs coupled as $\xi R h^2$ obeys the Mathieu equation:

$$h''+\left(k^{2}+\xi R\right)h=0\;\longrightarrow\;y''+\left(A-2q\cos 2z\right)y=0,\qquad z=mt,\quad A=\frac{k^{2}}{m^{2}}+\frac{X}{2},\quad q=-\tfrac34X,\quad X=\frac{\xi\Phi^{2}}{M^{2}}$$

At k = 0 the lowest mass squared in each cycle is $-X$: negative every cycle.

Table 51. Floquet growth rates, from the monodromy matrix over one period.

$X = \xi\Phi^2/M^2$	Growth at k = 0	Fastest growth over k
0.1	0 (held)	0.04
0.3	0 (held)	0.11
0.5	0 (held)	0.19
1.0	0.33	0.37
1.33	0.48	0.48
2	0.62	0.62
5	0 (held)	0.57
10	1.16	1.16
20	1.55	1.55

- **The drive holds it.** In Table 51, the uniform mode stays bounded for $X < 0.77$, and again for $4.85 < X < 5.42$, even though its mass squared goes negative every cycle. This is Kapitza's inverted pendulum, stable when $a^2\omega^2 > 2gl$ (Kapitza 1951).
- **Slow growth elsewhere.** At other k only slow, narrow growth survives, which the expansion tends to wash out.
- **Broad resonance.** It needs $|q| \gtrsim 1$, so $\xi \gtrsim (4/3)(M/\Phi)^2$: 1.33 for $\Phi = M$ and 0.67 for $\Phi = \sqrt{2}\cdot M$. This matches strong Higgs production for $\xi \gtrsim 1$ (Herranen, Markkanen, Nurmi and Rajantie 2015).

The window for ξ.

- **The bounds.** Actual vacuum decay needs the growth to reach the barrier. The lattice study of Figueroa, Rajantie and Torrenti (2018) finds instability for $\xi \gtrsim 4$ to 5 at $m_t \approx 173.3$ GeV, and for $\xi \gtrsim 11$ to 12 at $m_t \approx 172.1$ GeV.
- **The lower end.** $\xi \gtrsim 0.06$ is needed during inflation (Herranen, Markkanen, Nurmi and Rajantie 2014).
- **The interpolation.** Linear interpolation to 172.60 GeV gives $\xi \lesssim 8.6$, an estimate made here and not a published number. So the window is roughly $0.06 \lesssim \xi \lesssim 9$.
- **Under the balance rule.** The vacuum is metastable, so the deeper vacuum exists and the upper bound stands. Table 52 sets the two held states side by side.

Table 52. The two held states side by side.

	Ball lightning	The Higgs after inflation
drive	the 50 Hz field of a power line	the inflaton, oscillating at m
coupling	the glow goes as E^2	$\xi R h^2$, with R quadratic in φ
response	99.4 Hz, 2f	curvature at 2m, 2f
held?	yes, by stored energy; the line only modulates it, well under 1 W	yes, for $X < 0.77$ at $k = 0$
released?	faded after 1.64 s	broad resonance for $\xi \gtrsim 1$; decay for $\xi \gtrsim 4$ to 12, depending on m_t

11.4 Beside the standard theories

General relativity computes every classical number on this layer, and the calculator gives the same ones; the alternatives that history tried, Newton's half angle, Le Sage's push and Kaluza–Klein's extra circle, fail the same tests for both. Where the calculator speaks and general relativity is silent or different is the interior: a held foundation with no exit, and a fork at the centre, ends or turns, that section 13 carries to the cosmos.

Table 53. Gravity on one yardstick.

Quantity	Standard theory	The calculator	Measured	Standing
light bending at the Sun's edge	1.7512″, with $\gamma = 1$ (Einstein 1915)	the time leg plus the space leg, 1.7512″	$\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ (Bertotti, less and Tortora 2003)	same
Newton's and Soldner's half	0.8756″	the time leg alone	1.98 ± 0.12″ in 1919 (Dyson,	excluded for both

Quantity	Standard theory	The calculator	Measured	Standing
			Eddington and Davidson 1920)	
geodetic precession	6,606.1 mas/yr, Gravity Probe B's prediction	6,604.8 mas/yr	6,601.8 ± 18.3 (Everitt et al. 2011)	same
frame dragging	39.2 mas/yr	39.1 mas/yr	37.2 ± 7.2 (Everitt et al. 2011)	same
gravitational waves	quadrupole radiation at light speed	the squared field: twice the orbital frequency, two modes 45° apart	the binary pulsar at 0.9983 ± 0.0016 of the prediction (Weisberg and Huang 2016); GW170817 at light speed	same
the equivalence principle	exact	one mass on both legs	$\eta = (-1.5 \pm 2.3 \pm 1.5) \times 10^{-15}$ (Touboul et al. 2022)	same
antimatter's fall	down at g, from CPT and the equivalence principle	the inside-out keeps the face sphere	(0.75 ± 0.13 ± 0.16) g (Anderson et al. 2023)	same
the horizon	$r = 2GM/c^2$	the flip's fixed sphere	the shadows of M87* and Sgr A*; the area law at 4.4σ in GW250114	same mathematics
what can return	at most $1 - 1/\sqrt{2}$ of the mass, as spin energy (Christodoulou 1970)	the held part stays; the rest returns on the hypotenuse as jets	jets from spinning holes (Event Horizon Telescope Collaboration 2021)	same
the interior	a singularity; the Einstein–Rosen bridge pinches off	the held foundation; whether the centre ends or turns is open	no white hole seen	reading

Quantity	Standard theory	The calculator	Measured	Standing
held pairs	positronium bound by $m_e\alpha^2/4$ = 6.803 eV (QED)	$m \cdot g^2/4$ at the $\sqrt{4}$ level	6.803 eV; toponium near 1.5 GeV against the top's 1.32 GeV width	same
ball lightning's flicker	no standard theory	the 2f answer of a held state to a 50 Hz drive	99.4 Hz (Cen, Yuan and Xue 2014)	reading
the Higgs after inflation	the Mathieu equation, Floquet growth	held by the drive below X = 0.77	the lattice's safety window, roughly $0.06 \lesssim \xi \lesssim 9$	same

Origination. M for every number, each with its source above. **N** for the two-leg reading of light's double bending, the black hole as the flip with the interior held, and the held states as one family at $\sqrt{4}$.

12. The strong scale: the proton's thirds

The calculator reads the proton as three phases on three legs, so its spin, ½ = ½ΔΣ + ΔG + L, divides into equal sixths: the quark spin share ΔΣ = ⅓, the gluon spin ΔG = ⅙ and the orbital share L = ⅙. Against the measurements and the continuum lattice, at one resolution (2 GeV), the three pieces sit at −0.1σ (ΔΣ against HERMES), −0.2σ and −0.4σ. The proton's radius computes as four reduced Compton wavelengths, 0.841236 fm against 0.84075 ± 0.00064 fm. At this scale the per-cycle load is no longer slight, so the static values the cube gives, the quark model's, are loaded by amounts that only QCD computes.

12.1 The proton's thirds

Thirds. The proton's spin is ½ = ½ΔΣ + ΔG + L. Read on three legs, each is ⅙, so the quark spin share is ΔΣ = ⅓:

- **quark spin.** HERMES measured ΔΣ = 0.330 ± 0.039 (via Liu and Ma 2015), and the lattice gives ½ΔΣ = 0.190 ± 0.022 (Alexandrou et al. 2026);
- **gluon spin.** The lattice gives ΔG = 0.231 ± 0.047 at 10 GeV² (Zhao et al. 2025), or 0.158 ± 0.039 at 2 GeV, the scale the behavior rule fixes (section 17.1);
- **orbital motion.** Folded to one resolution, L = 0.152 ± 0.043 (section 12.2).

Beyond the spin legs.

- **The gluon momentum fraction.** The 2026 continuum lattice decomposition gives $\langle x \rangle_g$ = 0.372 ± 0.035 at 2 GeV, a third within 1.1σ (Alexandrou et al. 2026).
- **Mass is the difference.** $E^2 = (pc)^2 + (mc^2)^2$ is the sum, and the rest mass comes back from the difference: 938.2721 MeV for the proton. 99% of it is field energy. The quark masses add to about 9.0 MeV at 2 GeV, 1% of the total; the rest is the energy of the colour field and the quarks' motion, which QCD's trace anomaly assigns mostly to the gluon field.

The radius. $r_p = 4\lambda\!\!\!^{-}_C = 0.841236$ fm, against 0.84075 ± 0.00064 fm (CODATA 2022), so $r_p/\lambda\!\!\!^{-}_C = 3.9977 \pm 0.0030$, 0.8σ off.

The radius by method. Against $4\lambda\!\!\!^{-}_C = 0.84124$ fm:

- muonic hydrogen gives 0.84087 ± 0.00039 fm (-0.9σ);
- the 2026 hydrogen 2S–6P measurement gives 0.8406 ± 0.0015 fm (-0.4σ; Maisenbacher et al. 2026);
- the electron–hydrogen average gives 0.8439 ± 0.0013 fm ($+2.0\sigma$), with χ^2/dof 2.5 across its inputs.

The chance of an alphabet member this close is 1 in 6 (section 16.6), and the relation is prior art (Standing, below).

The statics, loaded.

- **The moment ratio.** SU(6) gives $\mu_p/\mu_n = -\frac{2}{3}$, and the measured ratio is 2.75% larger in size.
- **The axial coupling.** SU(6) gives $g_A = 5/3$, and the measured 1.27641 ± 0.00056 (Märkisch et al. 2019) is 23.4% smaller.

The SU(6) statics. In the quark model the proton's spin-flavour state is symmetric, and the magnetic moment sums the quarks' moments weighted by their spin projections: $\mu_p = (4\mu_u - \mu_d)/3$ and $\mu_n = (4\mu_d - \mu_u)/3$. With equal masses the moments scale with charge, $\mu_d = -\mu_u/2$, so $\mu_p = (3/2)\mu_u$ and $\mu_n = -\mu_u$, and $\mu_n/\mu_p = -\frac{2}{3}$ exactly. The same state gives the axial coupling as the expectation of $\Sigma\sigma_z\tau_z$, which is 5/3.

Why the loads are not slight here.

- **The size of the load.** At the proton's scale α_s is about 0.5, so $\alpha_s/2\pi \approx 0.08$ before any count, and the counts are of order one to ten. The series does not converge there, which is the statement that the strong force confines.
- **What the loads are.** The loads on the statics, -23.4% on g_A and $+2.75\%$ on $|\mu_n/\mu_p|$, are what the standard account calls the quarks' relativistic motion and the pion cloud. Lattice QCD gets $g_A = 1.27 \pm 0.02$ from first principles.
- **Their signs.** The loads run both ways, so the calculator predicts neither direction, only where the loads begin.

The hadrons are outside the rule's domain. A hadron's mass is 99% field energy and 1% Yukawa, so it is not a leg of the kind the balance rule uses. Near-degenerate triples give $Q \to \frac{1}{3}$ whatever the physics: the proton, Λ and Ξ give 0.335.

What decides. The Electron-Ion Collider measures ΔG to about 10% and $\Delta\Sigma$ to about 15% (Aschenauer, Sassot and Stratmann 2015), and the lattice's separate quark and gluon orbital pieces decide the third leg.

Standing.

- **M.** The statics are SU(6)'s. $r_p = 4\lambda\!\!\!^{-}_C$ is prior art, stated explicitly by Haramein and co-authors (Haramein et al. 2025, from Haramein 2013) and equivalent to Trinhammer and Bohr's $\pi r_p = 2\lambda\!\!\!^{-}_C$ (2019).

- **N.** Equal thirds of the spin, with the gluon share ⅛, is the calculator's own commitment.

12.2 Worked example: the proton as folded reactions

The author: “I suspect proton may be folded reactions.”

Unfolded, the proton misses.

- One reaction going round 2π(4ƛ□) at c gives 4 μ_N, 43.2% above the measured 2.79284734463(82) μ_N (Mohr et al. 2024).
- One Dirac reaction on its own Compton circle gives 1 μ_N, 64.2% below.

Folded, it narrows. The neutron shows the folds: it has a moment of −1.91304276(45) μ_N and no charge, and a reaction with no charge carries no current, i = qω = 0. Fold the proton into three g = 2 reactions with charges ⅔, ⅔ and −⅓ in the quark model's symmetric spin-flavour state, each a third of the proton's mass:

- μ□ = (4μ$_u$ − μ_d)/3 = 3 μ_N, 7.42% high;
- μ□ = (4μ_d − μ$_u$)/3 = −2 μ_N, 4.55% high in size;
- their ratio is −⅔, against the measured −0.68497934(16), which is 2.75% larger in size.

The ring's unfolded 4 μ_N, 43% high, narrows to 7%. The rest needs the masses the folded reactions carry: 335.955 MeV from μ□ and 326.974 MeV from μ□, against m□/3 = 312.757 MeV. No corner gives them.

The spin's legs, folded to one resolution. The proton's spin is ½ = ½ΔΣ + ΔG + L: quark spin, gluon spin and orbital motion. The calculator puts each leg at ⅛ (section 12.1).

- **The problem.** The lattice gluon spin, ΔG = 0.231 ± 0.047, is quoted at μ² = 10 GeV² (Zhao et al. 2025), and the lattice quark spin, ½ΔΣ = 0.190 ± 0.022, at μ = 2 GeV (Alexandrou et al. 2026). Their remainder, L = ½ − ½ΔΣ − ΔG = 0.079 ± 0.052, mixes two resolutions.
- **The evolution.** At leading order ΔΣ keeps its value while α_s(ΔG + 4ΔΣ/β$_0$) is invariant, so ΔG grows as the resolution rises and the orbital piece falls to match (Ji, Tang and Hoodbhoy 1996).
- **The coupling.** α_s is 0.2476 at 10 GeV² and 0.3004 at 4 GeV², run at four loops from α_s(m_Z) = 0.1179 (d'Enterria et al. 2022).
- **The result at 2 GeV.** ΔG folds to 0.158 ± 0.039 (−0.2σ from ⅛), L to 0.152 ± 0.043 (−0.4σ), and ½ΔΣ stays 0.190 ± 0.022 (+1.1σ).

That is weaker than three checks. Some resolution always has ΔG = ⅛. 2 GeV is the lattice's convention, which the behavior rule keeps because its consequences elsewhere match (section 17.1), not a scale the calculator's structure names; the quark leg is the one that holds at every resolution. Folding removes only the −1.7σ that mixing two resolutions produced. The totals hold: at 2 GeV the momentum fractions sum to 0.995 ± 0.067 and the spin pieces to 0.507 ± 0.078 (Alexandrou et al. 2026).

Standing.

- **M.** The ratio −⅔ is SU(6) (Bég, Lee and Pais 1964, within 3% of experiment, as described by Greenberg 2008). The evolution is standard.
- **N.** Reading the proton as folded reactions, and folding the legs to one resolution before comparing them with ⅛, are the author's.

12.3 Beside the standard theories

On this layer the standard theory computes rather than takes inputs: lattice QCD gives the spin pieces, the momentum fractions, the radius and the axial coupling from first principles, with errors of several percent. So the comparison is three-way, the calculator's fixed fractions against the lattice's computed values against the measurements.

Table 54. The strong scale on one yardstick.

Quantity	Standard theory (QCD, the lattice)	The calculator	Measured	Pull of the calculator	Standing
the quark spin share ΔΣ	0.380 ± 0.044 (Alexandrou et al. 2026)	⅓	0.330 ± 0.039 (HERMES, via Liu and Ma 2015)	−0.1σ; +1.1σ against the lattice	prediction; the Electron-Ion Collider decides
the gluon spin ΔG at 2 GeV	0.158 ± 0.039 (Zhao et al. 2025, evolved to 2 GeV)	⅛	no full moment yet; 0.20 +0.036/−0.043 truncated at x > 0.05 (de Florian et al. 2014)	−0.2σ against the lattice	prediction
the orbital share L at 2 GeV	0.152 ± 0.043, by difference at one resolution	⅛	—	−0.4σ against the lattice	prediction
the gluon momentum fraction	0.372 ± 0.035 (Alexandrou et al. 2026)	not claimed; a third	—	+1.1σ	noted, not counted
the proton's radius	0.820 ± 0.014 fm (Djukanovic et al. 2024), +1.5σ from the measurement	4ƛ□ = 0.841236 fm	0.84075 ± 0.00064 fm (CODATA 2022)	−0.8σ	reading; prior art
μ⍰/μ⍰	−⅔ in the SU(6) quark model, loaded	the static −⅔	−0.68497934(16)	load not predicted	same statics

Quantity	Standard theory (QCD, the lattice)	The calculator	Measured	Pull of the calculator	Standing
	by the pion cloud				
g_A	5/3 in SU(6); 1.27 ± 0.02 on the lattice	the static 5/3	1.27641 ± 0.00056 (Märkisch et al. 2019)	load not predicted	same statics; QCD supplies the load
the proton's mass	936 ± 34 MeV from the field (Dürr et al. 2008); 99% field energy	the difference table; no number	938.2721 MeV	—	same

The score on this layer. The lattice computes; the calculator commits to fixed fractions. Today both sit within about 1.5σ of the data and of each other, so the measurement that separates them is the Electron-Ion Collider's ΔG and the lattice's orbital pieces taken separately (section 17).

Origination. M: the statics are SU(6)'s; r⍰ = 4ǎ□ is prior art (Haramein et al. 2025, from Haramein 2013; equivalent to Trinhammer and Bohr 2019). **N**: equal thirds of the spin, with the gluon share ⅛ and the orbital share ⅛, read at one resolution; the proton as three folded reactions.

13. The cosmos: the hum, the held store and the share

On the calculator the cosmos carries three numbers and one commitment. The vacuum's residual energy, the hum, is $v^2/M_{23} = 2.02 \pm 0.20$ meV, against a measured fourth root of the dark-energy density of 2.24 meV (+1.1σ). A store that keeps its density while space grows has $w = -1$ exactly. The unit 4-ball fills $\pi^2/32 = 0.3084$ of the side-2 tesseract, read as the matter share: −1.2σ against the supernovae with the microwave background and the baryon acoustic oscillations, +0.9σ against Planck and −1.3σ against DESI's oscillations alone, with the acceleration today, $q_0 = -0.537$, following from it. The galaxies' extra pull is read as held mass. At a black hole's centre the geometry either ends or turns, and the sign of the curvature and the primordial gravitational waves decide which. ΛCDM takes the share and Λ as inputs and assumes $w = -1$, and quantum field theory's own estimate of the vacuum energy misses by about 10^{120}. The data that prefer a drifting w press on ΛCDM and the calculator alike.

Four things are called held in this paper, and they are four:

- the held store, the dark energy, whose density is fixed;
- held mass, what gravitates without shining;
- the held plasma of ball lightning (section 11.3);
- the held dimension √4 on which all three are read.

13.1 The hum

In Bass's picture the cosmological constant is a residual of size v^2/M for the scale M where the emergent symmetries dissolve. The calculator puts M at the weak = strong crossing, $M_{23} = 3.0 \times 10^{16}$ GeV:

```
\mu_{\rm hum}=\frac{v^{2}}{M_{23}}=\frac{(246.22\
\mathrm{GeV})^{2}}{3.0\times10^{16}\ \mathrm{GeV}}=2.02\pm0.20\ \mathrm{meV}
```

The measured fourth root of the dark-energy density is 2.24 meV (Planck Collaboration 2020), +1.1σ.

The hum's arithmetic. $(246.22 \text{ GeV})^2 = 60{,}624 \text{ GeV}^2$, divided by $M_{23} = 3.006 \times 10^{16}$ GeV, gives 2.017×10^{-12} GeV. The ±0.20 meV is the crossing's dependence on α_s. At $\alpha_s = 0.1171$, 1.0σ below the PDG value, the crossing falls at the 2.7×10^{16} GeV that the measured value needs, and the hum matches it exactly (section 5.4).

- **The measured value.** $\rho_\Lambda = \Omega_\Lambda \cdot 3H_0^2/(8\pi G)$, with $H_0 = 67.4$ km/s/Mpc and $\Omega_\Lambda = 0.685$, is 5.25×10^{-10} J/m³ $= 2.52 \times 10^{-47}$ GeV⁴, whose fourth root is 2.24 meV.
- **The overshoot.** The Higgs vacuum's own depth, $\lambda v^4/4 = m_H^2 v^2/8 = 1.19 \times 10^8$ GeV⁴, overshoots $(2.24 \text{ meV})^4$ by $10^{54.7}$: the cosmological-constant problem in the Higgs sector alone.
- **What each part is.**
    - The cancellation is the symmetry, exact for the same reason the fundamentals cancel in a balanced neutral whatever the voltage.
    - The residual is the leading operator that breaks it.
    - Its coefficient is of order one.
    - The one choice in the number is the crossing M_{23}, which the other two crossings would replace with 5.5 eV or 0.36 eV.

13.2 The held store and w = −1

The first law for an expanding volume is $\dot{\rho} = -3H(\rho + p)$. A store that keeps its density while space grows has $\dot{\rho} = 0$, so $p = -\rho$ and $w = -1$. Its gravitational source, $\rho + 3p = -2\rho$, is negative, so it pushes, and it gains energy with the volume, $dE = \rho_\Lambda\, dV$.

The held store and the expansion.

- **The excess.** A model-independent reconstruction of the expansion rate from DESI DR2 with Pantheon+ and DES-SN5YR finds E(z) 3 to 4% above the Planck ΛCDM curve at z = 0.35 to 0.55, at 5σ and more, and 7.2σ above the calculator's own curve at z = 0.349: 1.247 ± 0.006 against 1.204, where Planck's ΛCDM curve gives 1.208, 6.5σ (Mukherjee and Sen 2025).
- **Where it sits.** If it stands, the excess sits in the window between the two holds, z = 0.31 and 0.65, exactly where the handover happens.
- **What it rests on.** It rests on Planck's sound horizon and on the DES-SN5YR sample, whose low-redshift calibration is disputed (Efstathiou 2025; Vincenzi et al. 2025).
- **No oscillation.** An expansion that itself oscillated would print a periodic residual on the supernova Hubble diagram and on the BAO distances, and none is found, to the few-percent level. The one oscillation the universe carries is the baryon acoustic oscillation, a sound wave frozen at recombination at a comoving 147 Mpc.

13.3 The share and the two holds

The unit 4-ball fills $\pi^2/32 = 0.3084$ of the side-2 tesseract.

- **The split.** The calculator reads 0.3084 as the matter share and $1 - \pi^2/32 = 0.6916$ as dark energy.
- **The two holds.** Matter thins as a^{-3} while the store does not, so with the calculator's share matter and store were equal at $z = 0.31$ and acceleration began at $z = 0.65$ (0.30 and 0.63 on Planck's split). At $z = 1$ matter was 3.6 times the store.
- **The acceleration today.** $q_0 = 3\Omega_m/2 - 1 = -0.537$.

The share and time. In ΛCDM the split moves, because matter dilutes as a^{-3} while the store does not. The near-equality of the two today is the coincidence problem. A geometric split that held at all times would remove it, but it would need the two densities to scale together, which the calculator has not shown, so the share is stated for today (section 19.1). The split that can hold at all times is inside the matter sector, where both parts dilute together: dark to baryonic matter, 16/3 (below). Table 55 sets the curvature, and the shares the supernovae give on their own, against the calculator.

Table 55. Curvature and the supernovae alone.

Data	Value	Against the calculator	Note
Planck with BAO (2018)	$\Omega_k = 0.0007 \pm 0.0019$	$+0.4\sigma$	flat
microwave background alone, Planck with ACT lensing, ΛCDM	$\Omega_k = -0.011$ $+0.006/-0.005$	-1.7σ	closed
DESI DR2 with the CMB, ΛCDM (2025)	$\Omega_k = +0.0023 \pm 0.0011$	$+2.1\sigma$	open
the same with w free and supernovae	$\Omega_k = +0.0006$ to $+0.0011 \pm 0.0013$	within 1σ	flat
DES-SN5YR supernovae alone	$\Omega_\Lambda = 0.648 \pm 0.017$	-2.6σ	calibration disputed
Pantheon+ alone	$\Omega_\Lambda = 0.666 \pm 0.018$	-1.4σ	—
Union3 alone	$\Omega_\Lambda = 0.644$ $+0.026/-0.028$	-1.8σ	—

The two curvature leans run opposite ways, closed from the CMB alone and open from DESI with the CMB. The open lean goes away only with a drifting w, which $w = -1$ forbids, so Euclid decides this row. With $c/H_0 = 4.45$ Gpc and $|\Omega_k| < 0.0045$ at 2σ, the curvature radius exceeds 66 Gpc, fifteen Hubble radii. Table 56 lists ratios that are noted but not counted.

Table 56. Ratios noted, not counted. Each is an alphabet member within 2σ in a region where the alphabet is dense, so agreement at this level is expected by chance (section 16.7).

Ratio	Calculator	Measured	Pull
dark to baryonic matter, Ω_c/Ω_b	16/3 = 5.333	5.364 ± 0.065 (Planck); 5.199 ± 0.035 (Planck, ACT and DESI DR2)	+0.5σ; −3.8σ
galaxy acceleration scale a_0/cH_0	3/16	0.1833 ± 0.0031 at H_0 = 67.4; 0.1806 at 68.4 (±0.037 systematic)	−1.4σ; −2.3σ
spectral tilt n□	26/27 = 0.9630	0.9649 ± 0.0042 (Planck Collaboration 2020); 0.9690 ± 0.0035 (with DESI DR2); 0.9737 ± 0.0025 (SPT-3G, Planck and ACT with DESI DR2); 0.975 ± 0.003 (Planck, ACT and DESI DR2, PDG 2026)	+0.5σ; +1.7σ; +4.3σ; +4.0σ
age times Hubble rate H_0t_0	2(1 − π/6) = 0.9528	0.9510 ± 0.004 (Planck); 0.964 ± 0.004 (flat ΛCDM at Ω□ = 0.300)	−0.4σ; +2.8σ
baryonic Tully–Fisher slope	4	3.85 ± 0.09 (Lelli et al. 2019); 3.98 ± 0.06 (McGaugh 2012)	−1.7σ; −0.3σ

Adding DESI pushes every CMB set's spectral tilt upward, ACT's most (McDonough and Ferreira 2026). A bounce before the start gives n□ between 0.96 and 0.97, and inflation gives 1 − 2/N, with N = 54 giving 26/27 exactly; the Simons Observatory decides. The same high values press on the simplest plateau inflation, n□ = 1 − 2/N = 0.964 to 0.967 for N = 55 to 60, by 3.8σ and 2.8σ against the PDG value. The share's own consequence for the age, in flat ΛCDM with Ω□ = π²/32, is H_0t_0 = 0.9566, not the noted 2(1 − π/6), which is one more reason that row is not counted.

13.4 Held mass: the pull from within

The galaxies' extra pull is read as held mass. It fits four observations:

- no particle has been found in decades of searches: LZ's σ_SI < 2.2 × 10^{-48} cm² at 40 GeV from 4.2 tonne-years (LZ Collaboration 2025);
- the extra pull is fixed by the visible matter at every radius in 153 galaxies (McGaugh, Lelli and Schombert 2016);
- M87's inflow ring joins its jet (Lu et al. 2023);
- the Bullet Cluster's pull stays with the galaxies rather than the gas (Clowe et al. 2006).

The microwave background's third peak asks only that the held mass be cold, clustering and photon-blind, which it is (Planck Collaboration 2020). What it is made of stays open, and the data fence it:

- **Black holes.** If it is black holes, they can be all of it only between 10^{17} and 10^{23} g, formed when the plasma was between 30 TeV and 3×10^{7} GeV, where the calculator has no scale.
- **The Higgs transition.** The calculator's own event, the Higgs transition at 160 GeV, makes Earth-mass holes (horizon mass 10^{-5} $M\odot$).
- **Microlensing.** Twenty years of OGLE microlensing cap compact objects from 10^{-4} to 10^{3} $M\odot$ at about a percent of the dark matter (Mróz et al. 2024).

The reading also inherits what cold mass owes: the tightness of the radial-acceleration relation and the core–cusp problem.

13.5 The foundation, and ends or turns

The foundation.

- **No exit.** With no exit, the mass that falls into black holes is held. Inside, the inward direction becomes time; from outside, everything that fell in is frozen at the horizon, a graveyard that still gravitates.
- **The area law.** GW250114, the loudest merger yet recorded, at signal-to-noise 80, confirmed that the horizon area never decreases and resolved two ringdown tones consistent with a Kerr hole (LIGO, Virgo and KAGRA Collaborations 2025).
- **The feed.** It is the held store itself, $\rho_\Lambda\, dV$, not the black holes.

Ends or turns. Two facts frame the question.

- **The Hubble sphere.** In a flat universe the Hubble sphere sits exactly on its own horizon: $\rho = 3H^2/8\pi G$ makes the mass inside c/H equal to $c^3/2GH$, whose Schwarzschild radius is c/H, at every epoch, if and only if $\Omega_k = 0$ (Pathria 1972). Today that is 9.2×10^{52} kg inside 1.37×10^{26} m.
- **The inside of a hole.** In general relativity it is already a universe:
    - the interior of a collapsing dust star is a closed Friedmann universe run backwards (Oppenheimer and Snyder 1939);
    - it reaches the centre in at most $\pi GM/c^3$ of proper time: 40 μs for a 2.6 $M\odot$ hole, 67 s for Sgr A* and 28 hours for M87*;
    - its volume grows as $3\sqrt{3}\pi\, M^2 t$ (Christodoulou and Rovelli 2015), so a solar-mass hole a billion years old holds 2×10^{5} solar volumes behind a 3 km horizon.

The turned version has a literature of fifty years, each entry with a test:

- Frolov, Markov and Mukhanov (1989, 1990);
- Smolin's selection for the number of holes (1992), which needs no neutron star much above 2 $M\odot$, against 2.08 ± 0.07 and 2.35 ± 0.17 $M\odot$ measured (Fonseca et al. 2021; Romani et al. 2022);
- Popławski's torsion bounce (2010), with a preferred axis and a handedness;
- the Planck-star bounce (Rovelli and Vidotto 2014; Haggard and Rovelli 2015), black to white in a time $\propto M^2$ rather than Hawking's M^3;

- white-hole remnants as the dark matter (Rovelli and Vidotto 2018);
- Gaztañaga's bounce on the exclusion principle (Gaztañaga et al. 2025).

If the centre turns:

- the universe is closed and never open, $\Omega_k \leq 0$, with $|\Omega_k| < 0.0045$ putting the parent above 1.4×10^{54} kg;
- the entropy of the start is paid by a stiff contraction before the turn (Khoury, Ovrut, Steinhardt and Turok 2001), which leaves no primordial gravitational waves, $r \approx 0$. Inflation's simplest potentials give $r \approx 0.003$ to 0.01. BICEP/Keck has $r < 0.036$ at 95% (BICEP/Keck Collaboration 2021). LiteBIRD, planned for launch in Japan's fiscal year 2032, aims for $\delta r = 0.001$ (Ghigna et al. 2024); CMB-S4, designed for the same reach, lost its U.S. support on 9 July 2025 (CMB-S4 Project 2025).

No signal yet bears on the fork.

- **Echoes.** Searches for gravitational-wave echoes in GW150914 and GW190521 bound them below 0.15 to 0.21 of the main signal (Miani et al. 2023).
- **Fast radio bursts.** White-hole bursts cannot be their main source, whose rate is far above what returning holes could supply (Ewasiuk and Profumo 2026).

The fork's test is Ω_k and r. The ended version keeps the balance line at zero and owes nothing further. Penrose's objection (1979), that a hole's interior is gravity's state of highest entropy while a daughter universe needs a low-entropy start, stands against every turned version until the stiff phase is derived rather than assumed.

13.6 Beside the standard theories

On this layer the standard theories take most of their numbers as inputs: Λ, the matter share, the dark-to-baryonic ratio and H_0 are all fitted in ΛCDM. The one standard prediction for the vacuum's energy, quantum field theory's sum of zero-point energies, fails by about 10^{120}. Table 57 sets the two side by side.

Table 57. The cosmos on one yardstick.

Quantity	Standard theory	The calculator	Measured	Pull of the calculator	Standing
the vacuum's energy scale, $\rho_\Lambda^{1/4}$	quantum field theory: about 10^{120} too large with the reduced Planck mass as cutoff, 10^{123} with the full one (Weinberg 1989); the	$v^2/M_{23} = 2.02 \pm 0.20$ meV	2.24 meV (Planck Collaboration 2020)	$+1.1\sigma$	new number; the crossing is chosen, the other two give 5.5 eV and 0.36 eV

Quantity	Standard theory	The calculator	Measured	Pull of the calculator	Standing
	Higgs vacuum alone $10^{54 \cdot 7}$; ΛCDM takes Λ as an input				
the equation of state w	ΛCDM assumes −1; quintessence lets it drift	−1, from a held store	−0.995 ± 0.019 for constant w (Popovic et al. 2025); a drift preferred at 2.8 to 4.2σ (DESI Collaboration 2025), 3.2σ after DES's recalibration	+0.3σ for constant w	exposure, shared with ΛCDM
the matter share Ω□	an input, fitted	π²/32 = 0.3084	0.3045 ± 0.0032 (supernovae, CMB and BAO; Popovic et al. 2025); 0.3153 ± 0.0073 (Planck); 0.2975 ± 0.0086 (DESI BAO)	−1.2σ; +0.9σ; −1.3σ	new number; weak, since the alphabet is dense near 0.31 (section 16)
the acceleration today q_0	−0.527, from Planck's fitted share	−0.537	−0.537 ± 0.026 for constant w (Popovic et al. 2025); −0.51 ± 0.024 below z = 0.8 (Brout et al. 2022)	0.0σ; +1.1σ	follows from the share
the expansion rate at z = 0.349, E(z)	1.208, Planck ΛCDM	1.204	1.247 ± 0.006 (Mukherjee and Sen 2025), resting on Planck's sound horizon and	+7.2σ (Planck ΛCDM +6.5σ)	exposure, shared with ΛCDM

Quantity	Standard theory	The calculator	Measured	Pull of the calculator	Standing
			DES-SN5YR's calibration		
why Ω□ and Ω_Λ are close today	unexplained, the coincidence problem	the share stated for today only	—	—	open for both
curvature Ω_k	inflation: flat (Guth 1981); ΛCDM usually fixes 0	flat on the balance line; the turned version Ω_k ≤ 0	0.0007 ± 0.0019 (Planck with BAO); +0.0023 ± 0.0011 (DESI DR2 with the CMB)	+0.4σ; +2.1σ	same for flatness; prediction for the sign
primordial gravitational waves r	0.003 to 0.01 from the simplest potentials	r ≈ 0 in the turned version; no number in the ended one	r < 0.036 at 95% (BICEP/Keck Collaboration 2021)	—	prediction; LiteBIRD decides
the spectral tilt n□	1 − 2/N = 0.964 to 0.967 for N = 55 to 60	26/27 = 0.9630, inflation's value at N = 54; noted, not counted	0.975 ± 0.003 (Planck, ACT and DESI DR2; PDG 2026)	+4.0σ (inflation +3.8 to +2.8σ)	noted; pressed for both
the galaxies' extra pull	ΛCDM: a cold particle, its density fitted, none found (LZ Collaboration 2025); MOND: modified dynamics, Tully–Fisher slope 4 (Milgrom 1983)	held mass: cold, clustering, photon-blind, set by the visible matter; composition open	the radial acceleration relation in 153 galaxies (McGaugh, Lelli and Schombert 2016); the Bullet Cluster (Clowe et al. 2006)	—	reading; MOND fails the Bullet Cluster and the third peak, which held mass and cold dark matter pass

Quantity	Standard theory	The calculator	Measured	Pull of the calculator	Standing
a black hole's centre	a singularity in classical general relativity	ends or turns; open	echoes below 0.15 to 0.21 of the main signal (Miani et al. 2023)	—	reading; Ω_k's sign and r decide
H_0, S_8 and N_eff	H_0 and S_8 fitted or inferred; N_eff = 3.044 from three neutrinos	H_0 and S_8 not computed; three phases give three species	below	—	shared

What neither computes.

- **H_0.** The Hubble constant is a number of neither. The values are 73.17 ± 0.86 km/s/Mpc from the Cepheid ladder (Breuval et al. 2024), 70.4 ± 1.9 from the JWST tip-of-the-red-giant-branch and carbon-star ladders (Freedman et al. 2025), and 67.19 ± 0.38 from the three CMB experiments combined (Camphuis et al. 2025).
- **S_8.** The combined CMB value from Planck, ACT DR6 and SPT-3G, $S_8 = 0.836 +0.012/-0.013$ (DES Collaboration 2026, via Pantos and Perivolaropoulos 2026), and the KiDS-Legacy lensing value, $0.815 +0.016/-0.021$ (Wright et al. 2025), now agree within 1σ.
- **N_eff.** ACT DR6 finds 2.86 ± 0.13 relativistic species, three within 1.4σ (ACT Collaboration 2025).

The score on this layer.

- **Inputs against numbers.** ΛCDM fits the share and Λ; the calculator gives both as numbers, the hum at +1.1σ with the crossing chosen among three, and the share at −1.2σ, weak because cube numbers crowd near 0.31.
- **The standard failure.** Quantum field theory's vacuum energy misses by about 10^{120}, a failure scored as one (section 2.5).
- **Shared pressure.** The drift in w, at 2.8 to 4.2σ, and the expansion excess at z ≈ 0.35, at 6.5σ and 7.2σ, press on ΛCDM and the calculator alike, and both pressures rest on DES-SN5YR's disputed calibration and on Planck's sound horizon.
- **What separates them.** The sign of Ω_k and r: the turned version predicts Ω_k ≤ 0 and r ≈ 0, and the simplest inflation r ≈ 0.003 to 0.01. Euclid and the final DESI data measure the share and Ω_k to about ±0.003, and LiteBIRD r to ±0.001 (section 17).

Origination.

- **M.** w = −1 is ΛCDM's assumption, and the continuity equation is standard. The cancellation, with its residual v^2/M, is Bass's (Bass 2015 to 2026). The Hubble sphere on its own horizon is Pathria's (1972), the interior as a closed Friedmann universe Oppenheimer and Snyder's (1939), and each turned version is its authors' (section 13.5).

- **N.** w = −1 derived from a held store; the hum at the weak = strong crossing; π²/32 as the matter share; the galaxies' extra pull as held mass, with no dark-matter particle (the author's); the author's picture of the macro scale.
- **D.** w = −1 for all time diverges from the evolving dark energy that DESI's data with supernovae prefer at 2.8 to 4.2σ, and at 3.2σ after DES's recalibration (DESI Collaboration 2025; Popovic et al. 2025).

14. Atoms: the tables at human scale

At the scale of atoms the calculator's two tables return as a spectrum. Hydrogen's lines are the inverse difference table, $1/\lambda = R(n_2^2 - n_1^2)/(n_1^2 n_2^2)$: Lyman-α is the cell (2, 1)/4 at 121.57 nm, and H-α the cell (3, 2)/36 at 656.47 nm. A subshell holds a gnomon of orbitals, $2\ell + 1 = (\ell + 1)^2 - \ell^2$, and the shells hold $2n^2$ electrons, the diagonal sum cells (n, n). The cube's 27 points are the sites of crystal chemistry, and every element from hydrogen to oganesson sits in the sum table, in the difference table or one leg beyond them. Each number on this layer is the standard theories': Rydberg's and Bohr's, Fock's, quantum electrodynamics' and crystal-field theory's. What the calculator owns here is the single figure that holds them all, and the check that the figure predicts no chemistry (section 14.5).

14.1 Hydrogen's lines: the inverse difference table

The lines are the inverse difference table scaled by Rydberg's constant for hydrogen. Table 58 reads the first five as cells, in vacuum wavelengths.

Table 58. The hydrogen lines as the inverse difference table. $1/\lambda = R(n_2^2 - n_1^2)/(n_1^2 n_2^2)$, with $R_H = 1.0967758 \times 10^7\ m^{-1}$.

Line	Cell	Wavelength
Lyman-α	(2, 1)/4	121.57 nm
H-α	(3, 2)/36	656.47 nm
H-β	(4, 2)/64	486.27 nm
H-γ	(5, 2)/100	434.17 nm
H-δ	(6, 2)/144	410.29 nm

The corrections to these cells, the fine structure, the hyperfine line and the Lamb shift, are further powers of the per-cycle coupling α, and quantum electrodynamics computes them (section 5.3).

14.2 The shells, and the atom as the square of the oscillator

The shells.

- **Gnomons stacked.** A subshell with angular momentum ℓ holds $2\ell + 1 = (\ell + 1)^2 - \ell^2$ orbitals, a gnomon. Stacking the gnomons for ℓ = 0 to n − 1 gives n^2 orbitals, and two spins make $2n^2$ electrons.
- **The filling order.** The periodic table's rows run 2, 8, 8, 18, 18, 32, 32 because the subshells fill in the Madelung order, which the squares alone do not say.

- **By dimension.** The shell sizes are 1, 3, 5, 7 in two dimensions, 1, 4, 9, 16 in three and 1, 5, 14, 30 in four, each column the running sum of the one before. Three-dimensional hydrogen is a four-dimensional sphere (Fock 1935).
- **Parity.** $(-1)^{\ell}$ is inside-out: s and d are even, p and f odd.

The atom is the square of the oscillator.

- **In three dimensions.** The Coulomb problem is the four-dimensional isotropic oscillator projected by the Hopf map, whose fibres are circles each linked once with every other (Kustaanheimo and Stiefel 1965). Hydrogen's orbits upstairs are those Hopf circles, the same linked circles as the held torus, and the direction lost in the projection is a U(1) phase, the electron's charge coupling.
- **In two dimensions.** The map is Levi-Civita's $w = z^2$ (1920): the planar Kepler problem is the square of the planar oscillator. The $1/r$ and r^2 potentials are each other's conformal squares, the Newton–Hooke duality in which potentials r^a and r^b pair when $(a + 2)(b + 2) = 4$ (Arnold 1990).

The atom is the square of the oscillator, with one circle's worth of phase folded away.

14.3 Crystals and the 27 points

Crystal fields. For point charges, the splitting of the five d orbitals in a tetrahedral arrangement is $-4/9$ of the octahedral one, and in a cubic arrangement $-8/9$. These follow by direct computation of the field at the ion from charges on the cube's face points, its alternate corners and its corners. The octahedral ligands sit on the axes and the tetrahedral and cubic ones on the diagonals, so the orbital that points along the axes is raised in one case and lowered in the other. That is why cobalt chloride is pink with six waters and blue with four chlorides. The d orbitals $x^2 - y^2$ and xy are the difference table and the third table of one complex square.

Crystals and images.

- **Neighbours.** Face-centred cubic neighbours sit across a face diagonal at $a\sqrt{2}/2$ (Cu 2.556 Å, Al 2.863 Å) and body-centred ones across the body diagonal at $a\sqrt{3}/2$ (Fe 2.482 Å, W 2.741 Å).
- **The flip.** The reciprocal lattice of FCC is BCC and of BCC is FCC, the flip exchanging the two. The method of images places a charge's mirror at R^2/d from the centre of a grounded sphere of radius R: the inversion again.
- **Maxwell under inversion.** Maxwell's equations keep their form under inversion through a sphere (Bateman 1910; Cunningham 1910), and mass breaks that symmetry.

Table 59 lays everything that sits on the cube's 27 points over one another.

Table 59. The 27 points as one overlay.

Points	Distance	What sits there
the centre	0	the atom's site; three-phase zero; the T-bar's spin point; the hinge of the flip

Points	Distance	What sits there
6 face points	1	the octahedron's corners; octahedral ligands; p lobes; the directions of ± phase; the points the flip fixes
12 edge midpoints	√2	FCC neighbours; t_2g directions; the three-phase hexagon (side = radius = √2)
8 corners	√3	the delimiters; the drive's switch states; BCC neighbours; alternate corners at the tetrahedral angle 109.47°, carbon's bond angle
the body diagonal	—	the Clarke zero axis; the cipher's ’–× axis

The governing symmetry is the cube's full group O_h: 48 operations, the 24 rotations doubled by inside-out.

14.4 Everyday matches

The same cells and rungs turn up in ordinary measurements:

- **Orbits.** Escape over circular speed is √2 for the Earth, the Moon and the Sun, and 12 km/s from the Earth leaves at 4.344 km/s.
- **Relativity.** A muon with γ = 10 sits at the cell (10, 1), √99, and its lifetime stretches from 2.197 to 21.97 μs.
- **Lamps and the neutral.** Lamps flicker at 2f. With third harmonics the neutral current can exceed the phase current, because the three third harmonics add in the neutral (section 3.7.1). For short current pulses that do not overlap between phases, as computer power supplies draw, the neutral carries √3 times the phase current: the √3 rung.

14.5 Every element placed

Section 3.5 gives the rules and the tally; this subsection gives:

- the corrections to her two element pages (Table 60);
- why the noble gases split as they do;
- where the tables stop fitting the atom: the filling order and the superheavy elements;
- the ledger of all 118 elements (Table 93).

Her two pages.

- *Root Elements* tags each whole-number cell of Tables 3 and 4 with the element of that atomic number.
- *Calc Elements* gives each element from Z = 1 to 54:

- a tier (1D baseline, 2D sum, 2D difference, 3D nested or 4D);
- a coordinate form;
- a descriptor taken from chemistry.

Table 60. Corrections to the element pages. Checked in appA.py.

Page and entry	Z	As written	Correct
Root Elements, sum cell (3, 1)	10	He 10	Ne 10: $3^2 + 1^2 = 10$ is neon
Root Elements, sum cell (9, 8)	145	√146 (Elem 146)	√145, since 81 + 64 = 145
Calc Elements, V	23	4D	within legs of 10 or less, four squares, as (3, 3, 2, 1); with any legs, $12^2 - 11^2$, a difference
Calc Elements, Co	27	4D, (3, 3, 3, 0)	$6^2 - 3^2$, on the page; the vector (3, 3, 3) is three-dimensional
Calc Elements, Ga	31	3D, (3, 3, 3, 1) “= 31”	9 + 9 + 9 + 1 = 28. Within the cap 31 needs four squares, as (5, 2, 1, 1); with any legs, $16^2 - 15^2$
Calc Elements, Y	39	4D, (5, 3, 2, 1)	$8^2 - 5^2$, on the page
Calc Elements, Tc	43	4D, (5, 3, 3, 0)	25 + 9 + 9 = 43, three squares within the cap; with any legs, $22^2 - 21^2$
Calc Elements, Ru	44	unfinished	$6^2 + 2^2 + 2^2$ within the cap; with any legs, $12^2 - 10^2$
Calc Elements, Ag	47	(5, 4, 3, 1) “= 47”	25 + 16 + 9 + 1 = 51. Within the cap, (6, 3, 1, 1); with any legs, $24^2 - 23^2$
Calc Elements, In	49	2D degenerate, (7, 0)	7^2, a square; also $25^2 - 24^2$
Calc Elements, Be	4	3D degenerate, (2, 0, 0)	2^2, a square

Page and entry	Z	As written	Correct
Calc Elements, unmarked duals	5, 8, 13, 17, 25, 29, 37, 41, 52, 53	one tier each	each is both a sum and a difference of two squares

What the corrections show:

- **The 3D rows are right as written.** Her rows labelled 3D (C 6, Si 14, Ti 22, Zn 30, Sr 38, Mo 42, Pd 46 and Xe 54) are exactly the elements up to 54 that need three squares (section 3.5).
- **The 4D rows come from the leg cap.** Every 4D row is either a slip or an effect of capping the legs at 10, never of the numbers themselves:
    - V 23, Ga 31 and Ag 47 are of the form 8m + 7, so within the cap they need four squares;
    - with any legs, each is a difference.

The elements now being sought sit one leg off the page. $119 = 12^2 - 5^2$ and $120 = 11^2 - 1^2$, both differences only. The next closed shell is a difference either way:

- 168 by the Madelung order: $13^2 - 1^2$;
- about 172 in Pyykkö's relativistic table (Pyykkö 2011): $44^2 - 42^2$.

Why the noble gases split as they do (added).

- **Periods.** The period lengths are the diagonal sum cells, $(n, n) = 2n^2$. These are 2 more than a multiple of 4 for odd n, and a multiple of 8 for even n.
- **Running sums.** The noble-gas numbers are the running sums 2, 2 + 8, 2 + 8 + 8, …. So a noble gas lands in the difference table only when an even number of odd-n periods have closed: krypton at 36, and next at 168.
- **The heaviest three.** They need three squares because $54 = 2 \cdot 3^3$, $86 = 2 \cdot 43$ and $118 = 2 \cdot 59$ each carry a prime of the form 4k + 3 to an odd power.
- **Abundance.** The even–odd alternation of cosmic abundance, the Oddo–Harkins rule (Lodders 2021), shows the same parity, but its cause is nuclear pairing, not the tables.

Where the tables stop fitting the atom: the filling order. The Madelung order, by $n + \ell$, turns the shell sizes 2, 8, 18 and 32 into the period lengths 2, 8, 8, 18, 18, 32 and 32. That order is not in the tables, and in 2025 it was shown not to be exact in the atoms either:

- **Baez (2025)** shows that the bound states of the spin-½ Kepler problem are unitarily equivalent to solutions of the Weyl equation on $\mathbb{R} \times S^3$, Fock's sphere with spin added. He derives the Madelung rules by modifying that free field's Hamiltonian, and calls them “a reasonable approximation to the observed filling of subshells”.
- **Bjerg, Fournais and Hearnshaw (2025)** prove that for large Z the angular-momentum distribution of a neutral atom follows the Thomas–Fermi distribution and departs from the Madelung rule.
- **The measured exceptions** run from chromium (24) to lawrencium (103):
    - lawrencium's ground state is predicted to be $7s^2 7p^1$ rather than $6d^1$, the first element whose configuration the relativistic stabilisation of $7p_{1/2}$ changes outright;

- its first ionisation potential, 4.96 +0.08/−0.07 eV (Sato et al. 2015), agrees with relativistic coupled-cluster calculations (Guo et al. 2024).

Where the table's own columns stop fitting: the superheavy elements, observed.

- **Nihonium and moscovium.** Yakushev et al. (2024) measured nihonium (113) and moscovium (115) adsorbing on silicon oxide with $-\Delta H_ads$ = 58 +8/−3 and 54 +11/−5 kJ/mol. That is about 100 kJ/mol weaker than their homologues thallium and bismuth, yet more reactive than their closed-shell neighbours copernicium (112) and flerovium (114): “the large energy splitting in the 7p shell and the stabilization of the $7p_{1/2}^2$ subshell has been confirmed experimentally”.
- **Flerovium and copernicium.** Flerovium behaves as a volatile metal rather than a noble gas (Yakushev et al. 2022), and copernicium as a volatile metal (Eichler et al. 2007).
- **Oganesson (118).** It is predicted to have no shell structure left in its electron density (Jerabek et al. 2018), to be a semiconductor (Mewes et al. 2019) and to be a solid at room temperature (Smits et al. 2020). That ends “noble gas” as a property of a column.
- **The same relativity elsewhere.** It turns gold yellow and keeps mercury liquid (Pyykkö 1988; Calvo et al. 2013). The limits of the table are reviewed by Smits et al. (2024).

The search for elements 119 and 120.

- **Berkeley.** In 2024 Berkeley Lab made two atoms of ^{290}Lv with a titanium-50 beam on plutonium-244, at 0.44 +0.58/−0.28 pb (Gates et al. 2024). This is the beam that reaches element 120 on californium.
- **Other searches.** RIKEN's vanadium-51 on curium-248 search for element 119 has run since 2020, and JINR's chromium-54 on curium-248 search for element 120 is in preparation.
- **No atom yet.** None of these searches had reported an atom by September 2026, when Oak Ridge was fabricating the californium-249 target for Berkeley's search for element 120 (Myhre 2026). GSI's limits from 2007 to 2012 are 90 to 200 fb.

The tables say nothing about whether 119 or 120 will be found; they say only where those numbers sit.

The ledger.

- **Sum and difference cells.** These are listed with $a \geq b \geq 1$; for differences, the first three routes are given.
- **Three squares.** The three-square cell is given only where neither table reaches the element. It is written as the sum cell plus the extra leg of section 3.5.
- **The last column.** It gives the element's standing on her page as printed, with legs 1 to 10.

Table 93, all 118 elements in the three tables, is with the supplementary tables at the end of the paper.

14.6 Beside the standard theories

On this layer the standard theories compute everything, from the lines to the chemistry of the superheavy elements, and the calculator gives the same numbers as cells of its tables. Table 61 sets the two side by side, including where the tables stop.

Table 61. Atoms on one yardstick.

Quantity	Standard theory	The calculator	Measured	Standing
hydrogen's lines	Rydberg's formula (Rydberg 1888); Bohr's levels (Bohr 1913)	the inverse difference table	the same lines; the residue is the fine structure and the Lamb shift (section 5.3)	same
the corrections	quantum electrodynamics, to parts in 10^{15}	further powers of the per-cycle coupling α; no new digit	the 1S–2S interval, 2,466,061,413,187,035 ± 10 Hz (Parthey et al. 2011)	same; QED computes
the n^2 degeneracy	Fock's four-dimensional symmetry (Fock 1935)	gnomons stacked; the diagonal sum cell (n, n) = $2n^2$	shells of 2, 8, 18 and 32	same
the Coulomb problem	the four-dimensional oscillator under the Hopf map (Kustaanheimo and Stiefel 1965); $w = z^2$ in the plane (Levi-Civita 1920)	the atom as the square of the oscillator, one circle of phase folded away	—	same mathematics; reading
the filling order	the Madelung rule, approximate (Baez 2025), departing at large Z (Bjerg, Fournais and Hearnshaw 2025); configurations computed from first principles	not in the tables	exceptions from chromium (24) to lawrencium (103)	the standard theory computes; the calculator is silent
crystal-field splitting	point charges: tetrahedral −4/9 and cubic −8/9 of the octahedral splitting (Bethe 1929)	the same, from the face points, the alternate corners and the corners	cobalt chloride pink with six waters, blue with four chlorides	same

Quantity	Standard theory	The calculator	Measured	Standing
lattice neighbours	face-centred at $a\sqrt{2}/2$, body-centred at $a\sqrt{3}/2$	the edge midpoints and the corners	Cu 2.556 Å, Al 2.863 Å; Fe 2.482 Å, W 2.741 Å	same
the reciprocal lattice	FCC and BCC are each other's reciprocal	the flip	—	same; reading
the periodic table	ordered by Z (Mendeleev 1869), its chemistry from quantum mechanics	every Z in the sum table, the difference table or one leg beyond; the tiers carry no chemistry	—	reading; number theory
the superheavy elements	relativistic quantum chemistry: the split 7p shell, flerovium and copernicium as volatile metals, oganesson a semiconductor	not in the tables	nihonium and moscovium on silicon oxide (Yakushev et al. 2024)	the standard theory computes and is confirmed; the calculator is silent
the next closed shell	168 by the Madelung order, about 172 relativistically (Pyykkö 2011)	both are differences, $13^2 - 1^2$ and $44^2 - 42^2$; the tables do not choose	—	the standard theory predicts; the calculator is silent

The score on this layer.

- **The same numbers.** Every number here is the standard theories'; the calculator reads each as a cell and adds no digit.
- **Where the tables stop.** The filling order, the chemistry of the superheavy elements and the next closed shell are computed by quantum mechanics with relativity, and confirmed where they have been measured; the tables say only where the numbers sit.
- **Checked against chance.** The element tiers are a property of the integer Z, and three checks show that they carry no chemistry (section 3.5).

Origination.

- **M.** Rydberg (1888), Bohr (1913) and Fock (1935); Levi-Civita (1920) and Kustaanheimo and Stiefel (1965); the Newton–Hooke duality (Arnold 1990); the point-charge crystal field (Bethe 1929);

Maxwell's equations under inversion (Bateman 1910; Cunningham 1910); the theorems on sums of squares (section 3.5).

- **N.** Reading the lines, the shells, the crystals and the elements as cells of the two tables, the author's single figure (*Root Elements*; *Calc Elements*); the extension of her element page to all 118, with its corrections (added); the noble gases' split as parity (added).
- **D.** None: the tables predict no chemistry, as checked.

15. The scorecard: both theories on one yardstick

Every number the calculator commits to sits within 1.6σ of the primary measurement in its row (Table 62). For the same 22 quantities the standard theories mostly take inputs: eleven are free parameters or assumptions of the Standard Model or ΛCDM, one follows from them, six are predictions the calculator shares, three are lattice computations, and one, the vacuum's energy, is a standard prediction that fails by about 10^{120}. The exposures appear where the same data are read another way. The ΛCDM effective neutrino mass, the curvature from DESI with the microwave background and the preferred drift in w are one cosmological discrepancy, and it presses on ΛCDM as hard as on the calculator (section 15.3). Section 16 shows that agreement at the calculator's level is what an alphabet of its density produces by chance, while the standard theories' advance predictions are not, so this scorecard records the state of the data and is not, by itself, evidence. Pulls are (measured − computed)/σ; for an asymmetric error the side facing the computed value is used.

15.1 The committed numbers

Table 62 sets each number the calculator commits to beside the standard theories' value or status for the same quantity, and beside the measurement.

Table 62. Every committed number on one yardstick.

Number	Section	Standard theory	The calculator	Measured	Pull of the calculator
lepton angle to the diagonal	9.1	input: the three masses are free, and Koide's relation is unexplained	45°	44.99991° ± 0.00022°, from the PDG 2026 masses	−0.4
τ mass from e and μ	9.1	input	1,776.969 MeV	1,776.93 ± 0.09 MeV (PDG 2026)	−0.4
lepton phase δ	9.1	input	2/9 = 0.222222	0.2222248 ± 0.0000063	+0.4

Number	Section	Standard theory	The calculator	Measured	Pull of the calculator
neutrino mass sum, normal ordering	9.1	input; oscillations fix only the splittings	59.2 meV	below 64 meV at 95%; the ΛCDM effective mass −0.101 +0.047/−0.056 eV (Elbers et al. 2025)	inside the cap; −3.4 against the effective mass
top pole mass	9.4	input	172.57 ± 0.38 GeV	172.60 ± 0.27 GeV direct; 172.1 ± 0.6 GeV pole (PDG 2026)	+0.1; −0.7
Higgs mass	9.4	input; a vacuum exactly on its edge gave 126 GeV in 2009 at a top of 171.3 GeV (Shaposhnikov and Wetterich 2010), and gives 128.55 ± 0.96 GeV with today's top, −3.5σ	125.20 ± 0.92 GeV	125.13 ± 0.11 GeV (PDG 2026)	−0.1
$\sin^2\theta_{13}$	10.1	input	0.022329, from the 15° tilt	0.02200 ± 0.00056 (NuFIT 6.0, as used by Esteban et al. 2026)	−0.6
$\sin^2\theta_{12}$	10.1	input	0.318107, from the 15° tilt	0.3096 +0.0057/−0.0073 (Esteban et al. 2026); JUNO alone 0.3092 ±	−1.5; −1.0

Number	Section	Standard theory	The calculator	Measured	Pull of the calculator
				0.0087 (JUNO Collaboration 2025)	
unitarity-triangle angle α	10.1	input: the quark-mixing phase is free	90°	84.1 +3.7/−3.0°, all channels (PDG 2026); 92.6 +4.5/−4.7° from B → ρρ alone (Belle II Collaboration 2025)	−1.6; +0.6
light bending, γ = 1	11.1	the same, from general relativity	1.7512″	$\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ (Bertotti, Iess and Tortora 2003)	+0.9
geodetic precession	11.1	6,606.1 mas/yr	6,604.8 mas/yr	6,601.8 ± 18.3 (Everitt et al. 2011)	−0.2
frame dragging	11.1	39.2 mas/yr	39.1 mas/yr	37.2 ± 7.2 (Everitt et al. 2011)	−0.3
antihydrogen falls at g	11.2	the same, from CPT and the equivalence principle	1.00 g	0.75 ± 0.21 g (Anderson et al. 2023)	−1.2
black-hole return at maximal spin	11.1	$1 - 1/\sqrt{2}$, Christodoulou's bound (Christodoulou 1970)	29.3%	—	same
quark spin share ΔΣ	12.1	the lattice: 0.380 ± 0.044 (Alexandrou et al. 2026)	⅓	0.330 ± 0.039 (HERMES, via Liu and Ma 2015)	−0.1

Number	Section	Standard theory	The calculator	Measured	Pull of the calculator
gluon spin ΔG at 2 GeV	12.1	the lattice: 0.158 ± 0.039 (Zhao et al. 2025, evolved to 2 GeV)	⅛	no full moment yet; 0.20 +0.036/−0.043 truncated at x > 0.05 (de Florian et al. 2014)	−0.2 against the lattice
proton radius in reduced Compton wavelengths	12.1	the lattice: 3.90 ± 0.07, from 0.820 ± 0.014 fm (Djukanovic et al. 2024)	4	3.9977 ± 0.0030 (CODATA 2022)	−0.8
vacuum energy scale	13.1	quantum field theory: about 10^{120} too large (Weinberg 1989); Λ an input in ΛCDM	2.02 ± 0.20 meV	2.24 meV (Planck Collaboration 2020)	+1.1
curvature Ω_k	13.3	flat, from inflation	0	0.0007 ± 0.0019 (Planck with BAO); +0.0023 ± 0.0011 (DESI with the CMB)	+0.4; +2.1
matter share Ω□	13.3	input	$\pi^2/32$ = 0.308425	0.3045 ± 0.0032, supernovae with CMB and BAO (Popovic et al. 2025); 0.3153 ± 0.0073 (Planck); 0.2975 ± 0.0086 (DESI BAO)	−1.2; +0.9; −1.3

Number	Section	Standard theory	The calculator	Measured	Pull of the calculator
deceleration q_0 = 3Ω□/2 − 1	13.3	−0.527, from Planck's fitted share	−0.53736	−0.537 ± 0.026 for constant w (Popovic et al. 2025)	0.0
equation of state w	13.2	assumed −1 in ΛCDM	−1 for all time	−0.995 ± 0.019 for constant w (Popovic et al. 2025); a drift preferred at 2.8 to 4.2σ (DESI Collaboration 2025), 3.2σ after recalibration	+0.3 for constant w; the exposure, shared with ΛCDM

Independence. Counted honestly, Table 62 holds fewer independent numbers of the calculator's own than it has rows.

- **Standard results.** The gravity rows, the curvature, the black-hole return and the deceleration are standard results, or follow from other rows.
- **Followers.** The Higgs follows from the top, the τ from the angle, and $\sin^2\theta_{12}$ from the same 15° as $\sin^2\theta_{13}$.
- **Inherited.** The Koide angle and phase are Koide's and Brannen's. α = 90° is Fritzsch and Xing's value, and r□ = 4ƛ□ is prior art.
- **The calculator's own.** That leaves the 15° tilt, the δ_3 load, the spin thirds (ΔΣ and ΔG), the hum and $\pi^2/32$, with w = −1 shared with ΛCDM.

What the table cannot settle. Every one of these numbers sits within 1.6σ of its primary measurement, and section 16 asks whether that is more than chance. The standard theories score no pull on the eleven inputs, because a fitted number cannot miss.

15.2 New numbers, re-readings and deviations

Where the standard theories take an input. The Standard Model has 19 free parameters before neutrino masses and 26 after. General relativity and ΛCDM add G, Λ, the matter densities and the initial conditions. A theory that fits a number cannot miss it, so on the yardstick of section 2.5 those rows earn no pull for either theory; they score where the calculator gives a number and the standard theory does not, and section 16 asks whether those numbers beat chance.

Three kinds of statement appear in this paper and should not be confused:

- **A re-reading** keeps every standard number and changes the account.
- **A new number** stands where the standard theory has an input, and can be wrong.

- **A deviation** says something the standard theory denies or leaves open, with a consequence that can fail.

Table 63 sorts each committed statement into the three kinds, with the standard answer beside it and what would show it wrong.

Table 63. The committed statements, sorted.

Statement	Standard physics	The calculator	What would show it wrong	Status, October 2026
$w = -1$ exactly, for all time	ΛCDM assumes it; w_0w_aCDM frees it	derived: a held store keeps its density while space grows	any confirmed drift; a ΛCDM neutrino mass below the floor; open curvature in ΛCDM	DESI with supernovae prefers a drift at 2.8 to 4.2σ, 3.2σ after recalibration; Euclid and the final DESI decide (deviation)
the share $1 - \pi^2/32 = 0.6916$	fitted	a geometric number	Ω□ away from 0.3084 at ±0.003	−0.9σ (Planck); +1.6 to +2.1σ (Planck with DESI, and with ACT); supernovae alone 1.4 to 2.6σ (new number)
neutrino masses with no free parameter	three free masses	0.364, 8.66 and 50.13 meV, normal ordering	the inverted ordering; a sum above 64 meV; a cosmological preference for a vanishing mass that is not a systematic	inside the cap; 3.0σ from the ΛCDM effective mass (new number; the exposure is shared with $w = -1$)
lepton mixing from the 15° tilt	three free angles and a phase	$\sin^2\theta_{12} = 0.318$, $\sin^2\theta_{13} = 0.0223$, δ_{CP} tied to the octant	JUNO at ±0.0016 below 0.313	−1.5σ (NuFIT 6.1); the rival frame out at 5.5σ; decided by about 2031 (new number)
the top and the Higgs from the edge and one load	free	172.57 ± 0.38 and 125.20 ± 0.92 GeV	a top pole mass outside 172.57 ± 0.4 GeV	+0.1σ against the direct mass or −0.7σ against the pole mass, and

Statement	Standard physics	The calculator	What would show it wrong	Status, October 2026
				−0.1σ; the threshold scan decides (new number)
a right-angled unitarity triangle, α = 90°	free	the up and down legs perpendicular	α outside 90 ± 4° at Belle II and LHCb	+0.6σ (ρρ), −1.6σ (all channels); taken as a value, not derived (a reading with a number)
the proton's spin in sixths	computed on the lattice	½ΔΣ, ΔG and L each ⅙	ΔG or L away from ⅙ at 3σ	+1.1σ, −0.2σ and −0.4σ at 2 GeV (new number)
the hum v^2/M at the weak = strong crossing	Λ fitted	2.02 ± 0.20 meV	α_s to ±0.0002 moving the crossing	+1.1σ, with the coefficient fixed at 1 (new number)
the proton's radius 4ƛ□	the lattice to 3%	the held factor read twice	PRad-II at 0.43%	−0.4 to +2.0σ by method (a reading with a number, prior art)
Ω_k ≤ 0 and r ≈ 0 if the centre of a hole turns	inflation: flat, r of order 10^{-3} to 10^{-2}	the fork of section 13.5, the one reading here that predicts a sign	an open universe at ±0.001; r detected	the microwave background alone closed at 1.7σ; DESI with it open at 2.1σ; r < 0.036 (deviation, open)
antimatter falls at g; the equivalence principle exact	the same	inside-out keeps the face sphere	antihydrogen falling differently; a violation at 10^{-15}	0.75 ± 0.21 g; MICROSCOPE at 10^{-15} (no deviation)
three generations and no fourth light one	an input	three phases, the fourth level held	a fourth light neutrino	N_ν = 2.9963 ± 0.0074; N_eff = 2.86 ± 0.13 (a reading)
the Standard Model's structure: charges,	inputs or theorems	the 16-state register	a fourth colour or a fifth slot; a	exact reproduction (a

Statement	Standard physics	The calculator	What would show it wrong	Status, October 2026
anomalies, √(5/3), running counts, pairwise crossing			change in the running	re-reading with a reason)

What only the calculator says. The ten tables headed “on one yardstick” (Tables 27, 29, 30, 31, 44, 47, 53, 54, 57 and 61) set the standard answer beside the calculator's on every layer. Everything in them that is not listed here is the standard result reached by a geometric road. The calculator's own statements are:

- a reason for the Koide relation, and a value for its phase;
- the neutrino masses from the same equation;
- a value for the mixing tilt, and the reason the α axis survives;
- the top and the Higgs from the edge and one load;
- the three-phase start as the reason the forces cross in pairs;
- the anomaly conditions as a code's parity checks;
- $w = -1$, with the feed from a held store;
- a size for the hum and a number for the share;
- the gluon share;
- one picture in which the black hole's inversion, the torus, the held foundation and the pull from within are one object.

Of these, section 16 scores only the ones with a number. Table 80 holds the frozen ones.

15.3 Published tensions, cell by cell

Thirty-six published tensions and tests were placed in the calculator's cells. Scored for the calculator:

- **14 agree;**
- **5 expose it**, four of them through cosmology;
- **15 land in named cells where it commits to no number:** H_0, S_8, the matter dipole, R(D), wide binaries and the Higgs self-coupling among them;
- **2 have no cell:** X17 and the lithium abundance.

The same 36, scored for the standard theories: 13 agree, 15 are tensions of 2σ or more, 4 are inputs and 4 are open or experimental. The list was assembled from published tensions, so it is weighted against the standard theories by construction. It shows where each theory is exposed, not its overall record, which section 16.8 scores.

The exposures cluster:

- **Shared.** Four of the calculator's five exposures are tensions of ΛCDM in the same data: the floor neutrino mass, $w = -1$, the curvature and the expansion rate. Only $\sin^2\theta_{12}$ is the calculator's alone.
- **One discrepancy, several faces.** Flat curvature, $w = -1$ and the floor neutrino mass are one 3σ discrepancy with three faces. Elbers et al. (2025) find the ΛCDM effective neutrino mass 3.0σ below the normal-ordering floor, a tension that eases only if w evolves.

- **One edge.** All of them sit on one edge, the matter share. DESI's BAO gives Ω□ = 0.2975, the supernovae with the CMB give 0.317, and π²/32 = 0.3084 lies between them.
- **Recalibration.** DES's recalibrated supernovae lower the preference for evolving dark energy from 4.2σ to 3.2σ (Popovic et al. 2025).

A rate cell gives almost any rate a side, so the 15 placements without numbers are bookkeeping, not evidence.

Across the ten circles of the layers, 33 of the 40 positions hold a published measurement (section 4.10.2). The other seven are named empty cells, each with the measurement that would fill it: for example, an Aharonov–Bohm measurement with positrons, the antitop's width, and the cosmic jerk.

Table 64 counts the results by layer, for the calculator and for the standard theories.

Table 64. Results by layer, for both.

Layer	The calculator: agree	expose	cell, no number	no cell	The standard theories: agree	tension	input	open
fabric and weak	1	0	4	1	1	3	0	2
leptons	4	2	1	0	2	3	2	0
Higgs vacuum	2	0	2	0	3	0	1	0
strong scale	2	0	1	0	2	0	0	1
cosmos	1	3	4	1	0	8	1	0
gravity	4	0	2	0	5	0	0	1
the cube's own numbers	0	0	1	0	0	1	0	0
total (36)	**14**	**5**	**15**	**2**	**13**	**15**	**4**	**4**

15.4 The author's physical descriptions against observation

The author described eighteen physical behaviours in words before any of them was compared with data. This section scores each one. The derivations behind several items are in sections 5 to 14, and are not repeated here.

15.4.1 How each was tested

- **Predictions first.** Each description was turned into something it predicts we should see, stated before looking.
- **What counts.** Only measurements count: published observations, instrumented recordings and the paper's own runs on public collision data. Simulations and eyewitness reports are marked as such.
- **The verdicts.**
    - Confirmed: the prediction is observed.
    - Denied: an observation contradicts it.
    - Open: the effect is observed, but cannot yet tell the author's mechanism from the standard one.
- **Two readings.** Where a description has two readings, both are scored.

15.4.2 The scorecard

Table 65 scores each of the eighteen descriptions against the data, with the standard account of the same observation beside it.

Table 65. Eighteen descriptions against the data.

#	Description	If true, we should see	What is observed	Standard account	Verdict
1	Lightning radiates in all directions; on our legs one path looks zigzag	growth spreading in many directions, one zigzag path connecting, the scale set by the air	discharges spread from their source in all directions; one branch connects; the size follows the air's density, which the gravity well sets	stepped leaders branch in the storm's field (Dwyer and Uman 2014)	confirmed
2	Lightning picks one path, like a photon	many tries, one connection	many branches and upward leaders, only one connects	the first upward leader to meet a branch fixes the strike (Dwyer and Uman 2014)	confirmed
3	The three-phase diagonal oscillates both	triplen currents pile up in the neutral;	the neutral reaches √3 = 173% of the phase current	circuit theory: triplen	confirmed

#	Description	If true, we should see	What is observed	Standard account	Verdict
	ways; action at √2, the hold at √4	single-phase power pulses at 2f	for non-overlapping pulses; lamps and the 2012 ball lightning pulse at 2f	harmonics add in the neutral	
4	2D ⟶ 3D: the missing leg is filled by phases (the Higgs)	massive W and Z carry a third polarisation; the photon does not	longitudinal W and Z seen, up to 7.1σ; photon mass below 10^{-18} eV; the Higgs couples at the predicted strength	the Higgs mechanism: the Goldstone modes become the third polarisation (Englert and Brout 1964; Higgs 1964)	confirmed
5	Gravity is a perpendicular pull; energy moving in all directions is pulled differently	clocks slow lower down; light bends more than matter; vacuum energy pushes	all three measured	general relativity	confirmed
6	Ball lightning is plasma held on the √4 side; electricity is invisible until plasma	stored energy, 2f pulsing, glow only once ionised	soil elements; 99.4 Hz beside a 50 Hz line; under 1 W available from the line; a dark discharge below breakdown	no accepted theory; stored-energy and microwave models (Abrahamson and Dinniss 2000; Wu 2016)	confirmed
7	The fabric is dynamic	spacetime can ripple	gravitational waves detected	general relativity's gravitational waves	confirmed

#	Description	If true, we should see	What is observed	Standard account	Verdict
8	Our perspective carries the lensing distortion	what we see depends on our motion	our motion distorts the microwave sky; Terrell rotation photographed; muon clocks stretched	special relativity: aberration and Terrell rotation (Terrell 1959)	confirmed in part
9	T-bar flip; negative is not reverse	the flip happens; antimatter does not fall up	the flip shown in orbit; antihydrogen falls down	rigid-body dynamics, with the middle axis unstable; CPT and the equivalence principle for antimatter	confirmed
10	Balance: slightly past 45°, held	the Higgs vacuum just past the stability edge and long-lived; a top above 170.94 GeV and a Higgs below 128.55 GeV	λ dips to -0.011; lifetime about 10^{139} years; the direct top (172.60) and the Higgs (125.13) both on the predicted side	the Standard Model's running: metastable, with the distance past the edge not predicted	confirmed (the size is the δ_3 lead of section 9.4)
11	Gravity wells are perpendicular, so air circulates perpendicular, so lightning carries what the well holds	air thinning at a rate set by g; storms charging in vertical circulation; the discharge scale following the well	gravity-built air gives a 19 m minimum streamer at 80 km, the size of real sprites; flash rate tracks updraft volume ($r = 0.93$); lightning doubles over shipping lanes; different planets,	hydrostatics and convective charging (Saunders 2008)	confirmed

#	Description	If true, we should see	What is observed	Standard account	Verdict
			different lightning		
12	The fabric is massive filament pathways; the observable universe is in a path	matter in filaments around voids; we sit in a flow; at the largest scales, a preferred direction	the cosmic web mapped and imaged; the Milky Way flows inside Laniakea; the quasar dipole twice too big (4.9σ); the microwave sky even to 1 part in 100,000	the cosmic web from gravitational instability (Bond, Kofman and Pogosyan 1996); the dipole excess unexplained	confirmed in part
13	No dark matter: what falls in is held on the √4 torus; the pull appears from within	no separate particle; extra pull inside the matter; torus flows at black holes; the pull staying with the black holes	no particle found; the extra pull fixed by the visible matter at every radius in 153 galaxies; M87's inflow ring joins its jet; the Bullet Cluster's pull stays with the galaxies	a cold particle, unseen, or modified gravity	open (fits four observations; the microwave background is the hard test)
14	A mass can pull next to the perpendicular and spin	spinning masses drag space sideways; tilted discs wobble	Earth's spin drags orbits and gyroscopes; a white dwarf drags a pulsar's orbit; M87's tilted disc wobbles every 11 years	frame dragging (Lense and Thirring 1918)	confirmed
15	The fabric is a capacitive–ind	board constants	$c = 1/\sqrt{(\mu_0 \varepsilon_0)}$; 376.73 Ω;	the quantum vacuum, with	confirmed (the hum's size and

#	Description	If true, we should see	What is observed	Standard account	Verdict
	uctive board; particles are reactions; base ohms	setting light's speed and a base impedance; reactive zones around sources; masses mostly reaction	near-field impedances inverse around 376.73 Ω; the proton 99% field energy; g−2, Casimir, Lamb shift	ε_0, μ_0 and Z_0 as its constants	the base-ohm value are open)
16	The board is semiconductive before action	no current below a gap; pairs above it; recombination glow; an exciton; breakdown under an extreme field	pairs from photons above 1.022 MeV; the 511 keV line; positronium; pairs from laser light; pair freeze-out in the early universe	Dirac's sea and pair production (Dirac 1930)	confirmed
17	Weight in y is torus gravity; weight in x is the spin's cling	a gyroscope near a mass precesses two ways; one m in weight and in inertia	Gravity Probe B, 6,601.8 ± 18.3 and 37.2 ± 7.2 mas/yr (computed 6,604.8 and 39.1); MICROSCOPE, weight and inertia equal to 10^{-15}	general relativity's two precessions; the equivalence principle	confirmed
18	The roots √1 to √4 with the ½, ⅓, ¼ grids; the ⅓ grid on the ¼ grid	charge in thirds; two-state pairs in halves; one generation on the tesseract with every	charges 0, ⅓, ⅔, 1; isospin ±½; the 16 states anomaly-free, with inside-out = antimatter; quark spin	the Standard Model's representations, as inputs	confirmed (the spin third is a weak test on its own)

#	Description	If true, we should see	What is observed	Standard account	Verdict
		balance sum zero	share 0.330 ± 0.039		

The tally. Fifteen descriptions are confirmed, two are confirmed in part, and one is open. None, as stated, is contradicted. For every confirmed item the standard account describes the same observation, so the verdicts weigh the author's descriptions against nature, not one theory against another. They are not evidence that the calculator's numbers are more than numerology; that question belongs to section 16.

15.4.3 The evidence, item by item

1. Lightning radiates in all directions, and we see one zigzag. Where the field is even, a discharge spreads from its source in every direction and branches as it goes: Lichtenberg figures and laboratory streamer bursts. In a storm the field points mostly down, so leaders spread down and sideways.

- **The growth model.** Grow a discharge on a grid of legs by the dielectric-breakdown rule, each new piece where the field is strongest with probability ∝ E^η (Niemeyer, Pietronero and Wiesmann 1984). For η = 1 the pattern has fractal dimension 1.66 in the paper's runs, against the published 1.7.
- **The connecting path.** Each centre-to-tip path is 1.51 times its straight distance, 41% of its steps change direction, and the connecting branch points every way round the circle over 16 runs.
- **Each step is an event in the air.** 13 of 14 recorded leader steps fired an X-ray burst (Dwyer 2005), with steps every 10 to 50 μs.
- **No fixed grid.** Stereo cameras agree on one three-dimensional zigzag (Willett, Le Vine and Idone 2008). Real channels turn 17.3° per segment on average (Hill 1968), not 90°, so the legs cannot be a fixed grid in space; the folding of the turns is in sections 4.5 and 4.10.3.

2. Lightning picks one path.

- **One contact.** The downward leader branches and several upward leaders rise from the ground, but only one upward leader meets a branch. That contact fixes the strike point, and the return stroke follows it (Dwyer and Uman 2014).
- **The video record.** High-speed video of one flash caught seven upward leaders that never connected beside the one that did (Tran and Rakov 2017).
- **Not a photon.** The difference from a photon is that the winner is whichever leader reaches the other first, not a quantum measurement.

3. The three-phase diagonal.

- **Third harmonics.** The three third harmonics of a balanced set are in phase, pure (1, 1, 1), invisible to the Clarke view, and they add in the neutral (section 3.7.1).
- **The √3 ceiling.** For current pulses that do not overlap between phases, as rectifier loads draw, the cross terms vanish. Then $I_{N,rms}^2 = 3\, I_{p,rms}^2$, so the neutral carries √3 = 173% of the phase current (section 14.4).

- **The 2f pulse.** Single-phase power pulses at twice the line frequency. Lamps flicker at 100 Hz on 50 Hz grids and at 120 Hz on 60 Hz grids (Wilkins, Veitch and Lehman 2010; Pacific Northwest National Laboratory 2016), and the 2012 ball lightning pulsed at 99.4 Hz beside a 50 Hz line (section 11.3).
- **The reading.** "Absorbs at √2, holds at √4" is the author's reading of these exact ratios. The ratios themselves are measured.

4. The missing leg is filled by phases.

- **W and Z have the third leg.** Massive W and Z bosons carry the third, longitudinal, polarisation.
    - In top-quark decays its share is 0.684 ± 0.015, against 0.687 ± 0.005 predicted (ATLAS Collaboration 2023, W helicity).
    - Pairs with both legs filled are seen in WZ at 7.1σ (ATLAS Collaboration 2023, WZ polarisation) and in ZZ at 4.3σ (ATLAS Collaboration 2023, ZZ).
    - Same-sign WW pairs with at least one leg filled are seen at 3.3σ (ATLAS Collaboration 2025).
- **The photon stays two-legged.** Its mass is below 10^{-18} eV (PDG).
- **The Higgs fills it at the predicted strength.** Its coupling to W and Z is κ_V = 1.035 ± 0.031 (ATLAS Collaboration 2022). The Goldstone equivalence behind this is in section 3.7.1.

5. Gravity as the perpendicular pull.

- **Gravity acts through time.**
    - A clock raised 33 cm runs faster by $(4.1 \pm 1.6) \times 10^{-17}$ (Chou et al. 2010).
    - Light climbing a 22.5 m tower shifts as predicted, with a ratio of 1.05 ± 0.10 (Pound and Rebka 1960).
- **Light is pulled differently.** Light bends twice the Newtonian amount:
    - γ = 0.99983 ± 0.00045 from radio deflection (Shapiro et al. 2004);
    - $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ from the Cassini signal delay (Bertotti, Iess and Tortora 2003).
- **Vacuum energy pushes.** Its pressure-to-energy ratio is $w = -1.03 \pm 0.03$ (Planck Collaboration 2020), so it accelerates the expansion. DESI's second data release prefers an evolving w at 2.8 to 4.2σ (DESI Collaboration 2025).
- **Horizons exist.** The Event Horizon Telescope imaged M87*'s ring of 42 ± 3 μas, consistent with a spinning black hole, and Sgr A*'s ring within about 10% of prediction (Event Horizon Telescope Collaboration 2019, 2022).
- **The literal reading is denied.** Ordinary laboratory masses, with no black hole involved, attract by the inverse-square law down to 52 μm (Lee et al. 2020).

6. Ball lightning held on the √4 side. The one instrumented ball, its 2f pulse and the power bound are in section 11.3.

- **Laboratory balls.** Balls of vaporised silicon last up to 8 s on stored energy (Paiva et al. 2007). Microwave fireballs last only while the power is on (Dikhtyar and Jerby 2006).
- **Invisible until plasma.** Below breakdown, current in air is a dark discharge with no visible light. Light appears once electrons gain enough energy to excite atoms (Treumann, Kłos and Parrot 2008).

7. The fabric is dynamic. Ripples in spacetime were detected directly: GW150914, with peak strain 1.0×10^{-21}, above 5.1σ (LIGO Scientific and Virgo Collaborations 2016).

8. Our perspective carries the distortion.

- **Our motion.** Our motion, about 370 to 384 km/s, distorts the microwave sky. The aberration is seen at 4 to 5σ in each of four methods (Planck Collaboration 2014).
- **Terrell rotation.** The relativistic apparent rotation predicted by Terrell and Penrose was photographed in 2025 (Hornof et al. 2025).
- **Muon clocks.** Fast muons live 29.33 times longer, as relativity predicts, to 0.2% (Bailey et al. 1977).

9. The T-bar flip; negative is not reverse.

- **The flip.** The T-handle flip was shown in orbit in 1985; there is a historical report but no primary paper (Trivailo and Kojima 2017). Montgomery's formula describes it (Montgomery 1991). The flip time in sums and differences of squares is in section 4.4.1, and a phone's gyroscope record is in section 4.7.4.
- **Antimatter falls.** Antimatter, the "negative" of matter, still falls down: a_g = (0.75 ± 0.13 ± 0.16) g, and repulsive antigravity is ruled out (Anderson et al. 2023).

10. Balance: just past the edge, held. The full test, with the friction angles, is in section 9.4.

11. Lightning carries what the gravity well holds.

- **The well builds the air.** Each layer of air holds up the weight above it, $dp/dz = -\rho g$, so pressure falls on a scale $H = RT/(Mg) = 8.43$ km at the ground. Integrated with the standard atmosphere's temperatures, this gives 1.05 Pa at 80 km, about 1/96,000 of ground pressure; the U.S. Standard Atmosphere (1976) lists 1.052 Pa.
- **The air sets the scale.** A discharge depends on the field per molecule E/N and the number of molecules crossed, N·d (Townsend similarity). So when the density falls by a factor F, every length stretches by F, and streamer width times pressure stays at 0.20 ± 0.02 mm·bar (Briels et al. 2008; Ebert et al. 2010). At 80 km the air is about 10^5 times thinner, so a 0.2 mm streamer at the ground is 19 m wide, the width of real sprites.
- **A different well.** With the same temperatures, gravity 10% weaker would shrink the minimum sprite streamer to 6 m, and 10% stronger would grow it to 60 m. A local gravity anomaly of 1 part in 10^4 changes the pressure at 80 km by only 0.1%, so it is the planet's mass that counts.
- **The circulation is perpendicular.** Storms charge inside buoyant updrafts while heavier graupel falls through them, and the collisions sort charge into vertical layers (Saunders 2008). The flash rate tracks the volume of updraft faster than 5 to 10 m/s in the below-freezing zone, r = 0.93 (Deierling and Petersen 2008).
- **What the well holds matters.**
    - Ship-exhaust particles double the lightning over the busiest shipping lanes (Thornton et al. 2017).
    - Jupiter, at 2.5 times Earth's gravity with hydrogen air and water and ammonia clouds, has lightning, including shallow lightning high in ammonia–water clouds (Brown et al. 2018; Becker et al. 2020).

    - Mars, at 0.38 g with thin carbon dioxide and no rain clouds, gives only small sparks inside dust devils (Chide et al. 2025).

12. The fabric as massive filament pathways.

- **The pathways exist.** Galaxies lie along filaments and walls around empty voids, first mapped as the CfA "Great Wall" (Geller and Huchra 1989) and named the cosmic web (Bond, Kofman and Pogosyan 1996).
    - Lensing shows that a filament between the clusters Abell 222 and 223 pulls far harder than its visible matter can (Dietrich et al. 2012).
    - The glowing gas of a filament joining two quasars, seen as it was about 11.5 billion years ago, was imaged directly (Tornotti et al. 2025).
- **We sit in a flow.** The Milky Way lies inside Laniakea, a basin about 520 million light-years across defined by galaxy motions, in which galaxies flow inward toward the Great Attractor region (Tully et al. 2014).
- **The whole observable universe in one path is open.**
    - For: across 1.36 million quasars, the pile-up in our direction of motion is over twice what our motion explains, at 4.9σ (Secrest et al. 2021). The Giant Arc (3.3 billion light-years) and the Big Ring (1.3 billion) exceed the expected size limit of about 1.2 billion light-years (Lopez et al. 2022, 2024).
    - Against: once our motion is removed, the microwave background is the same in every direction to about 1 part in 100,000.
    - The observable universe is about 93 billion light-years across, 28 times the Giant Arc. A path wide enough to hold it would reach beyond what we can see, so only a direction could show.
- **The same shape as lightning.** Both are branching networks in which the flow keeps using paths already made.

13. No dark matter: the pull from within. The torus geometry, the Hopf picture and the share rule are in sections 4.6, 13.3 and 13.4. The observations:

- **No separate particle.** The most sensitive search found nothing, with a limit of 2.2×10^{-48} cm² at 40 GeV (LZ Collaboration 2025). In the author's picture there is nothing separate to find.
- **The extra pull sits inside the matter.** Across 2,693 points in 153 galaxies, the full gravity at every radius is fixed by the visible matter there (McGaugh, Lelli and Schombert 2016):
    - with one acceleration scale, 1.20×10^{-10} m/s²;
    - with a scatter of 0.13 dex, mostly measurement error.
- **Torus flows at black holes.** At 86 GHz, M87's black hole shows a 64 μas ring of inflowing matter that joins straight onto the base of its jet (Lu et al. 2023): in around the middle, out along the poles.
- **The pull stays with the black holes.** In the Bullet Cluster collision the lensing pull stayed with the galaxies, where the black holes are, and left behind the hot gas that holds most of the ordinary mass (Clowe et al. 2006).
- **The hard test.** The microwave background, 380,000 years after the start, already needs about 5.36 ± 0.06 times as much extra pull as ordinary matter, from something that does not oscillate

with the glowing gas (Planck Collaboration 2020). The author's picture must have the held flow in place from the start, and the share held at √4 must be stated before it is compared with 5.36.

14. A pull next to the perpendicular, with spin.

- **Measured around Earth.** Gravity Probe B's gyroscopes were dragged by Earth's spin at 37.2 ± 7.2 mas per year, against 39.2 predicted (Everitt et al. 2011). The LAGEOS orbits are dragged at 99% of the prediction, ±10% (Ciufolini and Pavlis 2004).
- **Measured in a star pair.** A white dwarf's spin drags the orbit of its pulsar partner, PSR J1141−6545 (Venkatraman Krishnan et al. 2020).
- **The disc on edge, at a black hole.** M87's jet swings round every 11 years, seen across 22 years of radio data: a tilted disc around the spinning black hole is dragged into a wobble (Cui et al. 2023).
- **The size.** In standard gravity this sideways pull is about $(v/c)^2$ of the ordinary pull, 5.4×10^{-7} for a galaxy's 220 km/s. Two analyses of 2023 found it about a million times too small to explain galaxy rotation (arXiv:2303.06115; arXiv:2303.16679). So spin adds the sideways, looping part, and the extra pull must still come from within.

15. The fabric as a capacitive–inductive board. The board's constants, base ohms and the near-field impedances are in section 5.1 and Table 23. The virtual-particle rows are in Table 25. The further checks:

- **Light on the board.** Light carries equal electric and magnetic energy, and E/H is the base ohms.
- **The T-bar flip on the fields.** $(E, cB) \rightarrow (cB, -E)$ swaps electric and magnetic energy, and twice gives −1. The empty-space Maxwell equations are unchanged by it: duality (Heaviside; Larmor).
- **A proton is a reaction.** The quark masses are 9.0 of 938.3 MeV, 0.96%. The lattice gets the nucleon at 936 ± 34 MeV from the field alone (Dürr et al. 2008).
- **Mass as an added reaction.** $\omega^2 = \omega_c^2 + c^2k^2$ gives a mass $\hbar\omega_c/c^2$. Radio reflects off the ionosphere below its plasma cutoff, waveguides have a cutoff, and the plasmon led to the Higgs mechanism (Anderson 1963).
- **No wind.** A static cloud everywhere must look the same to every steady observer. It does, to $\Delta c/c \approx 10^{-17}$ (Herrmann et al. 2009), which the board meets only if ε_0 and μ_0 are the same in every frame.
- **The hum.** A hum of spacetime is seen in pulsar timing at 3.5 to 4σ (NANOGrav Collaboration 2023). A summed board hum would be 10^{123} times too large, which is the problem section 13.1 addresses.
- **History.** Maxwell's vortex medium (1861); Michelson and Morley (1887); Einstein (1920), space with physical properties but no state of motion; Casimir (1948); Dirac (1951), “Is there an aether?”; Anderson (1963); Sakharov (1967), gravity as the vacuum's elasticity; Rañada (1989), light with linked field lines; Wilczek (2008), “the Grid”.

16. Base ohms, semiconductive before action. In the difference table the gap is the space between the two branches of $E^2 = (pc)^2 + (mc^2)^2$, $2mc^2$ wide. Inside it the momentum would be imaginary, so a reaction is held and the board only polarises. Table 66 reads the board as a semiconductor.

Table 66. The board as a semiconductor.

Semiconductor feature	The board	Number	Observed
band gap	$2mc^2$ for each kind of reaction	electron 1.0220 MeV (9.1×10^5 times silicon's 1.12 eV); muon 211.32 MeV; proton 1,876.54 MeV	pairs from photons above 1.022 MeV; the first positrons (Anderson 1933)
below the gap, polarisation only	vacuum polarisation	−27 MHz of hydrogen's 1,057.8 MHz Lamb shift (Uehling 1935)	measured
recombination glow	a pair falls back across the gap as two photons	2×510.999 keV	the Galaxy's 511 keV line; PET scanners
exciton	positronium	binding 6.8028 eV; 1S–2S at the hydrogen-like level, 1,233.691 THz	1,233.607 THz (Fee et al. 1993): the sub-reactions close the last 0.0068%
breakdown, as Zener breakdown in a diode	rate $\propto \exp(-\pi E_c/E)$	$E_c = 1.323 \times 10^{18}$ V/m; the strongest laser reaches 6.9×10^{-4} of it, a rate $\propto 10^{-1983}$	no spontaneous breakdown; pairs from light plus high-energy photons (Burke et al. 1997)
intrinsic carriers	pairs fill the board when $kT \approx mc^2$	5.93×10^9 K; freeze-out leaves neutrinos cooler by (4/11)^⅓ = 0.71377, 1.945 K	$N_{eff} = 2.99 \pm 0.17$ against 3.044 predicted (Planck Collaboration 2020)
today	a perfect insulator to steady current	the pair factor at 2.7 K is about $e^{(-2.2 \times 10^9)}$	the board carries waves at base ohms and nothing else

The laser figure uses the record intensity of 1.1×10^{23} W/cm² (Yoon et al. 2021). The history is Dirac (1930), Sauter (1931), Zener (1934), Heisenberg and Euler (1936) and Schwinger (1951): standard physics in the author's language, not a new number.

17. Gravity on a gyroscope. The geodetic and frame-dragging derivations are in section 11.1. The further rows:

- **Perpendicular response.** A pull in y turns a spinning axis in x: precession at 90°, the i-turn, not falling.
- **A static cling in constant motion.** The same m appears in weight and in F = ma, and inertia resists a change of speed as an inductor resists a change of current. MICROSCOPE finds weight and inertia equal to $(-1.5 \pm 2.7) \times 10^{-15}$ (Touboul et al. 2022).

- **Cling from the whole universe.** Sciama's inertia by induction needs $GM/(Rc^2) \approx 1$ (Sciama 1953), and a flat universe gives $\Omega/2 = 0.50$. Local gyroscopes do not turn against the distant universe, to below 4.7×10^{-11} of the expansion rate (Saadeh et al. 2016).
- **Held past the tipping point.** A sleeping top stays up while its spin exceeds $2\sqrt{(mgI_{\perp})}/I_3$. For a 100 g toy top with its centre of mass 3 cm up, $I_{\perp} = 4.0 \times 10^{-5}$ and $I_3 = 2.0 \times 10^{-5}$ kg m², that is 108 rad/s, 1,036 rpm. Tops sleep, then wobble and fall as friction slows them.
- **History.** Foucault (1852); Lense and Thirring (1918); Schiff (1960); Sciama (1953); Everitt et al. (2011).

18. The roots and the grids. The ladder, the moves as roots of 1, the 16-state generation and the hypercharge root are in sections 4, 7.1 and 8.1. The observations:

- **Charge in thirds.** Charges 0, ⅓, ⅔ and 1 sit on the cube's corners (Harari 1979; Shupe 1979). Quark charges of ⅓ and ⅔ are seen in deep inelastic scattering, Drell–Yan and the R ratio.
- **Pairs in halves.** Weak isospin and spin are ±½, in the doublets (u, d) and (ν, e).
- **One generation balances.** All the balance sums vanish over the 16 states (ΣY, ΣY^3, ΣQ, $SU(2)^2Y$, $SU(3)^2Y$), which is the anomaly cancellation the theory needs, and inside-out sends all 16 to $Q \to -Q$ (Georgi 1975; Fritzsch and Minkowski 1975).
- **The spin third.** Each leg carries ⅓ of the diagonal, and the quark spin share is $\Delta\Sigma = 0.330 \pm 0.039$ (HERMES Collaboration 2007), -0.09σ. It is consistent but not selective: 79 members of the alphabet sit within 2σ of it (section 16.7.4).
- **The ¼ grid's fourth slot.** Reading the lepton as the fourth colour (Pati and Salam 1974) is a reading, not a test.
- **The weak phase is one-handed** (Wu et al. 1957). This is where inside-out (-1) and the 120° turn (ω) differ: ω is a rotation, while -1 in 3D is a mirror.

15.4.4 What it means, and what would move the open items

What it means.

- **Motion.** The author's descriptions of motion hold up. Phases, legs and hypotenuse, rotation, time, gravity and the flip all describe what nature does. Several observed numbers land on the root ladder:
    - the neutral's ceiling is $\sqrt{3}$;
    - single-phase power and the ball pulse at 2f;
    - light bends twice as much as matter.
- **Gravity, air and lightning.** The chain holds end to end: gravity builds the air's pressure profile, the pressure sets the discharge's size, and the result matches real sprites.
- **Filaments.** The filament pathways fit, from lightning to the cosmic web. Whether the whole observable universe sits in one path is open.
- **No dark matter.** It fits four observations. The microwave background is the hard test.
- **Balance.** Balance fits the Higgs vacuum; its size is the δ_3 lead.

Tests that would move the open items.

- **Balance (10).** The e^+e^- top threshold scan, about ±0.035 GeV. A pole mass above 170.94 GeV keeps the vacuum just past the edge; one at or below it breaks the rule.

- **The whole-observable path (12).** The matter dipole from Euclid, Rubin and the SKA. If the excess holds and lines up with the giant structures, sitting inside a larger flow becomes a live explanation; if it shrinks to what our motion explains, the idea fades.
- **The extra gravity (13).** State the share held at √4 against the share acting on √2, and compare it with 5.36. Then check the microwave background's peaks. A second check: galaxies whose black holes have passed more matter should carry more pull from within at the same visible mass.

Standing.

- **M.** Every measurement and derivation here is standard, with its source given.
- **N.** The descriptions are the author's, as are the readings that join them.
- **D.** The no-dark-matter picture diverges from the particle reading of the extra pull. It is scored open, not confirmed.

15.5 Borrowed observations

Papers written for other purposes were read whole, not by their abstracts, for measurements that land on the calculator's numbers. Three add detail not given elsewhere.

The proton's spin, piece by piece. The continuum-limit lattice decomposition of Alexandrou et al. (2026) used four ensembles with physical up, down, strange and charm masses at four lattice spacings, with all renormalisation, including the quark-singlet–gluon mixing, done non-perturbatively. Their Table II (MS-bar, 2 GeV) gives:

- ½ΔΣ = 0.190(22), J_g = 0.209(26)(66) and L_q = 0.108(37)(24);
- the sums J_N = 0.507(43)(65) and ⟨x⟩_N = 0.995(60)(29), so both sum rules close.

Per flavour, the up quark carries +0.417 of spin and −0.177 of orbital, and the down quark −0.203 of spin and +0.228 of orbital. These cancel in J_d, so the proton's spin is mostly up-quark spin plus gluon. The same group's earlier single-ensemble result (Alexandrou et al. 2020: ½ΔΣ = 0.191(15), J_g = 0.187(46), L_q = 0.094(51)) is reproduced. ΔG = 0.231(17)(44) at 10 GeV² in the continuum limit is 46(9)% of the spin, and “clearly rules out a negative gluon helicity” (Zhao et al. 2025). Table 67 sets the thirds against the continuum lattice.

Table 67. The thirds against the continuum lattice.

Piece	The calculator	Lattice	Pull	Source
quark spin ½ΔΣ, 2 GeV	⅙	0.190 ± 0.022	+1.1σ	Alexandrou et al. 2026
gluon helicity ΔG, 2 GeV	⅙	0.158 ± 0.039	−0.2σ	Zhao et al. 2025, evolved (section 12.2)
gluon total J_g = ΔG + L_g, 2 GeV	⅙ if L_g = 0; ⅓ if the orbital sixth were all gluon	0.209 ± 0.071	+0.6σ; −1.8σ	Alexandrou et al. 2026

Piece	The calculator	Lattice	Pull	Source
quark orbital L_q, 2 GeV	at most ⅛	0.108 ± 0.044	−1.3σ against ⅛	Alexandrou et al. 2026
total orbital L = ½ − ½ΔΣ − ΔG, both folded to 2 GeV	⅛	0.152 ± 0.043	−0.4σ	section 12.2
ΔΣ, 2 GeV	⅓	0.380 ± 0.044	+1.1σ	Alexandrou et al. 2026; HERMES 0.330 ± 0.039, −0.1σ
gluon momentum ⟨x⟩_g, 2 GeV	⅓, not claimed	0.372 ± 0.035	+1.1σ	Alexandrou et al. 2026

Folded to one resolution, the orbital sixth is the closest of the three. As a zeroth-order split the thirds survive. As a prediction, the lattice central values say spin-heavy, and the three errors are not independent, since the orbital piece is what is left. The full paper also gives the gluon's share of the momentum as a third within errors, a thirds statement the calculator never made, sitting beside the one it did.

Table 68. The τ mass by method, against Koide's 1,776.969 MeV.

Measurement	Method	Value (MeV)	Pull
Belle II (2023)	pseudomass endpoint, 175 million pairs	1,777.09 ± 0.08 ± 0.11	+0.9σ
BESIII (2014)	threshold scan	1,776.91 ± 0.12 +0.10/−0.13	−0.4σ
KEDR (2023)	threshold scan	1,776.69 +0.17/−0.19 ± 0.15	−1.2σ
PDG 2026 average		1,776.93 ± 0.09	−0.4σ

The prediction sits between the two precise methods. A BESIII scan at the 2026 energy-calibration level, or Belle II's full sample, decides at ±0.05 MeV.

α from B → ρρ. Belle II measured on 387 million Υ(4S) decays (Belle II Collaboration 2025):

- $B(B^0 \to \rho^+ \rho^-) = (2.89\ ^{+0.23}_{-0.22}\ ^{+0.29}_{-0.27}) \times 10^{-5}$;
- the longitudinal fraction $f_L = 0.921\ ^{+0.024}_{-0.025}\ ^{+0.017}_{-0.015}$;
- $S = -0.26 \pm 0.19 \pm 0.08$ and $C = -0.02 \pm 0.12\ ^{+0.06}_{-0.05}$.

The isospin analysis of this result alone gives 91.5 +4.8/−5.2°. Combined with all other B → ρρ results it gives 92.6 +4.5/−4.7°, with the second solution, at 177.4°, excluded by the other angles and unitarity.

The ρρ channel has the smallest penguin contamination of the three, which makes it the cleanest reading of α. The all-channel average, 84.1 +3.7/−3.0° (PDG 2026), is the one that sits 1.6σ low.

What the borrowed observations change.

- **The sharpest new fact is cosmological, and it cuts.** DESI's neutrino analysis puts two of the calculator's commitments, w = −1 and the floor neutrino mass, on one scale and finds them 3σ from the data together. The same 3σ shows as the drift in w and the open curvature: one discrepancy with three faces (section 15.3).
- **The thirds gain a witness.** The continuum lattice gives all three sixths at once, each within 1.4σ once the orbital piece is folded to one resolution.
- **α = 90° gains its cleanest channel.** B → ρρ alone says 92.6°; only the all-channel average says 84°.

15.6 Against earlier theories

Table 69 sets each idea beside the earlier theory that held it, what observation decided and how the calculator reads it.

Table 69. Where each idea stands, beside the theories that held it first.

Idea	Earlier theory	What it said	What observation decided	The calculator
gravity's pull on light	Newton; Soldner (1801)	light bends 0.87″ at the Sun, half the later value (Soldner's own figure was 0.84″)	the 1919 eclipse, 1.98 ± 0.12″: Newton's half angle off by 9.2σ	time leg plus space leg on the 45° line: 1.75″, as Einstein (1915)
gravity as a push	Le Sage (1748)	a flux of particles pushes bodies together	rejected: it would heat and slow every body	a pull from within; no flux
gravity's magnetic side	Heaviside (1893); Lense and Thirring (1918)	a spinning mass drags space	Gravity Probe B, LAGEOS, a pulsar pair, M87's jet	the spin next to the perpendicular: the same effect
entrance and exit	Einstein and Rosen (1935); Kruskal (1960); Fuller and Wheeler (1962)	a bridge joins two sides; the full solution has a white hole; the bridge pinches off	horizons seen; no white hole seen	the bridge as the inversion, with the horizon as the fixed sphere; no exit
flux through two mouths	Misner and Wheeler (1957), “charge without charge”	field lines enter one mouth and leave the other	not observed	the same torus picture

Idea	Earlier theory	What it said	What observation decided	The calculator
a hidden circle at every point	Kaluza (1921); Klein (1926)	an extra dimension curled into a small circle	no extra particles at the LHC; gravity normal down to 52 µm	the perpendicular circle as the held side, not a new direction of space
circles woven into the 3-sphere	Hopf (1931); Penrose's twistors (1967)	the 3-sphere is made of linked circles; light rays twist the same way	geometry; twistors are used in modern amplitude calculations	the torus on √4 and the Hopf fibration of section 4.6
the extra pull	Zwicky (1933); Rubin and Ford (1970s); dark-matter particles (1980s on)	a hidden substance	no particle found in decades of searches	held mass, cold and clustering, its make-up open
	MOND, Milgrom (1983)	gravity changes at low acceleration	the pull follows visible matter (yes); the Bullet Cluster and the CMB (no)	a pull from within like MOND, but carried by mass that moves with the galaxies
	white-hole dark matter, Rovelli and Vidotto (2018)	long-lived white-hole remnants	untested	the closest published relative of the turned centre
the universe's shape	Friedmann (1922); Lemaître (1927); inflation, Guth (1981)	open, closed or flat; inflation predicts flat	$\Omega_k = 0.0007 \pm 0.0019$	flat at the balance line, by a different route
the matter share	standard cosmology (1990s on)	a free number, fitted	0.3153 ± 0.0073 (Planck)	$\pi^2/32 = 0.3084$, a number, not selective (section 16)
static against sliding	Amontons (1699); Coulomb (1785)	static friction exceeds sliding friction	universal; stick-slip faults; balanced rocks outlast quakes	balance: slightly past 45° to start the turn

Idea	Earlier theory	What it said	What observation decided	The calculator
nature at the edge	Froggatt and Nielsen (1996); Shaposhnikov and Wetterich (2010)	exactly on the edge: top 173 ± 5 GeV and Higgs 135 ± 9 GeV in 1995; Higgs 126 GeV in 2009	the vacuum sits slightly past the edge, metastable (Buttazzo et al. 2013)	slightly past, held, by δ_3 (section 9.4)
gravitational waves	Einstein (1916, 1918)	twice the orbital frequency, two modes, light speed	binary pulsar 0.9983 ± 0.0016; GW170817; tensor modes favoured	the squared field gives 2f and modes 45° apart: the same answer
lightning	Franklin (1752); Lichtenberg (1777); Schonland (1934); Niemeyer, Pietronero and Wiesmann (1984)	electric; branching; stepped leaders; growth driven by the field	all confirmed	radiating in all directions with one path connecting (section 15.4.3)
ball lightning	Kapitsa (1955); Abrahamson and Dinniss (2000)	fed from outside by microwaves, or stored energy in vaporised soil	the 2012 ball: soil elements, under 1 W available from the line (section 11.3)	held, pulsing at 2f, radiating
antimatter	Dirac (1928–31); Morrison (1958)	negative-energy states; antigravity ruled out by energy conservation	antihydrogen falls (Anderson et al. 2023)	negative is not reverse
the flip	Euler (1758); Poinsot (1834); Dzhanibekov in orbit, 1985 (Trivailo and Kojima 2017); Montgomery (1991)	the middle axis of a spinning body is unstable and flips	seen in orbit and in a phone (section 4.7.4)	the T-bar flip, its rate in sums and differences of squares (section 4.4.1)

Standing.

- **M.** Every derivation in sections 5 to 14 is standard, with its source given.

- **N.** The readings that join them are the author's, as each subsection marks.

16. Tests for chance, applied to both theories

Is the calculator how reality works? Not on present evidence. Decided on 10 October 2026 with five tests whose rules were fixed before they were run, the calculator:

- **is exact** where it restates established law;
- **is consistent but not discriminating** where it reads physics;
- **cannot be told from numerology** where it supplies numbers.

Nine frozen predictions can change the third verdict; nothing available today can. The same tests, applied to the standard theories, find their predictions of independent measurements confirmed far beyond chance, 102 bits against the calculator's 0.6 to 4.0 (section 16.8). They also find six failures, among them the vacuum's energy and the Hubble constant, which under the same rule reject the standard theories' claim to be complete (section 16.9).

16.1 The question, made testable

“Is this how reality works?” splits into four claims:

- **A, the encoding.** The calculator's corner values and turns satisfy established physics exactly.
- **B, the readings.** Its physical readings describe nature beyond restating standard physics.
- **C, the numbers.** Its matches to measured constants show more than chance.
- **D, the predictions.** Its frozen numbers will be borne out. This one is testable only later.

The standard theories face the same four claims, stated for them in section 16.9.

The alphabet. The alphabet was fixed on 3 October 2026, before any test. It has 37 base numbers:

- the levels and grids: 1, √2, √3, 2, 1/√2, 1/√3, ½, ⅓, ¼, ⅔, ¾, 3/2, √3/√2 and √3/2;
- the shares: π/4, π/6 and π²/32, their complements 1 − π/4, 1 − π/6 and 1 − π²/32, and 4/π, 2π/3, 1/(2π) and 1/(8π²);
- the golden numbers φ, 1/φ and √5;
- 3/8 and 5/3;
- the counts 3, 4, 6, 8, 12, 16, 24 and 27.

With every product and quotient of two, merged at 10^{-9}, it holds 998 numbers from 0.00016 to 2,132.

The outside list. It is every dimensionless parameter of five standard sets, 29 in all:

- the electroweak masses over v, α_s and 1/α;
- the lepton and light-quark mass ratios;
- the CKM and PMNS parameters;
- the nucleon's ratios;
- Planck's ΛCDM numbers.

The values are those of CODATA 2022, NuFIT 6.0 and 6.1, Planck 2018 and PDG 2026.

The nulls and the scripts.

- **Hit.** A hit is an alphabet member within 2σ of a target.
- **Primary null.** Alphabets built by the same recipe, with π, √2, √3 and φ replaced by random numbers of similar size.
- **Scripts.** blindtest.py and decide.py, with their hashes in section 20.
- **Decision rule.** The author's Quad-Core test (section 2.3).

16.2 Test 1: the identities, and where the calculator is exact

Every identity holds:

- **The turns.** J is unique, $J^4 = 1$, det J = +1, and all 16 corners and 32 edges are kept. J carries the Front string onto the Back string. $K^2 = (JK)^2 = -1$, JK = −KJ and TJT = J^3.
- **The electron's corners.** $qv = \varphi i = m_e c^2$, $q\varphi = h$, $v/i = \varphi/q = h/e^2$, and $i \cdot \pi \lambda\!\!\!^{-2} = \mu_B$ to 30 digits.
- **The board.** $Z_0/(h/e^2) = 2\alpha$, $r_e a_0 = \lambda\!\!\!^{-2}$, $C_e = 4\pi\varepsilon_0 r_e$ and $L_e = \pi\mu_0 a_0$.

Verdict: the encoding is exact. That makes the calculator a correct notation for established relations: circuit laws, the Planck–Einstein relation, the flux and resistance quanta, the Standard Model's charges. It does not make it new physics, because each identity is a known law written on a cube.

Where the calculator is exact. The statements in Table 70 are identities or proved statements. They are what the calculator runs on, and the verdict above applies to them: they make it a correct notation for established relations, not new physics.

Table 70. The exact statements.

Exact statement	Check
27 points at levels √k, with counts 1, 6, 12 and 8, plus the held centre	$C(3, k) \cdot 2^k$; 1 + 6 + 12 + 8 = 27 = 3^3; counts (1, 3, 3, 1) by set bits
the three moves are the roots of 1, and generate all 48 operations	$(-1)^2 = \omega^3 = i^4 = 1$; 6 permutations × 8 sign patterns = 48 = O_h, by enumeration
inside-out is a rotation in 2D and 4D and a mirror in 3D	determinants $(-1)^n$: +1, −1, +1
level k flips onto the cube of half-side 1/k; the grids are the inverse levels squared	$s\sqrt{k} = 1/\sqrt{k}$ gives s = 1/k for k = 1, 2, 3, 4
the ladder $(\sqrt{k})^2 + 1^2 = (\sqrt{(k+1)})^2$, with projections at j/k and binomial counts	leg angles 45°, 54.74° and 60°; (1, 2, 1), (1, 3, 3, 1), (1, 4, 6, 4, 1)
the alternate-corner tetrahedron is ⅓ of the cube	8 − 4 × 4/3 = 8/3
$(a + bi)^2 = (a^2 - b^2) + 2ab \cdot i$, with modulus $a^2 + b^2$	all 55 cells of the 10 × 10 tables, and the 200 printed cells of the author's tables
$1 + \omega + \omega^2 = 0$; three-phase power is flat; line = √3 × phase; third harmonics are pure (1, 1, 1)	proved symbolically, 25 statements (section 3.7.1)

Exact statement	Check
the Clarke transform's kernel is the diagonal; the Scott-T teaser is √3/2	proved (section 3.7.1)
the 16 states, every balance sum zero; Tr Y² = 10/3, Tr T_3^2 = 2, √(5/3), sin²θ_W = 3/8	the table of section 8.1; the even-parity code of length 5, 16 words at distance 2
inside-out of the four free slots sends Q → −Q and keeps the generation	checked for all 16; the five-slot inside-out is charge conjugation, 16 → 16-bar; determinant +1 in 4D
the one-loop running counts 41/10, −19/6 and −7 from the 16-state table	computed with weights ⅔, ⅓ and −11/3
the couplings cannot meet at a point with the measured starts and these counts	4.05 ≠ 5.51
$8\pi^2 = 4\pi \times 2\pi = 16 \times \pi^2/2 = (2\pi)^4/2\pi^2$; $\alpha/2\pi = e^2/8\pi^2 = e^2/(4\pi\varepsilon_0 hc)$	to 12 digits
the orthonormal Clarke rows are the tribimaximal columns	the same three vectors
Q = (1 + a²/2)/3, so Q = ⅔ ⇔ a = √2 ⇔ 45° to the diagonal	algebra; with the TM1 sum rule $\sin^2\theta_{12} = (1 - 3s_{13}^2)/(3(1 - s_{13}^2))$
the τ from e and μ in closed form	$x = 2s \pm \sqrt{(6s^2 - 3t)}$ (section 9.1)
three inside-outs in a ring oscillate at 0°, 120° and 240°; two hold	the ring oscillator and the latch (section 7.3)
a held store has w = −1, and pushes	$\dot{\rho} = -3H(\rho + p)$ with $\dot{\rho} = 0$; ρ + 3p = −2ρ
the Clifford algebras of 3- and 4-space are graded (1, 3, 3, 1) and (1, 4, 6, 4, 1)	the cube's and the tesseract's level counts
the mean direction change of a staircase is (2/√π)/√s rad	the binomial variance and the Gaussian mean absolute difference
Montgomery's T-bar formula; the flip rate λ = Ω√(cos 2θ_xy·cos 2θ_yz)	matched to 10^{-4} in three runs, 0.4804 against 0.4805 (section 4.4.1)
the hydrogen lines as the inverse difference table; the subshells as difference cells	Rydberg; $2\ell + 1 = (\ell + 1)^2 - \ell^2$ (section 14.2)
Schwarzschild in isotropic coordinates is unchanged by ρ → (M/2)²/ρ, with the horizon as the fixed sphere	checked symbolically (section 11.1)

Exact statement	Check
M_irr² = Mr₊/2, and the extractable fraction is at most 1 − 1/√2	A = 8πMr₊; the area theorem
escape speed = √2 × orbital speed; u_E = u_B; peak/RMS = √2; peak/average power = 2	Newton; Maxwell; the mean of sin²
the levels are the squared chords of the moves, |1 − e^{iθ}|² = 1, 2, 3, 4 at 60°, 90°, 120°, 180°; the unit cube's corner with k ones sits at level √(3 − k) of the side-2 cube	2(1 − cos θ); the corners of [0, 1]³ against the centre (1, 1, 1)
two perpendicular quarter turns compose to one 120° turn about a body diagonal; (1 + i)(1 + j)/2 = (1 + i + j + k)/2 is the tesseract corner on the unit sphere	R_y(90°)R_x(90°) has trace 0, angle 120° and axis (1, 1, −1); the 24 Hurwitz units close under multiplication and are the 24-cell
hydrogen is the Hopf projection of the 4D oscillator; the planar case is w = z²	Kustaanheimo and Stiefel (1965); Levi-Civita (1920); Fock's n² degeneracy
a view down n casts the unit cube's shadow A(n) = |n_x| + |n_y| + |n_z|: 1, √2 and √3 on the face, edge and corner axes, with mean 3/2 over all directions	a box's shadow is its face areas times the direction cosines; Cauchy's mean projection is a quarter of the surface, 6/4; Monte Carlo 1.49993
corner-on, level √k sits at depth k/√3 and radius 0, √(2/3), √(2/3), 0; the sequence tester with X_C = R reads (3 ± √3)/√10, and 2 + √3 = cot 15°	r² = k − k²/3 for a corner with k ones; the Clarke factor √(2/3); the floating star point solved symbolically (section 6.4)

16.3 Test 2: calibration on an outside list

The alphabet has a member within 2σ of 19 of the 29 targets, and its density alone predicts 18.3. Table 71 sets the alphabet against four nulls, and Figure 2 shows where the cube's 19 falls among the primary null's alphabets.

Table 71. The cube's alphabet against four nulls.

Alphabet or targets	Mean hits within 2σ	P(19 or more)
The cube's alphabet	19 (density predicts 18.3)	—
Same recipe, π, √2, √3 and φ random (primary)	19.32 ± 1.05	0.79
Every ingredient jittered by up to ±25%	19.90	0.92

Alphabet or targets	Mean hits within 2σ	P(19 or more)
Only the irrational ingredients jittered	19.91	0.94
The cube's alphabet, targets moved by up to ±25%	18.82	0.62

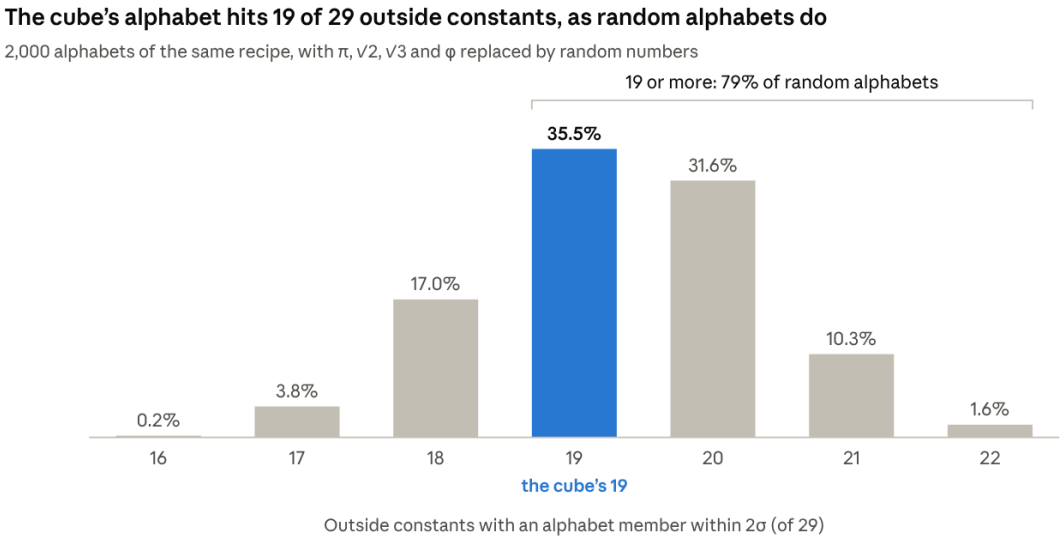


Figure 2 · the cube's alphabet among 2,000 random alphabets: 19 of 29 outside constants hit, as 79% of them do

The hits split by precision:

- **Imprecise targets.** All 18 targets known worse than 0.1% are hit, as the alphabet's density predicts.
- **Precise targets.** Of the 11 known better than 0.1%, ten are missed, by 5.7σ (g_A) to 5.3 × 10^{8} σ ($m_{\square}/m_e$). The eleventh, $r_{\square}/\lambda\square$, is hit by the integer 4.
- **Hits by numbers it never chose.** Today's $\sin^2\theta_{12}$ = 0.3096 lies nearer 1/(2φ) = 0.3090 than the calculator's 0.318, and Planck's $\Omega\square$ = 0.3153 lies nearer 1/√10 = 0.3162 than π²/32.

Verdict: on an outside list the alphabet lands on measured numbers as often as alphabets made from random ingredients.

16.4 Test 3: the precise constants

Table 72 sets each against the nearest member of the alphabet.

Table 72. The eleven constants known better than 0.1%, against the nearest alphabet member.

Constant	Relative σ	Nearest member	Pull
m_W/m_Z	9.9 × 10^{-5}	0.880540	+8.6

Constant	Relative σ	Nearest member	Pull
m_Z/v	2.3×10^{-5}	0.370240	+12.9
m_H/v	8.8×10^{-4}	0.504623	+8.0
1/α	1.5×10^{-10}	136.757	$+1.3 \times 10^{7}$
mμ/mₑ	2.2×10^{-8}	216	-2.0×10^{6}
mτ/mμ	5.1×10^{-5}	16.7926	+29.5
m□/mₑ	6.0×10^{-11}	1894.96	-5.3×10^{8}
μ□/μ_N	2.9×10^{-10}	2.792527	$+3.9 \times 10^{5}$
|μ□/μ□|	2.3×10^{-7}	0.688864	-2.4×10^{4}
g_A	4.4×10^{-4}	1.273240 = 4/π	+5.7
r□/ƛ□	7.5×10^{-4}	4	−0.8

The alphabet hits 1 of the 11. Alphabets of the same recipe with random ingredients average 1.83, and all 400 tried hit at least one. Two facts explain the gap:

- **The shared hit.** The one hit is the integer 4, which every alphabet of the recipe contains, so every random alphabet hits r□/ƛ□ too.
- **The hits it lacks.** 43% of random alphabets also hit m_H/v and 32% hit g_A. The cube's alphabet hits neither.

Good's judgement of 1990 is the historical parallel: of eight formulas for m□/mₑ he found only Lenz's $6\pi^5$ worth anything, and it now sits 3×10^5 standard deviations off.

Verdict: on the constants physics knows best, the calculator does worse than the average alphabet of random ingredients.

16.5 Test 4: with the load rule

The rule. The calculator's own rule moves a static value by its load: $k \times \alpha_i/2\pi$ for up to three phases, of either sign, of each coupling:

- $\alpha/2\pi = 0.001161$;
- $\alpha_2/2\pi = 0.005380$;
- $\alpha_s/2\pi = 0.018780$.

The result. The loaded alphabet holds 18,962 numbers and hits 3 of the 11 precise constants: m_H/v at +0.3σ, g_A at +0.1σ and r□/ƛ□ at −0.8σ. Random alphabets given the same loads hit 4.30 ± 0.73, and 99.8% of them hit at least 3.

Verdict: the load rule adds reach, not discrimination; any alphabet loaded this way reaches as far or farther.

16.6 Test 5: information

The chance probability of each committed match comes from the look-elsewhere test of 5 October 2026 (section 16.7.4), converted to bits as $-\log_2 p$ and combined by Fisher's method (Fisher 1932). Table 73 gives each match.

Table 73. Evidence against chance, in bits.

Match	Chance probability	Bits
15° tilt, strict set of 40 cube angles	0.12	3.1
15° tilt, generous set of 1,422 angles	1	0.0
Vacuum energy scale, coefficient fixed at 1	0.052	4.3
Vacuum energy scale, any order-one coefficient	0.57	0.8
Top and Higgs from one load	0.6	0.7
r□ = 4ƛ□ (prior art)	1/6	2.6
Ω□ = π²/32 (nine members within 2σ)	1	0.0
ΔΣ = ⅓; ΔG = ⅙	1 each	0.0
Koide's Q = ⅔ (Koide 1982)	1/850	9.7
The lepton phase 2/9 (Brannen 2006)	1/300	8.2
The calculator's own, strict: tilt, scale fixed at 1, top, r□	combined 0.064	**4.0** (3.6 without r□)
The calculator's own, generous	combined 0.68	**0.6**
The inherited lepton sector: Koide and Brannen	combined 5.3×10^{-5}	**14.2**

For scale, p = 0.01 is 6.6 bits, a factor of 100 and Jeffreys's grade “decisive”; 3σ is 8.5 bits and 5σ is 20.7 bits, both two-sided.

Verdict: the calculator's own numbers carry 0.6 to 4.0 bits, short of what even p = 0.01 needs. The real evidence against chance, in the lepton sector, belongs to relations Koide and Brannen found; the calculator explains them but did not find them.

16.7 The sweep of the alphabet, and the look-elsewhere test

Two searches were run before the five tests of sections 16.2 to 16.6:

- **The sweep (3 October 2026).** It set every number of the cube's alphabet against about 60 measured dimensionless numbers from ten fields, most of which no standard theory derives.
- **The look-elsewhere test (5 October 2026).** It asked how often the alphabet would match the committed numbers this well by accident.

Their rules were fixed before looking, and both are rerun here with PDG 2026 and NuFIT 6.1. The alphabet is the one of section 16.1. The sweep's own count was 971; rebuilt with near-duplicates merged at 10^{-9} it holds 998 numbers, and the densities below use the rebuild.

16.7.1 The rules

- **The targets.** About 60 measured dimensionless numbers:
    - mass ratios, mixing angles, cosmic shares and couplings;
    - nuclear and atomic ratios and critical exponents;
    - fluid constants, scaling exponents and bond angles;
    - the spin-½ period.
- **A nail** is within 1σ of the measurement, with no other alphabet member within 3σ, and with a reason for that number in the calculator, not just membership of the alphabet.
- **A break** is a reading the rules naturally give, excluded by the data at 5σ or more.
- **Not selective** means several alphabet members fit within 2σ: the measurement cannot tell them apart, and neither can the calculator.
- **No number** means nothing in the alphabet within 3σ, and no reading.

The score.

- one nail, a single fact with three faces (section 9.1);
- two leads with a reason: the tilted Clarke frame for lepton mixing (section 10.1), and, thinner, the proton's radius (section 12.1);
- eleven breaks;
- twelve not selective;
- two with no number;
- one check the same as standard physics;
- one prediction.

Eighteen operations were checked as well, fourteen of them repeating what standard theory derives.

16.7.2 Why the fits that work, work

A fit has a reason when it is one equation, not one number. The equation behind every exact fit in this paper is the same: two parts carry equal shares. Standard physics calls it the virial theorem, equipartition or resonance. Where that reason is present, a fit is a constraint and has to work. Where it is absent, a match is a coincidence until a reason is found. Table 74 lists the equation behind each fit.

Table 74. The equal-share equation behind each fit.

Fit	The equal-share equation	Standing
escape over orbit = √2	on a circle the kinetic energy is half the potential, and equal to it at escape: the virial 2 : 1 in energy is √2 in speed	same as Newton
light on the 45° line	electric energy equals magnetic energy at every instant	same as Maxwell
three-phase power is flat	three equal phases 120° apart sum to zero, 1 + ω + ω² = 0	same as engineering
the charged leptons at 45°	√m□ = μ[1 + a cos(δ + 120°k)]: DC power 1, AC power a²⁄2, equal when a = √2, which is Q = ⅔	holds to 1 part in 10^5; no standard derivation
the quark spin share ⅓	three equal legs of the proton's spin: ½ΔΣ = ΔG = L = ⅙	ΔΣ = 0.330 ± 0.039 (HERMES); ΔG = 0.231 ± 0.047 at 10 GeV² (+1.4σ) and 0.158 ± 0.039 folded to 2 GeV (−0.2σ); equal thirds hold at one resolution only (section 12.2)
the three-phase start	the load is the vector sum of the two previous phases, |y_t² + g²e^{i120°}|/8π², one loop	the Higgs to 0.1σ and the top to 0.1σ (direct) or 0.7σ (pole); the size of a load is the electron's own, e²⁄8π² (section 9.4)

16.7.3 Every operation, checked

Table 75 checks each operation of the calculator against what it reproduces, with its standing and what it adds or breaks.

Table 75. The operations of the calculator, each against what it reproduces.

Operation	What it does	Checked against	Standing	What it adds
the sum table a² + b²	a constant sum rides a circle	Pythagoras; E² = (pc)² + (mc²)²; the FCC neighbour distance a√2/2	same as standard	
the difference table a² − b²	a constant difference rides a hyperbola	mass from E and p; √3 = 2² − 1²; the interval	same as standard	

Operation	What it does	Checked against	Standing	What it adds
the third table 2ab	squaring doubles the angle	$(a^2 + b^2)^2 = (a^2 - b^2)^2 + (2ab)^2$; power at 2f	same as standard	
the inverse legs (1/a, 1/b)	series ↔ parallel; the angle mirrors across 45°	$1/a^2 + 1/b^2 = 1/h^2$; parallel resistors; the lens equation; the reduced mass	same as standard	
inside-out, p → −p	a mirror in 3D (det −1)	antimatter, Q → −Q on all 16 states; CPT within 1.3σ; antihydrogen falls	same as standard	antimatter falls, as the author's reading says (ALPHA-g)
the 24 rotations, 48 with inside-out	the point group m3̄m	the cipher's faces; the Hopf map 24 → 6; NaCl, Cu, Fe, diamond	same as standard	
the 120° turn about the zero axis	one phase step, the cube root of 1	$1 + \omega + \omega^2 = 0$; the colour cycle; the Clarke zero axis	same as standard	the leptons at √2 nail; the Higgs start is a lead
the 90° quarter turn	the fourth root of 1; two make the 180° turn, the 2D inside-out	$i^2 = -1$; the spin-½ 720° (716.8° ± 3.8° measured); E ↔ H duality	same as standard	
the diagonal inversion r → 1/r	levels ↔ inverse levels; centre ↔ infinity	the image charge R^2/d; the reciprocal lattice, FCC ↔ BCC; Maxwell's equations under inversion (Bateman 1910; Cunningham 1910)	same as standard	the v^2/M mirror of the hum is a lead
the dimension converter	each dimension adds one leg	exact; 3D hydrogen is the 4D sphere (Fock 1935); the 16	same as standard	

Operation	What it does	Checked against	Standing	What it adds
		states on the tesseract		
the L1 over L2 means, 4/π and 3/2	the step over the straight path	(a + b)/c = √2 sin(θ + 45°); lightning in kind	same as standard	
the shares π/4, π/6, π²/32	the sphere in the cube	exact		
the grids ½, ⅓, ¼	the inverse levels squared	charge thirds; isospin halves; the 16 sums zero; √(5/3); 3/8; the mean-field and 2D Ising exponents	same as standard	
the Fibonacci cell, φ	31.717°; φ ± 1/φ	exact; the golden angle 360°/φ² (Douady and Couder 1992)	same as standard	
balance, slightly past 45°	tan θ = 1 + δ	the Higgs vacuum past the edge; the leptons exactly at 45°		
bands and steady clocks	action and angle	Montgomery's T-bar formula; Kepler; the pendulum; chaos	same as standard	
the torus and the Hopf map on √4	two √2 circles linked once	exact		the no-dark-matter reading is open
the reactive board	ε_0, μ_0, Z_0; E ↔ H inverse	the speed of light; $Z_E \cdot Z_H = Z_0^2$; the $2mc^2$ gap	same as standard (Maxwell; Dirac; Bass)	the hum's size is a lead

16.7.4 The look-elsewhere test

For each committed number the test took three measures, given in Table 76:

- the alphabet's local density, as members per unit value within ±25% of the measurement;
- the count of members within 2σ;

- the chance that a random true value in that neighbourhood would have a member as close as the observed pull: density × 2|pull|σ, checked by Monte Carlo with 20,000 draws.

Table 76. How often the alphabet would match this well by accident.

Committed number	Pull	Members within 2σ	Chance of an accident this close	Standing
Koide's Q = ⅔	−0.4σ	1	1 in 850	selective; inherited (Koide 1982)
the lepton phase 2/9	+0.4σ	1	1 in 300	selective; inherited (Brannen 2006)
r⃝/ƛ□ = 4	−0.8σ	1	1 in 6 (1 in 2.3 for a 2σ window)	weakly selective; prior art
the dark share 1 − π²/32	−0.9σ	9	about 1	not selective
the quark spin share ⅓	−0.1σ	79	about 1	not selective
the gluon share ⅛	+1.4σ	115	about 1	not selective; the prediction is pending (the Electron-Ion Collider)
Ω_c/Ω_b = 16/3	+0.5σ	5	0.8	noted
$a_0/cH_0 = 3/16$	−1.4σ	8	about 1	noted
n⃝ = 26/27	+0.5σ	4	0.55 to 0.75	noted
$H_0t_0 = 2(1 - \pi/6)$	−0.5σ	2	0.6 to 0.9	noted
the Tully–Fisher slope 4	−1.7σ	11	about 1	noted

The 15° tilt. The measured tilt is 14.89° ± 0.19°. There are two definitions of “the cube's angles”:

- **Strict.** The angles between the 26 cube directions (35.26°, 45°, 54.74°, 60°, 70.53°, 90°), the ladder's step angles arctan(1/√k) (45°, 35.26°, 30°, 26.57°), and one sum or difference of two of them: 40 angles. 15° is the only one within 3σ. The density is 0.47 per degree between 5° and 45°, so the chance of an accident is 0.12.
- **Generous.** The arctan, arcsin or 90° × any alphabet member up to 1: 1,422 angles, 32 of them within 3σ. Not selective.

So “the only cube angle within 3σ” holds under the strict definition only.

The hum. The three crossing scales give v^2/M = 5.5 eV, 0.36 eV and 2.0 meV, against 2.24 meV.

- With the coefficient fixed at 1 beforehand, the chance that one of three candidates spread over five decades lands within ±10% is 0.052.
- With an order-one coefficient allowed, within a factor of 3, it is 0.57.

The top and the Higgs. There is one formula with a discrete load count. Two constructions land within 0.4σ, and the Higgs follows from the top along the edge, so the two hits are one. With about three constructions, a top sensitivity of about 1.7 GeV per 1% of load, a plausible load range of 0 to 3%, and a hit window of ±0.5 GeV, the chance is about 0.6. This estimate is rough.

Combined. Fisher's method (Fisher 1932) gives the results of Table 73:

- the inherited lepton sector at 5×10^{-5};
- the calculator's own numbers at 0.065 under the strict rules and 0.68 under the generous ones.

What it means.

- **The strong evidence is inherited.** The strong numerical evidence in the paper is the lepton sector, and it is Koide's and Brannen's. The calculator inherits it and supplies a reason, which carries no number of its own.
- **The calculator's own content.** Today it lies between a 1-in-15 surprise and none, depending on how much freedom its rules are read as having. That is what the pre-registration is for: the calculator's own evidence has to come from the pending measurements of Table 80, each written down first.
- **Not counted either way.** The gravity tests, $w = -1$, $\Omega_k = 0$ and the return fraction are the same as the standard theories. The neutrino sum cannot be told from the floor.

16.8 The same tests, applied to the standard theories

The standard theories pass the first test by construction and stand outside the second: their equations are the laws the calculator's identities restate, and the outside list is mostly their own inputs. On the tests that measure evidence, the third and the fifth, their predictions of independent measurements beat chance by a wide margin. They also carry failures that the calculator does not share. Table 78 sets the two side by side, test by test.

Test 1, the identities. The standard theories' equations are the laws the identities restate: Maxwell's for the board, Einstein's for the two tables, the Standard Model's charges for the register. They pass by construction and, as for the calculator, passing says nothing about new physics.

Test 2, the outside list. Most of the 29 targets are inputs of the Standard Model or ΛCDM: masses, mixings, couplings and densities, fitted to the very measurements the list holds. A fitted number cannot miss, so the standard theories score neither hit nor miss here. Where they compute a target rather than take it, they compute it with a stated error and agree within it: the W mass from the other inputs, and the nucleon's radius and axial coupling on the lattice (Table 54).

Test 3, the precise constants. Of the eleven constants known better than 0.1%, five are inputs. The electroweak theory computes m_W/m_Z from α, G_F, m_Z, m_t and m_H to 1.2×10^{-4}, within 0.6σ

(Table 77). The lattice computes the nucleon's mass to 4% (Dürr et al. 2008), its radius to 2% (Djukanovic et al. 2024) and its axial coupling to about 2% (Table 54): short of their measured precision, but within its own errors. Where the calculator hits one of the eleven, with an integer every alphabet contains, the standard theories predict at full precision the one precise ratio they do not take as input.

Test 4, the load rule. It has no counterpart. The standard theories have no alphabet and no rule applied quantity by quantity; each prediction is one computed number.

Test 5, information. Table 77 scores the standard theories' predictions of independent measurements on the convention of section 16.6, with one change that their form forces. A single computed number has no alphabet to search, so its chance probability is the width of its window, ±max(|pull|, 1)σ, against one decade of possible values. The floor of 1σ can only lower their score. Each number was computed from the theory and from other measurements, and none was fitted to the measurement it is compared with.

Table 77. The standard theories' predictions of independent measurements, scored for chance.

Prediction	Theory	Predicted	Measured	Pull	Chance	Bits
the speed of gravitational waves	general relativity (1916)	c	within -3×10^{-15} and $+7 \times 10^{-16}$ of c (LIGO Scientific and Virgo Collaborations 2017)	0	1.6×10^{-15}	49.1
the electron's moment against α from rubidium	quantum electrodynamics	1/α = 137.035999166(15), from g/2 (Fan et al. 2023)	137.035999206(11) (Morel et al. 2020)	+2.2	2.5×10^{-10}	31.9
light bending, γ	general relativity (1915)	1	$1 + (2.1 \pm 2.3) \times 10^{-5}$ (Bertotti, Iess and Tortora 2003)	+0.9	2.0×10^{-5}	15.6
the W mass	the electroweak theory, from α, G_F, m_Z, m_t and m_H	80,357 ± 6 MeV (CDF Collaboration 2022)	80,362.5 ± 7.7 MeV without CDF (PDG 2026)	+0.6	1.1×10^{-4}	13.2

Prediction	Theory	Predicted	Measured	Pull	Chance	Bits
the binary pulsar's orbital decay	the quadrupole formula (Einstein 1918)	1	0.9983 ± 0.0016 of it (Weisberg and Huang 2016)	−1.1	1.5×10^{-3}	9.4
geodetic precession	general relativity (de Sitter 1916)	6,606.1 mas/yr	6,601.8 ± 18.3 (Everitt et al. 2011)	−0.2	2.4×10^{-3}	8.7
helium-4, Yp	nucleosynthesis with the microwave background's baryon density	0.24691 ± 0.00018	0.2449 ± 0.0040 (Fields et al. 2020)	−0.5	0.014	6.1
deuterium, D/H	the same	$(2.57 \pm 0.13) \times 10^{-5}$	$(2.55 \pm 0.03) \times 10^{-5}$ (Fields et al. 2020)	−0.2	0.045	4.5
N_eff	three neutrinos in standard cosmology	3.044	2.86 ± 0.13 (ACT Collaboration 2025)	−1.4	0.053	4.2
frame dragging	general relativity (Lense and Thirring 1918)	39.2 mas/yr	37.2 ± 7.2 (Everitt et al. 2011)	−0.3	0.16	2.6
the Higgs mass	the electroweak fit before the discovery	91 +30/−23 GeV (Baak et al. 2012)	125.13 ± 0.11 GeV (PDG 2026)	+1.1	0.29	1.8
combined, Fisher's method					**1.8×10^{-31}**	**102**

Combined by Fisher's method, the eleven give $p = 1.8 \times 10^{-31}$, 102 bits. A narrower prior, ±10% instead of a decade, lowers each entry by 3.5 bits and the combination to 70 bits. Against caesium's α the electron's moment sits at −3.9σ instead, but the two atomic values disagree with each other at 5.5σ, and only the nearer is counted. The muon's moment is not counted: its 2025 prediction agrees at +0.6σ only after lattice input replaced the e^+e^- data, with which the 2020 prediction sat about 5σ from the measurement (Muon g−2 Theory Initiative 2025). Figure 3 sets each prediction beside the calculator's own numbers and the lepton relations it inherits.

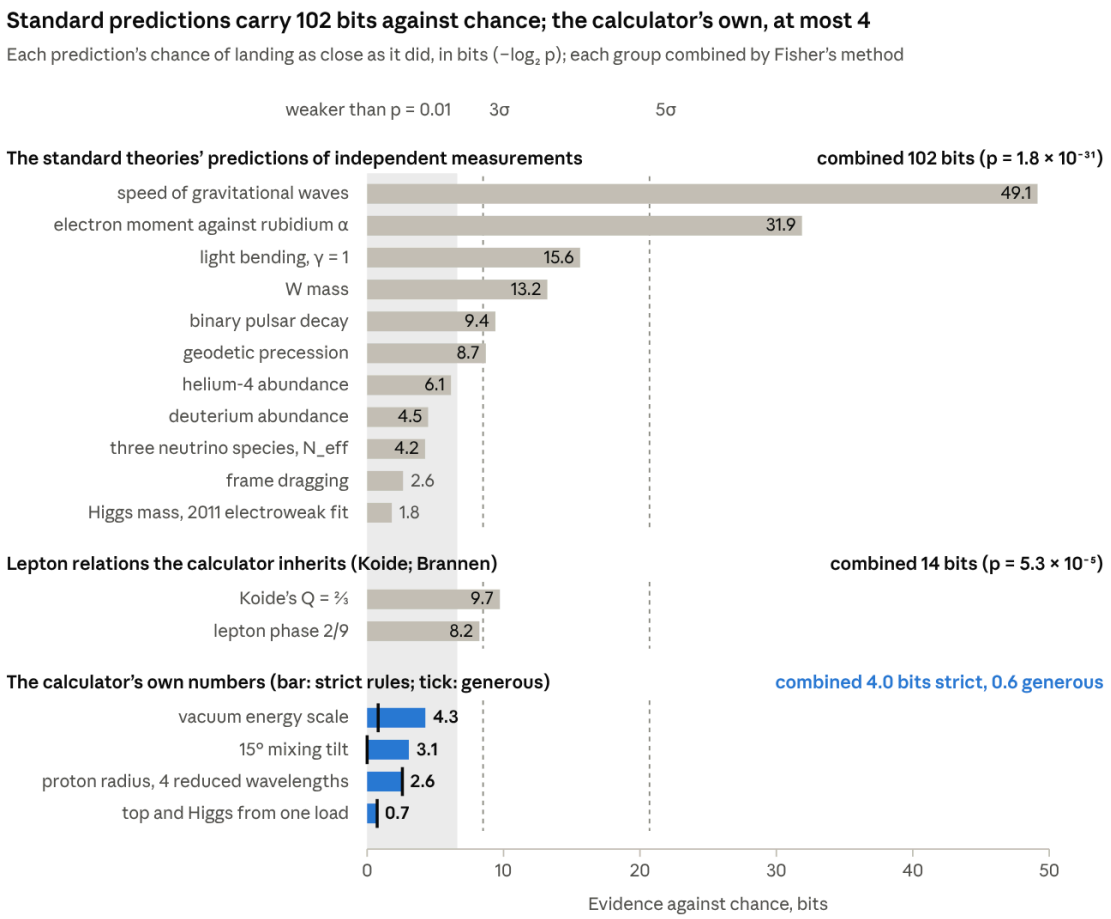


Figure 3 · evidence against chance, in bits: the standard theories' eleven predictions (Table 77), the lepton relations the calculator inherits and its own numbers (Table 73)

Failures. The rule of section 2.5 counts the standard theories' failures the same way as the calculator's:

- the vacuum's energy, about 10^{120} too large (Weinberg 1989);
- lithium-7 from nucleosynthesis, $(4.72 \pm 0.72) \times 10^{-10}$ predicted against $(1.6 \pm 0.3) \times 10^{-10}$, 4.0σ (Fields et al. 2020);
- the neutrino masses, zero in the minimal Standard Model and contradicted by oscillations (Fukuda et al. 1998);
- the Hubble constant from the microwave background in ΛCDM, 4.9 to 7.1σ from the ladders (Riess et al. 2022; H0DN Collaboration 2025; Freedman et al. 2025);

- first-row unitarity of the quark mixing matrix, 2.3σ short (PDG 2026);
- lepton universality in R(D) and R(D*), 3.7σ (HFLAV 2026).

The calculator shares none of these: it is silent on four, and gives a number for two (the hum, +1.1σ; the neutrino masses, inside the cap). Its own failures are of another kind. Among its committed numbers none misses at 2σ, but eleven readings its rules naturally give are excluded by the data (section 16.7.1). The standard theories made their failed predictions; the calculator did not commit to its breaks and kept only what survived them, which is why its survivors are weighed against chance in Table 73.

Table 78. The five tests, applied to both.

Test	The calculator	The standard theories
1. the identities	exact: a correct notation for established law	exact by construction: they are that law
2. calibration on an outside list	19 of 29 hit, as often as random alphabets (P = 0.79)	the targets are mostly their inputs; those they compute agree within stated errors
3. the precise constants	1 of 11 hit, below random alphabets' 1.83	m_W/m_Z at 1.2×10^{-4}, +0.6σ; the rest inputs, or lattice values to 2 to 4%
4. the load rule	adds reach, not discrimination (P = 0.998)	no counterpart
5. information	its own numbers 0.6 to 4.0 bits; the lepton relations it inherits 14.2 bits	eleven predictions, 102 bits
failures	none among committed numbers; eleven breaks among readings it did not commit to	six, among them the vacuum's energy and the Hubble constant

16.9 The verdict through the Quad-Core, for both

Table 79 puts each of the calculator's four claims, and the four matching claims of the standard theories, through the author's four cores: outcome, constraint, alternative and consequence.

Table 79. The four claims under the author's four cores.

Claim	Outcome	Constraint	Alternative	Consequence	State
A. Its identities are exact	not rejected: all hold to 30 digits; every electron	not rejected: they hold at every scale and frequency, being laws	not rejected: equivalent notations, bond graphs	not rejected: an error would show at once	**STABILIZED**

Claim	Outcome	Constraint	Alternative	Consequence	State
	corner meets a measurement		and phasors, do no better		
B. Its readings describe nature beyond standard physics	undetermined: they change no measured number	undetermined: carried into classical field theory the electron's half becomes ħ/3, a break if the reading is dynamical and none if it is bookkeeping	not rejected: standard physics gives the same numbers, so neither has fewer violations	undetermined: if wrong, nothing measurable changes, so the error is deferred	**UNSTABLE**
C. Its number matches show more than chance	not rejected: 19 of 29 within 2σ	**rejected**: at the precision of the best constants the matches vanish, 1 of 11 against a random mean of 1.83	**rejected**: random alphabets match as often (P = 0.79), and with the load rule more often (P = 0.998)	undetermined	**REJECTED**
D. Its frozen predictions will be borne out	undetermined: the data are not in	not rejected: each definition and scale is fixed, two of them by the author's behavior rule (section 17.1)	undetermined: today's data cannot yet separate its values from where the data sit	not rejected: each has a decider, a date and a refutation rule	**UNSTABLE**
A′. The standard theories' equations hold where tested	not rejected: the electron's moment to 0.13 parts per trillion, γ to 2 × 10^{-5}, the binary pulsar to 0.2%	not rejected within the tested range of scale and energy	not rejected: no rival matches as many measurements with fewer violations	not rejected: an error would show in the next digit	**STABILIZED**

Claim	Outcome	Constraint	Alternative	Consequence	State
B′. Their structures describe nature beyond what was known when they were written	not rejected: the positron (Anderson 1933), gravitational waves (LIGO Scientific and Virgo Collaborations 2016) and the Higgs boson's couplings at their predicted strength (ATLAS Collaboration 2022)	not rejected	not rejected: no other account predicted them	not rejected	**STABILIZED**
C′. Their predictions of independent measurements beat chance	not rejected: 102 bits over eleven predictions (Table 77)	not rejected: they hold at precisions from tens of percent to parts in 10^{15}	not rejected: no random procedure matches eleven independent measurements at these precisions	not rejected	**STABILIZED**
D′. They are complete: they account for every measurement	**rejected**: the vacuum's energy misses by about 10^{120}, and the minimal theory's massless neutrinos are contradicted by oscillations	**rejected**: general relativity's singularities and the vacuum's energy break at the extremes of scale	undetermined: no complete rival exists, the calculator included	**rejected**: the dark sector and the Hubble tension defer the error rather than remove it	**REJECTED**

The question this paper was asked to decide is the conjunction: is the calculator how reality works? Under the author's synthesis rule, with no core overriding another and no majority, a single rejection blocks stabilisation. The strong claim is therefore **rejected as of 10 October 2026**, because claim C is rejected. The same rule rejects the claim that the standard theories are the whole of how reality works,

because claim D′ is rejected, while it stabilises their claims to hold where tested, to describe nature beyond what was known, and to beat chance. For the calculator, stated plainly:

- **Notation.** Where it restates known law, the calculator is a correct notation for how reality works (A, stabilised).
- **Readings.** It is not shown to be how reality works beyond that (B, unstable).
- **Numbers.** Its numerical coincidences are what numerology produces at this density (C, rejected).
- **Predictions.** What can still change the verdict is D.

For the standard theories: their equations, their structures and their numbers stand (A′, B′ and C′, stabilised), and their completeness does not (D′, rejected). What separates the two theories today is C: the standard theories' numbers carry evidence that the calculator's do not yet.

This is not a finding that the calculator's readings are false. It finds that nothing measured today favours them over standard physics, and that the numbers offered in their support are what chance gives.

17. Predictions: what decides both theories

The calculator's own evidence has to come from measurements not yet made. Nine predictions are frozen with their definitions and a scoring rule, to be deposited unchanged before any of their data (section 17.1). If the calculator's numbers are right, the seven of its own give a median Bayes factor of 240, decisive with probability 0.68; if they are wrong, about 10^{-12}, decisive against with probability 0.92. JUNO's $\sin^2\theta_{12}$ (about 2031), the angle α (about 2030) and the Electron-Ion Collider's ΔG (the 2030s) carry most of that power. Four measurements of the coming decade bear on both theories at once: w, the curvature and the neutrino mass sum test commitments the two share, and r separates them (section 17.5).

17.1 Nine frozen predictions

The predictions below are frozen with their definitions and are to be deposited, unchanged, before any of their data. Table 80 lists them, with today's value, the decider and what a pass would weigh.

Table 80. The pre-registered predictions. “If wrong” means each value lies where today's data put it.

#	Prediction and definition	Value	Today	Pull	Decider and date	P(pass) if wrong	B if it passes
1	PMNS $\sin^2\theta_{12}$: the tribimaximal frame tilted 15° with its first column fixed	0.318107	0.3096 +0.0057/−0.0073	−1.5	JUNO, six years, ±0.0016, about 2031	0.14	6.9

#	Prediction and definition	Value	Today	Pull	Decider and date	P(pass) if wrong	B if it passes
2	PMNS $\sin^2\theta_{13} = \sin^2 15°/3$	0.022329	0.02200 ± 0.00056	−0.6	global fits, ±0.0004 (assumed), about 2030	0.70	1.4
3	CKM unitarity angle $\alpha = 90°$	90°	84.1 +3.7/−3.0°	−1.6	Belle II and LHCb, ±2°, about 2030	0.35	2.7
4	$\Omega_\Lambda = \pi^2/32$ in flat ΛCDM	0.308425	0.3045 ± 0.0032	−1.2	Euclid and final DESI, ±0.003, 2027–30	0.68	1.4
5	$\Delta G = 1/8$ at $\mu = 2$ GeV, MS-bar, the scale fixed by the behavior rule (0.236 at 10 GeV² at leading order)	0.166667	0.158 ± 0.039 (lattice, evolved to 2 GeV)	−0.2	Electron-Ion Collider, about 10%, evolved to 2 GeV, 2030s	0.54	1.8
6	$\Delta\Sigma = 1/3$, MS-bar	0.333333	0.352 ± 0.029 (HERMES and lattice combined)	+0.6	Electron-Ion Collider, about 15%, 2030s	0.90	1.1
7	Top pole mass from the δ_3 load (MS-bar 162.76 ±	172.57 ± 0.38 GeV	172.1 ± 0.6 (pole average)	−0.7	FCC-ee threshold scan, ±0.035 GeV, 2040s	0.67	1.4

#	Prediction and definition	Value	Today	Pull	Decider and date	P(pass) if wrong	B if it passes
	0.37 GeV, the same alignment)						
8	r□ = 4ħ/(m□c) (prior art)	0.841236 fm	0.84075 ± 0.00064	−0.8	hydrogen and muonic hydrogen, ±0.0002 fm (assumed), about 2030	0.35	2.7
9	mτ from Koide's relation (prior art)	1776.969 MeV	1776.93 ± 0.09	−0.4	Belle II, BESIII, STCF, ±0.05 MeV (assumed)	0.63	1.5

How the definitions are fixed. Two predictions can be stated in more than one alignment, and the behavior rule fixes both (rule 9 of section 4.1; step 8 of section 4.8). It was applied on 10 October 2026 to the data then published. Section 17.2 gives the scans, and section 17.3 applies the rule to the whole set.

- **The scale of ΔG: μ = 2 GeV.** Carried to 10 GeV², where the lattice gives the full value, the rule admits 1.50 to 3.04 GeV and is best met at 2.01 GeV. ⅛ at 10 GeV² misses the two lattice results combined by 2.2σ (1.4σ against Zhao et al. alone) and is not the best match, so it is left out. At the collider's 10 GeV², ⅛ at 2 GeV is 0.236 at leading order.
- **The top: the pole mass.** m_t(m_t) = 162.76 ± 0.37 GeV in MS-bar is the same alignment. Read as an unconverted MS-bar mass, 172.57 GeV would mean a pole mass of 182.9 GeV, 14.9σ from the pole data, so that reading is left out. The direct-mass reading misses nothing, but it differs from the pole reading by about 0.5 GeV, which only the threshold scan can resolve. The pole mass is kept because it is the mass the edge calculation takes, and row 7's value today is the pole average.

The lattice match that fixes ΔG's scale is the reason for the choice, not evidence for it; the deciders in Table 80 are the test.

The scoring rule.

- **The score.** A future result m ± σ□ on a prediction c ± σ_c scores the Bayes factor B = N(m; c, √(σ_c² + σ□²))/p₀(m), where p₀ is today's value taken as a split normal and spread by σ□.
- **The verdict.** The calculator's own seven are scored together by multiplying their B. A total above 100 is decisive for its numbers, below 1/100 decisive against, and anything between is open. The threshold of 100 is Jeffreys's grade “decisive” (Jeffreys 1961; Kass and Raftery 1995).
- **Refutation.** It is stated as a range. Each result's pull |m − c|/σ is taken on two yardsticks: σ = √(σ_c² + σ□²), with σ_c the calculation's precision as frozen in Table 80, and σ = √(σ_c′² + σ□²), with σ_c′ the same calculation at the input precision its decider brings. A pull above 5 on both refutes that prediction's rule, whatever the total; above 5 on the second alone, the rule is refuted at the decider's precision and stands at the frozen one; below 5 on both, it stands. Only the top carries a precision of its own: σ_c = 0.38 GeV, and σ_c′ = 0.15 GeV with α_s at ±0.0002, so at the scan's ±0.035 GeV its line runs from 0.77 to 1.91 GeV. For the other eight σ_c = σ_c′ = 0, and both lines are 5σ□.
- **Prior art.** The prior-art pair is scored separately.
- **What is not evidence:** matches to numbers known no better than about 0.1%, which the calibration shows the alphabet reaches by density; readings that reproduce standard physics; and anything fitted after the data.

What the set can decide.

- **If right.** If the calculator's numbers are right, the seven give a median B of 240, above 100 with probability 0.68.
- **If wrong.** If they are wrong, the median is about 10^{-12}, below 1/100 with probability 0.92.
- **The three that carry it.** JUNO's $\sin^2\theta_{12}$, the angle α and ΔG carry most of the power; passing all three by chance has probability 0.026 (1 in 38).
- **The rest.** ΔΣ, Ω□ and the top add little until their precision improves. The Electron-Ion Collider's projected 15% on ΔΣ (Aschenauer, Sassot and Stratmann 2015) is no better than the lattice today.

Recorded, not scored. Five numbers are recorded with the set and not scored, because standard physics shares them or because they wait on the calculator's own precision:

- Σmν = 59.2 meV with normal ordering;
- w = −1;
- Ω_k = 0;
- r ≈ 0;
- the vacuum energy scale, 2.02 ± 0.20 meV against 2.24 meV.

Their deciders are in Table 87 (section 17.4).

The deposit. The predictions, their definitions, the scoring rule and the script hashes are written into one pre-registration file, to be deposited unchanged on OSF before any of their data. Its SHA-256 is 624b271c504c25191447350a1ffbb90ade7f6da2ee7cf752dd109588ec8305df.

17.2 The behavior rule applied to the two definitions

Two definitions in Table 80, the scale of ΔG (prediction 5) and the top-mass scheme (prediction 7), are fixed by the author's behavior rule (section 4.1, rule 9): an alignment is kept only if nothing it touches

down-wind is missed. This subsection sets out every published observation under each alignment, including observations made for other purposes, and the scans that decide between them. Scripts: deltaG_scale.py, top_scheme.py, downstream_dg.py, downstream_top.py and choices_power.py.

The gluon spin and its scale.

- **ΔG runs.** At leading order ΔΣ stays fixed while ΔG grows with the scale and the orbital part falls to match (Ji, Tang and Hoodbhoy 1996). So ½ΔΣ, ΔG and L can all equal ⅛ at one scale only, and the calculator's structure does not yet name that scale (Table 90), so the behavior rule fixes it.
- **Where the data put ⅛.** Evolving the lattice values with the leading-order equation of section 12.2 and four-loop α_s (α_s(m_Z) = 0.1180):
    - from Zhao et al. (2025), ΔG = ⅛ at μ = 2.11 GeV, 1.69 to 2.79 GeV at 1σ;
    - from Yang et al. (2017), at 1.91 GeV, 1.55 to 2.49 GeV;
    - with HERMES's ΔΣ = 0.330 in place of the lattice's 0.380, both move down by 0.05 to 0.06 GeV.

  Both crossings lie at the lattice's conventional 2 GeV, where ½ΔΣ = 0.190 ± 0.022 and L = 0.152 ± 0.043 are quoted (section 12.2).

Table 81. The gluon spin by scale. The pull is (value − ⅛)/ σ .

Determination	Method	Scale	Value	Pull from ⅛
Zhao et al. 2025	lattice, continuum limit, MS-bar	$\mu^2 = 10$ GeV²	ΔG = 0.231 ± 0.047	+1.4σ
Yang et al. 2017	lattice, MS-bar	$\mu^2 = 10$ GeV²	ΔG = 0.251 ± 0.050	+1.7σ
DSSV14 (de Florian et al. 2014), as quoted by Aschenauer et al. (2015)	global fit, x > 0.05 only	$Q^2 = 10$ GeV²	0.20 +0.06/−0.07 at 90% C.L.	+0.8σ, truncated
NNPDF reweighted with STAR's 2009 jets, as quoted by Aschenauer et al. (2015)	global fit, 0.05 < x < 0.2 only	$Q^2 = 10$ GeV²	0.17 ± 0.06	+0.1σ, truncated
COMPASS Collaboration 2017	high-p_T hadrons, leading order	$\mu^2 = 3$ GeV², $x_g \approx 0.10$	Δg/g = 0.113 ± 0.052	a local ratio, not ΔG
Zhao et al., evolved (section 12.2)	lattice with leading-order evolution	μ = 2 GeV	ΔG = 0.158 ± 0.039	−0.2σ

- **The sign is settled.** The sign is fixed by two kinds of data:
    - PHENIX's direct photons reach the gluon through quark–gluon scattering at $0.02 < x < 0.08$ and disfavour a negative gluon spin by more than 2.8σ (PHENIX Collaboration 2023);
    - high-x DIS together with lattice data rules out the negative solutions of the global fits (Hunt-Smith et al. 2024).

  STAR's jets reach $x \approx 0.015$ (STAR Collaboration 2019).
- **The full moment is not settled by experiment.** Below $x \approx 0.05$ the fits cannot extrapolate:
    - an NNLO global analysis states that it cannot reliably determine the full first moments (Borsa et al. 2024);
    - inclusive and semi-inclusive DIS alone leave the polarised gluon largely unconstrained at NNLO (Bertone, Chiefa and Nocera 2025).

  The full ΔG therefore comes from the lattice today. In the 2030s it will come from the Electron-Ion Collider, whose projections are quoted at 10 GeV² (Aschenauer, Sassot and Stratmann 2015).

The rule applied to ΔG. Each row of Table 82 puts $\Delta G = \frac{1}{6}$ at one scale and carries it to 10 GeV², where the lattice gives the full value, with the leading-order evolution of section 12.2 and $\Delta\Sigma = 0.380$ (downstream_dg.py).

Table 82. The behavior rule on the scale of ΔG. Against the two lattice results combined as independent, 0.2405 ± 0.0342, and against Zhao et al. alone, 0.231 ± 0.047.

$\Delta G = \frac{1}{6}$ at	ΔG at 10 GeV²	Against both	Against Zhao et al.	Under the rule
1.2 GeV	0.383	$+4.2\sigma$	$+3.2\sigma$	left out
1.73 GeV (3 GeV², the COMPASS scale)	0.272	$+0.9\sigma$	$+0.9\sigma$	admitted
2 GeV, the lattice's scale	**0.241**	**$+0.0\sigma$**	**$+0.2\sigma$**	**kept: the best match**
2.5 GeV	0.201	-1.2σ	-0.6σ	admitted
3.16 GeV (10 GeV², the earlier definition)	0.167	-2.2σ	-1.4σ	left out
4 GeV	0.138	-3.0σ	-2.0σ	left out

- **The window.** The rule admits 1.50 to 3.04 GeV within 2σ and 1.72 to 2.42 GeV within 1σ, and is best met at 2.01 GeV. With Zhao et al.'s value alone, the 1σ window is 1.69 to 2.79 GeV, centred at 2.11 GeV.
- **The alignment kept.** $\Delta G = \frac{1}{6}$ at $\mu = 2$ GeV, where the lattice quotes the other two sixths, so all three are compared at one resolution. With the thirds exact ($\Delta\Sigma = \frac{1}{3}$), its value at the collider's 10 GeV² is 0.236.

- **What it costs.** The lattice match is the reason for the choice, not evidence for it. The Electron-Ion Collider remains the test: the alignment fails if the collider's full moment, evolved to 2 GeV, lands away from ⅛. Because the alignment predicts what the lattice already shows, a pass adds less: ΔG's chance of passing if the calculator is wrong is 0.54 (Table 80), against 0.34 under the earlier definition (Table 85).

The top mass and its scheme.

- **The mass the calculation uses.** The edge is computed with the pole mass as the input to the matching (Buttazzo et al. 2013), and δ_3 multiplies that edge.
- **MS-bar.** The pole and MS-bar masses differ by a nearly fixed factor at these masses, 1.0603 at four loops (Marquard et al. 2015, the conversion the PDG uses). So the rule is one prediction in both schemes: 172.57 ± 0.38 GeV pole is m_t(m_t) = 162.76 ± 0.37 GeV MS-bar, and the edge is 161.21 ± 0.36 GeV.
- **The conversion's own limit.** The four-loop term alone is 0.19 GeV. The terms beyond add about 0.3 GeV, with an intrinsic renormalon ambiguity of about 110 MeV in the pole mass itself (Beneke et al. 2017). A deposited MS-bar value should therefore be computed from the matching directly, not converted from the pole value.
- **The direct mass is a third quantity.** It is the simulation's mass parameter:
    - its difference from the pole mass is of order 0.5 GeV (PDG 2026);
    - the PDG's electroweak fit carries that difference as 0 ± 0.52 GeV;
    - one calibration, in e^+e^- event shapes, puts the pole mass 0.57 ± 0.28 GeV below the Pythia parameter, and the parameter within uncertainties of the MSR mass at 1 GeV (Butenschoen et al. 2016).

Table 83. The top mass by scheme, against the rule (172.57 ± 0.38 GeV) and the exact edge (170.94 ± 0.38 GeV). An asymmetric error is taken on the side that faces the prediction.

Determination	Scheme	Value (GeV)	Against the rule	Against the edge
PDG 2026, average of pole determinations	pole	172.1 ± 0.6	−0.7σ	+1.6σ
PDG 2026, direct average with the difference carried as 0 ± 0.52 GeV	pole-equivalent	172.60 ± 0.59	+0.0σ	+2.4σ
the direct average with the calibration of Butenschoen et al. (2016)	pole-equivalent	172.03 ± 0.39	−1.0σ	+2.0σ
CMS, normalised multi-differential	pole	170.5 ± 0.8	−2.3σ	−0.5σ

Determination	Scheme	Value (GeV)	Against the rule	Against the edge
cross sections, 13 TeV (CMS Collaboration 2020)				
CMS, $t\bar{t}$ + jet, 13 TeV (CMS Collaboration 2023)	pole	172.93 ± 1.36	+0.3σ	+1.4σ
ATLAS, $t\bar{t}$ + 1 jet, 13 TeV (ATLAS Collaboration 2025)	pole	170.7 ± 1.5	−1.2σ	−0.2σ
ATLAS and CMS, inclusive cross sections, 7 and 8 TeV (ATLAS and CMS Collaborations 2023)	pole	173.4 +1.8/−2.0	+0.4σ	+1.2σ
electroweak fit, indirect (PDG 2026)	pole	175.2 ± 1.8	+1.4σ	+2.3σ

- **Observations made for other purposes.** Two carry top-mass information without being measurements of it:
    - **The electroweak fit.** Built from Z-pole data and the W mass, it gives an indirect pole mass of 175.2 ± 1.8 GeV (PDG 2026), 1.4σ above the rule and 2.3σ above the edge.
    - **Toponium.** CMS set the ηt mass at 343 GeV, twice the top mass less a binding of about 2 GeV, with m_t fixed at 172.5 GeV and varied by ±1 GeV as a nuisance. With a resolution in m($t\bar{t}$) of about 15% at threshold, the peak's position is not used to extract the mass (CMS Collaboration 2025). Its position carries the mass in a threshold scheme, which is what the decider will measure.
- **The decider.** An e^+e^- threshold scan measures the potential-subtracted mass with a statistical error of 4.3 MeV, an experimental error of 6.8 MeV and a theory error of about 35 MeV. Converting it to MS-bar adds 7 to 23 MeV (Defranchis et al. 2025). Converting it to the pole mass inherits the pole mass's ambiguity of about 110 MeV.

The rule applied to the top. Each row of Table 84 reads the rule's 172.57 ± 0.38 GeV as one kind of mass and carries it to the pole-type observables (downstream_top.py). Every reading puts the pole mass above the edge, 170.94 GeV, on the metastable side the balance rule requires.

Table 84. The behavior rule on the top's alignment.

172.57 GeV read as	Pole mass implied (GeV)	Against the pole average, 172.1 ± 0.6	Against the electroweak fit, 175.2 ± 1.8	Under the rule
the pole mass (MS-bar 162.76, the same alignment)	**172.57 ± 0.38**	**−0.7σ**	**+1.4σ**	**kept: the mass the calculation takes**
an MS-bar mass, unconverted	182.88 ± 0.40	−14.9σ	−4.2σ	left out
the direct mass, calibrated as by Butenschoen et al. (2016)	172.00 ± 0.47	+0.1σ	+1.7σ	admitted; not separable today
the direct mass, with the difference carried as 0 ± 0.52 GeV (PDG 2026)	172.57 ± 0.64	−0.5σ	+1.4σ	admitted; not separable today

- **The alignment kept.** The pole mass, as written. Pole and MS-bar are one alignment in two rulers, so nothing down-wind separates them. The threshold scan reads it best in MS-bar, to about 0.04 GeV, against about 0.12 GeV as a pole mass.
- **Table 80.** Row 7 is compared with the pole average, 172.1 ± 0.6 GeV, at −0.7σ, and its chance of passing if the calculator is wrong is 0.67.

Table 85. What the rule's choices do to the set's power. Scored as in section 17.1, with 20,000 draws (choices_power.py); the first row is the earlier definitions, and the last the deposited ones, which blindtest.py scores in Table 80 within sampling error. At 2 GeV, ΔG today is Zhao et al. evolved, 0.158 ± 0.039, with the collider's 10%.

Choice	ΔG passes if wrong	Top passes if wrong	The seven if right: median B; P(B > 100)	The seven if wrong: median B; P(B < 1/100)
the earlier definitions: ΔG at 10 GeV², the top against the direct average	0.34	1.00	310; 0.72	2×10^{-13}; 0.93
ΔG at 2 GeV	0.54	1.00	160; 0.61	6×10^{-12}; 0.91
the top against the pole average	0.34	0.67	440; 0.77	6×10^{-14}; 0.94

Choice	ΔG passes if wrong	Top passes if wrong	The seven if right: median B; P(B > 100)	The seven if wrong: median B; P(B < 1/100)
both: the deposited definitions (Table 80)	0.54	0.67	230; 0.68	2×10^{-12}; 0.92

Standing. Both definitions are fixed by the behavior rule: ΔG = ⅙ at μ = 2 GeV, and the top as a pole mass, with m_t(m_t) = 162.76 ± 0.37 GeV in MS-bar the same alignment. The lattice match that fixes the scale is the reason for the choice, not evidence for it, and the Electron-Ion Collider tests it. The choices cost power: if the calculator is right, the set's median B falls from about 300 to 240, and the three predictions that carry it pass together by chance 1 in 38 rather than 1 in 60 (Table 80; Table 85).

17.3 The behavior rule across the set

Each scored prediction and each recorded number is carried to the observables it touches down-wind (behavior_audit.py; the values are those of sections 4.10 and 5 to 15). A miss that every alignment shows, the standard one included, is inherited: the rule cannot remove it, and it stays on the record.

Table 86. The frozen set under the behavior rule.

#	Alignment kept	Down-wind observables	Largest miss	Under the rule
1	$\sin^2\theta_{12}$ from the 15° tilt with the first column fixed	δ_CP tied to the θ_{23} octant, ±75° at $\sin^2\theta_{23} = 0.56$, against T2K and NOvA's −85° (section 10.1); the rival frame with the second column fixed	consistent; the rival frame misses by 5.5σ	kept; the rival frame left out
2	$\sin^2\theta_{13} = \sin^2 15°/3$	the same frame	consistent	kept
3	α = 90°	180° − β − γ: 91.0 ± 2.8° with HFLAV's γ, 89.4 ± 4.0° with PDG's	+0.4σ	kept
4	Ω□ = π²/32 in flat ΛCDM	the share 1 − π²/32 against Ω_Λ, −0.9σ; DESI's BAO, −1.3σ; with the	4.6σ against the distance ladder (Riess et al. 2022)	kept; inherited, since Planck's own H_0 misses by 4.9σ

#	Alignment kept	Down-wind observables	Largest miss	Under the rule
		CMB's $\Omega_m h^2$ = 0.1430 ± 0.0011 (Planck Collaboration 2020), H_0 = 68.09 ± 0.26 km/s/Mpc		
5	$\Delta G = \frac{1}{8}$ at 2 GeV	the lattice at 10 GeV² (Table 82); the sign, positive	+0.2σ	kept; chosen by the rule
6	$\Delta\Sigma = \frac{1}{3}$	$L = \frac{1}{8}$ at 2 GeV by the sum rule, against the folded 0.152 ± 0.043; HERMES; the lattice	+1.1σ	kept
7	the top's pole mass, 172.57 ± 0.38 GeV	the pole average; the electroweak fit; the edge, crossed on the metastable side; the Higgs from the top, −0.1σ	+1.4σ	kept; the unconverted MS-bar reading left out (Table 84)
8	$r_p = 4\hbar/(m_p c)$, prior art	CODATA 2022	−0.8σ	kept
9	m_τ from Koide's relation, prior art	PDG 2026	−0.4σ	kept
—	Σm_ν = 59.2 meV, normal ordering	the 64 meV cap, inside; the ΛCDM effective mass	−3.4σ	inherited: the ordering's floor, 58.8 meV, misses the same way
—	$w = -1$	DESI with supernovae prefers a drift	2.8σ to 4.2σ	inherited from ΛCDM; no alignment of the calculator avoids it

#	Alignment kept	Down-wind observables	Largest miss	Under the rule
—	Ω_k = 0	DESI with the CMB; the CMB alone	+2.1σ; −1.7σ	inherited; the two leans oppose
—	r ≈ 0	r < 0.036 (BICEP/Keck Collaboration 2021)	inside	kept
—	the vacuum energy scale, 2.02 ± 0.20 meV	ρ_Λ^¼ = 2.24 meV	+1.1σ	kept

- **The scored set.** All nine predictions pass. Two needed the rule to choose between alignments, ΔG and the top; the other seven have one alignment each, and nothing down-wind is missed.
- **What the rule cannot do.** The cosmological misses are shared with ΛCDM: the same data miss the standard model of cosmology the same way, so no alignment the calculator offers removes them. They stay on the record, and Euclid and the final DESI data decide them. Dropping them because the data lean against them would be fitting after the data, which the scoring rule counts as no evidence (section 17.1).

17.4 What decides the numbers outside the frozen set

Table 80 gives the deciders for the nine frozen predictions. Table 87 gives them for the rest.

Table 87. Deciders for the numbers outside Table 80.

Number	Current state	Decider	When
the neutrino sum, 59.2 meV, normal ordering	the cap is 64 meV (DESI with the microwave background, 95%); the ΛCDM effective mass sits 3.0σ below the floor, so the floor stands only if that preference is a systematic or w is not −1	the final DESI and Euclid with the microwave background on the sum; JUNO and DUNE on the ordering; KATRIN on the scale	this decade
the hum, 2.02 ± 0.20 meV	2.24 meV	α_s to ±0.0002, which moves the crossing; a reason for the crossing scale	with the top scan

Number	Current state	Decider	When
w = −1 and the share 0.6916	a drift preferred at 2.8 to 4.2σ; Ω_k = +0.0023 ± 0.0011 from DESI with the microwave background in ΛCDM: one datum read two ways	Ω□ to ±0.003, Ω_k to ±0.001, and w_0 and w_a (Euclid; the final DESI)	2027 to 2030
δ_CP by octant	the T2K and NOvA joint fit, best fit −85°, consistent with the frame's ±75° to ±90° (section 10.1)	DUNE; Hyper-Kamiokande	2030s
held mass in place of dark matter	fits four observations; the acoustic peaks closed (cold, clustering, photon-blind)	what it is made of: microlensing from 10^{17} to 10^{23} g (Subaru HSC; Roman), or a wave-like signature	open
the foundation, with no feed from black holes	the coupled holes, k = 3, disfavoured at about 2σ (Andrae and El-Badry 2023; Rodriguez 2023)	black-hole masses at fixed stellar mass across redshift (JWST)	this decade
Ω_k = 0 if the centre ends, ≤ 0 if it turns	0.0007 ± 0.0019 (Planck with BAO); −0.011 +0.006/−0.005 (the microwave background alone, closed at 1.7σ); +0.0023 ± 0.0011 (DESI with it, in ΛCDM, open at 2.1σ); flat with w free	Euclid and the final DESI to ±0.001	2027 to 2030
r ≈ 0 if the centre turns	r < 0.036 (BICEP/Keck Collaboration 2021)	LiteBIRD to ±0.001, launch planned for Japan's fiscal year 2032 (Ghigna et al. 2024); inflation's simplest potentials give 0.003 to 0.01	2030s

17.5 What decides the standard theories

The standard theories face deciders of their own in the same years, and four of them are the calculator's too. Table 88 lists them, with what would show each standard commitment wrong.

Table 88. Deciders for the standard theories' open commitments.

Commitment	Standard theory	Today	What would show it wrong	Decider	The calculator
w = −1 for all time	ΛCDM	a drift preferred at 2.8 to 4.2σ (DESI Collaboration 2025), 3.2σ after DES's recalibration	a confirmed drift	Euclid and the final DESI data, 2027 to 2030	the same commitment, derived from a held store
flatness	inflation	Ω_k = 0.0007 ± 0.0019 (Planck with BAO); +0.0023 ± 0.0011 (DESI with the CMB)	curvature beyond about ±0.001	Euclid and the final DESI data, 2027 to 2030	flat if the centre ends; Ω_k ≤ 0 if it turns
the neutrino mass sum above the oscillation floor	ΛCDM with three neutrinos	the effective mass 3.0σ below the normal-ordering floor (Elbers et al. 2025)	a sum confirmed below 58.8 meV	the final DESI and Euclid with the microwave background; KATRIN on the scale	the same floor, at 59.2 meV
primordial gravitational waves	inflation's simplest potentials, r ≈ 0.003 to 0.01	r < 0.036 (BICEP/Keck Collaboration 2021)	no r above 0.002	LiteBIRD, launch planned for Japan's fiscal year 2032 (Ghigna et al. 2024)	r ≈ 0 if the centre turns: the measurement that separates the two
the spectral tilt	plateau inflation, 1 − 2/N = 0.964 to 0.967	0.975 ± 0.003 (PDG 2026): 2.8 to 3.8σ high	the high value confirmed	the Simons Observatory	26/27, noted, pressed the same way

Commitment	Standard theory	Today	What would show it wrong	Decider	The calculator
the Hubble constant	ΛCDM with the microwave background, 67.19 ± 0.38 km/s/Mpc (Camphuis et al. 2025)	73.17 ± 0.86 from the Cepheid ladder (Breuval et al. 2024); 70.4 ± 1.9 from JWST ladders (Freedman et al. 2025)	a ladder value confirmed free of systematics	the JWST ladders	silent
the Higgs self-coupling	the Standard Model, $\kappa_\lambda = 1$	$-0.71 < \kappa_\lambda <$ 6.1 at 95% (ATLAS and CMS Collaborations 2026)	κ_λ away from 1	the high-luminosity LHC, 2030s	silent: too loose to test the balance
a cold dark-matter particle	ΛCDM's usual reading	none found: $\sigma_{SI} < 2.2 \times 10^{-48}$ cm² at 40 GeV (LZ Collaboration 2025)	none found down to the neutrino floor	the next xenon detectors	held mass; no particle

- **Shared deciders.** w, the curvature and the neutrino sum will move ΛCDM and the calculator together. A confirmed drift in w would break both.
- **The separating decider.** r separates the turned version of the calculator from inflation's simplest potentials; the ended version predicts no r.
- **The standard theories' own.** The Hubble constant, the Higgs self-coupling and the dark-matter particle are tests the calculator does not take, because it commits to no number there.

18. Origination ledger

Every piece of this paper is listed here with its originator and its closest prior work. The 61 pieces fall into four groups:

- **7 are classical results** reached independently (M), one of them, row 60, Claude's completion of her charts;
- **5 are new**, with no published counterpart found (N);
- **1 is a slip** (D);
- **48 are composite:** a classical result with a new reading, use or construction.

Twelve pieces carry a divergence:

- **six are slips or overstatements**, corrected or qualified here: rows 3, 6, 20, 21, 53 and 54;
- **one is a deliberate departure from teaching convention:** row 4;
- **five depart from published physics or from its rival proposals:** rows 34, 40, 41, 49 and 57.

Each row of Table 89 names the piece, its closest prior work and what that work says. “Not found” means not found in the searches of 9 and 10 October 2026.

Table 89. Every authored piece.

#	Piece	Closest prior work, and what it says	Verdict
1	The difference of squares as a triangle's imbalance; the virtual triangle (b, d, a)	Euclid II.5 (the gnomon); Fibonacci 1225 (three squares in equal steps); Wildberger 2005 (trigonometry in squared lengths, without roots)	M algebra; N reading and tables
2	$c^2 \pm d^2 = 2(\text{leg})^2$; $\frac{1}{2}c^2 - \frac{1}{2}d^2 = b^2$; $2c^2 - 2d^2 = (2b)^2$	elementary algebra; the congruum	M
3	The e construction and the (d, e) lock	no construction found; that two measurements fix a right triangle is classical; Mansfield and Wildberger 2017 (exact ratio trigonometry on Plimpton 322)	N construction; M fact; D: her $e^2 = (\frac{1}{2}b)^2$ contradicts the construction (section 3.2)
4	“Simplifying too soon loses info”: keep radicals factored	teaching convention reduces to simplest radical form	D, deliberately; N as a principle
5	The Tifinchi Trig Method: rotation before ratio, 30° and 60° normalised onto the 45° wave	Euler 1748 (sine on the circle); Wildberger 2005 (angles postponed)	M content; N sequence
6	“3 is the smallest integer that locks	any whole leg works at 90° steps, none at 45°	D as stated; N: the radius-3 sector of 40° equals the unit circle

#	Piece	Closest prior work, and what it says	Verdict
	increments of 45 degrees into integers”		
7	Root families	the square-free part, square classes of the rationals (standard number theory)	M mathematics; N name and grouping of triangles
8	What enters each table; repeats; triples	Girard 1625, Fermat 1640, Euler; Diophantus–Brahmagupta–Fibonacci identity; Euclid II.5, X.29	M
9	Sum, difference and three-square tiers; the 4D tier empty	Legendre 1798; Gauss 1801	M theorems; N reading and the empty-tier corollary
10	Root Elements: the elements placed by atomic number	none found	N; carries no chemistry (section 3.5)
11	The tables as a power calculator	Heaviside 1886; Kennelly 1893; Steinmetz 1893	M physics; N chart
12	Fibonacci read off the two tables	Lucas 1878; Vajda 1989	M identities; N teaching method
13	Centroids at ½√2, 1, ½√3, 1; “flip inside out → root families scaled down”	classical centroids; inversion in a sphere	M; N flip-grid reading
14	The Balance Chart	Archimedes, *On the Equilibrium of Planes*	M; N chart
15	The dimension converter, nested Pythagorean	Grassmann 1844; Cayley 1845; Schläfli 1852	M; N: m, n, o as motion
16	The converter's radians form	$\cos^2\theta + \sin^2\theta = 1$	M

#	Piece	Closest prior work, and what it says	Verdict
17	The root ladder 1, √2, √3, 2 as the dimensional ladder	diagonals of unit cubes	M; N reading
18	Manhattan and Euclidean distance as 90° out of phase; each dimension a new phase	Minkowski (taxicab metric); Krause 1975	M metrics; N phase reading
19	Dual solutions and the Möbius demonstration	the principal-root convention; Möbius 1865	M; N teaching use
20	“Every triangle has a counterpart that forms a perfect square when paired along the hypotenuse”	two copies make the a × b rectangle	D: a square only at 45°
21	The cube cut into identical pieces (bi-4, tri-3, tri-4); the tetrahedron as four cuts	Euclid XII.7; Kepler 1619	M; N labelling; D: the √5/2 label, corrected to √6/2 (section 3.9)
22	The ladder √8/2 → √5/2 → √2/2	sum cells (2,2), (2,1), (1,1), steps of 3	M; N reading
23	The phase circle: potential, container and change triangles	three-phase 120°	N construction; M parts
24	The Rosetta fold R(x, y) = (x + y, x − y) in seven scripts; six alignments at a = 2b	Sylvester 1867 (the Hadamard matrix); Dirac 1949 (light-cone coordinates)	M operation; N translation key
25	The delimiter cipher and its rules	oriented edges in algebraic topology; Leibniz 1703 (eight trigrams as binary 0 to 7)	M orientation; N notation
26	The fixed cube with tesseract turns	Steinmetz 1893 (derivative as iω);	M parts; N assembly

#	Piece	Closest prior work, and what it says	Verdict
		Hamilton 1843 (quaternions); the CPT theorem's complex Lorentz group (Greenberg 2006)	
27	J fixed by the electrical frame; Front carried onto Back	none found	N, Claude's derivation from her frame
28	Time-and-scale cells; K as z ⟶ z^i; the quaternion group	Cauchy–Riemann; Popławski 2010 (the isotropic flip); Braaten and Hammer 2006 (discrete scale invariance)	M parts; N assignment
29	Motion reversal TJT = J^3; folding, not reversing	time reversal flips current and flux (standard electrodynamics)	M; N reading
30	Folding with resolution, applied to Hill's lightning	Hill 1968, the data only	N reading; it fits shape, scaling and spectrum
31	The neighbours rule	none found as a method	N
32	q, i, φ, v inside the cube; dual corners as one-store circuits	Paynter's tetrahedron of state; Chua 1971; Vongehr 2012; Shang et al. 2015; Wang 2021	M tetrahedron; N cube placement
33	The electron's corners; resistance = memristance = h/e^2	Burke 2003 ($\sqrt{(L/C)} = h/e^2$ for quantum L and C)	M identities; N corner placement
34	The spin's half from the reaction's two stores	Sebens 2019 (charge circulates twice as fast as mass); Ohanian 1986; Blinder 2002	M factor; N store split; D: classical field theory gives ħ/3

#	Piece	Closest prior work, and what it says	Verdict
35	The proton as folded reactions	Bég, Lee and Pais 1964 (SU(6), $-\frac{2}{3}$); Ji, Tang and Hoodbhoy 1996	M; N reading and one-resolution folding
36	The vacuum as a reactive board, drawn on Bass's vacuum	Bass 2015–2024; Lamb 1995; Hobson 2013; Jaffe 2005	M; N circuit mapping
37	The balance rule and the load $\alpha_i/2\pi$ for masses	Schwinger 1948; the renormalisation group	M unit; N mass rule
38	Koide's relation as balance, DC = AC	Koide 1982; Foot 1994 (45°); Brannen 2006 (three phases, 2/9)	M relation; N reason
39	Neutrino masses from one flipped root	Brannen 2006; Rodejohann and Zhang 2011	M
40	Top and Higgs from δ_3 at 120°	Degrassi et al. 2012; Buttazzo et al. 2013; Froggatt and Nielsen 1996; Shaposhnikov and Wetterich 2010 (exactly on the edge)	N rule, about the twentieth tried; D: slightly past, not on the edge
41	The Clarke transform as the tribimaximal matrix; the 15° tilt	Clarke 1943; Harrison, Perkins and Scott 2002; Albright and Rodejohann 2009 (TM1)	M matrices and TM1; N identification, angle and reason; D against the rival tilt of Sharma and Katoch 2026
42	Quarks by magnitude; $\alpha = 90°$	Gatto, Sartori and Tonin 1968; Weinberg 1977; Fritzsch and Xing 2000	M
43	The start rule as the parity of the double cover; couplings crossing in pairs	Hamilton 1843; the Standard Model's running; Georgi and Glashow 1974	M; N reading
44	One generation as a 16-state register,	Wilczek and Zee 1982; Baez and Huerta 2010	M structure; N reading

#	Piece	Closest prior work, and what it says	Verdict
	anomalies as parity checks		
45	Gravity's two legs; the black hole as the flip; return at most 29.3%	general relativity; Popławski 2010; Christodoulou 1970	M; N reading
46	Thirteen lines of sight with shadows 1, √2, √3; the tester ratio cot 15°	Cauchy 1841; Fortescue 1918	M; N
47	The proton's spin in thirds, ΔG = ⅛	none found	N prediction
48	r⍰ = 4ƛ□	Haramein 2013 and Haramein et al. 2025 (stated explicitly); Trinhammer and Bohr 2019	M, not new
49	w = −1 from a held store; the scale at M_{23}; Ω□ = π²/32	ΛCDM (w = −1 assumed); Bass (the cancellation)	M w; N reason, scale and share; D against the evolving-w preference
50	The Quad-Core claim test	Popper 1959; Chamberlin 1890; Platt 1964	M parts; N four-core veto with an unstable state
51	Multimodal reasoning as a diagnosable style	multisensory perception; multimodal AI; Feng 2025	M uses of the term; N as a style to benchmark (not tested here)
52	The classifier of regimes: the sign of one quantity gives two branches, one boundary state, or none (section 3.14)	the quadratic discriminant; the damped oscillator's $c^2 - 4mk$; WKB turning points; the saddle-node bifurcation; Landau 1937 (order parameters)	M structure; N one classifier across domains, with the cube's corner as the critical surface after the fix in section 3.14
53	The notebook of axes and derivatives,	Newton 1687 (the laws of motion); the	M kinematics; N derivatives read as

#	Piece	Closest prior work, and what it says	Verdict
	February 2026 (section 3.13)	kinematics of constant acceleration	axes; D: the units and the first-law gloss
54	The Portable Reasoning Framework: a fixed pass order, prove-don't-prune, misread profiles, lossless reduction, failure retention (section 2.3)	Popper 1959 (falsification); Klein 2007 (the pre-mortem); red-teaming; the laboratory notebook; version control	M parts; N assembly; D: duplicated sections and conflicting pass orders (section 2.3)
55	Lightning radiating in all directions, one zigzag path seen on our legs, its scale set by the gravity well (section 15.4.3, items 1, 2 and 11)	Niemeyer, Pietronero and Wiesmann 1984 (dielectric breakdown); Townsend similarity; Briels et al. 2008; Dwyer and Uman 2014	M physics; N reading
56	Ball lightning as plasma held on the √4 side, pulsing at 2f and radiating rather than tunnelling (section 15.4.3, items 6 and 8; section 11.3)	Cen, Yuan and Xue 2014 (the second harmonic); Abrahamson and Dinniss 2000 (energy stored in vaporised soil); Kapitsa 1955	M the 2f and the store; N the held reading
57	Held mass in place of dark matter: what falls in is held on the √4 torus, and its pull appears from within (section 15.4.3, item 15; section 13.4)	Milgrom 1983 (MOND); McGaugh, Lelli and Schombert 2016 (the radial-acceleration relation); Rovelli and Vidotto 2018 (white-hole remnants); Farrah et al. 2023 (coupled black holes)	N reading, open; D from particle dark matter
58	The fabric as filament pathways and the observable universe in one path (section 15.4.3, item 12)	Bond, Kofman and Pogosyan 1996 (the cosmic web); Tully et al. 2014 (Laniakea)	M the web; N reading, open

#	Piece	Closest prior work, and what it says	Verdict
59	Gravity on a gyroscope: weight in y through the torus, in x through the spin's cling (section 15.4.3, item 19)	Schiff 1960 (the two precessions); Sciama 1953 (inertia by induction); Everitt et al. 2011	M; N reading
60	The completions of her unfinished charts: the Balance Chart's three columns, the phase circle's wedges and e column, the second Fibonacci option laid out (sections 3.4.1, 3.8.1 and 3.10.1)	classical perimeters, centroids and circle segments; Lucas 1878; Vajda 1989	M; Claude's, from her figures, marked as added
61	The one yardstick: the calculator and the standard theories scored by the same pull, standings and tests for chance, with a single-number chance convention for the standard theories' predictions and the four claims A′ to D′ (sections 2.5, 15, 16.8 and 16.9), at the author's direction	Good 1990 (judging numerology); Fisher 1932; Gross and Vitells 2010; Kass and Raftery 1995	M statistics; N assembly, Claude's, marked as added

Why the new pieces matter, and what they do not do.

- **What they do.** The mathematics (rows 1 to 25) is a coherent way of seeing classical results, in which one figure is read through sums and differences of squares. It is teachable and checkable, and its constructions are exact.
- **What they do not do.** The mechanics (rows 26 to 35) is a provenance ledger with a fixed cube, and organisation, not new measurement. The physical readings (rows 36 to 49) are new where they make numbers, and section 16 shows that those numbers do not yet carry evidence. Of rows 52 to 60, the classifier, the notebook and the framework are method; the descriptions of lightning, ball lightning, held mass, the filaments and the gyroscope are readings scored in section 15.4; and row 60 is Claude's completion of her charts. Row 61 is the method of this paper's comparison, built at the author's direction.

19. Limits of both theories

The verdicts of section 16 are limited by how they were reached. The calculator is limited by what it does not give, and the standard theories by what they leave open. This section sets out all three.

19.1 Of the calculator

- **Bookkeeping, not dynamics.** The quarter turn is a time derivative only for signals at one frequency: $J^2 = -1$ is $\ddot{q} = -\omega^2 q$. For other motion it is a reading.
- **Frames differ by layer.** The same turn is a time derivative in the electrical frame, a map between regions in gravity, and a step of growth in the cosmos; the ledger does not say why.
- **Rested corners.** The electrical frame, and so J, rests on three face meanings (load, derivative and rate). If those move, J moves.
- **Neighbours locate; they do not derive.** The neighbours rule places a misfit and gives a value only where an identity links the neighbours.
- **What the reading does not give.**
    - the fine-structure constant;
    - the mass scale of a family, and so $m_{\square}/m_e$ and m_μ/m_e;
    - a dynamical reason for the lepton phase $3\delta = Q$;
    - a rule, written before comparison, for which phases load which mass;
    - why the vacuum energy's scale is the weak = strong crossing;
    - whether the matter share stays fixed in time;
    - θ_{23} and the sign of the CP phase;
    - what the held mass is made of;
    - H_0, and why there are three generations;
    - the sizes of the strong-scale loads;
    - the irrational critical exponents of three-dimensional systems (the bootstrap's $\nu = 0.629971$ has no grid value);
    - whether a black hole's centre ends or turns.

19.1.1 What the calculator does not give

Table 90 gives each item of the list above with the exact missing step and what would supply it. Every rule of the calculator is an identity or standard physics except one, the load, which has two halves:

- **Equal shares.** The first half is exact.
- **Slightly past.** The second half, slightly past by one phase's coupling squared per sphere per cycle, has:
    - a geometric size;
    - one exact anchor, the electron's moment;
    - one law for the couplings, the one-loop counts;
    - one tested application to a mass, the top and the Higgs.

Nothing in Table 90 is filled with a chosen number.

Table 90. What is not given, why, and what would supply it.

Not given	Why	What would supply it
α = 1/137.035999	No theory derives it; it is an input of the Standard Model. The alphabet has nothing within 3σ, and any candidate must also give its running to 1/127.95 at the Z	a formula, written first, for α at zero momentum and for its running
the mass scale μ and the hierarchy: m□/m_e = 1,836.15 and mμ/m_e = 206.77	balance fixes a family's shape (a = √2) and its phase (δ = Q/3). The DC level is free: μ = 17.7158 √MeV, so μ^2 = 313.85 MeV, within 0.35% of m□/3 but not on it. So is the ratio of μ to v or to Λ_QCD	a rule tying μ to v or to Λ_QCD, written before the comparison
the dynamics of the phase	3δ = Q holds to 0.6σ for the one family inside the domain whose three members are measured. There are exact constructions of 2/9 on the cube, but no equation of motion produces it	a reason the three-phase vector's azimuth equals a third of its balance ratio
which phases load which mass	For the couplings the law is the one-loop count. For a mass, a phase whose carrier is much heavier than the state decouples, its load suppressed by the mass ratio squared, so the loading phases depend on the state's scale: the electron is loaded by the photon alone, the top by the phases it touches. Adding the colour phase to δ_3 as a third phasor at 240° gives 1.03% and a top of 172.69 GeV, −0.2σ, so the present data do not separate the two	one rule, written before the next comparison, that picks the phases and their angles from the state's scale; then the threshold scan at ±0.035 GeV
the coefficient of the hum	v^2/M takes the factor in front as 1 and M as the weak = strong crossing, the only one of the three crossings that gives the measured order of magnitude. Bass's cancellation is the	a reason the Higgs balances at the weak = strong crossing rather than at another scale; α_s to ±0.0002 for the crossing

Not given	Why	What would supply it
	symmetry and fixes that a residual exists; naturalness puts its coefficient at order one, as with the seesaw's $y^2 \approx 1$ and the top's y_t = 0.99. The scale is the calculator's one choice here (section 13.1)	
the constancy of the share	$1 - \pi^2/32$ is stated for the present split. A split fixed in time would remove the coincidence problem, but it needs the matter and held densities to scale together, which is not shown: the two holds dilute differently and were equal at z = 0.31. The split that can be constant is Ω_c/Ω_b = 16/3, which is noted, not counted	Ω□ to ±0.003 at several redshifts (Euclid)
θ_{23} and the sign of δ_CP	the tilted frame gives θ_{12} and θ_{13} and ties δ_CP to θ_{23}, but has no second angle, and the alphabet is dense near $\sin^2\theta_{23} = 0.56$. The joint T2K and NOvA fit's −85° is consistent with the frame's near-maximal phase; the negative sign it picks, the frame does not give	a second angle with a reason; DUNE and Hyper-Kamiokande
what the held mass is made of	the acoustic peaks are closed: cold, clustering, photon-blind held mass reproduces them whatever it is made of. Black holes can be all of it only from 10^{17} to 10^{23} g, formed at 30 TeV to 3×10^7 GeV, where the calculator has no scale; compact objects from 10^{-4} to 10^3 M⊙ are capped near a percent of the dark matter (Mróz et al. 2024)	a scale on the ladder for the lump, or a reason it is not a lump: a wave-like field, or Planck-mass remnants

Not given	Why	What would supply it
the hadrons	outside the domain: hadron masses are field energy, not Yukawa legs, so the balance rule does not apply. Near-degenerate triples give Q → ⅓ by arithmetic (p, Λ, Ξ: 0.335)	nothing; this is the boundary of the rule's domain, seen from the strong side
H_0	the expansion rate today is an initial condition, not a constant of nature. No theory derives it, and the measurements disagree: 67.4 from the microwave background, about 73 from the distance ladder	—
three generations	the calculator reads three free phases with a fourth level held. The Z width allows three light neutrino kinds (2.9963 ± 0.0074; Janot and Jadach 2020), and a heavy fourth generation is excluded by Higgs production	a reason the held level cannot be a free generation, one that predicts something not already known
the strong-scale loads	g_A and μ□/μ□ are the SU(6) statics loaded by −23.5% and +2.75%. The per-cycle load at that scale is α_s/2π ≈ 0.08 times a count of order one to ten, so no small-load law exists there, and the sizes belong to lattice QCD	nothing; this is the boundary of the rule's second half, derived
critical exponents in three dimensions	In two dimensions the Ising exponents are rational (⅛, 1, 7/4; Onsager 1944), because the plane has an infinite conformal symmetry, and above four dimensions mean field gives ½. In three dimensions the bootstrap gives irrational values: Δσ = 0.5181489 and Δε = 1.412625 (Kos et al. 2016; Chang et al. 2025), so β =	nothing; derived as a boundary

Not given	Why	What would supply it
	0.326419 and v = 0.629971. The grids hold where the exponents are rational, and cannot hold in 3D	
ends or turns at the centre of a hole	General relativity permits both. The ended version keeps $\Omega_k = 0$ exactly; the turned version predicts $\Omega_k \leq 0$ and $r \approx 0$, and owes the entropy of the start. The 2025 data pull both ways: the microwave background alone leans closed at 1.7σ, and DESI with it leans open at 2.1σ in ΛCDM, the lean $w = -1$ cannot absorb (section 13.5)	Euclid's Ω_k to ±0.001; LiteBIRD on r; a derived, not assumed, stiff phase for the entropy
two formulas at the meV scale	the hum is $v^2/M_{23} = 2.0$ meV, the seesaw's own form; the neutrino masses come from the flipped Koide branch, with scale $(\Sigma \pm \sqrt{m})^2/3 = 29.6$ meV. Both are meV numbers from different rules, a factor 14.7 apart with no cube number in it	one formula that gives both, or a reason they must differ
the right angle in the quark sector	the up and down legs are perpendicular within errors, and α = 90° is taken as the value (section 10.1). The calculator reads the right angle and does not derive it; V_{cb} and V_{ub} have no leg value, because the top's leg is v	a reason the two textures' phases are perpendicular; Belle II and LHCb to ±2° on α
the scale of the spin thirds	At leading order ΔΣ stays fixed while ΔG grows with the resolution and L falls to match, so the three can equal ⅙ together at one scale only, and the calculator's structure does not yet name it. The behavior rule fixes it at 2 GeV, where the	a derivation of 2 GeV from the calculator's structure, which would turn the rule's choice into a prediction

Not given	Why	What would supply it
	lattice quotes the other two (section 17.2)	

Selectivity and the domain. The alphabet's reach is in section 16, and the balance rule's domain in section 9.1. Both are limits of the same kind as Table 90's: they say where a match cannot count, and where the rule does not apply.

19.2 Of the standard theories

The standard theories are limited too, and the same yardstick records where. Table 91 lists what they do not give, why, and what would supply it. None of these is a failed prediction counted in section 16.8 unless it says so; most are places where the theories take an input or are silent.

Table 91. What the standard theories do not give.

Not given	Why	What would supply it
the values of the 19 parameters (26 with neutrino masses)	they are inputs of the Standard Model: the masses, the mixings, three couplings and the strong phase are fitted	a deeper theory that computes them; none is confirmed
the vacuum's energy	quantum field theory's sum of zero-point energies is about 10^{120} too large (Weinberg 1989); ΛCDM fits Λ	a mechanism for the cancellation; Bass's emergent vacuum is one proposal (Bass 2015 to 2026)
the neutrino masses and their kind	the minimal Standard Model has none; whether neutrinos are their own antiparticles is open	neutrinoless double-beta decay; KATRIN on the scale
what dark matter is	ΛCDM's cold component is unidentified; no particle has been found (LZ Collaboration 2025)	a detection, or a reason there is none
the excess of matter over antimatter	the Standard Model does not produce the observed amount	new sources of CP violation, or a different history
why the strong phase is so small	$\bar{\theta} < 10^{-10}$ is a fitted near-zero with no reason in the theory (PDG 2026)	an axion, or another mechanism
gravity with quantum theory	general relativity and quantum field theory do not combine; general relativity's own singularities end it	a quantum theory of gravity

Not given	Why	What would supply it
what drove inflation, and why the start was smooth	the inflaton is not identified, and the low entropy of the start is assumed (Penrose 1979)	primordial gravitational waves; the spectral tilt
the Hubble constant	ΛCDM fitted to the microwave background gives 67.19 ± 0.38 km/s/Mpc, 4.9 to 7.1σ from the ladders (Riess et al. 2022; H0DN Collaboration 2025; Freedman et al. 2025)	the ladders free of systematics, or new physics before recombination
three generations	an input of the Standard Model	a reason; the calculator's three phases are a reading, not a derivation
why Ω□ and Ω_Λ are close today	the coincidence problem; ΛCDM fits both	a reason the two are comparable now; the calculator states its share for today only

19.3 Of the method, and of the author's pages

- **One team.** The author and the model built the calculator and also designed and ran the tests. The rules were fixed in code before the runs, but by the people who wanted the answer. A blind replication is offered and has not been done (section 20.2).
- **Selection before the test.** The alphabet was fixed on 3 October 2026, after many of the calculator's numbers had already been compared with data. The calibration measures the alphabet's reach, not the history of the choices made with it; the δ_3 rule, for one, was about the twentieth candidate tried.
- **One outside list.** The 29 targets are one reasonable choice. Another list would give another count, although the nulls are judged on the same list and so share its bias.
- **Statistics.** Pulls combine errors in quadrature, and correlations, among the proton's spin pieces or the cosmological combinations, are not modelled. The chance probabilities of test 5 are estimates of each search's trial space, strict and generous.
- **AI assistance.** Every number was computed in code. Errors of judgment can still pass both author and model.
- **Novelty.** “Not found” means not found in searches run on 9 and 10 October 2026. The delimiter notation, the logical cube of opposition, and point groups used as ledgers were not searched exhaustively.
- **Evenhandedness.** The standard theories' predictions were scored with a convention that differs in form from the calculator's, because a single computed number has no alphabet to search (section 16.8). The floor on the window and the decade of possible values were chosen to be conservative toward the standard theories; a narrower prior lowers their total from 102 to 70 bits, and no conclusion changes.

- **The standard theories' failures.** Which standard predictions count as failures depends on judgments the field has not settled: the lithium deficit may be stellar, and the Hubble gap may sit in the ladders. They are counted as failures here because the rule of section 2.5 counts the calculator's misses that way.

Of the author's pages. Her notes are working pages. Notation varies between them, e for one: on one page it is the doubled area of a sliver, on another the leftover a − b. Several contain slips, which are corrected where the paper uses them. Two items she left unfinished, Option 2 of her Fibonacci methods and the blank columns of her Balance Chart, are completed from her own figures in sections 3.4.1 and 3.8.1, and marked as added.

20. Verification

Every number was computed by script, and every test can be rerun from its hash (section 20.1). Every source in the reference list says what it is cited for.

20.1 Calculations

Constants are CODATA 2022 with the exact SI values of h, e and c, PDG 2026, and NuFIT 6.0 and 6.1; identities were checked at 25 to 30 significant digits. The 39 scripts of Table 92 were rerun on the evening of 10 October 2026 and reproduced their recorded outputs. Two further scripts were written for this paper: cosmos_check.py computes the numbers of section 13, and std_bits.py the standard theories' scores of section 16.8. A port of the engine to JavaScript, built for the calculator's live mode, recomputes every committed number from the same inputs; it reproduces the scripts' values.

Table 92. Scripts and what they check.

Script	Checks	SHA-256
tesseract.py	J unique, $J^4 = 1$, det J = +1; 16 corners and 32 edges kept; the four layers; the torus; the 24-cell; Kruskal's regions	3dd09fae35613661f932db42463ec0a34408fcf19a1516d1e666ee171be1f7f1
ring.py	the cells and their shared faces; the strings; the circles; the electron's corners; the inverse-cell pulls	2c448838cab98b62962a2b567e97aa1d19caf7049c63bb2f82dc9aad1603245b
timescale.py	K, JK and T; the two-variable half turns; z → z^i; the electron's two scales; the isotropic flip	4d4df814fbbbeec7dbb8fc65dc708d7d55a9cd0e4742e62e43346ad2d5d6aa2a
group_check.py	the strings' orientations and inversions; the dual corners	cea17254f1e523f019353d54bbdadfc2efa32c8e9c45346e0ce9a5ee42307b3c

Script	Checks	SHA-256
sphere_check.py	the levels 1, √2, √3, 2; the ball shares π/6 and π²/32	aacd6dc93564836068d580a3f1211d6a5124feab77a8dc4acda93c66aeb14242
reaction.py	the electron's two stores; ħ/2 and g = 2; ħ/3 in the classical field; the tank's 2π	79b0fa64933f114eed8d2241dc88344c9586eaf3dc33c771e7718331c702704c
neighbours.py	the spin from the half turn; the misfits' pulls	f260decc3a9b8cc8a7d0a6c3dc9e5d0afa385aab8ae757e866294c922b86e164
folded.py	the folded proton; four-loop α_s; ΔG and L at 2 GeV	bac21d219729d107fd483b955a9fc3e85ef6cc6055e78ad65bdba88c231d8b71
tables.py	Tables 3 and 4 with their elements; the tier tally of all 118 elements	71265574d9c28475278cd8a49464c4718f0d501d796589ac5efbee723cf71f69
blindtest.py	the calibration (test 2) and the predictions' power (section 17.1), with the definitions fixed by the behavior rule; run 2026-10-10T22:36:17Z	b86d926aceaab119098aa47e8bf7c2d4b489dc2b5e930562a27e04b5d837bc74
n3hist.py	the distribution in Figure 2: mean 19.316, P = 0.79	b9156cf2b0d33c8afcef8e6e3ee18d8115b94047519889225a2596dbb06c9832
decide.py	tests 3 to 5	7e0117ca95b5092c58fa7b7f6518eeeaede1e216db67101c2d9e1478a6de6024
precise_who.py	which precise constants random alphabets hit	be02f6bf62478c93c360c926229304b8a91f03526fc91d605169f76e668a97eb
higgs2.py	the Standard Model running: two-loop couplings and three-loop QCD from the matching of Buttazzo et al. (2013); used by every Higgs script	2a111872f90a11ddf010d04b2292573097e09669729db844864fb9636e2525aa

Script	Checks	SHA-256
appA.py	the Balance Chart completed, the phase circle, Option 2 and the corrections to the element pages (sections 3.4.1, 14.5, 3.8.1 and 3.10.1)	e3623e8a8e92c85fe252461d55b46b9d5bdc86f5535da09184c07952808638ae
ledger118.py	every one of the 118 elements placed (section 14.5)	7d19eddd7647a8e2c316d2a40c9ac81de47d922cec1bc50e9894b66ee5ed62bb
appB.py	the checks of the framework and its engine (section 2.3)	9ad7aba7b9e847f53f35fcd4b8c7c0eb45242b075cd0fc32e1d59c8a0c651f44
appC.py	the complex cells, the converter, the dimension steps, the flip rate and the bands (sections 3.4.2, 3.7.1, 3.7.2, 4.4.1 and 4.9.1)	b19a7d28343721a5145212bb242ff9a8e0467218a09d40864f85759dc9707109
numbers.py	the four structures' numbers (section 4.7)	b78bd1e947eead172a6a1152a55e899d4fd793ebd1514a06220718640d301523
inversion.py	the inversion overlap volumes in 2D, 3D and 4D (section 4.7.1)	9f97f4f463914c37f34aac114f69f0c95a4de25dd68d5114ec4e07004aafb7fc
tbar.py	the T-bar periods and flip times (section 4.7.4)	115e0b7f11b52083d2da34ba71b8c9db1637a50fc78ac7ced36cc4e7e35fb118
edge2026.py	the edge with PDG 2026, and the loads at 0°, 90°, 120° and 180°	513f2a901ad7b4a4a76e49b894ee6282ea41da8f38ddb3908afadac80d9f9839
higgs_rule2026.py	the balance rule's top and Higgs, with the error budget and the matching's theory errors	ab0687bc92be3e960a6a26bc91140e9d39da148fa39f2ace45ebbcbb90cc3263
higgs_angle2026.py	the phase read from the masses, and every load tried	c3941342656717a993602deaa663b6f812fc1a80a8e8137fe11adff8db5403e8

Script	Checks	SHA-256
appG.py	λ against energy; the one-loop pieces; the code check; the gravity boundary; the double root; the bounce	6b060b71669f84cd917b5ff98454601024de48c2ebf5c1551cb23c5658df6009
appG2.py	the fifteen offsets tried as the vacuum's; gravity's strength at the boundary	76d2a4d6d7eacb891919b6c0d0110b71573c104618e4a1942f5775d2e719611c
appG3.py	the top-mass results against the rule and the edge; the open-data pulls	a3cc4d0df6b018939a3b0d23dea6fa9ef4fb60fe5eb7efd1a67788f3523d1110
higgs_decays2026.py	the Higgs decays computed as reactions	ae03d4a4f7afc9a1997bd49b00143326cdf91a6d41eab91e1533f3a26e418f65
ball_check.py	the most power a line can deliver to the 2012 ball	51e875de25a71dd27bac91aa6eef89dd688fe86265afc50d66353b864bbc061e
floquet_check.py	the Floquet growth of the Higgs after inflation	964cd61733bd722649a4edfb49e0c252d37d6dd08800b5283e36e863c72e79ca
appK.py	the Koide numbers with PDG 2026; the mixing tilt; TM1 and TM2	d5d52b2662eabcf0ba53834f8e27daacfad0022659c8b693596777ed99de3d56
alphabet_counts.py	the 998-number alphabet rebuilt, and its members near each target	4e9a37531b3957cc3c7dc6779d120099ef2c70805106c28c8c576bb2ccdf18fb
deltaG_scale.py	the scale at which ΔG = ⅛, from the lattice values (section 17.2; Table 81)	587d15ef79362181a14a99a20dbd6128c4dc50ebd60691ce07370241a923c511
top_scheme.py	the top mass by scheme, and the rule in MS-bar (section 17.2; Table 83)	7e57f4d67a2f58b6e051c3b1871bf6138085abb9e44ea44fd81a1f626039749a
choices_power.py	what the rule's choices do to the set's power (section 17.2; Table 85)	5d196150668d82b159518f30a3c8fefa5cfae99bae256cf5fbdf918f083c1a44

Script	Checks	SHA-256
downstream_dg.py	the behavior rule on the scale of ΔG (section 17.2; Table 82)	a119be14809c96a044e287eaa576671561ad78785c029bf92c81ab2ae924d118
downstream_top.py	the behavior rule on the top's alignment (section 17.2; Table 84)	dcaae9f60f0925a7be25f5a6ddbb268f6581f057d62ed0d032fb2f3964359f49
behavior_audit.py	the behavior rule across the set (section 17.3; Table 86)	feb3fadc7ad1ca9cce41e77015e1889143d905d315de08fcb3e1b5533290e503
alphabet_dg2.py	the alphabet near ΔG at 2 GeV (section 16.7.4)	3fbacd14bcab76d1e9df3ad9fbfa5ca84533257f00379c822d7e9607bd8c14eb
cosmos_check.py	the hum, the dark-energy density and the overshoot; the share, the two holds, q_0 and E(z); H_0t_0; the Hubble sphere; the collapse times and the interior volume; the spectral tilt against inflation (section 13)	f6e6d4b6badf967bd843f3e45f5a5f77ad76c78fe1c4ae651b3a3ea3ceb32cf4
std_bits.py	the standard theories' predictions scored for chance, Fisher's combination and the narrower prior (section 16.8)	fca390c00cf4a93a09273393851db1b318e817bea54bb30f7979c16af85f536a

20.2 What an independent check would add

The tests were designed and run by the same pair, author and model, that built the calculator. Three checks would remove that dependence:

- a blind replication, in which someone who has not seen the results rebuilds the alphabet from section 16.1, the targets and the nulls, and reruns tests 2 to 5;
- an independent derivation of J from the electrical frame of section 4.3;
- an independent choice of the prior and the window for the standard theories' predictions (section 16.8), with Table 77 recomputed from it.

All three are offered and none has been done.

21. Conclusion

Computed with the Tesseract Logic Calculator, reality has a definite structure: a reactive board whose states sit on the 27 points of a side-2 cube and change by quarter and half turns through a tesseract. The

calculator rebuilds four established structures exactly: the generation of matter as a 16-state register, the two tables of special relativity, the classical tests of gravity and the shells of the atom. Where the standard theories take inputs it gives numbers: Koide's relation as balance with its phase 2/9, the τ mass, the neutrino masses, the top and the Higgs from the vacuum's edge and one load, the mixing angles from a 15° tilt, the proton's spin in sixths, the matter share $\pi^2/32$ and the vacuum's energy scale. Every committed number sits within 1.6σ of its primary measurement.

What the evidence says, on one yardstick.

- **The structure.** Where the calculator restates established law it is exact, as the standard theories are by construction.
- **The numbers.** The calculator's own numbers are what an alphabet of its density produces by chance, 0.6 to 4.0 bits. The standard theories' predictions of independent measurements carry 102 bits. The strong evidence in the calculator's lepton sector, 14.2 bits, belongs to relations Koide and Brannen found, which the calculator explains but did not find.
- **The failures.** The standard theories carry six, among them the vacuum's energy, where the calculator gives a number within 1.1σ, and the Hubble constant, where it is silent. None of the calculator's committed numbers misses.
- **The verdict.** Under the author's Quad-Core rule, the claim that the calculator is how reality works is rejected as of 10 October 2026, and so is the claim that the standard theories are the whole of how it works. The standard theories' equations, structures and numbers stand. The calculator's notation stands, its readings are unstable, and its numbers have yet to beat chance.

What decides it. Nine frozen predictions decide the calculator's numbers between 2027 and the 2040s. The first and sharpest is JUNO's six-year $\sin^2\theta_{12}$:

- today's central value, 0.3096, sits 5.3 of JUNO's projected standard deviations below the calculator's 0.318;
- a final result near 0.318 would be the calculator's first evidence beyond chance;
- a result near today's value would refute the 15° tilt outright.

In the same years Euclid and the final DESI data decide $w = -1$ and the curvature for both theories at once, and LiteBIRD's r separates the turned version of the calculator from inflation's simplest potentials.

What stands either way. A teachable mathematics of imbalance; root families with their tables; a notation that tracks where every datum came from; eighteen descriptions of motion scored against observation, none contradicted; and a decision rule, the Quad-Core test, that judged her own system and the standard theories on the same terms. All of it is the author's, stated here in full.

Supplementary tables

These tables belong to the main text and keep its numbering; the section each serves is named in its title.

Table 93. All 118 elements (section 14.5).

Z	Element	Sum cells $a^2 + b^2$	Difference cells $a^2 - b^2$	Three squares	Tier	On the 10 × 10 page

1	H	–	–	–	square (1^2)	square
2	He	(1,1)	–	–	sum	sum
3	Li	–	(2,1)	–	difference	difference
4	Be	–	–	–	square (2^2)	square
5	B	(2,1)	(3,2)	–	both	both
6	C	–	–	(2,1,1)	three squares	off the page (three legs)
7	N	–	(4,3)	–	difference	difference
8	O	(2,2)	(3,1)	–	both	both
9	F	–	(5,4)	–	square (3^2)	square
10	Ne	(3,1)	–	–	sum	sum
11	Na	–	(6,5)	–	difference	difference
12	Mg	–	(4,2)	–	difference	difference
13	Al	(3,2)	(7,6)	–	both	both
14	Si	–	–	(3,2,1)	three squares	off the page (three legs)
15	P	–	(4,1), (8,7)	–	difference	difference
16	S	–	(5,3)	–	square (4^2)	square
17	Cl	(4,1)	(9,8)	–	both	both
18	Ar	(3,3)	–	–	sum	sum
19	K	–	(10,9)	–	difference	difference
20	Ca	(4,2)	(6,4)	–	both	both
21	Sc	–	(5,2), (11,10)	–	difference	difference
22	Ti	–	–	(3,3,2)	three squares	off the page (three legs)
23	V	–	(12,11)	–	difference	off the page (four legs)
24	Cr	–	(5,1), (7,5)	–	difference	difference
25	Mn	(4,3)	(13,12)	–	square (5^2)	square
26	Fe	(5,1)	–	–	sum	sum
27	Co	–	(6,3), (14,13)	–	difference	difference
28	Ni	–	(8,6)	–	difference	difference
29	Cu	(5,2)	(15,14)	–	both	sum
30	Zn	–	–	(5,2,1)	three squares	off the page (three legs)
31	Ga	–	(16,15)	–	difference	off the page (four legs)
32	Ge	(4,4)	(6,2), (9,7)	–	both	both
33	As	–	(7,4), (17,16)	–	difference	difference
34	Se	(5,3)	–	–	sum	sum
35	Br	–	(6,1), (18,17)	–	difference	difference

36	Kr	—	(10,8)	—	square (6^2)	square
37	Rb	(6,1)	(19,18)	—	both	sum
38	Sr	—	—	(5,3,2)	three squares	off the page (three legs)
39	Y	—	(8,5), (20,19)	—	difference	difference
40	Zr	(6,2)	(7,3), (11,9)	—	both	both
41	Nb	(5,4)	(21,20)	—	both	sum
42	Mo	—	—	(5,4,1)	three squares	off the page (three legs)
43	Tc	—	(22,21)	—	difference	off the page (three legs)
44	Ru	—	(12,10)	—	difference	off the page (three legs)
45	Rh	(6,3)	(7,2), (9,6), (23,22)	—	both	both
46	Pd	—	—	(6,3,1)	three squares	off the page (three legs)
47	Ag	—	(24,23)	—	difference	off the page (four legs)
48	Cd	—	(7,1), (8,4), (13,11)	—	difference	difference
49	In	—	(25,24)	—	square (7^2)	square
50	Sn	(7,1), (5,5)	—	—	sum	sum
51	Sb	—	(10,7), (26,25)	—	difference	difference
52	Te	(6,4)	(14,12)	—	both	sum
53	I	(7,2)	(27,26)	—	both	sum
54	Xe	—	—	(5,5,2)	three squares	off the page (three legs)
55	Cs	—	(8,3), (28,27)	—	difference	difference
56	Ba	—	(9,5), (15,13)	—	difference	difference
57	La	—	(11,8), (29,28)	—	difference	off the page (three legs)
58	Ce	(7,3)	—	—	sum	sum
59	Pr	—	(30,29)	—	difference	off the page (three legs)
60	Nd	—	(8,2), (16,14)	—	difference	difference
61	Pm	(6,5)	(31,30)	—	both	sum
62	Sm	—	—	(6,5,1)	three squares	off the page (three legs)
63	Eu	—	(8,1), (12,9), (32,31)	—	difference	difference
64	Gd	—	(10,6), (17,15)	—	square (8^2)	square
65	Tb	(8,1), (7,4)	(9,4), (33,32)	—	both	both
66	Dy	—	—	(5,5,4)	three squares	off the page (three legs)
67	Ho	—	(34,33)	—	difference	off the page (three legs)
68	Er	(8,2)	(18,16)	—	both	sum
69	Tm	—	(13,10), (35,34)	—	difference	off the page (three legs)
70	Yb	—	—	(6,5,3)	three squares	off

```
the page (three legs)
 71  Lu       —            (36,35)                    —        difference     off
the page (four legs)
 72  Hf       (6,6)        (9,3), (11,7), (19,17)     —        both           both
 73  Ta       (8,3)        (37,36)                    —        both           sum
 74  W        (7,5)        —                          —        sum            sum
 75  Re       —            (10,5), (14,11), (38,37)   —        difference
difference
 76  Os       —            (20,18)                    —        difference     off
the page (three legs)
 77  Ir       —            (9,2), (39,38)             —        difference
difference
 78  Pt       —            —                          (7,5,2)  three squares  off
the page (three legs)
 79  Au       —            (40,39)                    —        difference     off
the page (four legs)
 80  Hg       (8,4)        (9,1), (12,8), (21,19)     —        both           both
 81  Tl       —            (15,12), (41,40)           —        square (9²)    square
 82  Pb       (9,1)        —                          —        sum            sum
 83  Bi       —            (42,41)                    —        difference     off
the page (three legs)
 84  Po       —            (10,4), (22,20)            —        difference
difference
 85  At       (9,2), (7,6) (11,6), (43,42)            —        both           sum
 86  Rn       —            —                          (6,5,5)  three squares  off
the page (three legs)
 87  Fr       —            (16,13), (44,43)           —        difference     off
the page (four legs)
 88  Ra       —            (13,9), (23,21)            —        difference     off
the page (three legs)
 89  Ac       (8,5)        (45,44)                    —        both           sum
 90  Th       (9,3)        —                          —        sum            sum
 91  Pa       —            (10,3), (46,45)            —        difference
difference
 92  U        —            (24,22)                    —        difference     off
the page (four legs)
 93  Np       —            (17,14), (47,46)           —        difference     off
the page (three legs)
 94  Pu       —            —                          (7,6,3)  three squares  off
the page (three legs)
 95  Am       —            (12,7), (48,47)            —        difference     off
the page (four legs)
 96  Cm       —            (10,2), (11,5), (14,10) …  —        difference
difference
 97  Bk       (9,4)        (49,48)                    —        both           sum
 98  Cf       (7,7)        —                          —        sum            sum
 99  Es       —            (10,1), (18,15), (50,49)   —        difference
difference
100  Fm       (8,6)        (26,24)                    —        square (10²)   square
101  Md       (10,1)       (51,50)                    —        both           sum
102  No       —            —                          (7,7,2)  three squares  off
the page (three legs)
103  Lr       —            (52,51)                    —        difference     off
the page (four legs)
104  Rf       (10,2)       (15,11), (27,25)           —        both           sum
```

```
105  Db       –            (11,4), (13,8), (19,16) …  –        difference     off
the page (three legs)
106  Sg       (9,5)        –                          –        sum            sum
107  Bh       –            (54,53)                    –        difference     off
the page (three legs)
108  Hs       –            (12,6), (28,26)            –        difference     off
the page (three legs)
109  Mt       (10,3)       (55,54)                    –        both           sum
110  Ds       –            –                          (7,6,5)  three squares  off
the page (three legs)
111  Rg       –            (20,17), (56,55)           –        difference     off
the page (four legs)
112  Cn       –            (11,3), (16,12), (29,27)   –        difference     off
the page (four legs)
113  Nh       (8,7)        (57,56)                    –        both           sum
114  Fl       –            –                          (7,7,4)  three squares  off
the page (three legs)
115  Mc       –            (14,9), (58,57)            –        difference     off
the page (three legs)
116  Lv       (10,4)       (30,28)                    –        both           sum
117  Ts       (9,6)        (11,2), (21,18), (59,58)   –        both           sum
118  Og       –            –                          (9,6,1)  three squares  off
the page (three legs)
```

References

Sources are cited in the text by author and year and are listed here alphabetically; the author's own works are listed under Taylor. Each entry ends with what it is cited for.

1. S. Abachi et al. (DØ), Phys. Rev. Lett. 74, 2632 (1995). The top quark observed, at 199 ± 30 GeV.

2. G. Abbiendi et al. (OPAL), Eur. Phys. J. C 45, 1 (2006). The running of α measured in small-angle Bhabha scattering.

3. F. Abe et al. (CDF), Phys. Rev. Lett. 74, 2626 (1995). The top quark observed, at 176 ± 13 GeV.

4. K. Abe et al. (Belle), Phys. Rev. Lett. 99, 011801 (2007), arXiv:hep-ex/0608046. The τ^+ and τ^- masses agree within 2.8×10^{-4}.

5. J. Abrahamson and J. Dinniss, Nature 403, 519 (2000). Ball lightning as burning networks of silicon nanoparticles made when lightning strikes soil.

6. ACT Collaboration: T. Louis et al., arXiv:2503.14452 (2025), and E. Calabrese et al., JCAP 11 (2025) 063, arXiv:2503.14454. The DR6 power spectra and ΛCDM parameters; the extended models, among them N_eff and the spectral tilt.

7. G. Agazie et al. (NANOGrav), Astrophys. J. Lett. 951, L8 (2023), arXiv:2306.16213. Evidence for a gravitational-wave background in pulsar timing.

8. M. Ahmadi et al. (ALPHA), Nature 548, 66 (2017). Antihydrogen's hyperfine splitting, 1,420.4 ± 0.5 MHz.

9. M. Ahmadi et al. (ALPHA), Nature 557, 71 (2018), doi:10.1038/s41586-018-0017-2. Antihydrogen's 1S–2S line agrees with hydrogen's to a couple of parts in 10^{12}.

10. A. Airapetian et al. (HERMES), Phys. Rev. D 75, 012007 (2007), arXiv:hep-ex/0609039. The quark spin share, $\Delta\Sigma = 0.330 \pm 0.039$.

11. C. H. Albright and W. Rodejohann, Eur. Phys. J. C 62, 599 (2009), arXiv:0812.0436. The TM1 sum rule; the CP phase tied to θ_{23}.

12. S. Alekhin, A. Djouadi and S. Moch, Phys. Lett. B 716, 214 (2012), arXiv:1207.0980. A top pole mass of 173.3 ± 2.8 GeV from the cross section; full stability needs m_H ≥ 129.8 ± 5.6 GeV.

13. C. Alexandrou et al. (ETMC), *Complete flavor decomposition of the spin and momentum fraction of the proton*, Phys. Rev. D 101, 094513 (2020), arXiv:2003.08486. The earlier lattice decomposition of the proton's spin.

14. C. Alexandrou et al. (ETMC), *Spin and momentum fraction carried by partons in the nucleon*, arXiv:2607.21230 (2026). $\frac{1}{2}\Delta\Sigma = 0.190(22)$ at $\mu = 2$ GeV; quark orbit 0.108(37)(24); gluon 0.209(26)(66); momentum and spin sums 0.995 and 0.507.

15. R. Aliberti et al. (Muon g−2 Theory Initiative), Phys. Rep. 1143, 1 (2025), arXiv:2505.21476. The Standard Model a_μ, $116\,592\,033(62) \times 10^{-11}$ (WP25).

16. D. W. Allan, Proc. IEEE 54, 221 (1966). The two-sample (Allan) variance.

17. ALPHA Collaboration, Nature 578, 375 (2020), doi:10.1038/s41586-020-2006-5. Antihydrogen's fine structure within 2% and Lamb shift within 11% of hydrogen's.

18. U. Amaldi, W. de Boer and H. Fürstenau, Phys. Lett. B 260, 447 (1991). The couplings meet at one point only with supersymmetric particles added.

19. G. Amontons, *De la résistance causée dans les machines*, Mémoires de l'Académie Royale des Sciences (1699). Friction proportional to load; static friction exceeds sliding friction.

20. F. P. An et al. (Daya Bay), Phys. Rev. Lett. 108, 171803 (2012). θ_{13} observed to be non-zero.

21. V. V. Anashin et al. (KEDR), Phys. Part. Nucl. 54, 185 (2023). The τ mass at threshold, 1776.69 +0.17/−0.19 ± 0.15 MeV, as listed by the Particle Data Group (ANASHIN 2023A).

22. C. D. Anderson, Phys. Rev. 43, 491 (1933). The positive electron.

23. P. W. Anderson, Phys. Rev. 130, 439 (1963). A gauge field gains mass in a medium: plasmons, gauge invariance and mass.

24. E. K. Anderson et al. (ALPHA-g), Nature 621, 716 (2023), doi:10.1038/s41586-023-06527-1. Antihydrogen falls at [0.75 ± 0.13 ± 0.16] g.

25. R. Andrae and K. El-Badry, arXiv:2305.01307 (2023). The Gaia black holes against cosmological coupling with k = 3.

26. A. Andreassen, W. Frost and M. D. Schwartz, Phys. Rev. D 97, 056006 (2018), arXiv:1707.08124. The Standard Model vacuum's lifetime, of order 10^{139} years, and above 10^{58} years at 95% confidence.

27. Anthropic, *Tracing the thoughts of a large language model* (2025), page. For 36 + 59, parallel internal paths, an approximate magnitude and an exact last digit, beside a described method that carries the 1.

28. A. Antognini et al., Science 339, 417 (2013). The proton radius from muonic hydrogen, 0.84087(39) fm.

29. Archimedes, *On the Equilibrium of Planes*, Book I; *Measurement of a Circle*, Prop. 1. A triangle's centroid on its medians; a circle's area as half its circumference times its radius.

30. V. I. Arnold, *Huygens and Barrow, Newton and Hooke* (Birkhäuser, 1990). The Newton–Hooke duality: potentials r^a and r^b pair when $(a + 2)(b + 2) = 4$.

31. E. C. Aschenauer, R. Sassot and M. Stratmann, *Unveiling the proton spin decomposition at a future Electron-Ion Collider*, arXiv:1509.06489 (2015). Projected uncertainties at 10 GeV²: about 10% on ΔG and about 15% on ΔΣ.

32. E.-C. Aschenauer, A. Bazilevsky, M. Diehl et al. (RHIC Spin Collaboration), *The RHIC SPIN program: achievements and future opportunities*, arXiv:1501.01220 (2015). Quotes DSSV14's truncated gluon spin, 0.20 +0.06/−0.07 at 90% C.L. for $x > 0.05$ at $Q^2 = 10$ GeV², and NNPDF reweighted with STAR's 2009 jets, 0.17 ± 0.06 for $0.05 < x < 0.2$.

33. ATLAS Collaboration, Eur. Phys. J. C 79, 290 (2019). The Run 1 top-mass combination, 172.69 ± 0.48 GeV.

34. ATLAS Collaboration, Phys. Rev. Lett. 123, 052001 (2019). Light-by-light scattering observed at 8.2σ, with a fiducial cross section of 78 ± 13 ± 7 nb.

35. ATLAS Collaboration, *ATLAS 13 TeV open data for education and outreach* (2020), doi:10.7483/OPENDATA.ATLAS.NQ31.Y1OO. The 2016 collisions and simulations from which the peaks of section 9.6 are rebuilt.

36. ATLAS Collaboration, Eur. Phys. J. C 80, 528 (2020). The top pole mass from the $t\bar{t}$ cross section with 36 fb^{-1}, 173.1 +2.0/−2.1 GeV.

37. ATLAS Collaboration, Nature 607, 52 (2022), arXiv:2207.00092. The Higgs boson's couplings ten years after its discovery.

38. ATLAS and CMS Collaborations, *Combination of inclusive top-quark pair production cross-section measurements using ATLAS and CMS data at $\sqrt{s}$ = 7 and 8 TeV*, JHEP 07 (2023) 213, arXiv:2205.13830. $m_t(\text{pole}) = 173.4$ +1.8/−2.0 GeV and $\alpha_s(m_Z) = 0.1170$ +0.0021/−0.0018.

39. ATLAS Collaboration, JHEP 06 (2023) 019. The top mass from a soft-muon invariant mass with 36 fb^{-1}, 174.41 ± 0.81 GeV.

40. ATLAS Collaboration, JHEP 11 (2023) 112, copy. Magnetic monopoles excluded below 2.6 TeV (spin ½, Drell–Yan production, one Dirac charge).

41. ATLAS Collaboration, JHEP 12 (2023) 107, arXiv:2310.04350. Longitudinally polarised ZZ pairs, evidence at 4.3σ.

42. ATLAS Collaboration, Phys. Lett. B 843, 137829 (2023), arXiv:2209.14903. The longitudinal W share in top-quark decays, 0.684 ± 0.015.

43. ATLAS Collaboration, Phys. Lett. B 843, 137895 (2023), arXiv:2211.09435. WZ pairs with both bosons longitudinally polarised, observed at 7.1σ.

44. ATLAS Collaboration, Phys. Lett. B 846, 138223 (2023). The Higgs width from off-shell production, 4.5 +3.3/−2.5 MeV.

45. ATLAS and CMS Collaborations, Phys. Rev. Lett. 132, 261902 (2024). The combined Run 1 top mass, 172.52 ± 0.33 GeV.

46. ATLAS Collaboration, Eur. Phys. J. C 84, 1309 (2024), arXiv:2403.15085. The W mass, 80,366.5 ± 15.9 MeV.

47. ATLAS Collaboration, *Measurement of the top-quark pole mass in dileptonic $t\bar{t}$ + 1-jet events at $\sqrt{s}$ = 13 TeV*, JHEP 12 (2025) 023, arXiv:2507.02632. m_t(pole) = 170.7 ± 0.3 (stat) ± 1.4 (syst) ± 0.3 (scale) ± 0.2 (PDF and α_s) GeV.

48. ATLAS Collaboration, Phys. Rep. 1116, 127 (2025), arXiv:2404.10674. The Run 2 review of top-quark measurements: the $t\bar{t}$ cross section, 829.0 ± 1.0 ± 13.2 ± 8.0 pb, and its prediction.

49. ATLAS Collaboration, Phys. Rev. Lett. 135, 111802 (2025), arXiv:2503.11317. Same-sign WW pairs with at least one longitudinal boson, at 3.3σ.

50. ATLAS and CMS Collaborations, arXiv:2602.23991 (2026). Combined Higgs-pair searches: −0.71 < κ_λ < 6.1 at 95%.

51. ATLAS Collaboration, arXiv:2601.11780 (2026). The $t\bar{t}$ threshold enhancement, 9.3 +1.4/−1.3 pb.

52. M. Baak et al. (Gfitter), Eur. Phys. J. C 72, 2003 (2012), arXiv:1107.0975. The global electroweak fit, with its predicted W, top and Higgs masses.

53. J. C. Baez and J. Huerta, *The algebra of grand unified theories*, Bull. Amer. Math. Soc. 47, 483 (2010). The five-slot binary exposition of the SO(10) spinor.

54. J. C. Baez, arXiv:2507.11428 (2025). The bound states of the spin-½ Kepler problem as solutions of the Weyl equation on $\mathbb{R} \times S^3$; the Madelung rules from a modified Hamiltonian.

55. J. Bailey et al., Nature 268, 301 (1977). Muons at γ = 29.33 in a storage ring live longer by the predicted factor, to 0.2%.

56. S. D. Bass, papers on the emergent Standard Model vacuum: arXiv:1503.05483 (2015), arXiv:2001.01705 (2020), arXiv:2402.14719 (2024). The Standard Model as the long-range tail of a critical system; particles as collective excitations; the cosmological constant as a residual of cancelling energies.

57. S. D. Bass and J. Krzysiak, arXiv:2001.01706 (2020). The Casimir force computable without zero-point energies.

58. S. D. Bass, further papers on the emergent vacuum: arXiv:2004.05489 (2020), arXiv:2508.15289 (2025) and lectures, arXiv:2606.19403 (2026). Gauge symmetries that “dissolve” above an emergence

scale $M \approx 10^{16}$ GeV, with an effective theory below it; particle masses from interaction with the condensates; the vacuum energy as a residual of order v^2/M.

59. H. Bateman, Proc. London Math. Soc. (2) 8, 223 (1910). Maxwell's equations are unchanged by inversion and the wider conformal group.

60. H. N. Becker et al., Nature 584, 55 (2020). Small, shallow lightning flashes high in Jupiter's ammonia–water clouds.

61. M. A. B. Bég, B. W. Lee and A. Pais, Phys. Rev. Lett. 13, 514 (1964). SU(6) gives $\mu_p/\mu_n = -\tfrac{2}{3}$, within 3% of experiment.

62. Belle II Collaboration, *Measurement of the τ lepton mass with the Belle II experiment*, arXiv:2305.19116 (2023). The τ mass by the pseudomass endpoint, from 175 million pairs, 1777.09 ± 0.08 ± 0.11 MeV.

63. Belle II Collaboration, Phys. Rev. D (2025), arXiv:2412.19624. The angle α from $B \to \rho\rho$: 92.6 +4.5/−4.7°.

64. M. Beneke, P. Marquard, P. Nason and M. Steinhauser, *On the ultimate uncertainty of the top quark pole mass*, Phys. Lett. B 775, 63 (2017), arXiv:1605.03609. The renormalon ambiguity of the pole mass, about 110 MeV; the terms beyond four loops, about 300 MeV.

65. C. H. Bennett, *Logical reversibility of computation*, IBM J. Res. Dev. 17, 525 (1973). Reversible computation.

66. V. Bertone, A. Chiefa and E. R. Nocera (MAP), *Helicity-dependent parton distribution functions at next-to-next-to-leading order accuracy from inclusive and semi-inclusive deep-inelastic scattering data*, Phys. Lett. B (2025), arXiv:2404.04712. The polarised gluon largely unconstrained by DIS and SIDIS alone.

67. B. Bertotti, L. Iess and P. Tortora, Nature 425, 374 (2003). Cassini: $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$.

68. BESIII Collaboration, *Precision measurement of the mass of the τ lepton*, Phys. Rev. D 90, 012001 (2014), arXiv:1405.1076. The τ mass from a threshold scan, 1776.91 ± 0.12 +0.10/−0.13 MeV.

69. H. Bethe, *Termaufspaltung in Kristallen*, Ann. Phys. (Leipzig) 395, 133 (1929). The crystal-field splitting of atomic levels.

70. H. A. Bethe, Phys. Rev. 72, 339 (1947). The first estimate of the Lamb shift, 1040 MHz.

71. N. Bezginov et al., Science 365, 1007 (2019). The hydrogen n = 2 Lamb shift to 3.2 kHz, and a proton radius of 0.833(10) fm.

72. BICEP/Keck Collaboration, Phys. Rev. Lett. 127, 151301 (2021). Primordial gravitational waves: $r < 0.036$ at 95%.

73. Bjerg, Fournais and Hearnshaw, arXiv:2506.14227 (2025). For large Z the angular-momentum distribution of a neutral atom follows the Thomas–Fermi distribution and departs from the Madelung rule.

74. R. D. Blandford and R. L. Znajek, Mon. Not. R. Astron. Soc. 179, 433 (1977). Jets powered by a spinning hole's rotational energy.

75. S. M. Blinder, arXiv:physics/0208072 (2002). A classical electron whose field carries ħ/2 when its length parameter is set at $2r_0/3$.

76. N. Bohr, *On the constitution of atoms and molecules*, Phil. Mag. 26, 1 (1913). Hydrogen's levels.

77. J. R. Bond, L. Kofman and D. Pogosyan, Nature 380, 603 (1996). How filaments of galaxies are woven into the cosmic web.

78. D. Bor, J. Billington and S. Baron-Cohen, *Savant memory for digits in a case of synaesthesia and Asperger syndrome is related to hyperactivity in the lateral prefrontal cortex*, Neurocase 13, 311–319 (2007). No advantage for structured digit sequences in the savant studied; cited by the author, whose list gives 2008, 14(5), 390–398.

79. M. J. Borchert et al. (BASE), Nature 601, 53 (2022). Antiproton and proton charge-to-mass ratios equal within 16 parts per trillion.

80. I. Borsa, M. Stratmann, W. Vogelsang, D. de Florian and R. Sassot, *Next-to-next-to-leading order global analysis of polarized parton distribution functions*, Phys. Rev. Lett. 133, 151901 (2024), arXiv:2407.11635. The full first moments cannot be reliably determined, because of the low-x extrapolation.

81. E. Braaten and H.-W. Hammer, *Efimov physics in cold atoms*, arXiv:cond-mat/0612123 (2006). Discrete scale invariance, $e^{(\pi/s_0)} \approx 22.7$.

82. W. F. Brace and J. D. Byerlee, *Stick-slip as a mechanism for earthquakes*, Science 153, 990 (1966).

83. C. A. Brannen, *The Lepton Masses* (2006), pdf. The three-phase form of Koide's relation, the phase near 2/9, the neutrino solution with one flipped root.

84. L. Breuval et al., *Small Magellanic Cloud Cepheids observed with the Hubble Space Telescope provide a new anchor for the SH0ES distance ladder*, Astrophys. J. 973, 30 (2024), arXiv:2404.08038. $H_0 = 73.17 \pm 0.86$ km/s/Mpc.

85. T. M. P. Briels, E. M. van Veldhuizen and U. Ebert, J. Phys. D 41, 234008 (2008), arXiv:0805.1364. Streamer width times pressure stays near 0.20 ± 0.02 mm·bar.

86. M. Brignone et al., *An efficient method for the computation of electromagnetic fields associated with tortuous lightning channels*, IEEE Trans. Electromagn. Compat. (2022), doi:10.1109/TEMC.2022.3186726, copy. Hill's 17.3°, Idone and Orville's 17.0°, and a Gaussian model of 17.0° ± 11.0°.

87. D. Brout et al. (Pantheon+), Astrophys. J. 938, 110 (2022), arXiv:2202.04077. Supernova-only $\Omega_{\square} = 0.334 \pm 0.018$; $q_0 = -0.51 \pm 0.024$ below z = 0.8, from SH0ES.

88. S. Brown et al., Nature 558, 87 (2018). Juno's detection of prevalent lightning on Jupiter.

89. J. N. Brune, *Precariously balanced rocks and ground-motion maps for Southern California*, Bull. Seismol. Soc. Am. 86, 43 (1996).

90. D. L. Burke et al., *Positron production in multiphoton light-by-light scattering*, Phys. Rev. Lett. 79, 1626 (1997), pdf. 106 ± 14 positrons from a laser photon backscattered to GeV energy colliding with photons of the same laser; the pair needs at least four of them, and on average 4.7 are absorbed.

91. P. J. Burke, *An RF circuit model for carbon nanotubes*, arXiv:cond-mat/0207222 (IEEE Trans. Nanotechnol., 2003). Quantum inductance and capacitance, with $\sqrt{(L/C)}$ on the scale h/e^2.

92. M. Butenschoen, B. Dehnadi, A. H. Hoang, V. Mateu, M. Preisser and I. W. Stewart, *Top quark mass calibration for Monte Carlo event generators*, Phys. Rev. Lett. 117, 232001 (2016), arXiv:1608.01318. For Pythia 8.205 at 173 GeV: the MSR mass at 1 GeV is 172.82 ± 0.22 GeV and the pole mass 172.43 ± 0.28 GeV (N^2LL).

93. D. Buttazzo et al., JHEP 12, 089 (2013), arXiv:1307.3536. The Higgs vacuum's metastability at next-to-next-to-leading order.

94. F. Calvo, E. Pahl, M. Wormit and P. Schwerdtfeger, Angew. Chem. Int. Ed. 52, 7583 (2013). Relativity keeps mercury liquid.

95. E. Camphuis et al. (SPT-3G Collaboration), *SPT-3G D1: CMB temperature and polarization power spectra and cosmology from 2019 and 2020 observations of the SPT-3G Main field*, arXiv:2506.20707 (2025). H_0 = 66.66 ± 0.60 (SPT-3G) and 67.19 ± 0.38 km/s/Mpc (SPT, ACT and Planck).

96. E. Capote et al., *Advanced LIGO detector performance in the fourth observing run*, Phys. Rev. D 111, 062002 (2025), arXiv:2411.14607. Frequency-dependent squeezing cuts quantum noise by up to 5.2 dB at Hanford and 6.1 dB at Livingston.

97. Capozzi, Lisi, Marcone, Marrone and Palazzo, Phys. Rev. D 114, 016026 (2026), arXiv:2511.21650. A global oscillation fit with JUNO's first result: $\sin^2\theta_{12}$ = 0.3085 ± 0.0073.

98. H. B. G. Casimir, Proc. K. Ned. Akad. Wet. 51, 793 (1948). The attraction between two conducting plates.

99. A.-L. Cauchy, *Mémoire sur la rectification des courbes et la quadrature des surfaces courbes* (1841). A convex body's mean shadow is a quarter of its surface.

100. A. Cayley, *Chapters in the analytical geometry of (n) dimensions* (1845). Geometry in n dimensions.

101. CDF Collaboration, Science 376, 170 (2022), record. The W mass, 80,433.5 ± 9.4 MeV.

102. J. Cen, P. Yuan and S. Xue, Phys. Rev. Lett. 112, 035001 (2014). Ball lightning flickering at 99.4 Hz near a 50 Hz line, which the authors suggest drove it at the second harmonic; 1.64 s.

103. T. C. Chamberlin, *The method of multiple working hypotheses*, Science 15, 92 (1890).

104. C.-H. Chang, V. Dommes, R. S. Erramilli, A. Homrich, P. Kravchuk, A. Liu, M. S. Mitchell, D. Poland and D. Simmons-Duffin, *Bootstrapping the 3d Ising stress tensor*, JHEP 03 (2025) 136, arXiv:2411.15300. The three-dimensional Ising exponents.

105. M. S. Chanowitz and M. K. Gaillard, Nucl. Phys. B 261, 379 (1985). The equivalence theorem: longitudinal W and Z bosons scatter as the Higgs field's Goldstone modes.

106. S. M. Chester, W. Landry, J. Liu, D. Poland, D. Simmons-Duffin, N. Su and A. Vichi, *Carving out OPE space and precise O(2) model critical exponents*, JHEP 06 (2020) 142, arXiv:1912.03324. Δ_s = 1.51136(22), so ν = 0.67175(10).

107. K. G. Chetyrkin and M. F. Zoller, JHEP 06 (2012) 033, arXiv:1205.2892. Three-loop running of the top Yukawa and Higgs self-couplings.

108. B. Chide et al., *Detection of triboelectric discharges during dust events on Mars*, Nature (26 November 2025). Small sparks inside Martian dust devils.

109. C. W. Chou, D. B. Hume, T. Rosenband and D. J. Wineland, Science 329, 1630 (2010). A clock raised 33 cm runs faster by $(4.1 \pm 1.6) \times 10^{-17}$.

110. D. Christodoulou, Phys. Rev. Lett. 25, 1596 (1970). The irreducible mass; at most $1 - 1/\sqrt{2}$ of a hole's mass can be extracted.

111. D. Christodoulou and R. Ruffini, Phys. Rev. D 4, 3552 (1971). The mass formula $M^2 = M_{irr}^2 + (J/2M_{irr})^2$.

112. M. Christodoulou and C. Rovelli, Phys. Rev. D 91, 064046 (2015). A black hole's interior volume grows as $3\sqrt{3}\pi M^2 t$.

113. L. O. Chua, *Memristor — the missing circuit element*, IEEE Trans. Circuit Theory 18, 507 (1971). The four circuit variables and their six relations; memristance turned into a resistance law.

114. I. Ciufolini and E. C. Pavlis, Nature 431, 958 (2004). The LAGEOS orbits dragged at 99% of the prediction, ±10%.

115. A.-C. Clairaut, *Recherches sur les courbes à double courbure* (1731). Coordinates in three dimensions.

116. E. Clarke, *Circuit Analysis of A-C Power Systems* (Wiley, 1943). The orthonormal $\alpha\beta 0$ transform.

117. R. Clausius, Ann. Phys. 141, 124 (1870). The virial theorem.

118. D. Clowe et al., Astrophys. J. Lett. 648, L109 (2006). The Bullet Cluster: the pull follows the galaxies, not the gas.

119. CMB-S4 Project, statement of the Department of Energy and the National Science Foundation, 9 July 2025, cmb-s4.org. "DOE and NSF have jointly decided that they can no longer support the CMB-S4 Project."

120. CMS Collaboration, Phys. Lett. B 770, 50 (2017), arXiv:1610.09551. $m(t) - m(\bar{t}) = -0.15 \pm 0.19 \pm 0.09$ GeV.

121. CMS Collaboration, *Measurement of $t\bar{t}$ normalised multi-differential cross sections in pp collisions at $\sqrt{s}$ = 13 TeV, and simultaneous determination of the strong coupling strength, top quark pole mass, and parton distribution functions*, Eur. Phys. J. C 80, 658 (2020), arXiv:1904.05237. $m_t(\text{pole}) = 170.5 \pm 0.8$ GeV.

122. CMS Collaboration, Nature 607, 60 (2022), arXiv:2207.00043. The Higgs boson's couplings ten years after its discovery.

123. CMS Collaboration, Nature Phys. 18, 1329 (2022), arXiv:2202.06923. The Higgs width from off-shell production, 3.2 +2.4/−1.7 MeV.

124. CMS Collaboration, *Measurement of the top quark pole mass using $t\bar{t}$+jet events in the dilepton final state in proton-proton collisions at $\sqrt{s}$ = 13 TeV*, JHEP 07 (2023) 077, arXiv:2207.02270. m_t(pole) = 172.93 ± 1.36 GeV with ABMP16NLO.

125. CMS Collaboration, arXiv:2412.13872 (2024); Nature 652, 321 (2026). The W mass, 80,360.2 ± 9.9 MeV.

126. CMS Collaboration, *Observation of a pseudoscalar excess at the top quark pair production threshold*, Rep. Prog. Phys. 88, 087801 (2025), arXiv:2503.22382. The threshold enhancement, 8.8 +1.2/−1.4 pb, above 5σ; the ηt mass set at 343 GeV with m_t fixed at 172.5 GeV; m($t\bar{t}$) resolution about 15% at threshold.

127. CMS Collaboration, JHEP 08 (2025) 006, arXiv:2412.15413. Light-by-light scattering, 107 ± 24 ± 13 nb, above 5σ; the Breit–Wheeler cross section, 263.5 ± 1.8 ± 17.8 μb.

128. CMS Collaboration, arXiv:2607.28396 (2026). The Higgs mass from two photons, 125.13 ± 0.15 GeV.

129. CMS Collaboration, arXiv:2608.19546 (2026). The threshold enhancement in single-lepton events, 5.1 ± 0.9 pb, 6.1σ.

130. Cominotti et al., *Observation of temperature effects on false vacuum decay in atomic quantum gases*, Phys. Rev. Lett. 135, 183401 (2025), arXiv:2504.03528. Decay time against temperature, as finite-temperature instanton theory predicts.

131. COMPASS Collaboration, *Leading-order determination of the gluon polarisation from semi-inclusive deep inelastic scattering data*, Eur. Phys. J. C 77, 209 (2017), arXiv:1512.05053. $\langle \Delta g/g \rangle$ = 0.113 ± 0.038 ± 0.036 at $\langle x_g \rangle \approx 0.10$ and μ^2 = 3 GeV^2.

132. V. Cooray, Atmosphere 9, 205 (2018), article. Hill's mean turn of lightning channels, 17.3°.

133. J. M. Cornwall, D. N. Levin and G. Tiktopoulos, Phys. Rev. D 10, 1145 (1974). The equivalence theorem and tree-level unitarity.

134. R. Cotes, *Harmonia Mensurarum* (1722, posthumous). The radian.

135. C.-A. Coulomb, *Théorie des machines simples*, Mém. Math. Phys. Acad. R. Sci. 10, 161 (1785). Static friction exceeds sliding friction.

136. H. S. M. Coxeter, *Regular Polytopes*, 3rd ed. (Dover, 1973). The 24-cell as three 16-cells.

137. N. Craig, D. Green, J. Meyers and S. Rajendran, *No νs is good news*, JHEP (2024). A negative cosmological neutrino mass read as new physics unrelated to neutrinos.

138. C. J. Crompton et al., *Double empathy: why autistic people are often misunderstood*, Frontiers for Young Minds 9, 554875 (2021). Information passed as accurately in autistic as in non-autistic groups, less so in mixed groups.

139. Y. Cui et al., *Precessing jet nozzle connecting to a spinning black hole in M87*, Nature 621, 711 (2023). An 11-year wobble of M87's jet, from a tilted disc dragged by the hole's spin.

140. E. Cunningham, Proc. London Math. Soc. (2) 8, 77 (1910). The electromagnetic equations unchanged by inversion.

141. R. S. Decca et al., Eur. Phys. J. C 51, 963 (2007). The Casimir force measured to 0.2%.

142. M. M. Defranchis, J. de Blas, A. Mehta, M. Selvaggi and M. Vos, *A detailed study on the prospects for a $t\bar{t}$ threshold scan in e^+e^- collisions*, arXiv:2503.18713 (2025). The potential-subtracted mass to 4.3 MeV (statistical), 6.8 MeV (experimental) and about 35 MeV (theory); 7 to 23 MeV more on conversion to MS-bar.

143. G. Degrassi et al., JHEP 08, 098 (2012). Higgs mass and vacuum stability at NNLO.

144. W. Deierling and W. A. Petersen, J. Geophys. Res. 113, D16210 (2008). Total lightning tracks the storm's updraft volume faster than 10 m/s in the below-freezing zone, r = 0.93.

145. D. d'Enterria et al., *The strong coupling constant: state of the art and the decade ahead*, arXiv:2203.08271 (2022). α_s(m_Z) = 0.1179 ± 0.0009.

146. DES Collaboration, Astrophys. J. Lett. 973, L14 (2024), arXiv:2401.02929. DES-SN5YR: supernova-only $\Omega_m = 0.352 \pm 0.017$.

147. DES Collaboration, Year 6 cosmology, arXiv:2601.14559 (2026). $S_8 = 0.789 \pm 0.012$ and $\Omega_m = 0.333$.

148. R. Descartes, *La Géométrie* (1637). Coordinates.

149. DESI Collaboration, Phys. Rev. D 112, 083515 (2025), arXiv:2503.14738. The neutrino-mass cap, the evolving-w preference, Ω_k and Ω_m from the second data release.

150. J. P. Dietrich et al., *A filament of dark matter between two clusters of galaxies*, Nature 487, 202 (2012). Abell 222 and 223.

151. V. Dikhtyar and E. Jerby, Phys. Rev. Lett. 96, 045002 (2006). Fireballs made by microwaves, which last only while the power is on.

152. Ding, Li, Lu and Petcov, arXiv:2512.03809 (2025). Discrete mixing patterns against JUNO, with the six-year projection and the TM2 exclusion.

153. Diophantus (3rd century); Brahmagupta (7th century). A product of two sums of two squares is a sum of two squares in two ways.

154. P. A. M. Dirac, Proc. R. Soc. A 117, 610 (1928). The relativistic electron.

155. P. A. M. Dirac, Proc. R. Soc. A 126, 360 (1930). Holes in the negative-energy sea as positive electrons.

156. P. A. M. Dirac, *Forms of relativistic dynamics*, Rev. Mod. Phys. 21, 392 (1949). Light-front coordinates.

157. P. A. M. Dirac, *Is there an aether?*, Nature 168, 906 (1951).

158. A. Djouadi, *The anatomy of electroweak symmetry breaking*, Phys. Rep. 457, 1 (2008), arXiv:hep-ph/0503172. The Higgs decay formulas.

159. D. Djukanovic et al., Phys. Rev. Lett. 132, 211901 (2024), arXiv:2309.07491. The proton radius on the lattice, 0.820(14) fm.

160. S. Douady and Y. Couder, Phys. Rev. Lett. 68, 2098 (1992). Phyllotaxis as self-organised growth: the golden angle $360°/\varphi^2 = 137.51°$.

161. S. Dürr et al., Science 322, 1224 (2008). Light hadron masses from lattice QCD: the nucleon at 936 ± 34 MeV from the field alone.

162. P. Dutta and P. M. Horn, Rev. Mod. Phys. 53, 497 (1981). 1/f noise from activated barriers.

163. J. R. Dwyer et al., Geophys. Res. Lett. 32, L01803 (2005). X-ray bursts from stepping lightning leaders.

164. J. R. Dwyer and M. A. Uman, *The physics of lightning*, Phys. Rep. 534, 147 (2014).

165. R. S. Van Dyck, P. B. Schwinberg and H. G. Dehmelt, Phys. Rev. Lett. 59, 26 (1987). The electron's and the positron's g-factors compared.

166. F. W. Dyson, A. S. Eddington and C. Davidson, Phil. Trans. R. Soc. A 220, 291 (1920). The 1919 eclipse: starlight bent by 1.98 ± 0.12″.

167. U. Ebert et al., J. Geophys. Res. 115, A00E43 (2010), arXiv:1002.0070. Streamers and sprites: similarity with air density.

168. G. Efstathiou, *Evolving dark energy or supernovae systematics?*, Mon. Not. R. Astron. Soc. (2025), arXiv:2408.07175.

169. R. Eichler et al., Nature 447, 72 (2007). Copernicium behaves as a volatile metal.

170. A. Einstein, *Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig?*, Ann. Phys. 18, 639 (1905). A body's mass measures its energy content.

171. A. Einstein, Sitzungsber. Preuss. Akad. Wiss. 844 (1915); Ann. Phys. 49, 769 (1916). General relativity; light bent by 1.75″ at the Sun's edge.

172. A. Einstein, Sitzungsber. Preuss. Akad. Wiss. 688 (1916) and 154 (1918). Gravitational waves and the quadrupole formula.

173. A. Einstein, *Äther und Relativitätstheorie* (Springer, 1920). Space with physical properties but no state of motion.

174. A. Einstein and N. Rosen, Phys. Rev. 48, 73 (1935). The bridge joining two copies of the Schwarzschild exterior.

175. W. Elbers et al. (DESI), Phys. Rev. D 112, 083513 (2025), arXiv:2503.14744. The ΛCDM effective neutrino mass, −0.101 +0.047/−0.056 eV, 3.0σ below the normal-ordering floor.

176. F. Englert and R. Brout, Phys. Rev. Lett. 13, 321 (1964). Mass for gauge bosons from a broken symmetry.

177. L. Essen et al., Nature 229, 110 (1971). The hydrogen maser frequency.

178. I. Esteban et al. (NuFIT), arXiv:2601.09791 (2026). NuFIT 6.1 with JUNO: $\sin^2\theta_{12} = 0.3096$ +0.0057/−0.0073; uses NuFIT 6.0's $\sin^2\theta_{13} = 0.022 \pm 0.00056$ as an input.

179. Euclid, *Elements* (c. 300 BCE): I.41, I.47, II.5, VI.4, X.29 Lemma 1, XII.7.

180. L. Euler, *Introductio in analysin infinitorum* (1748), sine and cosine on the unit circle; and the proof of the two-squares theorem (1747–55).

181. L. Euler, *Du mouvement de rotation des corps solides autour d'un axe variable*, Mém. Acad. Sci. Berlin 14, 154 (1758; published 1765). Rigid rotation; the middle axis is unstable.

182. Event Horizon Telescope Collaboration, Astrophys. J. Lett. 875, L1 (2019) and 930, L12 (2022). The shadows of M87* (42 ± 3 μas) and Sgr A* (51.8 ± 2.3 μas).

183. Event Horizon Telescope Collaboration, Astrophys. J. Lett. 910, L13 (2021). M87*'s polarised ring: a magnetically arrested flow with a jet power of at least 10^{42} erg/s.

184. C. W. F. Everitt et al. (Gravity Probe B), Phys. Rev. Lett. 106, 221101 (2011). Geodetic precession 6,601.8 ± 18.3 and frame dragging 37.2 ± 7.2 mas/yr.

185. Ewasiuk and Profumo, *Quantum tunneling of primordial black holes to white holes*, arXiv:2603.22516 (2026). The fast-radio-burst rate is far above what returning holes could supply.

186. X. Fan, T. G. Myers, B. A. D. Sukra and G. Gabrielse, Phys. Rev. Lett. 130, 071801 (2023), arXiv:2209.13084. The electron's g/2 = 1.00115965218059(13).

187. D. Farrah et al., Astrophys. J. Lett. 944, L31 (2023), arXiv:2302.07878. Black holes coupled to the expansion, with $k \approx 3$.

188. M. S. Fee et al., Phys. Rev. Lett. 70, 1397 (1993). Positronium's 1S–2S interval, 1,233.607 THz.

189. W. Feng, interviewed in *International Business Times* (2025). “Multi-modality thinking” as a weaving of disciplines.

190. P. de Fermat, letter to Mersenne, 25 December 1640 (two squares); and the theorem that the congruum is never a square.

191. R. P. Feynman, *Simulating physics with computers*, Int. J. Theor. Phys. 21, 467 (1982).

192. B. D. Fields et al., JCAP 03, 010 (2020), arXiv:1912.01132. Lithium 2.95 times its predicted abundance while D/H agrees.

193. D. G. Figueroa, A. Rajantie and F. Torrenti, Phys. Rev. D 98, 023532 (2018). On the lattice, the Higgs goes unstable after inflation for $\xi \gtrsim 4$ to 5 at $m_t \approx 173.3$ GeV.

194. T. Fincke, *Geometria Rotundi* (1583). The names tangent and secant.

195. R. A. Fisher, *Statistical Methods for Research Workers*, 4th ed. (Oliver and Boyd, 1932). The method of combining p-values.

196. D. de Florian, R. Sassot, M. Stratmann and W. Vogelsang, *Evidence for polarization of gluons in the proton*, Phys. Rev. Lett. 113, 012001 (2014), arXiv:1404.4293. DSSV14: the gluon spin positive for $0.05 < x < 0.2$ at $Q^2 = 10\ \mathrm{GeV}^2$; its truncated value as quoted by Aschenauer et al. (2015).

197. D. de Florian et al. (LHC Higgs Cross Section Working Group), *Handbook of LHC Higgs cross sections: 4. Deciphering the nature of the Higgs sector*, CERN Yellow Report (2016), arXiv:1610.07922. The Higgs decay widths and branching ratios.

198. V. Fock, Z. Phys. 98, 145 (1935). Hydrogen's four-dimensional symmetry.

199. E. Fonseca et al., Astrophys. J. Lett. 915, L12 (2021). The heaviest well-measured neutron star, PSR J0740+6620.

200. R. Foot, arXiv:hep-ph/9402242 (1994). Koide's relation as a 45° angle.

201. C. L. Fortescue, *Method of symmetrical co-ordinates applied to the solution of polyphase networks*, Trans. AIEE 37, 1027 (1918).

202. L. Foucault, C. R. Acad. Sci. 35 (1852). The gyroscope.

203. B. Franklin, letter to P. Collinson (1752). The electrical kite and the lightning rod.

204. E. Fredkin and T. Toffoli, *Conservative logic*, Int. J. Theor. Phys. 21, 219 (1982).

205. W. L. Freedman et al. (CCHP), Astrophys. J. 985, 203 (2025), arXiv:2408.06153. $H_0 = 70.39 \pm 1.94$ km/s/Mpc.

206. A. Friedmann, Z. Phys. 10, 377 (1922). The expanding solutions.

207. H. Fritzsch and P. Minkowski, Ann. Phys. 93, 193 (1975). The 16 of SO(10).

208. H. Fritzsch, Phys. Lett. B 70, 436 (1977). Quark textures: the Cabibbo angle from mass ratios.

209. H. Fritzsch and Z.-z. Xing, Prog. Part. Nucl. Phys. 45, 1 (2000). Quark mass textures; a right-angled unitarity triangle.

210. C. D. Froggatt and H. B. Nielsen, Phys. Lett. B 368, 96 (1996). The multiple-point principle: the Higgs exactly at the stability edge.

211. V. P. Frolov, M. A. Markov and V. F. Mukhanov, Phys. Lett. B 216, 272 (1989) and Phys. Rev. D 41, 383 (1990). A de Sitter core in place of the singularity.

212. Y. Fukuda et al. (Super-Kamiokande Collaboration), *Evidence for oscillation of atmospheric neutrinos*, Phys. Rev. Lett. 81, 1562 (1998). Neutrinos have mass.

213. R. W. Fuller and J. A. Wheeler, Phys. Rev. 128, 919 (1962). The Einstein–Rosen bridge pinches off.

214. J. M. Gates et al., Phys. Rev. Lett. 133, 172502 (2024). Two atoms of ^{290}Lv from a titanium-50 beam on plutonium-244, 0.44 +0.58/−0.28 pb.

215. R. Gatto, G. Sartori and M. Tonin, Phys. Lett. B 28, 128 (1968). sin θ_C ≈ √(m_d/m_s).

216. C. F. Gauss, *Disquisitiones Arithmeticae* (1801), sums of three squares and unique factorisation; *Theoria residuorum biquadraticorum II* (1831), +1, −1 and √−1 as the direct, inverse and lateral units.

217. E. Gaztañaga, K. S. Kumar, S. Pradhan and M. Gallagher, Phys. Rev. D 111, 103537 (2025). A bounce on the exclusion principle, with Λ set by the parent's horizon.

218. M. J. Geller and J. P. Huchra, *Mapping the Universe*, Science 246, 897 (1989). The Great Wall.

219. H. Georgi and S. L. Glashow, Phys. Rev. Lett. 32, 438 (1974). Grand unification.

220. H. Georgi, in *Particles and Fields — 1974*, ed. C. E. Carlson (AIP, 1975), p. 575. SO(10) unification.

221. T. Ghigna et al. (LiteBIRD Collaboration), *The LiteBIRD mission to explore cosmic inflation*, Proc. SPIE 13092 (2024), arXiv:2406.02724. Launch aimed for Japan's fiscal year 2032; δr = 0.001.

222. A. Girard (1625). The statement of which numbers are sums of two squares.

223. L. T. Glöggler et al. (AEgIS), Phys. Rev. Lett. 132, 083402 (2024). Positronium laser-cooled from 380 K to 170 K.

224. J. Goldstone, Nuovo Cimento 19, 154 (1961); J. Goldstone, A. Salam and S. Weinberg, Phys. Rev. 127, 965 (1962). One massless mode for each broken direction.

225. I. J. Good, *A quantal hypothesis for hadrons and the judging of physical numerology*, in *Disorder in Physical Systems* (eds G. R. Grimmett and D. J. A. Welsh, Oxford University Press, 1990), pdf. Bayesian judging of numerological formulas, accuracy against complexity; “There is no precise demarcation between numerology and a scientific theory”; of eight formulas for m□/m$_e$ only Lenz's scores positively.

226. H. Grassmann, *Die lineale Ausdehnungslehre* (1844). Spaces of any dimension.

227. GRAVITY Collaboration, Astron. Astrophys. 636, L5 (2020), arXiv:2004.07187. The Schwarzschild precession of S2, f = 1.10 ± 0.19.

228. O. W. Greenberg, *Why is CPT fundamental?*, Found. Phys. 36, 1535 (2006), arXiv:hep-ph/0309309. Jost: spacetime inversion is connected to the identity in the complex Lorentz group.

229. O. W. Greenberg, arXiv:0803.0992 (2008). Describes Bég, Lee and Pais's SU(6) moment ratio.

230. E. Gross and O. Vitells, *Trial factors for the look elsewhere effect in high energy physics*, Eur. Phys. J. C 70, 525 (2010), arXiv:1005.1891.

231. Guo et al., arXiv:2405.20572 (2024). Relativistic coupled-cluster ionisation potentials of lawrencium.

232. G. S. Guralnik, C. R. Hagen and T. W. B. Kibble, Phys. Rev. Lett. 13, 585 (1964).

233. A. H. Guth, Phys. Rev. D 23, 347 (1981). Inflation.

234. H0DN Collaboration (S. Casertano et al.), arXiv:2510.23823 (2025). H_0 = 73.50 ± 0.81 km/s/Mpc, 7.1σ from the microwave background.

235. Ḥabash al-Ḥāsib (c. 860). The first tables of tangents.

236. H. M. Haggard and C. Rovelli, Phys. Rev. D 92, 104020 (2015). Black-hole fireworks: tunnelling to a white hole.

237. W. R. Hamilton, *On quaternions* (1843; Phil. Mag. 1844). i·j = k.

238. K. Hamraoui, L. Van Damme, P. Mardešić and D. Sugny, *Classical and quantum rotation numbers of asymmetric top molecules*, Phys. Rev. A 97, 032118 (2018), arXiv:1803.07356. Time near the separatrix grows as the log of one over the distance.

239. N. Haramein, O. Alirol and C. Guermonprez, Preprints.org (2025), doi:10.20944/preprints202509.1835.v2. r⍰ = 4ƛ□ stated explicitly, following Haramein (2013).

240. H. Harari, Phys. Lett. B 86, 83 (1979). Quarks and leptons from two preons.

241. P. F. Harrison, D. H. Perkins and W. G. Scott, Phys. Lett. B 530, 167 (2002), arXiv:hep-ph/0202074. Tribimaximal mixing.

242. S. W. Hawking, Nature 248, 30 (1974); Commun. Math. Phys. 43, 199 (1975). Black holes radiate at T = ħκ/2πck.

243. O. Heaviside (1886), the word impedance; A. E. Kennelly (1893), impedance as a right triangle.

244. O. Heaviside, *A gravitational and electromagnetic analogy*, The Electrician 31, 281 and 359 (1893). Gravitomagnetism by analogy with Maxwell's equations.

245. W. Heisenberg and H. Euler, Z. Phys. 98, 714 (1936). The vacuum's nonlinear response to strong fields.

246. H. Hellwig et al., IEEE Trans. Instrum. Meas. 19, 200 (1970), as quoted by M. Diermaier et al., arXiv:1610.06392. Hydrogen's hyperfine splitting, 1,420,405,751.768(2) Hz.

247. M. Herranen, T. Markkanen, S. Nurmi and A. Rajantie, Phys. Rev. Lett. 113, 211102 (2014). Spacetime curvature and the Higgs vacuum during inflation.

248. M. Herranen, T. Markkanen, S. Nurmi and A. Rajantie, Phys. Rev. Lett. 115, 241301 (2015), arXiv:1506.04065. Strong Higgs production after inflation for $\xi \gtrsim 1$.

249. S. Herrmann et al., Phys. Rev. D 80, 105011 (2009), arXiv:1002.1284. The isotropy of light to about 10^{-17}.

250. HFLAV, R(D) and R(D*) average, ICHEP 2026, page. 3.7σ from the Standard Model.

251. P. W. Higgs, Phys. Rev. Lett. 13, 508 (1964).

252. R. D. Hill, *Analysis of irregular paths of lightning channels*, J. Geophys. Res. 73, 1897 (1968). Channel turns averaging 17.3°, Gaussian about zero; mean turn as section length to the power 0.1; Fourier amplitudes as 1/√n.

253. C. T. Hill, Phys. Rev. D 24, 691 (1981). The top Yukawa coupling held near an infrared fixed point.

254. C. H. Hinton, *A New Era of Thought* (1888). The word tesseract.

255. A. H. Hoang, Annu. Rev. Nucl. Part. Sci. 70, 225 (2020), arXiv:2004.12915. The direct and pole top masses may differ by about 0.5 GeV.

256. A. Hobson, *There are no particles, there are only fields*, Am. J. Phys. 81, 211 (2013).

257. G. 't Hooft, *The Cellular Automaton Interpretation of Quantum Mechanics* (Springer, 2016).

258. H. Hopf, Math. Ann. 104, 637 (1931). The Hopf map and its linked circles.

259. D. Hornof et al., Commun. Phys. 8, 161 (2025). Terrell rotation photographed.

260. G. W. Housner, *The behavior of inverted pendulum structures during earthquakes*, Bull. Seismol. Soc. Am. 53, 403 (1963). A rocking block needs more than its static tipping push to fall.

261. J. Hu, L. Feng, Z. Zhang and C. Chin, *Quantum simulation of coherent Hawking–Unruh radiation*, Nature Phys. (2019), doi:10.1038/s41567-019-0537-1, arXiv:1807.07504. Temperature and entropy as Unruh predicts.

262. N. T. Hunt-Smith, C. Cocuzza, W. Melnitchouk, N. Sato, A. W. Thomas and M. J. White (JAM), *On the resolution of the sign of gluon polarization in the proton*, arXiv:2403.08117 (2024). Negative gluon-spin solutions cannot account for high-x DIS together with lattice and jet data.

263. A. Hurwitz, *Vorlesungen über die Zahlentheorie der Quaternionen* (Springer, 1919). The 24 unit quaternions of the Hurwitz order.

264. IBM, *What are large language models (LLMs)?*, *What is multimodal AI?* and *AI hallucinations*, IBM Think (n.d.), ibm.com/think/topics. Language models as pattern detectors; hallucination defined.

265. V. P. Idone and R. E. Orville, Geophys. Res. Lett. 15, 645 (1988), record. Lightning channel tortuosity; 17.0° as computed by Brignone et al.

266. G. Isidori, V. S. Rychkov, A. Strumia and N. Tetradis, *Gravitational corrections to Standard Model vacuum decay*, Phys. Rev. D 77, 025034 (2008), arXiv:0712.0242.

267. R. L. Jaffe, *The Casimir effect and the quantum vacuum*, Phys. Rev. D 72, 021301 (2005). The Casimir force without zero-point energy.

268. P. Janot and S. Jadach, Phys. Lett. B 803, 135319 (2020). The number of light neutrinos from the Z width, 2.9963 ± 0.0074.

269. H. Jeffreys, *Theory of Probability*, 3rd ed. (Oxford, 1961). Bayes factors above 100 graded “decisive”.

270. U. D. Jentschura and I. Nándori, *Attempts at a determination of the fine-structure constant from first principles: a brief historical overview*, arXiv:1411.4673 (2014). Eddington's 1931 conjecture; Wyler's formula and its critics, Robertson (1971) and Pease (1977); “Most probably, this observation is a numerical coincidence”.

271. P. Jerabek, B. Schuetrumpf, P. Schwerdtfeger and W. Nazarewicz, Phys. Rev. Lett. 120, 053001 (2018). Oganesson's electron density loses its shell structure.

272. X. Ji, J. Tang and P. Hoodbhoy, Phys. Rev. Lett. 76, 740 (1996), arXiv:hep-ph/9510304. Leading-order evolution of the spin pieces.

273. Ji et al., *Experimentally probing the quantum physics in the inverted harmonic oscillator*, arXiv:2606.05125 (2026, preprint). A condensate's trap flipped into an inverted oscillator; squeezing 10.6(1.3) dB below vacuum.

274. W. Jones, *Synopsis Palmariorum Matheseos* (1706). The symbol π.

275. J-PET Collaboration, arXiv:2511.20098 (2025). A CPT asymmetry in positronium decay of −0.00029 ± 0.00022.

276. J-PET Collaboration, arXiv:2607.19495 (2026). The three-photon Dalitz plot of ortho-positronium against QED, to about 3%.

277. JUNO Collaboration, arXiv:2511.14593 (2025; Nature, 2026). $\sin^2\theta_{12} = 0.3092 \pm 0.0087$; $\Delta m^2_{21} = (7.50 \pm 0.12) \times 10^{-5}$ eV².

278. Th. Kaluza, Sitzungsber. Preuss. Akad. Wiss. 966 (1921). A fifth dimension for electromagnetism.

279. KamLAND-Zen Collaboration, arXiv:2406.11438 (2024). $T_{1/2} > 3.8 \times 10^{26}$ yr; $m_{\beta\beta}$ below 28 to 122 meV.

280. P. L. Kapitsa, Dokl. Akad. Nauk SSSR 101, 245 (1955). Ball lightning fed by microwaves from the storm.

281. P. L. Kapitza, Zh. Eksp. Teor. Fiz. 21, 588 (1951). A pendulum held upside down by shaking its pivot, stable when $a^2\omega^2 > 2gl$.

282. P. Kargupta et al., *Cognitive foundations for reasoning and their manifestation in LLMs*, arXiv:2511.16660 (2025). Models deploy learned behavioural strategies rather than adapting to the problem.

283. R. E. Kass and A. E. Raftery, *Bayes factors*, J. Am. Stat. Assoc. 90, 773 (1995). The scale of evidence; above 100, decisive.

284. KATRIN Collaboration, Science 388, 180 (2025), arXiv:2406.13516. $m_\nu < 0.45$ eV at 90%, from tritium β decay.

285. E. C. Kemble, Phys. Rev. 48, 549 (1935). Transmission through a parabolic barrier.

286. J. Kepler, *Harmonices Mundi* (1619). The stella octangula: two tetrahedra in a cube.

287. R. P. Kerr, Phys. Rev. Lett. 11, 237 (1963). The rotating black hole.

288. J. Khoury, B. A. Ovrut, P. J. Steinhardt and N. Turok, Phys. Rev. D 64, 123522 (2001). A slow, stiff contraction before the hot start, leaving $r \approx 0$.

289. Muḥammad ibn Mūsā al-Khwārizmī, *al-Kitāb al-mukhtaṣar fī ḥisāb al-jabr wa'l-muqābala* (c. 820). Restoring and balancing: doing the same to both sides.

290. H. J. Kimble, Y. Levin, A. B. Matsko, K. S. Thorne and S. P. Vyatchanin, Phys. Rev. D 65, 022002 (2001). Frequency-dependent squeezing for interferometers.

291. O. Klein, Z. Phys. 37, 895 (1926). The fifth dimension curled into a circle.

292. G. Klein, *Performing a project premortem*, Harvard Business Review 85(9), 18 (2007). The pre-mortem: imagine that the project has failed, then list why.

293. Y. Koide, Lett. Nuovo Cimento 34, 201 (1982); Phys. Lett. B 120, 161 (1983). Q = ⅔ for the charged leptons.

294. F. Kos, D. Poland, D. Simmons-Duffin and A. Vichi, JHEP 08 (2016) 036. The three-dimensional Ising and O(N) exponents.

295. E. F. Krause, *Taxicab Geometry* (Addison-Wesley, 1975). The taxicab metric's geometry.

296. M. D. Kruskal, Phys. Rev. 119, 1743 (1960). The maximal extension of the Schwarzschild geometry.

297. P. Kustaanheimo and E. Stiefel, J. reine angew. Math. 218, 204 (1965). The Coulomb problem as the four-dimensional oscillator.

298. J.-L. Lagrange (1770). Every number is a sum of four squares.

299. C. S. Lam, Phys. Rev. Lett. 101, 121602 (2008), arXiv:0804.2622. The symmetry behind tribimaximal mixing.

300. W. E. Lamb, *Anti-photon*, Appl. Phys. B 60, 77 (1995).

301. S. K. Lamoreaux, Phys. Rev. Lett. 78, 5 (1997). The Casimir force measured to 5%.

302. L. D. Landau, Zh. Eksp. Teor. Fiz. 7, 19 (1937). The theory of phase transitions: mean-field exponents.

303. L. D. Landau and E. M. Lifshitz, *Electrodynamics of Continuous Media* (Pergamon, 1960), §59. The eddy-current response of a conducting sphere.

304. J. Larmor (1897), with O. Heaviside: the electric–magnetic duality of the source-free field equations.

305. B. W. Lee, C. Quigg and H. B. Thacker, Phys. Rev. D 16, 1519 (1977). The unitarity bound on the Higgs mass.

306. J. G. Lee, E. G. Adelberger, T. S. Cook, S. M. Fleischer and B. R. Heckel, *New test of the gravitational $1/r^2$ law at separations down to 52 μm*, Phys. Rev. Lett. 124, 101101 (2020), arXiv:2002.11761.

307. LEGEND Collaboration, arXiv:2509.21166 (2025). First LEGEND-200 results; with GERDA and MAJORANA, $T_{½} > 1.9 \times 10^{26}$ yr.

308. A.-M. Legendre, *Essai sur la théorie des nombres* (1798). Sums of three squares.

309. G. W. Leibniz, *Explication de l'arithmétique binaire*, Mémoires de l'Académie Royale des Sciences (1703). Binary arithmetic and the eight trigrams.

310. F. Lelli, S. McGaugh and J. Schombert, Astron. J. 152, 157 (2016), arXiv:1606.09251. SPARC: rotation curves of 175 galaxies.

311. F. Lelli et al., Mon. Not. R. Astron. Soc. 484, 3267 (2019). The baryonic Tully–Fisher slope from SPARC, 3.85 ± 0.09.

312. G. Lemaître, Ann. Soc. Sci. Bruxelles A 47, 49 (1927). The expanding universe and its rate.

313. J. Lense and H. Thirring, Phys. Z. 19, 156 (1918). Frame dragging by a rotating mass.

314. F. Lenz, Phys. Rev. 82, 554 (1951). $m_?/m_e \approx 6\pi^5$.

315. Leonardo of Pisa (Fibonacci), *Liber Abaci* (1202) and *Liber Quadratorum* (1225). The sequence; three squares in equal steps; the product of sums of two squares.

316. T. Levi-Civita, Acta Math. 42, 99 (1920). The planar Kepler problem as a squared oscillator, $w = z^2$.

317. G. C. Lichtenberg, Novi Comment. Soc. Reg. Sci. Gotting. 8 (1777). Branching discharge figures.

318. LIGO Scientific and Virgo Collaborations (B. P. Abbott et al.), Phys. Rev. Lett. 116, 061102 (2016). GW150914, the first observed gravitational wave.

319. LIGO Scientific and Virgo Collaborations, with Fermi-GBM and INTEGRAL (B. P. Abbott et al.), Astrophys. J. Lett. 848, L13 (2017). GW170817 with GRB 170817A: gravity travels at the speed of light.

320. LIGO, Virgo and KAGRA Collaborations, *GW250114*, Phys. Rev. Lett. 135, 111403 (2025), arXiv:2509.08054. The horizon-area law confirmed at 4.4σ.

321. LIGO, Virgo and KAGRA Collaborations, *Tests of general relativity with GWTC-3*, Phys. Rev. D 112, 084080 (2025), arXiv:2112.06861. No post-merger echoes.

322. J. Liouville, J. Math. Pures Appl. 20, 137 (1855); V. I. Arnold, *Mathematical Methods of Classical Mechanics* (Springer, 1978). Action-angle variables for integrable motion.

323. J. A. Lipa, J. A. Nissen, D. A. Stricker, D. R. Swanson and T. C. P. Chui, *Specific heat of liquid helium in zero gravity very near the lambda point*, Phys. Rev. B 68, 174518 (2003), arXiv:cond-mat/0310163. $\alpha = -0.0127 \pm 0.0003$, so $\nu = 0.6709(1)$.

324. X. Liu and B.-Q. Ma, Phys. Rev. D 92, 096003 (2015), arXiv:1510.07783, citing HERMES, Phys. Rev. D 75, 012007 (2007). $\Delta\Sigma = 0.330 \pm 0.039$.

325. R. Liu et al., *Mind your step (by step): chain-of-thought can reduce performance on tasks where thinking makes humans worse*, ICML 2025, arXiv:2410.21333.

326. S. Lloyd, Nature 406, 1047 (2000). The ultimate physical limits to computation.

327. S. Lloyd, *Computational capacity of the universe*, Phys. Rev. Lett. 88, 237901 (2002).

328. K. Lodders, Space Sci. Rev. 217, 44 (2021). Solar-system abundances; the Oddo–Harkins even–odd alternation.

329. A. M. Lopez, R. G. Clowes and G. M. Williger, Mon. Not. R. Astron. Soc. 516, 1557 (2022), the Giant Arc; and the Big Ring, presented at AAS 243 (2024), slides. Structures larger than the expected limit of about 1.2 billion light-years.

330. R.-S. Lu et al., *A ring-like accretion structure in M87 connecting its black hole and jet*, Nature 616, 686 (2023).

331. É. Lucas, Am. J. Math. 1, 184 and 289 (1878). The Fibonacci and Lucas identities.

332. Lumen Learning, *Waymaker Psychology*, “multi-modal perception”; IBM, “What is multimodal AI?”. The two senses of “multimodal” the author combines.

333. M. Luo, H. Wang and Y. Xiao, Phys. Rev. D 67, 065019 (2003), arXiv:hep-ph/0207271. Two-loop running of the Standard Model couplings.

334. LZ Collaboration, Phys. Rev. Lett. 135, 011802 (2025), arXiv:2410.17036. σ_SI < 2.2 × 10^{-48} cm² at 40 GeV from 4.2 tonne-years.

335. Y. G. Ma, S. Sahebdivan, C. K. Ong, T. Tyc and U. Leonhardt, *Perfect imaging without negative refraction for microwaves*, New J. Phys. 13, 033016 (2011), arXiv:1007.2530. Maxwell's fish eye resolves outlets 0.2λ apart.

336. Y. Ma, K. E. Khosla, B. A. Stickler and M. S. Kim, *Quantum persistent tennis racket dynamics of nanorotors*, Phys. Rev. Lett. 125, 053604 (2020), arXiv:2007.10065.

337. L. Maisenbacher et al. (2026), hydrogen 2S–6P: r_p = 0.8406 ± 0.0015 fm, as collected by Miller (2026), Table 1.

338. J. Maldacena, S. H. Shenker and D. Stanford, *A bound on chaos*, JHEP 08 (2016) 106, arXiv:1503.01409. λ_L ≤ 2πkT/ħ.

339. D. F. Mansfield and N. J. Wildberger, *Plimpton 322 is Babylonian exact sexagesimal trigonometry*, Historia Mathematica 44, 395 (2017). A trigonometry of exact ratios that “does not need the familiar modern concept of angles”.

340. P. Mardešić, L. Van Damme, G. J. Gutierrez Guillen and D. Sugny, *Geometric origin of the tennis racket effect*, Phys. Rev. Lett. 125, 064301 (2020), arXiv:2003.13539. The twist from a pole of a Riemann surface.

341. N. Margolus and L. B. Levitin, Physica D 120, 188 (1998). The maximum speed of dynamical evolution.

342. B. Märkisch et al. (PERKEO III), Phys. Rev. Lett. 122, 242501 (2019), arXiv:1812.04666. g_A = 1.27641(56).

343. P. Marquard, A. V. Smirnov, V. A. Smirnov and M. Steinhauser, *Quark mass relations to four-loop order in perturbative QCD*, Phys. Rev. Lett. 114, 142002 (2015), arXiv:1502.01030. The pole–MS-bar relation at four loops.

344. J. C. Maxwell, *On physical lines of force*, Phil. Mag. (4) 21, 161 (1861). The vortex medium.

345. E. McDonough and E. G. M. Ferreira, arXiv:2512.05108 (2026). The spectral tilt across microwave-background data sets, with and without DESI.

346. S. S. McGaugh, Astron. J. 143, 40 (2012). The baryonic Tully–Fisher relation of gas-rich galaxies.

347. S. McGaugh, F. Lelli and J. Schombert, Phys. Rev. Lett. 117, 201101 (2016), arXiv:1609.05917. The radial acceleration relation: observed and baryonic pulls match at every radius in 153 galaxies, with $g† = 1.20 \times 10^{-10}$ m/s².

348. A. L. McWhorter, in *Semiconductor Surface Physics*, ed. R. H. Kingston (Univ. Pennsylvania Press, 1957), p. 207. 1/f noise from a spread of trapping times.

349. D. Mendeleev (Mendelejeff), Z. Chem. 12, 405 (1869). The periodic system.

350. J.-M. Mewes, O. R. Smits, P. Jerabek and P. Schwerdtfeger, Angew. Chem. Int. Ed. 58, 14260 (2019). Oganesson is a semiconductor.

351. A. Miani et al., Phys. Rev. D 108, 064018 (2023), arXiv:2302.12158. Limits on gravitational-wave echo amplitudes, 0.13 to 0.23.

352. A. A. Michelson and E. W. Morley, Am. J. Sci. 34, 333 (1887). No ether wind.

353. M. Milgrom, Astrophys. J. 270, 365 (1983). Modified Newtonian dynamics.

354. M. Milgrom, arXiv:2001.09729 (2020). $2\pi a_0 \approx cH_0$.

355. E. A. Milne (1935) and H. Bondi, *Relativity and Common Sense* (1964). The factor-k argument for the invariant interval.

356. H. Minkowski, *Raum und Zeit* (1908; Phys. Z. 10, 104, 1909), the invariant interval; *Geometrie der Zahlen* (1910), the taxicab metric.

357. C. W. Misner and J. A. Wheeler, Ann. Phys. 2, 525 (1957). Charge as field lines threading a wormhole.

358. A. F. Möbius, Ber. Sächs. Ges. Wiss. 17, 31 (1865). The one-sided strip.

359. U. Mohideen and A. Roy, Phys. Rev. Lett. 81, 4549 (1998). The Casimir force measured to 1%.

360. P. J. Mohr, D. B. Newell, B. N. Taylor and E. Tiesinga, *CODATA recommended values of the fundamental physical constants: 2022*, arXiv:2409.03787 (2024). $m_e c^2$, μ⍰, μ⍰, m⍰/m_e, r⍰ and the exact SI constants.

361. R. Montgomery, *How much does the rigid body rotate?*, Am. J. Phys. 59, 394 (1991). The phase gained by a tumbling body.

362. L. Morel, Z. Yao, P. Cladé and S. Guellati-Khélifa, Nature 588, 61 (2020). α from rubidium recoil.

363. P. Morrison, Am. J. Phys. 26, 358 (1958). Antimatter and gravity: the perpetual-motion argument.

364. P. Moskal et al. (J-PET), Nature Commun. 12, 5658 (2021). No CPT violation in positronium at a precision of 10^{-4}.

365. P. Mróz et al., Nature 632, 749 (2024). Twenty years of OGLE microlensing: compact objects from 10^{-4} to 10^{3} $M_\odot$ capped near a percent of the dark matter.

366. P. Mukherjee and A. A. Sen, *New expansion rate anomalies at characteristic redshifts geometrically determined using DESI-DR2 BAO and DES-SN5YR observations*, arXiv:2505.19083 (2025). Deviations of about 4 to 5σ from Planck ΛCDM at z ≈ 0.35 to 0.55.

367. J. Müller et al., arXiv:0903.3197 (2009). The Dutta–Dimon–Horn relations, restated.

368. J. R. Muñoz de Nova, K. Golubkov, V. I. Kolobov and J. Steinhauer, Nature 569, 688 (2019). Thermal Hawking radiation at the temperature set by the surface gravity.

369. Muon g−2 Collaboration, Phys. Rev. Lett. 135, 101802 (2025), arXiv:2506.03069. The final a_μ, 116 592 070.5 ± 14.8 × 10^{-11} (127 ppb).

370. K. Myhre (Oak Ridge National Laboratory), *Fabrication of californium-249 targets to enable the search for element 120 and other superheavy nuclides*, talk, RIKEN, 30 August to 4 September 2026, indico. No atom of a new element yet; the ^{249}Cf target for Berkeley's element-120 search.

371. I. Newton, *Philosophiæ Naturalis Principia Mathematica* (1687). The laws of motion.

372. L. Niemeyer, L. Pietronero and H. J. Wiesmann, *Fractal dimension of dielectric breakdown*, Phys. Rev. Lett. 52, 1033 (1984). Discharge growth with probability ∝ E^η.

373. B. A. Nosek, C. R. Ebersole, A. C. DeHaven and D. T. Mellor, *The preregistration revolution*, Proc. Natl. Acad. Sci. 115, 2600 (2018), doi:10.1073/pnas.1708274114.

374. H. C. Ohanian, *What is spin?*, Am. J. Phys. 54, 500 (1986). Spin and moment as flows of energy and charge in the wave field.

375. L. Onsager, Phys. Rev. 65, 117 (1944). The two-dimensional Ising model solved.

376. OpenStax, *College Physics*, “Friction”, Table 5.1, chapter. Static and kinetic friction coefficients.

377. J. R. Oppenheimer and H. Snyder, Phys. Rev. 56, 455 (1939). Continued gravitational collapse.

378. Pacific Northwest National Laboratory, *Characterizing photometric flicker*, PNNL-25135 (2016), report. Lamps flicker at twice the line frequency.

379. G. S. Paiva et al., Phys. Rev. Lett. 98, 048501 (2007). Luminous balls made by discharges into silicon.

380. I. Pantos and L. Perivolaropoulos, *Status of the S_8 tension: a 2026 review of probe discrepancies*, arXiv:2602.12238 (2026). The combined CMB S_8 = 0.836 +0.012/−0.013, as derived by DES.

381. K. Pardo, M. Fishbach, D. E. Holz and D. N. Spergel, *Limits on the number of spacetime dimensions from GW170817*, JCAP 07 (2018) 048, arXiv:1801.08160.

382. R. H. Park, Trans. AIEE 48, 716 (1929). The two-reaction transformation.

383. R. H. Parker, C. Yu, W. Zhong, B. Estey and H. Müller, Science 360, 191 (2018). α from caesium recoil.

384. C. G. Parthey et al., Phys. Rev. Lett. 107, 203001 (2011), arXiv:1107.3101. Hydrogen's 1S–2S interval, 2,466,061,413,187,035(10) Hz.

385. Particle Data Group, electron and muon listings (2020). Particle–antiparticle comparisons, among them R. S. Van Dyck, P. B. Schwinberg and H. G. Dehmelt, Phys. Rev. Lett. 59, 26 (1987): the e± g-factors equal within $(-0.5 \pm 2.1) \times 10^{-12}$.

386. Particle Data Group (S. Navas et al.), *Review of Particle Physics*, Phys. Rev. D 110, 030001 (2024). The electroweak review's fit values; the Higgs review's coupling modifiers (Table 11.3).

387. Particle Data Group (Takahashi et al.), *Review of Particle Physics*, Int. J. Mod. Phys. A 41, 2630011 (2026). Masses, couplings, the CKM fit, and the τ mass 1776.93 ± 0.09 MeV.

388. R. K. Pathria, Nature 240, 298 (1972). The universe as a black hole.

389. J. C. Pati and A. Salam, Phys. Rev. D 10, 275 (1974). Lepton number as the fourth colour.

390. H. M. Paynter, the tetrahedron of state of bond graphs (1959–61), as described in the bond-graph literature (Bond graph).

391. B. Pendleton and G. G. Ross, Phys. Lett. B 98, 291 (1981). The top Yukawa coupling's infrared fixed point.

392. R. Penrose, Proc. Camb. Phil. Soc. 55, 137 (1959), the apparent shape of a moving sphere; J. Math. Phys. 8, 345 (1967), twistors; and the Weyl curvature hypothesis, in *General Relativity: An Einstein Centenary Survey*, ed. S. W. Hawking and W. Israel (Cambridge, 1979).

393. S. Perlmutter et al., Astrophys. J. 517, 565 (1999), arXiv:astro-ph/9812133. $\Lambda > 0$ at 99%.

394. PHENIX Collaboration (N. J. Abdulameer et al.), *Measurement of direct-photon cross section and double-helicity asymmetry at $\sqrt{s}$ = 510 GeV in polarized p+p collisions*, Phys. Rev. Lett. 130, 251901 (2023). Gluon sensitivity at $0.02 < x < 0.08$; a negative gluon spin disfavoured by more than 2.8σ.

395. Planck Collaboration, *Planck 2013 results. XXVII. Doppler boosting of the CMB*, Astron. Astrophys. 571, A27 (2014). Our motion's aberration of the microwave sky.

396. Planck Collaboration, *Planck intermediate results. XIII. Constraints on peculiar velocities*, Astron. Astrophys. 561, A97 (2014). Cluster flows capped below 254 km/s at 95%.

397. Planck Collaboration, Astron. Astrophys. 641, A6 (2020), arXiv:1807.06209. Ω□, H_0 and the dark-energy density.

398. J. R. Platt, *Strong inference*, Science 146, 347 (1964).

399. Plimpton 322, Columbia University (c. 1800 BCE). A table of right-triangle ratios; read by Mansfield and Wildberger (2017).

400. L. Poinsot, *Théorie nouvelle de la rotation des corps* (1834). The geometry of free rotation.

401. E. Poisson and W. Israel, Phys. Rev. D 41, 1796 (1990). Mass inflation at a rotating hole's inner horizon.

402. B. van der Pol and J. van der Mark, Phil. Mag. (7) 6, 763 (1928). The heartbeat as a relaxation oscillation.

403. N. J. Popławski, Phys. Lett. B 687, 110 (2010), arXiv:0902.1994. The isotropic Schwarzschild metric is unchanged by an inversion whose fixed sphere is the horizon.

404. N. J. Popławski, Phys. Lett. B 694, 181 (2010), arXiv:1007.0587. Cosmology with torsion: a bounce in place of inflation.

405. B. Popovic et al. (DES), arXiv:2511.07517 (2025). Recalibrated DES supernovae: $\Omega_m = 0.330 \pm 0.015$; evolving w at 3.2σ; constant w = −0.995 ± 0.019 with $q_0 = -0.537 \pm 0.026$; $\Omega_m = 0.3045 \pm 0.0032$ with the CMB and BAO.

406. K. R. Popper, *The Logic of Scientific Discovery* (Hutchinson, 1959; *Logik der Forschung*, 1934). Falsification.

407. R. V. Pound and G. A. Rebka, Phys. Rev. Lett. 4, 337 (1960). The gravitational redshift over 22.5 m, ratio 1.05 ± 0.10.

408. P. Pyykkö, *Relativistic effects in structural chemistry*, Chem. Rev. 88, 563 (1988).

409. P. Pyykkö, Phys. Chem. Chem. Phys. 13, 161 (2011). A periodic table to Z = 172 from Dirac–Fock calculations.

410. A. F. Rañada, Lett. Math. Phys. 18, 97 (1989). Light whose field lines are linked Hopf circles.

411. H. Rauch, A. Zeilinger, G. Badurek, A. Wilfing, W. Bauspiess and U. Bonse, Phys. Lett. A 54, 425 (1975). The neutron's 4π spinor rotation verified.

412. Rhind Mathematical Papyrus, problem 51 (c. 1550 BCE). The area of a triangle.

413. A. G. Riess et al., Astron. J. 116, 1009 (1998), arXiv:astro-ph/9805201. $q_0 < 0$ at 2.8σ and 3.9σ.

414. A. G. Riess et al. (SH0ES), Astrophys. J. Lett. 934, L7 (2022), arXiv:2112.04510. $H_0 = 73.04 \pm 1.04$ km/s/Mpc.

415. W. Rodejohann and H. Zhang, arXiv:1101.5525 (2011). Koide-type neutrino masses.

416. C. L. Rodriguez, arXiv:2302.12386 (2023). Old stellar-mass black holes in NGC 3201 against cosmological coupling.

417. C. Röken, *The construction and application of Penrose diagrams, with a focus on the maximally analytically extended Schwarzschild spacetime*, arXiv:2507.23514 (2025). The four regions of the extended Schwarzschild geometry related by discrete reversal isometries.

418. R. W. Romani et al., Astrophys. J. Lett. 934, L17 (2022). PSR J0952−0607, the heaviest neutron star yet weighed.

419. T. S. Roussy et al., Science 381, 46 (2023), paper. The electron's EDM, $(-1.3 \pm 2.0 \pm 0.6) \times 10^{-30}$ e·cm; below 4.1×10^{-30} at 90%.

420. C. Rovelli and F. Vidotto, Int. J. Mod. Phys. D 23, 1442026 (2014). Planck stars.

421. C. Rovelli and F. Vidotto, *White-hole dark matter and the origin of past low-entropy*, arXiv:1804.04147 (2018). White-hole remnants as the dark matter.

422. V. C. Rubin and W. K. Ford, Astrophys. J. 159, 379 (1970). The flat rotation curve of Andromeda.

423. D. Rubin et al. (Union3), Astrophys. J. 986, 231 (2025), arXiv:2311.12098. Supernova-only Ω_Λ = 0.356 +0.028/−0.026.

424. J. R. Rydberg (1888–90). The hydrogen line formula.

425. D. Saadeh, S. M. Feeney, A. Pontzen, H. V. Peiris and J. D. McEwen, Phys. Rev. Lett. 117, 131302 (2016), arXiv:1605.07178. No rotation of the universe, below 4.7×10^{-11} of the expansion rate.

426. G.-L. Le Sage (1748; *Lucrèce Newtonien*, 1784). Gravity as a push of streaming corpuscles.

427. A. D. Sakharov, Dokl. Akad. Nauk SSSR 177, 70 (1967). Gravity as the vacuum's elasticity.

428. M. Sassoli de Bianchi, Found. Sci. 22, 627 (2017), arXiv:1601.07053. The 4π periodicity seen by Rauch et al. and Werner et al. (1975).

429. T. K. Sato et al., Nature 520, 209 (2015). Lawrencium's first ionisation potential, 4.96 +0.08/−0.07 eV.

430. C. Saunders, *Charge separation mechanisms in clouds*, Space Sci. Rev. 137, 335 (2008).

431. F. Sauter, Z. Phys. 69, 742 (1931). Pairs from the vacuum in a strong field.

432. L. I. Schiff, Phys. Rev. Lett. 4, 215 (1960). The geodetic and frame-dragging precessions of a gyroscope.

433. L. Schläfli, *Theorie der vielfachen Kontinuität* (written 1852, published 1901). Regular figures in any dimension.

434. A. Schmitt et al., *Mesoscopic Klein–Schwinger effect in graphene*, Nature Phys. (2023), doi:10.1038/s41567-023-01978-9, arXiv:2207.13400. Pair creation at the Schwinger rate; G_0 = 0.18 ± 0.02 mS against 0.186 predicted.

435. G. Schneider et al., Science 358, 1081 (2017). The proton's moment, 2.79284734462(82) μ_N.

436. B. F. J. Schonland and H. Collens, Proc. R. Soc. A 143, 654 (1934), and later papers. The stepped leader.

437. J. Schwinger, Phys. Rev. 73, 416 (1948). The electron's anomaly, α/2π.

438. J. Schwinger, Phys. Rev. 82, 664 (1951). Pairs from the vacuum in a constant field; the critical field 1.32×10^{18} V/m.

439. D. W. Sciama, Mon. Not. R. Astron. Soc. 113, 34 (1953). Inertia by induction from distant matter.

440. C. F. Scott (Westinghouse), the T connection between two- and three-phase systems (1890s).

441. C. T. Sebens, *How electrons spin*, Stud. Hist. Philos. Mod. Phys. 68, 40 (2019), arXiv:1806.01121. In the Dirac field the charge rotates twice as fast as the mass, which gives g = 2.

442. N. J. Secrest et al., Astrophys. J. Lett. 908, L51 (2021), arXiv:2009.14826. The quasar dipole, twice our motion's, 4.9σ.

443. Shang et al., arXiv:1508.03661 (2015). A relational graph of circuit elements.

444. S. S. Shapiro et al., Phys. Rev. Lett. 92, 121101 (2004). Light bending by radio interferometry, γ = 0.99983 ± 0.00045.

445. M. Shaposhnikov and C. Wetterich, Phys. Lett. B 683, 196 (2010). Asymptotic safety puts the Higgs at the stability edge.

446. Sharma and Katoch, arXiv:2608.19691 (2026). A rigid deformation of tribimaximal mixing giving $\sin^2\theta_{12} = 0.301$.

447. K. Shu et al., *Cooling positronium to ultralow velocities with a chirped laser pulse train*, Nature 633, 793 (2024). Positronium cooled from 610 K to 1 K.

448. M. A. Shupe, Phys. Lett. B 86, 87 (1979). Quarks and leptons from two preons.

449. W. de Sitter, Mon. Not. R. Astron. Soc. 77, 155 (1916). The geodetic precession of a gyroscope in orbit.

450. O. R. Smits, P. Mewes, P. Jerabek and P. Schwerdtfeger, Angew. Chem. Int. Ed. 59, 23636 (2020). Oganesson predicted to be a solid at room temperature.

451. O. R. Smits, C. E. Düllmann, P. Indelicato, W. Nazarewicz and P. Schwerdtfeger, Nat. Rev. Phys. 6, 86 (2024). The limits of the periodic table.

452. L. Smolin, *Did the universe evolve?*, Class. Quantum Grav. 9, 173 (1992). Cosmological natural selection: universes born in black holes, no neutron star much above 2 $M_\odot$.

453. C. Smorra et al. (BASE), Nature 550, 371 (2017). The antiproton's moment, −2.7928473441(42) μ_N.

454. J. G. von Soldner, Berliner Astronomisches Jahrbuch 1804, 161 (written 1801). Light bent by 0.84″ at the Sun on a Newtonian reckoning.

455. STAR Collaboration, Phys. Rev. D 100, 052005 (2019), arXiv:1906.02740. Jet and dijet double-spin asymmetries at 510 GeV, reaching the gluon down to $x \approx 0.015$.

456. STAR Collaboration, Phys. Rev. Lett. 127, 052302 (2021). The Breit–Wheeler process, with its polarisation-dependent modulation.

457. J. Steinhauer, Nature Phys. 12, 959 (2016). Analogue Hawking radiation in a condensate.

458. C. P. Steinmetz, *Complex quantities and their use in electrical engineering*, Proc. International Electrical Congress, Chicago (1893). Phasors: a time derivative multiplies by iω.

459. Stewart, Dinh Thi et al., Nature (5 August 2026). Vacuum birefringence.

460. Super-Kamiokande Collaboration, *Solar neutrino measurements using the full data period of Super-Kamiokande-IV*, Phys. Rev. D 109, 092001 (2024), arXiv:2312.12907. Solar $\Delta m^2_{21} = 6.10\ ^{+0.95}_{-0.81} \times 10^{-5}$ eV².

461. J. J. Sylvester, Phil. Mag. 34, 461 (1867). Matrices of ±1 with orthogonal rows, the Hadamard matrices.

462. T2K and NOvA Collaborations, *Joint neutrino oscillation analysis from the T2K and NOvA experiments*, Nature 646, 818 (2025), doi:10.1038/s41586-025-09599-3. The CP phase near $-85°$; $\sin^2\theta_{23} = 0.56\ {}^{+0.03}_{-0.05}$.

463. R. Taverna et al., Science 378, 646 (2022). X-ray polarisation from a magnetar, read as vacuum birefringence.

464. T. Taylor, *Small Pattern, Big Deal*, unpublished manuscript (November 2025). Tifinchi Trig, the difference-of-squares tables and root families; the author's.

465. T. Taylor, *Tifinchi's Diagnostic and Reasoning Framework*, unpublished specification of 7,817 lines (2026a; Part II dated 1 January 2026). The Quad-Core claim test and the framework around it; the author's.

466. T. Taylor, *Rosetta Octants*, unpublished notebook of 17 pages (2026b; 24 February 2026). The notebook of axes and derivatives; the author's.

467. T. Taylor, *Identifying Multimodal Reasoning in LLMs and Humans*, research proposal, Boise State University (2026c; 15 March 2026). Multimodal reasoning as a style to diagnose; the author's.

468. T. Taylor, *Atypical Reasoning Relationships: Multimodal Thinking*, research article, Boise State University (2026d; 12 April 2026). Structure before sequence; the author's.

469. T. Taylor, *Root Patterns*, unpublished manuscript (2026e; September 2026). Root families and their tables; the author's.

470. T. Taylor, *Tifinchi Rosetta Trig Proofs*, unpublished (2026f; 1 October 2026). The Rosetta fold and its proofs; the author's.

471. T. Taylor, *Fib Fam Fun*, unpublished (2026g; 2 October 2026). The Fibonacci options read off the two tables; the author's.

472. T. Taylor, *Revisions 2.0*, with *Lateral Structures* and *Stuff*, unpublished (2 October 2026). One sign rule for every domain: the classifier of regimes; the author's.

473. T. Taylor, *ascii geometry* ("Spatial Comprehension"), unpublished notes (undated). The delimiter cipher and the cube it writes; the author's.

474. T. Taylor, *Tifinchi's Cypher Notes*, unpublished notebook of nine pages (undated): the cipher, the pyramid page, the converter page, the Balance Chart (p. 8) and the phase circle (p. 9); the author's.

475. T. Taylor, *Root Elements*, unpublished chart (undated). The elements tagged on the cells of the two tables; the author's.

476. T. Taylor, *Calc Elements*, unpublished chart (undated). Tiers and coordinate forms for the elements to Z = 54; the author's.

477. A. Tchekhovskoy, R. Narayan and J. C. McKinney, Astrophys. J. 711, 50 (2010), arXiv:0911.2228. The jet power of a spinning hole, $P \approx (\kappa/4\pi c)\Phi^2\Omega_H^2$ with $\kappa \approx 0.05$.

478. J. Terrell, Phys. Rev. 116, 1041 (1959). The apparent rotation of fast-moving bodies.

479. W. Thomson (Lord Kelvin), letter to J. Liouville, J. Math. Pures Appl. 12, 256 (1847). Inversion carries harmonic functions to harmonic functions with a weight (the Kelvin transform).

480. J. Thomson (1873). The name radian.

481. J. A. Thornton, K. S. Virts, R. H. Holzworth and T. P. Mitchell, Geophys. Res. Lett. 44, 9102 (2017). Ship exhaust doubles the lightning over the busiest shipping lanes.

482. D. Tornotti et al., Nature Astron. (2025), arXiv:2406.17035. A filament between a close quasar pair, imaged directly.

483. P. Touboul et al. (MICROSCOPE), Phys. Rev. Lett. 129, 121102 (2022), arXiv:2209.15487. The equivalence principle to about 10^{-15}.

484. J. S. Townsend, *Electricity in Gases* (Clarendon, 1915). Breakdown set by the field per molecule and the molecules crossed.

485. M. D. Tran and V. A. Rakov, Sci. Rep. 7, 15761 (2017). High-speed video of upward leaders that never connected, beside the one that did.

486. R. A. Treumann, Z. Kłos and M. Parrot, Space Sci. Rev. 137, 133 (2008). Physics of electric discharges in the atmosphere.

487. O. L. Trinhammer and H. G. Bohr, *On proton charge radius definition*, EPL 128, 21001 (2019). πr□ = 2λ□.

488. P. M. Trivailo and H. Kojima, ISSFD 2017, paper. The tumbling T-handle seen in orbit by V. Dzhanibekov in 1985, a historical report with no primary paper.

489. R. B. Tully, H. Courtois, Y. Hoffman and D. Pomarède, *The Laniakea supercluster of galaxies*, Nature 513, 71 (2014), arXiv:1409.0880.

490. E. A. Uehling, Phys. Rev. 48, 55 (1935). Vacuum polarisation in the Coulomb field.

491. W. G. Unruh, Phys. Rev. D 14, 870 (1976). An accelerated observer sees a thermal bath at ħa/2πck.

492. *U.S. Standard Atmosphere, 1976* (NOAA, NASA and USAF, 1976), record. 1.052 Pa at 80 km.

493. S. Vajda, *Fibonacci & Lucas Numbers, and the Golden Section* (Ellis Horwood, 1989).

494. V. Venkatraman Krishnan et al., Science 367, 577 (2020). Frame dragging by a white dwarf in PSR J1141−6545.

495. M. Vincenzi et al. (DES), *Comparing the DES-SN5YR and Pantheon+ SN cosmology analyses*, Mon. Not. R. Astron. Soc. (2025), arXiv:2501.06664.

496. S. Vongehr, arXiv:1205.6129 (2012). The q, i, v, φ tetrahedron with six relations.

497. T. Wada and J. Shimoda, *The Dzhanibekov effect as a possible source of magnetar activity*, arXiv:2401.16783 (2024).

498. F. Z. Wang, arXiv:2109.03041 (2021). A six-element star of circuit elements.

499. R. A. Webb, S. Washburn, C. P. Umbach and R. B. Laibowitz, Phys. Rev. Lett. 54, 2696 (1985). h/e Aharonov–Bohm oscillations in gold rings.

500. K. Weierstrass, *Zur Theorie der Potenzreihen* (written 1841). The absolute-value bars.

501. S. Weinberg, Trans. N.Y. Acad. Sci. 38, 185 (1977). Quark mass ratios and the Cabibbo angle.

502. S. Weinberg, *The cosmological constant problem*, Rev. Mod. Phys. 61, 1 (1989). The vacuum-energy estimate, about 10^{120} too large.

503. J. M. Weisberg and Y. Huang, Astrophys. J. 829, 55 (2016), arXiv:1606.02744. The Hulse–Taylor pulsar's orbital decay against general relativity.

504. T. A. Welton, Phys. Rev. 74, 1157 (1948). The Lamb shift as the electron jittered by vacuum fluctuations.

505. S. A. Werner, R. Colella, A. W. Overhauser and C. F. Eagen, Phys. Rev. Lett. 35, 1053 (1975). An early observation of the neutron's 4π spinor period.

506. M. S. Wheatland, T. Murphy, D. Naoumenko, D. van Schijndel and G. Katsifis, *The mobile phone as a free-rotation laboratory*, Am. J. Phys. (2021), doi:10.1119/10.0003380, arXiv:2011.02024. One 360° turn plus a 180° flip, the spin's sign reversed.

507. J. A. Wheeler, *Information, physics, quantum: the search for links*, in Proc. 3rd Int. Symp. Foundations of Quantum Mechanics (Tokyo, 1989). It from bit.

508. F. Wilczek and A. Zee, Phys. Rev. D 25, 553 (1982). The SO(10) spinor's binary structure.

509. F. Wilczek, *The Lightness of Being* (Basic Books, 2008). The vacuum as “the Grid”.

510. N. J. Wildberger, *Divine Proportions: Rational Trigonometry to Universal Geometry* (Wild Egg, 2005). Trigonometry in quadrances and spreads, without square roots or angles.

511. N. J. Wildberger, *Chromogeometry*, arXiv:0806.2789 (2008). The three quadratic forms $a^2 + b^2$, $a^2 - b^2$ and $2ab$.

512. Wilkins, Veitch and Lehman, IEEE ECCE 2010, 171, doi:10.1109/ECCE.2010.5618050. LED flicker at twice the line frequency.

513. J. C. Willett, D. M. Le Vine and V. P. Idone, J. Geophys. Res. 113 (2008), doi:10.1029/2006JD008116. Stereo photographs agree on one three-dimensional channel.

514. E. R. Williams, J. E. Faller and H. A. Hill, Phys. Rev. Lett. 26, 721 (1971). Coulomb's inverse-square law tested to about 3×10^{-16}.

515. A. H. Wright et al. (KiDS-Legacy), arXiv:2503.19441 (2025). $S_8 = 0.815 +0.016/-0.021$.

516. C. S. Wu, E. Ambler, R. W. Hayward, D. D. Hoppes and R. P. Hudson, Phys. Rev. 105, 1413 (1957). Parity violation in β decay.

517. H.-C. Wu, Sci. Rep. 6, 28263 (2016), arXiv:1411.4784. Ball lightning from a microwave bubble, after Kapitsa.

518. A. Wyler, C. R. Acad. Sci. Paris A 269, 743 (1969) and A 272, 186 (1971). α from the volumes of symmetric domains; with Robertson's critique, Phys. Rev. Lett. 27, 1545 (1971), and Pease (1977); and Eddington's conjecture, Proc. Camb. Phil. Soc. 27, 15 (1931).

519. A. Yakushev et al., Front. Chem. 10, 976635 (2022). Flerovium behaves as a volatile metal.

520. A. Yakushev et al., Front. Chem. 12, 1474820 (2024), doi:10.3389/fchem.2024.1474820. Nihonium and moscovium adsorbed on silicon oxide.

521. Y.-B. Yang, R. S. Sufian, A. Alexandru, T. Draper, M. J. Glatzmaier, K.-F. Liu and Y. Zhao (χQCD), *Glue spin and helicity in the proton from lattice QCD*, Phys. Rev. Lett. 118, 102001 (2017), arXiv:1609.05937. S_G = 0.251(47)(16) at $\mu^2 = 10$ GeV², MS-bar.

522. YBC 7289, Yale Babylonian Collection (c. 1800–1600 BCE). $\sqrt{2}$ to about six decimal places.

523. J. W. Yoon et al., Optica 8, 630 (2021). Laser intensity of 1.1×10^{23} W/cm².

524. P. Żenczykowski, arXiv:1210.4125 (2012). The Koide phase indistinguishable from 2/9.

525. C. Zener, Proc. R. Soc. A 145, 523 (1934). Electrical breakdown in solids by tunnelling.

526. A. Zenesini et al., *Observation of false vacuum decay via bubble formation in ferromagnetic superfluids*, Nature Phys. 20, 558 (2024), arXiv:2305.05225.

527. D.-J. Zhao et al., *Total gluon helicity contribution to the proton spin from lattice QCD*, arXiv:2512.24315 (2025; v2 2026). $\Delta G = 0.231(17)(44)$ at $\mu^2 = 10$ GeV².

528. Zhao Shuang, commentary on the *Zhoubi Suanjing* (3rd century CE). The rearrangement proof of the right-angle theorem.

529. K. Zuse, *Rechnender Raum* (Vieweg, 1969). The universe as a computing space.

530. F. Zwicky, Helv. Phys. Acta 6, 110 (1933). The Coma cluster's missing mass.